MINIMAL OBSTRUCTIONS FOR THE SIZE RAMSEY NUMBER OF STAR FORESTS, II: SOURCE STRUCTURE AND THE EXACT COST LADDER

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Abstract. We prove the algebraic and matching implications that assemble fifteen structural statements for minimal certificate obstructions associated with star forests. The statements describe source graphs, their exact initial costs, native rows, deletion processes, and compatible matching prefixes in five even degrees. In degree four, the cost comparison identifies a positive correction term, and the matching constructions exclude pentagons and heptagons from supplied saturated terminals. The assembly identifies the technical inputs whose complete proofs remain to be inserted, and it keeps the sufficient reduction to the star-forest size Ramsey formula conditional on all five source-class exclusions. The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system.

1. Introduction

Background. The size Ramsey problem for star forests asks for the least number of edges that force one of two prescribed forests under every red and blue edge coloring. Decompositions that separate degree bounds from matching bounds provide a way to study this problem. The star-forest setting and its even-capacity decompositions are discussed in [Dav+21]; the matching and coloring background includes [LP86, Bro41]. Domination estimates provide related tools for bounding independent cuts and source order; relevant approaches appear in [BK14, Buj20]. These references supply context here. None replaces an absent technical input in the proofs below.

The series separates three tasks. The main paper uses the emptiness of five obstruction classes to obtain the size Ramsey formula. Companion I develops the certificate framework and the sufficient reduction. The present paper assembles the structural restrictions and exact cost comparisons used by the five degree exclusions.

[cite/blocker] The requested series references [Main] and [I] have no entries in the supplied reference ledger. Their authors, bibliographic metadata, and precise result numbers must be supplied before those references can be cited. No bibliography entries have been invented for them.

The sufficient reduction. Definition 2.2 introduces the certificate costs. Definition 2.3 defines the obstruction classes $\mathcal{M}_d$ and the assertions N_d that these entire classes are empty. The reduction retains proper-subgraph minimality, including edge deletions.

Theorem A (Sufficient five-degree reduction)**.** *Assume $N_4, N_6, N_8, N_{10}, N_{12}$, with the entire classes defined in Definition 2.3. Let s_f, t_f be positive integers, and let*

$$n_1 \geq \cdots \geq n_{s_f} \geq 1, \qquad m_1 \geq \cdots \geq m_{t_f} \geq 1$$

be integer lists. Let

$$F_1 = \dot{\bigcup}_{i=1}^{s_f} K_{1,n_i}, \qquad F_2 = \dot{\bigcup}_{j=1}^{t_f} K_{1,m_j}.$$

Here $K_{1,h}$ is the star with one center, h leaves, and an edge from the center to each leaf. Define $\widehat{R}(F_1, F_2)$ to be the least nonnegative integer number of edges in a finite simple graph whose every

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red and blue edge coloring contains a red copy of F_1 or a blue copy of F_2. Then

$$\widehat{R}(F_1, F_2) = \sum_{j=2}^{s_f+t_f} L_j, \qquad L_j = \max_{\substack{1\le u\le s_f\\ 1\le v\le t_f\\ u+v=j}} (n_u + m_v - 1).$$

The five exclusions suffice. No converse is asserted.

Theorem A is the reduction included in Proposition 8.1(VII). Its reduction input is identified in Section 3.1. None of the five hypotheses is discharged by the numerical source restrictions alone.

Source structure and exact costs. The central distinction is between a numerical cost identity and an edge partition that attains a proposed total. Proposition 8.1 gives simultaneous necessary restrictions in degrees $4, 6, 8, 10, 12$. Proposition 8.2 separately treats induced-minimal failures in degrees six and eight. The remaining propositions retain the original vertices, induced edges, and previously fixed matching edges throughout their constructions.

The sextic results describe every native row and complete process, then identify a physical odd-cycle residual after four compatible matchings. The octic results construct an actual triangle selector and supply its first three matching stages. The degree-ten statements distinguish strict and equality sectors. The degree-twelve statements derive a shortest-cycle reserve before applying the matching-prefix argument.

In degree four, Proposition 8.12 identifies the correction between a proposed debit and the exact second cost. Proposition 8.13 gives the port criterion and the cost of repairing its deficient paths. Propositions 8.14 and 8.15 use that criterion to exclude whole pentagons and heptagons from every supplied saturated terminal. These conclusions concern the stated templates and frames; they do not exclude the entire quartic source class.

Proof architecture and status. Section 2 fixes the definitions shared by the fifteen propositions. Section 3 identifies the eight technical developments required by their assembly. Section 8 states every supplied result and gives its assembly argument, retaining the calculations and the original quantifiers. Section 9 records the remaining separation between structural results, matching prefixes, and degree exclusions.

[prime/blocker] The complete proofs cannot be certified from the supplied material because the technical dependencies identified in Section 3 are absent.

[author/blocker] No author, affiliation, or email was supplied. The author block contains placeholders.

[metadata/blocker] The Mathematics Subject Classification remains to be supplied.

[length/blocker] The requested twenty-page frame limit and fifty-page assembled limit have not been verified. Preserving all supplied statements and proof steps takes precedence over silently shortening the mathematics.

[compile/blocker] Compilation and the required compile-verification gate were not run. No claim of a checked compilation is made.

Acknowledgements. The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system.

Remark 1.1 (Automated assistance). The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts. [ack/blocker] No independent verification statement was supplied.

2. Preliminaries

We use finite simple graphs throughout, with matching terminology as in [LP86], and define the certificate quantities needed for the assembly explicitly.

Notation 2.1 (Graphs and integer domains)**.** For a graph G, write $V(G), E(G), e(G), \nu(G), \alpha(G), \Delta(G)$ for its vertex set, edge set, edge count, matching number, independence number, and maximum degree. Write $d_G(v)$ for vertex degree and $N_G(v)$ for the open neighborhood. For a vertex set A, $N_G(A)$ is the union of its open neighborhoods.

Paths P_j, cycles C_j, and complete graphs K_j are indexed by vertex order. Graph sums mean vertex-disjoint unions. A whole component is an entire connected component. Vertex deletion is induced. Subtracting a matching removes its edges and retains all vertices. Edge-set degrees and matching numbers refer to the corresponding spanning graphs.

Capacities, budgets, least universal totals, cost indices, degrees, orders, cardinalities, matching sizes, component counts, phase lengths, omissions, and finite choice domains are nonnegative integers unless an explicit different integer range is given. Gaps and parameter differences may be signed integers. A displayed cardinality condition always refers to actual vertices of the stated graph.

Definition 2.2 (Certificate costs)**.** For nonnegative integers A_c, B_c, let $c_{(A_c,B_c)}(G)$ be the least nonnegative integer T such that every pair of nonnegative integers x_c, y_c with $x_c + y_c = T$ permits an edge partition

$$E(G) = Q_R \,\dot\cup\, Q_B \,\dot\cup\, P_R \,\dot\cup\, P_B$$

satisfying

$$\Delta(Q_R) \le A_c, \qquad \Delta(Q_B) \le B_c, \qquad \nu(P_R) \le x_c, \qquad \nu(P_B) \le y_c.$$

For every nonnegative integer i, put

$$c_i(G) = \max_{\substack{A_c,B_c\ge 0\\ A_c+B_c=i}} c_{(A_c,B_c)}(G), \qquad c_0 = c_0,$$

where the maximum is over integer capacities. Define

$$\mathrm{ex}(G) = c_0(G) - \nu(G), \qquad \mathrm{ps}(G) = e(G) - \nu(G) - \sum_{i\ge 1} c_i(G),$$

and

$$\mathrm{gap}(G) = \sum_{i\ge 0} c_i(G) - e(G).$$

The sums are finite, and

$$\mathrm{gap}(G) = \mathrm{ex}(G) - \mathrm{ps}(G).$$

If $c_0(G) = 0$, put $\mathrm{tr}(G) = 0$. Otherwise $\mathrm{tr}(G)$ is the greatest nonnegative integer length of a consecutive run of integers $x \in \{0, \dots, c_0(G) - 1\}$ for which $E(G)$ has no partition into two parts with matching numbers at most x and $c_0(G) - 1 - x$. Such an allocation is called a missing pair.

Definition 2.3 (Minimal sources)**.** Put

$$h_-(D) = \max\{4, \lfloor D/2 \rfloor + 1\}.$$

Property T_- is the implication

$$\mathrm{gap}(G) > 0 \quad\Longrightarrow\quad \mathrm{gap}(G) = 1 \ \text{ and } \ \mathrm{tr}(G) + \mathrm{ex}(G) \le h_-(\Delta(G)).$$

For $d \in \{4, 6, 8, 10, 12\}$, let $\mathcal{M}_d$ be the entire class of graphs F with maximum degree exactly d that violate T_-, while every proper subgraph satisfies T_-. Proper subgraphs allow arbitrary vertex deletions, edge deletions, or both. Such a graph is called a source. Let N_d assert $\mathcal{M}_d = \varnothing$.

Setup 2.4 (Source parameters)**.** For a source F, put

$$n = |V(F)|, \quad k = \nu(F), \quad a = \alpha(F), \quad E = \mathrm{ex}(F),$$

and

$$\lambda = 2E - 3 - (k - a), \quad \delta = c_0(F) - c_2(F), \quad \epsilon = c_0(F) - c_1(F),$$
$$q_{\mathrm{def}} = \delta - \epsilon, \qquad p = E - 2 - \lambda, \qquad r = k - \lambda - 1 + \delta,$$
$$s = 2\lambda + 3 - 2\delta.$$

The equality sector is $\delta = 0$, and the strict sector is $\delta > 0$. When $d = 8$, also put

$$q_8 = c_4(F) - r, \qquad \eta = \lambda - \delta.$$

In particular, q_8 and q_{def} are different parameters. We use

$$u = r + s, \qquad m = p + \delta, \qquad \beta = p + \delta - \epsilon$$

where the relevant statement invokes them.

Definition 2.5 (Rows and processes)**.** For a source with the parameters in Setup 2.4, a native row is an induced graph A satisfying

$$|V(A)| = r, \quad \Delta(A) \le 2, \quad \mathrm{ps}(A) = 0, \quad \nu(A) = p + \delta, \quad c_1(A) = p + q_{\mathrm{def}}, \quad c_2(A) = p.$$

A complete source process first deletes any maximal independent set and then deletes arbitrary current-maximum-degree vertices while the maximum degree exceeds two. An actual prefix is any remaining induced graph after the initial independent-set deletion. An entry precedes the entire indicated degree phase. A full degree-j phase consists of all consecutive deletions at current degree j. A complete continuation on an induced graph follows the same deletion rule until maximum degree is at most two. Its degree word records the current degree at each deletion, with exponents denoting phase lengths.

A graph is factor-critical if deleting any one vertex leaves a perfect matching. It is well-covered if every maximal independent set is maximum. An exposed port v in a component C satisfies $\nu(C - v) = \nu(C)$.

Definition 2.6 (Small component types)**.** For a graph B, $\mathrm{cor}(B)$ attaches one distinct private leaf to each vertex of B. The graph B_{ij} consists of disjoint cycles of orders i, j joined by exactly one bridge. The graph S_6, also denoted $L_{\square}$, is a square with distinct private leaves at two adjacent square vertices. An ordinary cycle or path is a whole component, rather than a constituent shore of a barbell.

Setup 2.7 (Quartic frames)**.** In the quartic statements, the standing hypothesis may be broader than membership in $\mathcal{M}_4$. Let F be connected, four-regular, and factor-critical, with

$$\mathrm{gap}(F) \ge 1, \qquad E = \mathrm{ex}(F) \ge 5.$$

Put

$$a = \alpha(F), \qquad \sigma = 2E - 2 - (\nu(F) - a).$$

For a maximal independent set S, put

$$H = F - S, \quad v = |V(H)|, \quad b = \nu(H) - c_1(H), \quad m = E - 1 - \sigma + b.$$

A saturated frame is a supplied pair (S, U), where U is an independent set of exactly m vertices of H-degree three. Put

$$W_{\mathrm{fr}} = S \cup U, \qquad R = F - W_{\mathrm{fr}}.$$

When $F[W_{\mathrm{fr}}]$ is the frame matching specified below, write P_{fr}, Z for its endpoints and isolates, and $L_{\mathrm{fr}} = E(F[W_{\mathrm{fr}}])$.

3. Required technical developments

This section identifies the missing technical inputs in the prescribed chapter order; it does not replace them with unsupported theorem statements.

3.1. **Universal structure.** The universal development must establish the source structure, well-coveredness, exact costs, numerical restrictions, and sufficient reduction used in Proposition 8.1. It must also supply the generic-certificate reduction and criticality results used in Proposition 8.2, and the missing-budget bounds used later.

The complete development is given in Section 4.

3.2. **Degree six.** The sextic development must establish native-row existence, the initial cost ladder, canonical replay, the entire complement catalogue, the exclusion of odd coronas, and the ownership and free-component assignment theorem. The rank-forcing and restoration inputs must apply to the same retained graph and the same fixed matching edges.

The complete development is given in Section 5.

3.3. **Degree eight: rows.** The octic row development must establish native replay, the full degree word, the cubic-entry catalogue, the complete local terminal counts, and the exclusions of the additional cubic-entry types.

The complete development is given in Section 6.

3.4. **Degree eight: matchings.** The octic matching development must derive the permitted retained-set range from the original missing budgets, construct a matching with the prescribed retained owners, and prove the simultaneous rank statement for every contained native row.

The complete development is given in Section 7.

3.5. **Degree ten.** The degree-ten development must keep the strict and equality sectors separate. It must establish the strict terminal selector and its omission range, the equality continuation word and cost ladder, and the generic missing-budget matching input.

The complete development is given in Section 8.

3.6. **Degree twelve.** The degree-twelve development must establish strictness, the native-row and owner structure, the terminal selector identities, and the two exceptional-boundary restrictions used to prove the shortest-cycle reserve.

The complete development is given in Section 9.

3.7. **Quartic frames.** The quartic-frame development must establish the uniform complement identities, the saturated reflection theorem, the restriction on terminal components, and the canonical neutral-selector identities.

The complete development is given in Section 6.

3.8. **Quartic odd-cycle contacts.** The quartic contact development must establish the parent-pair restrictions in every original and reflected saturated frame. Those restrictions are the nonlocal inputs to the pentagon and heptagon exclusions.

The complete development is given in Section 7.

4. Universal structure of minimal obstructions: well-coveredness, cuts and generic matching interfaces

This chapter establishes the degree-uniform matching interfaces and the source structure used in subsequent specializations. We first state the prerequisites, then prove the matching, cut, well-coveredness, and phase results; the final assembly records the sharper residual restrictions with their original conditional scope.

4.1. **Setup and preliminaries.** We assume, established earlier in this series, the certificate framework and the explicitly recorded inputs below.

Remark 4.1 (Editorial status). [addressed-partial: placeholders retained; identifying data remain unsupplied.]

[cite/blocker] The requested references of the form [I, Theorem X] have no supplied theorem numbers or bibliography entry. The assumed inputs below record the available statements; they do not supply fabricated citations to Companion I or to the earlier development.[deferred: to reference_verify]

[addressed-partial: the process conclusions now state their coverage hypotheses; independent residual inputs and the exposure-attainer definition remain missing.][addressed-partial: missing inputs remain flagged; the three requested mathematical repairs require changes to formal statements or proof steps outside the permitted scope of this revision.]

[acknowledged: standalone front matter, statements, proofs, and examples retained.]

[acknowledged: compilation remains subject to the external gate; no compile result is asserted.]

Notation 4.2. All graphs are finite and simple. We write $e(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for the edge count, matching number, independence number, and maximum degree of G. For an edge set, these quantities refer to its spanning graph. Vertex deletion takes an induced subgraph; subtraction of a matching deletes only its edges unless vertex deletion is explicitly indicated.

All capacities, budgets, row indices, orders, cardinalities, component counts, matching sizes, omissions, and finite choice parameters are nonnegative integers. A parameter required to be positive is a positive integer. Explicit differences are integers and are not presumed nonnegative. Paths P_j and cycles C_j have j vertices, with $j \geq 1$ for paths and $j \geq 3$ for cycles. Graph sums denote vertex-disjoint unions.

Definition 4.3 (Certificates and costs)**.** Let $A, B, x, y \in \mathbb{Z}_{\geq 0}$. An $(A, B; x, y)$-certificate for G is a partition

$$E(G) = Q_R \,\dot\cup\, Q_B \,\dot\cup\, P_R \,\dot\cup\, P_B$$

such that

$$\Delta(Q_R) \leq A, \qquad \Delta(Q_B) \leq B, \qquad \nu(P_R) \leq x, \qquad \nu(P_B) \leq y.$$

The cost $c_{(A,B)}(G)$ is the least $T \in \mathbb{Z}_{\geq 0}$ for which every split $x + y = T$ has such a certificate. Set

$$c_i(G) = c_i^*(G) = \max_{\substack{A,B \in \mathbb{Z}_{\geq 0} \\ A+B=i}} c_{(A,B)}(G), \qquad c_0(G) = c_{0,0}(G) = c_{(0,0)}(G).$$

We use

$$\Sigma(G) = \sum_{i \geq 0} c_i(G), \qquad \mathrm{ex}(G) = c_0(G) - \nu(G),$$

$$\mathrm{ps}(G) = e(G) - \nu(G) - \sum_{i \geq 1} c_i(G), \qquad \mathrm{gap}(G) = \Sigma(G) - e(G) = \mathrm{ex}(G) - \mathrm{ps}(G).$$

The sums are finite by Assumed input 4.5.

If $c_0(G) = 0$, set $\mathrm{tr}(G) = 0$. Otherwise, $\mathrm{tr}(G)$ is the longest consecutive run of integers $x \in \{0, \ldots, c_0(G) - 1\}$ for which the pair

$$\big(x, c_0(G) - 1 - x\big)$$

has no $(0, 0; x, c_0(G) - 1 - x)$-certificate. At zero capacities the degree-bounded parts are empty, so this is exactly a missing two-part matching allocation.

The property T_- is the implication

$$\mathrm{gap}(G) > 0 \quad \Longrightarrow \quad \mathrm{gap}(G) = 1 \ \text{ and } \ \mathrm{tr}(G) + \mathrm{ex}(G) \leq \max\left(4, \left\lfloor \frac{\Delta(G)}{2} \right\rfloor + 1\right).$$

A graph is *LP-good* when $\Sigma(G) \leq e(G)$.

Definition 4.4 (Sources, good graphs, and processes)**.** For $d \in \{4,6,8,10,12\}$, let $\mathcal{M}_d$ be the entire class of graphs F with $\Delta(F) = d$ which fail T$_-$ and whose every proper subgraph satisfies T$_-$. Proper subgraphs may delete vertices, edges, or both, including spanning edge deletions. A member of $\mathcal{M}_d$ is a *source*. Write N_d for the assertion $\mathcal{M}_d = \varnothing$.

A graph is factor-critical if deletion of any vertex leaves a perfect matching. It is well-covered if every maximal independent set has cardinality $\alpha(G)$.

For $\theta \in \mathbb{Z}_{\geq 1}$, a graph is θ-good if its maximum degree is at most θ and one matching covers every vertex of degree θ. Let $h_\theta(F)$ be the maximum order of an induced θ-good subgraph of F.

A complete source process first deletes a specified maximal independent set S, then repeatedly deletes an arbitrary vertex of the current maximum degree while that degree is at least three. It stops at maximum degree at most two. A deletion parent is *covered* if one matching covers every vertex of its actual maximum degree.

Setup 4.5 (Certificate framework)**.** We assume the following consequences of the certificate framework of Companion I. Vertex insertion gives, for every $W \subseteq V(G)$,

$$c_{(A,B)}(G) \leq |W| + c_{(A,B)}(G - W), \qquad c_i(G) \leq |W| + c_i(G - W).$$

A zero row cost means that every capacity split at that row has a zero-budget certificate. A θ-good graph has $c_\theta = 0$. If $\Delta(G) < \theta$, then $c_\theta(G) = 0$; if $\Delta(G) = \theta$, then $c_\theta(G) = 0$ exactly when its degree-θ vertices have simultaneous matching coverage. In particular, all rows above the maximum degree vanish. Also

$$c_0(G) \leq \tau(G) = |V(G)| - \alpha(G),$$

where τ is minimum vertex-cover size. If $\mathrm{gap}(G) = 1$ and $c_0(G) \geq 4$, then $\mathrm{tr}(G) \geq 1$. At its least feasible total a cost permits every allocation; at a positive smaller total immediately below it, at least one allocation is missing.

Setup 4.6 (The unchanged obstruction class)**.** The class $\mathcal{L}_{\mathrm{rig}}$ is the exact class defined in Companion I, with every original condition retained. Its degree-d slice equals $\mathcal{M}_d$ and the corresponding degree export class $\mathcal{E}_d$. Each member F, with

$$d = \Delta(F), \quad k = \nu(F), \quad E = \mathrm{ex}(F), \quad a = \alpha(F),$$

is connected, d-regular, factor-critical, and of order $2k+1$. It has gap one and every proper subgraph satisfies T$_-$. The possible degrees are $4,6,8,10,12$, with respective initial excess floors $4,5,6,7,9$.

The earlier source assembly strengthens these floors to $5,5,6,11,10$, retains the same source classes, and supplies its rank and sector restrictions, fixed-rank bounds, and the same five sufficient emptiness inputs. Stronger floors recorded later do not change these classes.

[math/blocker] [GAP: the complete original definition of $\mathcal{L}_{\mathrm{rig}}$, the export definitions of $\mathcal{E}_d$, and all inherited same-class clauses were not supplied. These names retain their earlier meanings; the displayed consequences are not substitute definitions.][out-of-scope: thm-body-math; env=setup; label=setup:class]

Setup 4.7 (Independent-cut inputs)**.** For $F \in \mathcal{L}_{\mathrm{rig}}$ and a nonempty independent set I, put

$$j = |I|, \qquad J = F - I, \qquad b = j + \nu(J) - k.$$

In degrees $6,8,10,12$, the earlier all-cut theorem gives

$$\mathrm{gap}(J) = 0, \qquad \mathrm{ex}(J) = \mathrm{ps}(J) = E - b, \qquad c_\theta(J) = c_\theta(F) - j \quad (0 \leq \theta < d),$$

$$1 \leq b \leq \min\left(E, \left\lceil \frac{j}{2} \right\rceil\right), \qquad c_d(F) = 1, \qquad c_d(J) = 0,$$

and vanishing higher rows. It imposes no bound on j, E, or the connectedness of J.

The original quartic cut theorem allows excess-one positive-gap complements only at cut sizes a or $a-1$, and gives zero gap whenever $j \leq a-2$ or $j \leq 2E-2$. Its strengthened quartic version

gives well-coveredness and all the displayed cut identities for every nonempty independent I, now with $0 \le \theta \le 3$. For every maximal independent S it gives

$$\operatorname{ex}(F-S) = \operatorname{ps}(F-S) = \operatorname{gap}(F-S) = 0, \qquad |S| + \nu(F-S) - k = E.$$

The earlier result without an explicit minimality premise covered $4 \le E \le 8$.

Setup 4.8 (Contained obstructions and the conditional formula)**.** Let F_1, F_2 be nonempty star forests with positive nonincreasing integer component edge-size lists $n_1, \dots, n_s$ and $m_1, \dots, m_t$. Put

$$M = s + t - 1, \qquad \ell_l = \max_{\substack{1 \le i \le s,\ 1 \le j \le t \\ i+j=l}} (n_i + m_j - 1), \qquad L_q = \sum_{l=q+2}^{s+t} \ell_l \quad (0 \le q < M).$$

Let $\mathcal{A}_q$ contain every ordered pair of nonempty actual suffixes obtained by deleting legal leading components with total number q.

The contained-minimal-obstruction theorem of Companion I says that a graph H with $e(H) < L_q$ which arrows every entire prescribed-color member of $\mathcal{A}_q$ contains a subgraph $F \in \mathcal{L}_{\mathrm{rig}}$, with $\Delta(F) \le \Delta(H)$. The subgraph need not be induced. If no such subgraph exists, some actual member has a coloring avoiding both entire prescribed targets. Every such coloring extends over the canonical q-star pool and over every specified actual outer q-vertex set with residual H, preserving all H-edge colors and avoiding the unchanged full originals.

Under $N_4, N_6, N_8, N_{10}, N_{12}$, the earlier conditional assembly gives

$$\widehat{R}(F_1, F_2) = \sum_{l=2}^{s+t} \ell_l.$$

[math/blocker] [GAP: the canonical star pool and the two original-list pullback constructions were not supplied. Their full operational statements are assumed here and are not reconstructed.][out-of-scope: thm-body-math; env=setup; label=setup:extraction]

Setup 4.9 (Proper quartic subgraphs)**.** If F is connected and four-regular, every proper subgraph H with positive gap satisfies

$$\operatorname{gap}(H) = 1, \qquad \operatorname{ex}(H) = 1, \qquad 1 \le \operatorname{tr}(H) \le 2.$$

Every proper H with $\operatorname{ex}(H) \ge 2$ is LP-good.

Setup 4.10 (Covered processes and terminal accounting)**.** For sources of degrees $6, 8, 10, 12$, every complete process starting from the complement of a maximum independent set is covered at every deletion parent. The earlier all-degree accounting theorem applies to such covered processes, including degree four.

If the initial independent set has size a, then $c_{d-1}(F) = a$. A deletion at actual degree $D \ge 3$ loses one in every positive row below D and zero in the actual top row. Each terminal R has $\operatorname{ps}(R) = 0$. For every actual prefix W, including the initial independent set,

$$c_0(F) = |W| + \nu(F - W) + \operatorname{ps}(F - W).$$

The possible terminal components are

$$K_1,\ P_2,\ P_3,\ P_4,\ P_6,\ C_3,\ C_4,\ C_5,\ C_7.$$

If $o(R)$ counts odd-cycle components and $b(R)$ counts positive-even-order path components, then

$$c_2(R) = o(R), \qquad c_1(R) = \nu(R) - b(R).$$

For $d = 6, 8, 10, 12$, the earlier row-optimum theorem gives all assertions of Theorem 4.24(II) under its maximum-independent-set and covered-process hypotheses, including the capacity pair chosen before the optimal deletion and every subset of it.

Setup 4.11 (Higher-degree well-coveredness inputs)**.** Sources of degrees ten and twelve are well-covered. In degrees eight, ten, and twelve,

$$c_{d-1}(F) = \alpha(F).$$

For an octic source, a smaller maximal independent set has size $w = a - 1$, and its complement R satisfies

$$\Delta(R) = 7, \quad c_7(R) = 1, \quad \operatorname{gap}(R) = 0, \quad s = \operatorname{ps}(R) \in \{2, 3\}, \quad c_0(F) = w + \nu(R) + s.$$

For an octic source with a covered process from a maximum independent complement,

$$k - a \le 2E - 3.$$

The connected near-perfect degree-eight bound gives

$$k \ge 3E - 1 + 2\left\lfloor \frac{2E-3}{6} \right\rfloor.$$

For the actual host and independent outer set above, an uncovered degree-seven complement with hereditary T_-, slack three, and $w \ge 8$ gives

$$c_0(F) \le w + \nu(R) + 2.$$

At slack two, any maximum-cardinality degree-seven exposure attainer X, of size $b \ge 1$, gives exactly $b + 1$ required odd factor-critical components $\mathcal{Q}$ of $R - X$. If P is the union of the other components, then

$$\nu(R) = b + \sum_{Q \in \mathcal{Q}} \nu(Q) + \nu(P), \qquad \operatorname{ps}(P) = 0, \qquad E_R(X, V(P)) = \varnothing,$$

and

$$\nu(Q) \le \alpha(R) - b + 3 \qquad (Q \in \mathcal{Q}).$$

For this actual-pool configuration, if $w + b \ge 2$, the companion P and all its induced subgraphs have gap at most one, and $c_0(F) = w + \nu(R) + 2$, then

$$\max_{Q \in \mathcal{Q}} \nu(Q) \ge w + b + \nu(P) + 5.$$

Setup 4.12 (Sextic and odd-carrier inputs)**.** Every maximal independent set of a sextic source has size a or $a - 1$, and

$$|V(F)| - 5a \le 18 - 2E.$$

In the actual independent-complement setting $G = F - I$, write $w = |I|$ and assume

$$\Delta(G) = 5, \quad \operatorname{gap}(G) = 0, \quad s = \operatorname{ex}(G) = \operatorname{ps}(G), \quad c_0(F) = w + \nu(G) + s,$$

with hereditary T_- and $w \ge \max(2, \alpha(G) - 1)$. If $s \le 1$, the degree-five vertices have simultaneous matching coverage. If $s \in \{2, 3\}$ and they are uncovered, the smaller-pool theorem gives

$$c_0(F) \le w + \nu(G) + s - 1.$$

If instead $s \ge 4$, $w \ge 2$, the degree-five vertices are uncovered, and

$$|V(G)| < 4w + 6s + 7,$$

the marked-carrier theorem gives a whole uncovered component C, the companion $P = G - V(C)$, and a nonempty maximum-cardinality marked-exposure attainer $X \subseteq V(C)$, of size $b \ge 1$. The graph $C - X$ has exactly $b + 1$ odd factor-critical components, all of whose vertices have degree five in C. A selected component Z of largest matching number has boundary one and

$$\operatorname{ex}(Z) = \operatorname{ps}(Z) = s, \qquad \operatorname{gap}(Z) = 0, \qquad \nu(Z) \ge w + b + \nu(P) + 3s - 1.$$

For such a connected factor-critical Z, with nonnegative degree-five deficits summing to one and hereditary gap at most one, the odd-cap rank bound gives

$$\nu(Z) \le \alpha(Z) + 3s - 1.$$

[math/blocker] [GAP: the definitions of degree exposure attainters and marked-exposure attainters, and the original full statements surrounding these specialized carrier inputs, were not supplied. Only the precise consequences used here have been recorded.][out-of-scope: thm-body-math; env=setup; label=setup:sextic]

Setup 4.13 (Full-boundary absorption)**.** Let H be an induced subgraph of F, with $N = |V(F) \setminus V(H)|$. Partition $E(H)$ into a paid graph P and an unpaid graph L. Let $D \in \mathbb{Z}_{\ge 1}$, assume that L has maximum degree at most D, and assume that one matching covers every degree-D vertex of L. If

$$N \ge \max(0, \nu(P) - 1),$$

the generic full-boundary absorption theorem gives

$$c_D(F) \le N + \nu(P).$$

It restores every omitted-incident edge, including edges with two omitted endpoints, for every capacity split totaling D and every matching-budget split totaling $N + \nu(P)$.

Setup 4.14 (Cover-assisted three-matching input)**.** We assume the strengthened version of Theorem 4.17(I)–(III) in which whole P_3 components are also allowed in the initial and alternative rows and the omission hypothesis is

$$N \ge \max(0, b + h - 2).$$

Its other hypotheses remain

$$\nu(H) \ge m + h, \qquad c_1(F) \ge N + h + b, \qquad c_2(F) \ge N + h + p.$$

For arbitrary sequential maximum matchings, it gives the same three ranks, simultaneous row restrictions, clearing of all row edges, survival of exterior edges, and complete certificates with the exact unpaid matching unions. This is an assumed result of the earlier development; Theorem 4.17 below proves the version with the larger omission margin directly.

[math/blocker] The opening paragraph of Assumed input 4.15 and its item (II) specify prerequisites through forward references to Theorem 4.25, whose proof invokes that same assumed input. These references need independent, precise earlier-series statements before they can support the final proof. The supplied material does not independently identify which clauses are earlier prerequisites and which are consequences of the final theorem.][addressed-partial: the original opening import and original item (II) have been removed; the missing independent earlier statements are now recorded explicitly.][out-of-scope: thm-body-math; env=setup; label=setup:later]

Setup 4.15 (Inputs to the final residual assembly)**.** The earlier residual assembly has initial excess floor five in degree four and excess floors $5, 7, 12, 10$ in degrees six, eight, ten, and twelve, respectively.

The following additional statements are supplied as earlier results:

(I) No connected four-regular factor-critical graph of gap one has excess five.

(II) In the octic excess-seven residual, all epsilon-zero rows with a whole P_3 are excluded. The remaining epsilon-zero row is $3C_3 + C_5 + 9K_1$, and the epsilon-one row is $4C_3 + P_2 + 9K_1$.

(III) If a well-covered graph G contains an induced $A = mC_3 + P_3 + zK_1$, where $m, z \in \mathbb{Z}_{\ge 0}$, then, with $u_{\mathrm{out}} = |V(G) \setminus V(A)|$,

$$u_{\mathrm{out}} \ge 1, \qquad c_2(G) \le u_{\mathrm{out}} + m - 1.$$

These inputs are assumed, not re-proved.

[math/blocker] [GAP: independent earlier statements are missing for the remaining source restrictions, process contracts, supported-child minimum, degree-twelve stages, octic excess-seven parameters and singleton identities, sextic conditional estimates, budgeted-state deficiency bounds, and fixed triangle-interval assertions. No independent statement of the octic four-type stage classification or its degree-four neighbor property is supplied either. These claims are not assumed by reference to the final theorem.][addressed-partial: the circular imports have been removed; the affected deductions still require independent prerequisites.]

[math/blocker] [GAP: the full inherited rank and sector contracts, the formal row-complement-frame definition, the native contact patterns, the admissible batch families, and the forced-edge triangle-interval data and physical ledgers were not supplied. Their exact preservation requires the earlier statements. In particular, the final assembly's references to those contracts are conditional references, not newly established definitions or constructions.][addressed-partial: the missing definitions and contracts remain explicit; no final-assembly conclusion is imported as a prerequisite.][out-of-scope: thm-body-math; env=setup; label=setup:later]

Acknowledgements.

4.2. **Generic matching interfaces.** In this section, we prove the matching statements that apply independently of source degree, regularity, and minimality.

Theorem 4.16 (Missing allocations and exposed ports)**.** *Let F have a vertex partition $V(F) = V(R) \dot\cup U$, where R is induced and every component C of R satisfies*

$$\nu(C) = \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

Assume

$$c_{0,0}(F) = |U| + \nu(R).$$

Let $x, y \in \mathbb{Z}_{\geq 0}$ satisfy $x + y = c_{0,0}(F) - 1$, assume that F has no $(0, 0; x, y)$-certificate, and put $j = |U| - x \geq 1$. Let $\mathcal{O}$ be the family of odd-order components of R, put $o = |\mathcal{O}|$, and define

$$P(C) = \{v \in V(C) : \nu(C - v) = \nu(C)\} \qquad (C \in \mathcal{O}).$$

For $\mathcal{C} \subseteq \mathcal{O}$, let $N_U^{\mathrm{port}}(\mathcal{C})$ consist of the vertices of U adjacent to at least one vertex of $\bigcup_{C \in \mathcal{C}} P(C)$. Then:

(I) *For every $S \subseteq U$ with $|S| = j$,*

$$\nu(F[V(R) \cup S]) = \nu(R) + j = y + 1.$$

Every maximum matching consists of a maximum matching inside each component of R and exactly one edge from each vertex of S to R. It has no edge inside S. The R-endpoints lie at exposed ports of j distinct odd-order components.

(II) *We have $j \leq o$. For every $T \subseteq U$ with $|T| \leq j$, there is a matching taking every vertex of T to an exposed port in a distinct odd component, compatible with a maximum matching in every component of R. Hence $F[V(R) \cup T]$ has a matching of size $\nu(R) + |T|$.*

(III) *For every $\mathcal{C} \subseteq \mathcal{O}$ of size ℓ with $o - j + 1 \leq \ell \leq o$,*

$$|N_U^{\mathrm{port}}(\mathcal{C})| \geq |U| - o + \ell.$$

No regularity, minimality, component factor-criticality, sector, or degree bound is assumed. Exposability is a condition on the actual vertex, not a consequence inferred from odd order.

Proof. Since $x \geq 0$, we have $j \leq |U|$, so a set $S \subseteq U$ of size j exists. Fix such an S and put $W = U \setminus S$; then $|W| = x$. If $\nu(F - W) \leq y$, assign every edge incident with W to the first matching-bounded part and all remaining edges to the second. Every matching in the first part uses a distinct vertex of W on each edge, so has size at most x. The second part is $F - W$, with matching number at most y. Empty degree-bounded parts would give the forbidden certificate. Therefore

$$\nu(F - W) \geq y + 1 = c_{0,0}(F) - x = \nu(R) + j.$$

For any matching M of $F - W$, let M_R be its edges internal to R. There are at most $\nu(R)$ such edges. Every other edge meets S, so there are at most j other edges. Thus $|M| \leq \nu(R) + j$, proving equality.

For every maximum M, both bounds are attained. Its j edges meeting S must each use exactly one vertex of S; otherwise they would use more than j distinct vertices of S. They saturate S and none is internal to it. Matching numbers add across disjoint components, because restrictions give an upper bound and component maximum matchings can be united. Hence M_R is maximum in every component. It covers every even-order component and leaves exactly one vertex uncovered in each odd-order component. The crossing endpoints must be these uncovered vertices in distinct odd components. If v is the uncovered vertex in C, then the restriction of M_R gives $\nu(C - v) \geq \nu(C)$; inclusion gives the reverse inequality. Thus $v \in P(C)$. This proves (I) and $j \leq o$.

For (II), extend T to a set $S \subseteq U$ of size j. Retain the entire internal part of a maximum matching supplied by (I), together with only the crossing edges incident with T. The resulting matching has size $\nu(R) + |T|$, with the asserted component and port properties. This also covers $T = \varnothing$.

For (III), put $A = U \setminus N_U^{\mathrm{port}}(\mathcal{C})$. Suppose $|A| \geq o - \ell + 1$. The integer $o - \ell + 1$ is positive and at most j, so choose $T \subseteq A$ of that size. Part (II) matches T to exposed ports in distinct odd components. None can lie in $\mathcal{C}$, by the definition of A. There are only $o - \ell$ other odd components, a contradiction. Therefore $|A| \leq o - \ell$, which gives the claimed neighborhood bound. □

Theorem 4.17 (Three matchings forced by two tight costs)**.** *Let H be an induced subgraph of F, and let A be an induced subgraph of H. Every whole component of A is an isolate, a positive-even-order path, or a cycle. Put*

$$r = |V(A)|, \quad m = \nu(A), \quad t = \#\{\textit{positive-even-order path components of } A\},$$

$$p = \#\{\textit{odd-cycle components of } A\}, \quad b = m - t, \quad h = |V(H)| - r, \quad N = |V(F)| - |V(H)|.$$

Assume

$$\nu(H) \geq m + h, \qquad N \geq \max(0, b + h - 1),$$

$$c_1(F) \geq N + h + b, \qquad c_2(F) \geq N + h + p.$$

There is no bound on path or cycle length. Set $P_0 = H$. Choose any maximum matching M_0 of P_0, any maximum matching M_1 of $P_1 = P_0 - M_0$, and any maximum matching M_2 of $P_2 = P_1 - M_1$. Put $P_3 = P_2 - M_2$. All these graphs retain $V(H)$. Then $0 \leq p \leq b$, and:

(I)

$$\nu(P_0) = m + h, \qquad \nu(P_1) = b + h, \qquad \nu(P_2) = p + h.$$

Both assumed cost lower bounds are equalities.

(II) *These same matchings satisfy the following simultaneously for every induced $R \subseteq H$ with order r, matching number m, exactly t positive-even-order path components, exactly p odd-cycle components, and no other component types besides isolates, positive-even-order paths, and cycles.*

Put $W_R = V(H) \setminus V(R)$. Each M_i has exactly h edges saturating W_R into R, and no edge inside W_R. The restriction of M_0 to each whole component of R is maximum. Its crossing

endpoints are exposed vertices in distinct odd whole components, where exposed means $\nu(C-v)=\nu(C)$*. The entire internal residuals are*

$$R_1=(V(R),E(R)\setminus M_0)=pP_3+(b-p)P_2+\textit{isolates},$$

$$R_2=(V(R),E(R)\setminus(M_0\cup M_1))=pP_2+\textit{isolates}.$$

The restriction of M_1 *to* R_1 *is maximum, and* M_2 *contains every edge of* R_2*. Thus the three matchings contain every edge of every such* R*, while every original edge of* $H[W_R]$ *survives in* P_3*. Every vertex of* W_R *has at least three distinct original neighbors in* R*. If it has exactly three, the prefix deletes all three incident edges into* R*. The path notation in these unions does not refer to the residual graphs* P_i*.*

(III) *For* $i=1,2$*, put*

$$L_i=\left(V(H),\bigcup_{j=0}^{i-1}M_j\right),\qquad \rho_i=\nu(P_i).$$

For every capacity split $A_c+B_c=i$ *and every budget split* $x+y=N+\rho_i$*, the entire* F *has a certificate with* $Q_R\cup Q_B=E(L_i)$ *exactly. Every original edge with an omitted endpoint, including an edge with two omitted endpoints, lies in* $P_R\cup P_B$*. The full totals are*

$$N+\rho_1=N+h+b=c_1(F),\qquad N+\rho_2=N+h+p=c_2(F).$$

(IV) *For any nonempty finite family* $(R_j)_{j\in\mathcal{I}}$ *of the rows in (II), put*

$$C_{\mathrm{common}}=\bigcap_{j\in\mathcal{I}}V(R_j),\qquad X_{\mathrm{common}}=V(H)\setminus\bigcup_{j\in\mathcal{I}}V(R_j).$$

Each of the same three matchings saturates X_{common} *into* C_{common}*. Every vertex of* X_{common} *has at least three distinct neighbors in* C_{common}*, and every* $X\subseteq X_{\mathrm{common}}$ *satisfies*

$$3|X|\le\sum_{v\in C_{\mathrm{common}}}\min\bigl(3,|N_H(v)\cap X|\bigr).$$

(V) *For any* $p_0,h_0,N_0\in\mathbb{Z}_{\ge0}$ *with* $N_0\ge\max(0,p_0+h_0-1)$*, take*

$$F=p_0C_3+(h_0+N_0)K_{1,3}.$$

Retain h_0 *star centers and omit the other* N_0 *centers. Let* H *contain every vertex except those omitted centers, and let* A *contain all triangle vertices and all star leaves. Then*

$$r=3p_0+3(h_0+N_0),\quad m=b=p=p_0,\quad t=0,\quad h=h_0,\quad N=N_0,$$

$$c_1(F)=c_2(F)=p_0+h_0+N_0.$$

These data satisfy all hypotheses, allowing arbitrary nonnegative retained exterior size.

No source degree, regularity, minimality, or well-coveredness assumption is imposed. The theorem neither proves its hypotheses for a source nor excludes a source class.

Proof. We use Assumed input 4.13 only in its generic form.

Step 1. We determine the residual of a component maximum matching.

An even path on $2l$ vertices, $l\ge1$, has matching number l. Its perfect matching is unique: its first endpoint forces its edge, and removal of the first two vertices repeats the argument. Deleting that matching leaves $l-1$ disjoint edges and two isolates. An even cycle on $2l$ vertices, $l\ge2$, has matching number l; every perfect matching alternates, and its complement is another perfect matching. An odd cycle on $2l+1$ vertices, $l\ge1$, has matching number l. A maximum matching exposes one vertex v; on deleting v, it is the unique perfect matching of the remaining even path. Its complement in the cycle is one P_3 centered at v and $l-1$ disjoint edges. An isolate remains isolated. Alternating matchings attain the half-order upper bounds in each case.

Consequently,

$$b = \sum_{\text{even paths}} (l-1) + \sum_{\text{even cycles}} l + \sum_{\text{odd cycles}} l \geq p \geq 0.$$

Deleting any componentwise maximum matching leaves exactly $pP_3 + (b-p)P_2$ and isolates, of matching number b.

Step 2. Tight matching and cost bounds force the first two residual ranks.

Fix an allowed row R; the original A ensures that such rows exist. Every matching of H has at most m internal edges and at most h other edges, because each other edge meets W_R. The assumed matching gives $\nu(H) = m + h$.

For any maximum M_0, both bounds are attained. Its h noninternal edges cannot use two exterior vertices on any edge, since they would then use more than h distinct exterior vertices. They therefore saturate W_R into R. The m internal edges are maximum in every component. Even components are fully matched internally. Each odd cycle and each isolate has exactly one internally exposed vertex, so crossing endpoints lie in distinct odd components at exposed vertices. This argument applies to the same M_0 for every permissible R.

Step 1 gives the entire row R_1, so $\rho_1 = \nu(P_1) \leq b + h$. The unpaid graph M_0 has maximum degree at most one and is covered by itself. The omission hypothesis gives $N \geq \max(0, \rho_1 - 1)$. Assumed input 4.13 yields

$$N + h + b \leq c_1(F) \leq N + \rho_1 \leq N + h + b.$$

Hence $\rho_1 = b + h$ and $c_1(F) = N + h + b$.

Fix any maximum M_1 of this same P_1. For every R, it attains the internal bound b and exterior bound h. The exterior edges again saturate W_R into R. Internally it takes one edge from each P_3 and every whole P_2 edge. Thus the entire R_2 is pP_2 and isolates.

Now $\rho_2 = \nu(P_2) \leq p + h$. The unpaid graph $M_0 \cup M_1$ has maximum degree at most two; every degree-two vertex is covered by M_0. Since $p \leq b$, the omission condition implies $N \geq \max(0, \rho_2 - 1)$. Therefore

$$N + h + p \leq c_2(F) \leq N + \rho_2 \leq N + h + p,$$

and $\rho_2 = p + h$, $c_2(F) = N + h + p$.

Step 3. The third matching clears every row simultaneously.

Any maximum M_2 attains both the internal bound p and exterior bound h, relative to every allowed row. It contains all edges of R_2 and saturates W_R into R. None of the three matchings deletes an edge inside W_R. Their edge sets are disjoint by construction, so an exterior vertex uses three distinct edges into R. Simplicity gives three distinct neighbors. If its original row degree is three, these exhaust all its row edges. This proves (I) and (II).

Step 4. We construct the complete certificates and the common-row matchings.

For $i = 1, 2$, assign A_c of the i fixed matchings in L_i to Q_R, and the remaining B_c to Q_B. Their degree bounds hold and their union is exactly $E(L_i)$.

Order the N omitted vertices. Assign each omitted-incident edge to its earliest omitted endpoint. This partitions all such edges into N possibly empty stars, including each edge with two omitted endpoints exactly once. These stars and P_i are edge-disjoint.

Put $\rho = \rho_i$ and let $x + y = N + \rho$. If $\rho = 0$, then P_i is empty; assign x stars to P_R and y to P_B. If $\rho \geq 1$, then $N \geq \rho - 1$, so $x + y \geq 2\rho - 1$; at least one budget is at least ρ. If $x \geq \rho$, assign all of P_i and $x - \rho$ stars to P_R, and the remaining y stars to P_B. Here $0 \leq x - \rho \leq N$. A matching in P_R has at most ρ edges from P_i and at most one from each assigned star. Thus its size is at most x, and the other part has matching number at most y. Interchange the colors if $y \geq \rho$. This includes zero budgets and empty parts and proves (III).

A vertex outside every row has, in each fixed M_i, its unique mate inside every row by (II). Its mate lies in their intersection. Restricting the three matchings gives three edge-disjoint matchings

from X_{common} to C_{common}. For any $X \subseteq X_{\mathrm{common}}$, their union has exactly $3|X|$ edges incident with X. At a vertex v of the common intersection, it has at most three edges and at most $|N_H(v) \cap X|$ edges. Summation proves (IV).

Step 5. We verify the examples directly.

Put $z = p_0 + h_0 + N_0$. The indicated A consists of p_0 triangles and $3(h_0 + N_0)$ isolates, giving the stated parameters. The graph H consists of p_0 triangles, h_0 full stars, and the isolated leaves of omitted stars. Taking one edge from each retained nontrivial component gives a matching of size $p_0 + h_0 = m + h$, which is maximal in size componentwise. The omission inequality is the assumed one.

For $i = 1, 2$, any capacity split has a positive capacity. Put one edge from each triangle into that degree-bounded part. These edges form a matching. The paid remainder is a disjoint union of z nonempty stars. For every $x + y = z$, assign x components to one paid part and y to the other. Hence $c_i(F) \le z$.

If $z \ge 1$, choose capacities $(1, 0)$ at row one and $(1, 1)$ at row two. For any $T < z$, take budgets $(0, T)$. The zero-budget part is empty. The degree parts remove at most one edge per original component at row one and at most two at row two, because a matching has at most one edge in a triangle or a three-leaf star. Each component originally has three edges, so at least one remains in every one of the z disjoint components. The other paid part has matching number at least $z > T$, a contradiction. Thus $c_i(F) \ge z$. For $z = 0$, both costs are zero directly. This proves (V). □

Theorem 4.18 (Simultaneous confinement on one retained graph)**.** *Let H be induced in F, and let R_0, R_1 be equal-order induced subgraphs of H. Fix one matching M of H, not necessarily maximum. Put*

$$P = E(H) \setminus M, \quad B_i = (V(R_i), E(R_i) \setminus M), \quad A_i = V(H) \setminus V(R_i),$$
$$t = |A_0| = |A_1|, \quad u = |V(F) \setminus V(R_0)|, \quad N = |V(F) \setminus V(H)| = u - t.$$

Assume $\nu(B_0) = \nu(B_1) = m$ and $N \ge \max(0, m + t - 1)$. Set

$$\rho = \nu((V(H), P)), \qquad \widehat{D}_1 = N + \rho.$$

Then

$$c_1(F) \le \widehat{D}_1 \le u + m.$$

For every capacity split totaling one and every budget split totaling $\widehat{D}_1$, there is a certificate on the entire original graph, including all omitted-incident edges. The unpaid graph is exactly M with isolated vertices.

If the actual equality $c_1(F) = u + m$ holds, then:

(I) *$\rho = m + t$ and $\widehat{D}_1 = u + m$. Every maximum P-matching L consists, simultaneously for $i = 0, 1$, of a maximum internal B_i-matching of size m and exactly t edges saturating A_i into R_i. Their row endpoints are distinct. No maximum P-matching has an edge inside either A_i.*

(II) *Put*

$$V_{\mathrm{common}} = V(R_0) \cap V(R_1), \qquad A_{\mathrm{common}} = V(H) \setminus (V(R_0) \cup V(R_1)).$$

Every maximum P-matching pairs every vertex of A_{common} with a vertex of V_{common}. Hence there is an injection $\tau_{\mathrm{common}} \colon A_{\mathrm{common}} \to V_{\mathrm{common}}$ using only edges of P. Every $X \subseteq A_{\mathrm{common}}$ has at least $|X|$ paid neighbors in V_{common}. No edge from A_{common} to either opposite row difference occurs in a maximum P-matching. These conclusions do not replace any fixed edge of M.

(III) *Assume additionally that each entire B_i consists of $pP_3 + P_2$ and isolates, with $p \ge 0$, so $m = p + 1$. Let E_i contain the middle vertices of its P_3 components and both endpoints of*

its unique P_2. Let $\mathcal{D}_i$ consist of the leaf pair of each P_3, as one block, and each isolate, as a singleton block. These blocks partition $V(R_i) \setminus E_i$.

For every maximum P-matching, its exterior endpoints avoid E_i and meet each block of $\mathcal{D}_i$ at most once, simultaneously for both rows. The common injection may therefore have image in $V_{\mathrm{common}} \setminus (E_0 \cup E_1)$, meeting each block of either family at most once.

For a fixed i, an injective map $\tau_i \colon A_i \to V(R_i)$ using actual P-edges extends by an internal B_i-matching to a maximum P-matching if and only if its image avoids E_i and meets every block of $\mathcal{D}_i$ at most once. The extension chooses in each P_3 an edge avoiding an assigned leaf and includes the unique P_2 edge. Arbitrarily prescribed maps for the two rows are not asserted to extend simultaneously.

The statement is conditional on the stated equality and omission margin. It asserts no prescribed-row existence, strict payment, optimization over different retained sets or unpaid matchings, source exclusion, or Ramsey formula.

Proof. A P-matching has at most m internal edges relative to either row, and at most t other edges because each meets A_i. Thus $\rho \le m + t$. The unpaid graph is exactly M, of maximum degree at most one, covered by that same matching. Since $N \ge \max(0, \rho - 1)$, Assumed input 4.13 gives

$$c_1(F) \le N + \rho \le u - t + m + t = u + m.$$

For the exact unpaid assertion, put M in the degree part of capacity one. Order the omitted vertices and assign each omitted-incident edge to its first omitted endpoint. The resulting N stars and the retained paid graph admit the budget allocation constructed in Step 4 of the proof of Theorem 4.17. This includes edges with two omitted endpoints exactly once and leaves precisely M unpaid.

Under $c_1(F) = u + m$, every inequality is an equality. Let L be any maximum matching of this fixed paid graph. For either i, its internal and noninternal edge counts attain their bounds m and t. The t noninternal edges use at least t exterior vertices, so each uses exactly one, and they saturate A_i. Its internal part is maximum in B_i. The argument applies to the same L for both rows, proving (I).

If $w \in A_{\mathrm{common}}$, its unique mate under L belongs to both rows by (I). It therefore lies in V_{common}. Restriction of L gives the common injection and the paid-neighborhood inequality for every subset. An edge from A_{common} to a row difference has both endpoints outside the other row and is excluded by (I). All these edges belong to P, so none replaces an edge of M. This proves (II).

For (III), a maximum matching of a whole P_3 covers its middle vertex and one leaf, leaving exactly the other leaf exposed. The unique P_2 contributes its edge and exposes neither endpoint. Isolates contribute no edges. Componentwise choices give every maximum B_i-matching, of size $p + 1$. Hence all vertices of E_i are covered internally, and at most one leaf of each P_3 is available as an exterior endpoint. Injectivity handles singleton blocks. This proves the necessity and its simultaneous application to both rows.

Conversely, fix i and a map satisfying the stated conditions. In a P_3 whose leaf pair meets its image, choose the edge to the other leaf. If neither leaf is assigned, choose either edge. Include the unique P_2 edge. These choices give m internal edges avoiding all assigned endpoints. Together with the t exterior edges, they form a P-matching of size $m + t = \rho$. Thus the extension is maximum. The necessity already proved gives the equivalence. □

Theorem 4.19 (A missing allocation supplies a compatible prefix)**.** *Let A be induced in F, with every whole component an isolate, a P_3, a positive-even-order path, or a cycle. Put*

$$r = |V(A)|, \quad m = \nu(A), \quad t = \#\{\textit{positive-even-order path components}\}, \quad p = \#\{\textit{odd-cycle components}\},$$

$$b = m - t, \qquad U = V(F) \setminus V(A), \qquad u = |U|.$$

Assume

$$c_0(F) = u + m, \qquad c_1(F) \ge u + b, \qquad c_2(F) \ge u + p.$$

Fix a missing pair $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = c_0(F) - 1$, and put

$$j_{\mathrm{hi}} = u - \min(x, y), \qquad j_{\mathrm{lo}} = u - \max(x, y).$$

Choose any $W \subseteq U$, put $h = |W|$, and assume

$$h \leq j_{\mathrm{hi}}, \qquad 2h \leq u - b + 2,$$

equivalently $h \leq \min(j_{\mathrm{hi}}, \lfloor (u - b + 2)/2 \rfloor)$. Set

$$H = F[V(A) \cup W], \qquad S = V(F) \setminus V(H), \qquad N = u - h.$$

The quantities $j_{\mathrm{hi}}, j_{\mathrm{lo}}, b$ are initially signed integers.

Set $P_0 = H$, and successively choose any maximum matching M_i of P_i, with $P_{i+1} = P_i - M_i$, for $i = 0, 1, 2$, retaining $V(H)$. Then:

(I) *All choices exist, $0 \leq p \leq b$, and*

$$\nu(P_0) = m + h, \quad \nu(P_1) = b + h, \quad \nu(P_2) = p + h, \qquad c_1(F) = u + b, \quad c_2(F) = u + p.$$

Moreover,

$$N \geq \max(0, b + h - 2),$$
$$N - (b + h - 2) = |x - y| + 2(j_{\mathrm{lo}} - h) + t + 1 \geq 0.$$

The last identity does not require $j_{\mathrm{lo}} - h \geq 0$.

(II) *For every induced $R \subseteq H$ with the same r, m, t, p, and only the permitted component types, the same three matchings saturate $W_R = V(H) \setminus V(R)$ into R, with no edge inside W_R. The number of whole P_3 components may vary.*

The restriction of M_0 is maximum on every whole component of R, with crossing endpoints exposed in distinct odd components. The entire first internal residual is $pP_3 + (b - p)P_2$ and isolates; M_1 is maximum on it. The entire second residual is pP_2 and isolates, and M_2 contains all its edges. Thus every original row edge is cleared, every edge of $H[W_R]$ survives, and every exterior vertex has at least three distinct original row neighbors. All assertions hold simultaneously for every such row and every successive maximum choice.

(III) *If $v \in W_R$ has exactly one neighbor o which is exposed in an odd whole component of R, then the actual edge vo belongs to every allowed M_0. If v has exactly three original row neighbors, all three corresponding edges are deleted. Under both conditions, $vo \in M_0$, and the other two edges occur one in M_1 and one in M_2.*

For any independent $X \subseteq W_R$ consisting of vertices with exactly three row neighbors, $V(R) \cup X$ is independent in P_3, and $W_R \setminus X$ is a vertex cover. The chosen prefix and residual remain fixed.

(IV) *For $i = 1, 2$, let $L_i = \bigcup_{0 \leq j < i} M_j$. For every capacity split totaling i and every budget split totaling $N + \nu(P_i)$, the entire original F has a certificate with unpaid edge set exactly L_i. Every edge meeting S, including an edge with both endpoints in S, is paid. The totals are neutral:*

$$N + \nu(P_1) = c_1(F), \qquad N + \nu(P_2) = c_2(F).$$

(V) *The permitted retained range can exceed the smaller coordinate range. Take $F = 4K_{1,3}$, let A consist of all twelve leaves, and retain any three centers as W. Then*

$$r = 12, \quad m = t = p = b = 0, \quad u = 4, \quad h = 3, \quad N = 1,$$
$$c_0(F) = c_1(F) = c_2(F) = 4.$$

The pair $(0, 3)$ is missing, so $j_{\mathrm{hi}} = 4$, $j_{\mathrm{lo}} = 1$, $h > j_{\mathrm{lo}}$, and $2h = u - b + 2$.

No source degree, regularity, minimality, sector, or native-process assumption is imposed. The conclusions are a generic prefix and neutral certificates, not strict payment, source exclusion, or a Ramsey formula.

Proof. Every permitted component C satisfies $\nu(C) = \lfloor |V(C)|/2 \rfloor$. For an isolate this is zero; for P_3, all edges meet its middle vertex. Even paths and even cycles have alternating perfect matchings, and odd cycles have alternating matchings exposing one vertex. In every case half the order is an upper bound. Summing component ranks gives $b = m - t \geq p \geq 0$; a whole P_3 contributes one to b.

Put $x_{\min} = \min(x, y)$, $x_{\max} = \max(x, y)$. The ordered pair $(x_{\min}, x_{\max})$ is missing, since swapping the parts of a certificate would otherwise give one for (x, y). If $h > 0$, then $j_{\mathrm{hi}} \geq 1$, and Theorem 4.16(II) applies with row A, exterior U, and first budget $x_{\min}$. Its hypotheses are the component rank equality and $c_0(F) = u + m$. It gives a matching of the same H with $m + h$ edges, maximum inside every A-component and crossing from W into distinct exposed odd-component ports. If $h = 0$, a maximum matching of $A = H$ already has $m + h$ edges.

Since $N = u - h \geq 0$, the second retained-size inequality gives

$$N - (b + h - 2) = u - b - 2h + 2 \geq 0.$$

Also, using $x_{\min} + x_{\max} = u + m - 1$ and $b = m - t$,

$$\begin{aligned} |x - y| + 2(j_{\mathrm{lo}} - h) + t + 1 &= x_{\max} - x_{\min} + 2u - 2x_{\max} - 2h + t + 1 \\ &= 2u - (x_{\max} + x_{\min}) - 2h + t + 1 \\ &= u - m - 2h + t + 2 \\ &= u - b - 2h + 2. \end{aligned}$$

The nonnegativity follows from the scalar inequality, not from separate signs of its summands.

Apply Assumed input 4.14 to this same F, H, A. Its component list includes whole P_3's, its initial matching has just been established, and its omission condition follows from the preceding display. Its two cost hypotheses are exactly the assumptions because $N + h = u$. It gives the ranks, cost equalities, simultaneous row conclusions, and complete certificates in (I), (II), and (IV). Successive edge deletion makes the three matchings edge-disjoint.

For (III), part (II) forces the M_0-mate of v to be exposed in an odd whole component. Under uniqueness it must be o, so vo is compulsory. The three matchings use three distinct row edges at v. If its original row degree is three, these are all its row edges. Under both conditions, the other two occur in the two later matchings. The prefix also clears every edge inside R. If X is as stated, it clears every edge from X to R, and there was no edge inside X. Thus $V(R) \cup X$ is independent in P_3. Its complement $W_R \setminus X$ is a vertex cover.

For (V), four disjoint three-leaf stars have sixteen vertices and matching number four. Their leaf row has twelve isolates, giving the stated parameters. For $i = 0, 1, 2$, every capacity split and every budget split totaling four is realized by assigning whole star components to the paid parts and leaving both degree parts empty. Hence $c_i(F) \leq 4$.

For the reverse inequality, take capacities $(i, 0)$ and budgets $(0, T)$ with $T < 4$. The zero-capacity and zero-budget parts are empty. A degree-i part leaves at least one edge of every star outside it because each center has degree $3 > i$. The remaining paid part has a matching of size four, a contradiction. Thus $c_i(F) = 4$. The pair $(0, 3)$ is missing because its nonempty paid part would have to contain all of F. Finally $j_{\mathrm{hi}} = 4$, $j_{\mathrm{lo}} = 1$, and $2h = 6 = u - b + 2$. All hypotheses have been checked, so the universal conclusions apply. □

4.3. **Independent cuts and automatic quartic minimality.** In this section, we apply the earlier cut inputs to the unchanged source classes and remove the explicit minimality premise in degree four.

Theorem 4.20 (Universal cuts in the unchanged obstruction class)**.** *Use the exact class $\mathcal{L}_{\mathrm{rig}}$ of Assumed input 4.6, retaining every original condition and same-class consequence. Let $F \in \mathcal{L}_{\mathrm{rig}}$, and put*

$$d = \Delta(F), \quad k = \nu(F), \quad E = \mathrm{ex}(F), \quad a = \alpha(F).$$

Then F is connected, d-regular, and factor-critical, with order $2k+1$, gap one, and

$$\operatorname{tr}(F) + E \geq \max(4, d/2+1) + 1.$$

Every proper subgraph, including every spanning edge deletion, satisfies T_-*. The possible degrees are $4, 6, 8, 10, 12$, with respective excess floors $4, 5, 6, 7, 9$.*

For every nonempty independent I, set

$$j = |I|, \qquad J = F - I, \qquad b = j + \nu(J) - k.$$

If $d \geq 6$, then

$$\operatorname{gap}(J) = 0, \qquad \operatorname{ex}(J) = \operatorname{ps}(J) = E - b,$$

$$c^*_\theta(J) = c^*_\theta(F) - j \quad (0 \leq \theta < d), \qquad 1 \leq b \leq \min\left(E, \left\lceil \frac{j}{2} \right\rceil\right),$$

$$c^*_d(F) = 1, \qquad c^*_d(J) = 0,$$

and all higher rows vanish. There is no cut-size, excess, or complement-connectedness restriction. At degree four, all original cut assertions remain valid, including the possible excess-one positive complements at cut sizes a or $a-1$, and zero-gap cuts whenever $j \leq a-2$ or $j \leq 2E-2$.

Adding these clauses leaves $\mathcal{L}_{\mathrm{rig}}$ unchanged. For each allowed d, its degree slice remains exactly $\mathcal{M}_d = \mathcal{E}_d$; its emptiness is precisely N_d. The five inputs N_d remain unproved here.

For the full star lists, $\mathcal{A}_q$, and L_q of Assumed input 4.8, every H with $e(H) < L_q$ which arrows every entire prescribed-color member of $\mathcal{A}_q$ contains a subgraph F in this unchanged class satisfying all the preceding restrictions, with $d \leq \Delta(H)$. Inducedness is not required. In the absence of such a subgraph, some actual member admits a coloring avoiding both entire prescribed targets. Every such coloring has both unchanged extensions stated in Assumed input 4.8, preserving every residual edge color and both full original lists.

Under $N_4, N_6, N_8, N_{10}, N_{12}$,

$$\widehat{R}(F_1, F_2) = \sum_{l=2}^{s+t} \ell_l.$$

No degree exclusion, class emptiness, or unconditional Ramsey formula is asserted.

Proof. Assumed input 4.6 gives the regular factor-critical structure, degree list, floors, gap one, and proper-subgraph property. At gap one, failure of T_- says

$$\operatorname{tr}(F) + E > \max(4, d/2+1).$$

All terms are integers, which gives the displayed threshold.

For $d \in \{6, 8, 10, 12\}$, Assumed input 4.7 gives every independent-cut conclusion for every nonempty independent set, without an excess restriction. Its original quartic conclusions remain intact. Hence every newly listed clause already holds for every member of the original $\mathcal{L}_{\mathrm{rig}}$. Imposing those clauses neither adds nor removes a graph. The degree-slice equalities and their equivalence to the inputs N_d therefore remain the equalities in Assumed input 4.6.

Now take the full positive lists, a legal q, and H as stated. Assumed input 4.8 supplies an actual subgraph F of the old $\mathcal{L}_{\mathrm{rig}}$, with $d \leq \Delta(H)$. The preceding argument applies all the additional restrictions to this same F. It asserts neither inducedness nor degree equality.

If no such subgraph is contained, the contrapositive of the all-family arrowing statement gives a member of $\mathcal{A}_q$ and a coloring avoiding its two entire prescribed targets. Both extensions are exactly those assumed in Assumed input 4.8; they preserve the original lists and all residual colors.

Finally, under the five N_d, the conditional formula of Assumed input 4.8 applies to these arbitrary full positive lists. The added restrictions have not established any N_d; they describe each graph in its unchanged export class. □

Theorem 4.21 (Automatic quartic minimality and all independent cuts)**.** *Let $\mathcal{Q}_4$ be the class of all connected four-regular factor-critical graphs F with*

$$\operatorname{gap}(F) \geq 1, \qquad \operatorname{ex}(F) \geq 4.$$

Let $\mathcal{L}_4$ be the degree-four slice of the exact unchanged $\mathcal{L}_{\mathrm{rig}}$. Then

$$\mathcal{Q}_4 = \mathcal{L}_4.$$

No proper-subgraph, criticality, or top-run condition needs to be added to the definition of $\mathcal{Q}_4$.

For $F \in \mathcal{Q}_4$, put $k = \nu(F)$, $E = \operatorname{ex}(F)$, and $a = \alpha(F)$. Then $\operatorname{gap}(F) = 1$, F is well-covered, every proper subgraph satisfies T$_-$*, and every proper subgraph H with $\operatorname{ex}(H) \geq 2$ is LP-good.*

For every nonempty independent I, put $j = |I|$, $J = F - I$, and $b = j + \nu(J) - k$. Then

$$\operatorname{gap}(J) = 0, \qquad c_\theta(J) = c_\theta(F) - j \quad (0 \leq \theta \leq 3),$$

$$c_4(F) = 1, \qquad c_4(J) = 0, \qquad \operatorname{ex}(J) = \operatorname{ps}(J) = E - b,$$

$$1 \leq b \leq \min\left(E, \left\lceil \frac{j}{2} \right\rceil\right),$$

and all higher rows vanish. Every maximal independent S satisfies

$$\operatorname{ex}(F - S) = \operatorname{ps}(F - S) = \operatorname{gap}(F - S) = 0, \qquad |S| + \nu(F - S) - k = E.$$

Thus the conclusion without an explicit minimality premise extends from $4 \leq E \leq 8$ to all $E \geq 4$. The quartic exclusion remains the unproved assertion $\mathcal{Q}_4 = \varnothing$.

Proof. Let $F \in \mathcal{Q}_4$. By Assumed input 4.9, every positive-gap proper subgraph H has $\operatorname{gap}(H) = 1$, $\operatorname{ex}(H) = 1$, and $1 \leq \operatorname{tr}(H) \leq 2$. Therefore

$$\operatorname{tr}(H) + \operatorname{ex}(H) \leq 3 \leq \max\left(4, \left\lfloor \frac{\Delta(H)}{2} \right\rfloor + 1\right).$$

Such an H satisfies T$_-$. If its gap is nonpositive, the implication defining T$_-$ is vacuous. The same input gives LP-goodness whenever $\operatorname{ex}(H) \geq 2$.

The graph F fails T$_-$. Its positive gap is an integer. If it is at least two, it fails the required gap-one conclusion. If it is one, then $c_0(F) \geq E \geq 4$, and Assumed input 4.5 gives $\operatorname{tr}(F) \geq 1$. Hence

$$\operatorname{tr}(F) + E \geq 5 > 4 = \max\left(4, \left\lfloor \frac{4}{2} \right\rfloor + 1\right).$$

Thus $F \in \mathcal{M}_4$, and Assumed input 4.6 gives $F \in \mathcal{L}_4$.

Conversely, the exact degree-four export description in Assumed input 4.6 gives connected four-regular factor-critical structure, gap one, and excess at least four for every $F \in \mathcal{L}_4$. These are the defining conditions of $\mathcal{Q}_4$. This proves equality and shows that every initial member of $\mathcal{Q}_4$ has gap exactly one.

Now apply the strengthened quartic cut input in Assumed input 4.7 using this exact class membership. It gives well-coveredness, every independent-cut identity and bound, the top-row and higher-zero clauses, and the maximal-complement assertions. None has an upper bound on E. This is the stated extension of the earlier interval $4 \leq E \leq 8$. Since N_4 is emptiness of $\mathcal{L}_4$, the class equality identifies it with emptiness of $\mathcal{Q}_4$, without proving it. □

4.4. **Well-coveredness and covered processes.** In this section, we prove that every maximal independent set is eligible for the covered-process inputs.

[math/blocker] The degree-seven exposure attainer used in the proof of Theorem 4.22 is undefined in the supplied inputs. Assumed input 4.11 states consequences for an attainer X with $|X| \geq 1$, but supplies neither its defining objective and admissible sets nor an existence assertion for the uncovered degree-seven complement with slack two. Choosing such an X requires an exact earlier-series definition and a result guaranteeing a nonempty attainer under these hypotheses.][addressed-partial: the undefined term and missing nonempty-existence premise are marked explicitly at the attainer choice; neither is supplied by the embedded proof.][out-of-scope: thm-body-math; env=proof; label=thm:high-wellcovered]

Theorem 4.22 (Well-coveredness in degrees eight, ten, and twelve)**.** *Let $d \in \{8, 10, 12\}$ and $F \in \mathcal{M}_d$. Then F is well-covered and*

$$c_{d-1}(F) = \alpha(F).$$

For every maximal independent set I, every complete process from $F - I$ is covered at every deletion parent down to its terminal maximum degree at most two.

There is no rank-boundary, excess, order, or independence-number restriction; no extremal process or covered-process hypothesis is imposed. Proper-subgraph minimality includes edge deletions. No whole-degree exclusion or unrestricted size-Ramsey formula follows here.

Proof. The degree-ten and degree-twelve assertions of well-coveredness are included in Assumed input 4.11. We prove the remaining degree-eight assertion.

Put $E = \mathrm{ex}(F)$, $k = \nu(F)$, and $a = \alpha(F)$. Assumed input 4.6 gives connected eight-regular factor-critical structure, order $2k+1$, and $E \geq 6$. Choose a maximum independent set S. Assumed input 4.10 supplies a covered process from $F - S$, so Assumed input 4.11 gives $k - a \leq 2E - 3$. Its near-perfect degree-eight bound gives

$$k \geq 3E - 1 + 2u, \qquad u = \left\lfloor \frac{2E-3}{6} \right\rfloor.$$

Since $E \geq 6$, $u \geq 1$. Subtracting yields

$$a \geq E + 2 + 2u \geq 10. \tag{1}$$

Suppose that I is a smaller maximal independent set. Assumed input 4.11 gives

$$w = |I| = a - 1 \geq 9, \quad R = F - I, \quad \Delta(R) = 7, \quad c_7(R) = 1,$$

$$\mathrm{gap}(R) = 0, \qquad s = \mathrm{ps}(R) \in \{2, 3\}, \qquad c_0(F) = w + \nu(R) + s.$$

Thus no matching covers all degree-seven vertices of R, by Assumed input 4.5. Every induced subgraph of R, including R, is a proper subgraph of F, hence satisfies T_-.

If $s = 3$, the slack-three actual-pool input in Assumed input 4.11 applies: $w \geq 9 \geq 8$, the actual maximum degree is seven, the slack is three, hereditary T_- holds, and the degree-seven vertices are uncovered. It gives $c_0(F) \leq w + \nu(R) + 2$, contradicting the exact identity.

It remains to treat $s = 2$. [math/blocker] [GAP: the degree-seven exposure attainer has no supplied definition of its objective or admissible sets. The supplied proof also gives no argument ensuring a nonempty maximum-cardinality attainer in this slack-two complement. The following choice, and hence the remaining slack-two argument, requires both missing inputs.][addressed-partial: the missing definition and existence premise are explicit; no substitute definition or existence argument is asserted.] Choose a maximum-cardinality degree-seven exposure attainer X, and put $b = |X| \geq 1$. By Assumed input 4.11, the required family $\mathcal{Q}$ has exactly $b + 1$ odd factor-critical components, and the union P of the others satisfies

$$\nu(R) = b + \sum_{Q \in \mathcal{Q}} \nu(Q) + \nu(P). \tag{2}$$

Moreover, $\mathrm{ps}(P) = 0$, there are no edges between X and P, and every $Q \in \mathcal{Q}$ satisfies

$$\nu(Q) \le \alpha(R) - b + 3 \le a - b + 3 = w - b + 4. \tag{3}$$

The middle bound holds because R is induced in F.

Let C be induced by X and all required components. Components of $R - X$ have no edges between them, and there are no edges from X to P. Thus $R = C + P$. The components of $C - X$ are exactly $\mathcal{Q}$, with orders $2\nu(Q) + 1$. Additivity and (2) give

$$\nu(C) = b + \sum_{Q \in \mathcal{Q}} \nu(Q).$$

The graph P and all its induced subgraphs have gap at most one because they are proper subgraphs of F satisfying T_-. The actual-pool lower bound in Assumed input 4.11 applies with exact surplus two and $w + b \ge 2$. Thus

$$m := \max_{Q \in \mathcal{Q}} \nu(Q) \ge w + b + \nu(P) + 5.$$

Together with (3), this would give $2b + \nu(P) \le -1$, impossible for $b \ge 1$. Both slack possibilities are excluded, so F is well-covered.

The equality $c_{d-1}(F) = a$ in all three degrees is Assumed input 4.11. Every maximal independent set is now maximum. Assumed input 4.10 therefore applies to every such set and every complete process, giving coverage at every deletion parent, not merely at the initial parent. No complement is assumed to inherit source minimality. □

Theorem 4.23 (Unrestricted sextic well-coveredness)**.** *Let $F \in \mathcal{M}_6$. Then F is well-covered and*

$$c_5(F) = \alpha(F).$$

For every maximal independent set I, every complete process from $F - I$ is covered at every deletion parent.

There is no excess ceiling, rank slice, order or independence-number bound, extremal-process choice, or pre-existing coverage assumption. The assertion extends the maximum-independent-complement covered-process theorem to all maximal independent complements. It retains proper-subgraph minimality, including edge deletions, and asserts no whole-degree exclusion or unrestricted Ramsey formula.

Proof. Put $a = \alpha(F)$, $k = \nu(F)$, $n = |V(F)|$, and $E = \mathrm{ex}(F)$. Theorem 4.20 identifies F with the exact sextic slice of $\mathcal{L}_{\mathrm{rig}}$ and gives connected six-regular factor-critical structure,

$$n = 2k + 1, \qquad \mathrm{gap}(F) = 1, \qquad E \ge 5.$$

For every nonempty independent I, with $w = |I|$ and $G = F - I$, it gives

$$\mathrm{gap}(G) = 0, \qquad c_i(G) = c_i(F) - w \quad (0 \le i < 6). \tag{4}$$

Every induced subgraph of G is proper in F and satisfies T_-.

Take a maximum independent set S. Maximality gives $\Delta(F - S) \le 5$. If equality holds, Assumed input 4.10 covers every degree-five vertex at the initial parent. A complete process exists because successive vertex deletions terminate. Assumed input 4.5 then gives $c_5(F - S) = 0$. If the degree is at most four, the same conclusion follows from the degree bound. Therefore (4) gives $c_5(F) = a$.

By Assumed input 4.12, a smaller maximal independent set must have size $w = a - 1$. Suppose one exists and put $G = F - I$. It is nonempty as a set I, since F is nonempty. In fact $w \ge 2$. If $w = 1$, its unique vertex dominates F; its degree six gives $n \le 7$, so $k \le 3$. The vertex-cover bound yields

$$E = c_0(F) - k \le n - a - k = k + 1 - a \le k \le 3,$$

contradicting $E \ge 5$.

Again $\Delta(G) \le 5$, and (4) gives $c_5(G) = 1$. Thus the actual degree is five and its degree-five vertices lack simultaneous matching coverage. Put $s = \mathrm{ps}(G) = \mathrm{ex}(G)$. The zero row gives

$$c_0(F) = w + \nu(G) + s. \tag{5}$$

Since $\alpha(G) \le a = w + 1$, we have $w \ge \max(2, \alpha(G) - 1)$. Assumed input 4.12 contradicts uncoveredness when $s \le 1$. For $s = 2$ or 3, its smaller-pool bound contradicts (5). Therefore $s \ge 4$.

The uniform sextic order bound gives

$$\begin{aligned} |V(G)| = n - w &= 4w + (n - 5w) \\ &\le 4w + 23 - 2E \\ &\le 4w + 13 < 4w + 6s + 7. \end{aligned}$$

All marked-carrier hypotheses in Assumed input 4.12 now hold for the actual host F, outer set I, and residual G: the pool has $w \ge 2$, hereditary T_- holds, gap is zero, excess and slack are $s \ge 4$, the cost is (5), the strict order bound holds, and the degree-five vertices are uncovered.

We obtain a carrier C, whole companion $P = G - V(C)$, and a nonempty maximum-cardinality marked-exposure attainer X of size $b \ge 1$. The graph $C - X$ consists of $b + 1$ odd factor-critical components whose vertices all have degree five in C. Its selected largest-matching component Z has boundary one, $\mathrm{ex}(Z) = \mathrm{ps}(Z) = s$, and gap zero. Writing $m = \nu(Z)$, we have

$$m \ge w + b + \nu(P) + 3s - 1. \tag{6}$$

The degree-five deficits inside Z are nonnegative and sum to its boundary one. The connected factor-critical Z inherits the hereditary gap bound from F and is LP-good. The odd-cap rank bound gives

$$m \le \alpha(Z) + 3s - 1. \tag{7}$$

The other b components of $C - X$ are nonempty and have no edges to Z or to one another. A maximum independent set of Z, together with one vertex from each, gives

$$\alpha(Z) + b \le \alpha(G) \le a = w + 1.$$

Thus (7) gives $m \le w + 3s - b$. Comparing with (6) yields $2b + \nu(P) \le 1$, impossible because $b \ge 1$.

Every maximal independent set therefore has size a, and $c_5(F) = a$ was already proved. Assumed input 4.10 now applies to every maximal set and every complete process from its complement. All uses of the source hypothesis retain proper-subgraph minimality, including edge deletion. □

4.5. **The all-degree positive-row and phase interface.** In this section, we derive exact row optima and phase restrictions for sources of degrees six, eight, ten, and twelve. In degree four, the process conclusions are conditional on coverage.

[math/blocker] Assumed input 4.10 guarantees coverage only for source degrees six, eight, ten, and twelve. Quartic well-coveredness does not establish the coverage prerequisite for applying its accounting to a quartic process. Neither $c_3(F - S) = 0$ for every maximal independent S nor coverage throughout every such quartic process is established by the supplied argument. The quartic conclusions of Theorem 4.24(I)–(IV) that depend on coverage therefore remain unproved. A conditional correction must specify coverage for all processes used in those arguments, including processes obtained by choosing alternative phases.][addressed: quartic accounting now assumes coverage of the chosen process; quartic assertions (II)–(IV) assume coverage for every complete process from every maximal independent complement, including alternative phases.][out-of-scope: thm-body-math; env=thm; label=thm:phases]

Theorem 4.24 (All maximal sets and exact positive-row phases)**.** *Let $d \in \{4, 6, 8, 10, 12\}$, $F \in \mathcal{M}_d$, $n = |V(F)|$, and $a = \alpha(F)$. Choose any maximal independent set S and any complete process from $F - S$.*

For $d = 4$, assertions (II)–(IV) *below require the additional hypothesis that every complete process from the complement of every maximal independent set of F is covered at every deletion parent. This includes every process obtained by choosing alternative legal phases.*

(I) *Every source is well-covered. Thus $|S| = a$. For $d \in \{6, 8, 10, 12\}$, every complete process from every maximal independent complement is covered at every deletion parent, and the exact accounting of Assumed input 4.10 holds for all these processes.*

For $d = 4$, if the chosen quartic process is covered at every deletion parent, then the same accounting holds for that process. Under the applicable coverage conclusion or hypothesis,

$$c_{d-1}(F) = a,$$

every positive row below a deletion degree loses one, the actual top row loses zero, the terminal slack is zero, and

$$c_0(F) = |W| + \nu(F - W) + \mathrm{ps}(F - W)$$

for every actual prefix W, including S. The source classes, earlier floors $5, 5, 6, 11, 10$, rank and sector partitions, fixed-rank bounds, and five sufficient emptiness inputs remain unchanged.

(II) *For every integer $1 \le \theta \le d$,*

$$c_\theta(F) = n - h_\theta(F) = \min\{|W| : W \subseteq V(F),\ c_\theta(F - W) = 0\}.$$

For $3 \le \theta < d$, let W_θ consist of S and all roots deleted at degree greater than θ. Then

$$|W_\theta| = c_\theta(F), \qquad F - W_\theta \text{ is } \theta\text{-good.}$$

Every terminal contains an induced θ-good subgraph of order $h_\theta(F)$ for $\theta = 1, 2$. At $\theta = d$, any singleton is an optimal deletion.

For any optimal W, any $A, B \in \mathbb{Z}_{\ge 0}$ with $A + B = \theta$ and $c_{(A,B)}(F) = c_\theta(F)$, and every $Z \subseteq W$,

$$c_{(A,B)}\big(F[(V(F) \setminus W) \cup Z]\big) = |Z|.$$

The maximizing pair may be chosen before W and Z. Every allocation totaling $|Z|$ is feasible; if $Z \neq \varnothing$, at least one allocation totaling $|Z| - 1$ is infeasible.

(III) *For $3 \le D \le d - 1$, put*

$$J_D = F - W_D, \quad X_D = \{v \in V(J_D) : \deg_{J_D}(v) = D\}, \quad K_D = J_D[X_D].$$

Then

$$c_i(J_D) = c_i(F) - c_D(F) \qquad (1 \le i \le D).$$

Full degree-D phases have vertex sets exactly the maximal independent sets of K_D, and every ordering of each is legal. For $D \ge 4$, the graph K_D is well-covered with

$$\alpha(K_D) = c_{D-1}(F) - c_D(F).$$

For every independent $I \subseteq V(K_D)$, $K_D - N_{K_D}[I]$ is well-covered of independence number

$$c_{D-1}(F) - c_D(F) - |I|.$$

Empty phase graphs are allowed. These assertions include the sextic degree-four and degree-five phases.

For $D = 3$ and any maximal independent T of K_3, put $R_T = J_3 - T$. Its components belong to

$$K_1,\ P_2,\ P_3,\ P_4,\ P_6,\ C_3,\ C_4,\ C_5,\ C_7.$$

With $o(R_T)$ counting odd cycles and $b(R_T)$ counting positive-even-order paths,

$$|T| + o(R_T) = c_2(F) - c_3(F),$$
$$|T| + \nu(R_T) - b(R_T) = c_1(F) - c_3(F).$$

No unweighted well-coveredness assertion is made for K_3.

(IV) *Let $\varnothing \neq Y \subseteq V(J_D)$ satisfy $\Delta(J_D[Y]) \leq D-1$. Let*

$$B_Y = N_{J_D}(Y) \setminus Y$$

be its entire external neighborhood. If B_Y is an independent subset of X_D, then $J_D[Y]$ is $(D-1)$-good for $D \geq 4$. For $D = 3$, each component of $J_3[Y]$ belongs to the nine-item terminal list.

In particular, for odd $D \geq 5$, this boundary condition forbids an odd-order $(D-1)$-regular induced graph $J_D[Y]$.

In degrees six, eight, ten, and twelve, no coverage premise is imposed. In degree four, the accounting in (I) *requires coverage of the chosen process, and* (II)–(IV) *require the additional universal coverage hypothesis above. No rank-slice, excess-ceiling, or order-bound premise is imposed. No process remainder is assumed minimal. No strict-sector transport, whole-degree exclusion, or unconditional Ramsey formula is asserted.*

Proof. The source classes are those in Assumed input 4.6, with their exact $\mathcal{L}_{\mathrm{rig}}$ identifications. Theorem 4.21 gives quartic well-coveredness, Theorem 4.23 gives sextic well-coveredness, and Theorem 4.22 gives the other three degrees. Hence every maximal S is maximum. For $d \in \{6,8,10,12\}$, Assumed input 4.10 guarantees coverage for every specified process from $F - S$, so its accounting applies. For $d = 4$, that input gives accounting if the specified process is covered at every deletion parent. This proves (I) with the stated distinction between degrees. The class definitions and emptiness inputs do not change.

For the quartic assertions (II)–(IV), assume throughout that every complete process from every maximal independent complement is covered at every deletion parent.

For $d = 6,8,10,12$, the row-optimum input in Assumed input 4.10 gives (II), including its common-capacity-pair quantifier. Its maximum-set and coverage hypotheses hold by (I). We prove the remaining quartic endpoints.

For any finite G and $\theta \geq 1$, $c_\theta(G) = 0$ implies that G is θ-good. The zero-budget certificate at capacities $(\theta, 0)$ forces $\Delta(G) \leq \theta$. The certificate at $(\theta - 1, 1)$ partitions its edges into a graph of degree at most $\theta - 1$ and a matching; every degree-θ vertex meets that matching. This includes $\theta = 1$. The converse is Assumed input 4.5. Vertex insertion therefore gives

$$c_\theta(F) \leq |W| \quad \text{whenever } c_\theta(F - W) = 0.$$

For $d = 4$, Theorem 4.21 gives

$$c_i(F - S) = c_i(F) - a \quad (0 \leq i < 4), \qquad c_4(F) = 1.$$

The maximality of S gives $\Delta(F - S) \leq 3$: every vertex outside S has a neighbor in S, and F is four-regular. If $\Delta(F - S) = 3$, the additional coverage hypothesis supplies a matching covering all degree-three vertices of this initial deletion parent. If $\Delta(F - S) \leq 2$, the empty matching covers all degree-three vertices. Thus $F - S$ is three-good. Assumed input 4.5 gives $c_3(F - S) = 0$, and the cut identity yields $|W_3| = |S| = c_3(F)$. Let t be the number of cubic roots and R the terminal. Exact accounting gives

$$c_i(F) = a + t + c_i(R) \quad (i = 1, 2), \qquad \mathrm{ps}(R) = 0.$$

The terminal input gives $c_2(R) = o(R)$ and $c_1(R) = \nu(R) - b(R)$.

Delete one vertex from every odd-cycle component of R. The remainder consists of paths and even cycles and has a matching covering every degree-two vertex. Even paths and even cycles have alternating perfect matchings; an odd path has an alternating matching leaving only an endpoint uncovered. This deletes exactly $c_2(R)$ vertices to a two-good graph.

For row one, label path vertices consecutively and cycle vertices cyclically. On

$$K_1,\ P_2,\ P_3,\ P_4,\ P_6,\ C_3,\ C_4,\ C_5,\ C_7$$

respectively, delete

$$\varnothing,\ \varnothing,\ \{2\},\ \{2\},\ \{2,5\},\ \{1\},\ \{1,3\},\ \{2,4\},\ \{2,4,6\}.$$

Every retained graph has maximum degree at most one. The deletion counts $0,0,1,1,2,1,2,2,3$ equal the component matching number minus one for positive-even-order paths and equal the matching number otherwise. They sum to $c_1(R)$.

Including the already deleted S and cubic roots gives deletions of exactly $c_1(F)$ and $c_2(F)$. The insertion lower bound proves optimality and the terminal-containment assertion. At $\theta = 4$, factor-criticality gives a perfect matching of $F - v$ for every v. Hence $F - v$ is four-good, and $c_4(F) = 1$ proves singleton optimality. Since zero row cost is equivalent to goodness, these minimum deletions also equal $n - h_\theta(F)$.

For the common-pair assertion, fix an optimal W, a maximizing pair A, B, and $Z \subseteq W$. Put

$$H = F[(V(F) \setminus W) \cup Z].$$

Insertion into $F - W$ gives $c_{(A,B)}(H) \le |Z|$. Inserting $W \setminus Z$ into H gives

$$c_\theta(F) = c_{(A,B)}(F) \le c_{(A,B)}(H) + |W| - |Z|.$$

Since $|W| = c_\theta(F)$, the reverse inequality follows. The exact allocation conclusions follow from the definition of the least feasible total. This completes (II).

For (III), every root before J_D has deletion degree greater than D. The independent-cut identities from Theorems 4.20 and 4.21, followed by the unit losses in (I), give

$$c_i(J_D) = c_i(F) - |W_D| \quad (1 \le i \le D).$$

Part (II) gives $|W_D| = c_D(F)$.

Let T be a maximal independent set of K_D. Every chosen vertex retains degree D in any ordering of T, since no earlier chosen vertex is its neighbor. Every unchosen vertex of X_D loses a neighbor by maximality, while every vertex outside X_D already had smaller degree. Thus each ordering is a full legal phase.

Conversely, a degree-D root must lie in X_D. Two chosen roots cannot be adjacent because the later one's degree would have fallen below D. At termination, each unchosen member of X_D has a chosen neighbor. Hence full phases correspond exactly to maximal independent sets of K_D.

The original prefix, any such phase, and arbitrary later maximum-degree deletions form a complete source process. Under the additional quartic hypothesis, every process obtained in this way is covered at every deletion parent. For $D \ge 4$, its before- and after-phase prefixes are optimal at thresholds D and $D - 1$ by (II). Their difference is T, so

$$|T| = c_{D-1}(F) - c_D(F).$$

This proves well-coveredness. If I is independent in K_D and T_0 is maximal independent in $K_D - N_{K_D}[I]$, then $I \cup T_0$ is maximal independent in K_D: I dominates its removed neighborhood, and T_0 dominates the remaining unchosen vertices. Therefore every such T_0 has the stated size, including an empty remainder.

For $D = 3$, every maximal T completes the source process to R_T, whose slack is zero by (I). The terminal input supplies its component list and the formulas for $c_1(R_T)$ and $c_2(R_T)$. Both rows lose $|T|$ during the cubic phase. Substituting $c_i(J_3) = c_i(F) - c_3(F)$ gives the two weighted identities. They do not force $|T|$ alone to be constant.

For (IV), extend B_Y to a maximal independent set T of K_D by successive additions. Finiteness makes this terminate. We prove $Y \cap T = \varnothing$. A vertex of $Y \setminus X_D$ is outside T. A vertex of $Y \cap X_D$ has total degree D but at most $D - 1$ neighbors in Y, so has an outside neighbor in $B_Y \subseteq T$. Independence excludes that vertex from T.

Every outside neighbor of Y belongs to B_Y and is deleted. Hence $J_D[Y]$ is a union of whole components of $J_D - T$. For $D \ge 4$, part (II) makes the latter graph $(D-1)$-good immediately after

the phase. Restrict its covering matching to these whole components. For $D = 3$, the latter graph is a terminal, giving the stated component list. Finally, an odd-order $(D-1)$-regular graph would require a perfect matching to be $(D-1)$-good, which is impossible on odd order. All arguments use actual adjacency, without assigning source minimality to a remainder. □

4.6. **Residual restrictions and the remaining degree inputs.** In this section, we assemble the additional earlier residual results with their exact conditional and operational scope.

Theorem 4.25 (Sharper residual structure and five remaining inputs)**.** *Use the source classes and costs of Definitions 4.3 and 4.4. For $F \in \mathcal{M}_d$, put*

$$k = \nu(F), \quad a = \alpha(F), \quad E = \mathrm{ex}(F), \quad \lambda = 2E - 3 - (k - a),$$

$$\delta = c_0(F) - c_2(F), \qquad \epsilon = c_0(F) - c_1(F), \qquad p = E - 2 - \lambda.$$

The quantities defined by differences are integers. The terms concerning inherited sectors and frames retain the meanings specified in Assumed input 4.15. Write $K_4 - K_4$ for two disjoint copies of K_4 joined by exactly one edge.

(I) *Every mathematical implication and conditional operational contract of the earlier residual assembly retains its exact hypotheses. Every source is connected, d-regular, factor-critical, and well-covered, of order $2k+1$, with gap one. The excess floors in degree order $4, 6, 8, 10, 12$ are*

$$6,\ 5,\ 7,\ 12,\ 10.$$

Degrees four and six are entirely strict. The entire octic excess-six and rank-zero slices are empty, and every octic source has

$$1 \le \lambda \le E - 3.$$

Both sectors of each of $\mathcal{M}_{10}$ and $\mathcal{M}_{12}$ have $\lambda \ge 3$ and $p \ge 5$. Every octic source-process prefix preserves independence number a. No entire N_d is concluded.

(II) *Let $F \in \mathcal{M}_8$ have $\delta = 0$. Choose any actual row-complement frame (A, T_5, T_4, T_3, R) of word $5^h 4^b 3^h$, where*

$$b = 2\lambda + 3 - 2h > 0,$$

with rows $pC_3 + (a-p)K_1$. For either

$$H = F[A \cup T_5 \cup T_4] \quad \text{or} \quad H = F[R \cup T_3 \cup T_4],$$

every degree-four vertex has a degree-four neighbor, and

$$H \cong (h-b)K_4 + b(K_4 - K_4) + (p-h-b)C_3 + (a-p)K_1.$$

Thus $b \le h$, $h + b \le p$, and the degree-four vertices induce exactly bK_2. A full quartic phase selects one endpoint of the joining edge in each of these b pairs. Its following cubic phase selects one vertex from each of the h surviving copies of K_4. Every component has at most eight vertices and every ordinary cycle has length at most four. The two stages have the same unlabeled isomorphism type; no ambient automorphism or prescribed matching transport is asserted.

For any spanning retained child K with $E(K) \subseteq E(H)$, its complementary batch has full unpaid cost at least one. The minimum over this entire supported family is exactly one. Thus the classification gives no budgeted child supported on the stage.

Separately, let $F \in \mathcal{M}_{12}$ have $\delta = 0$, and let $A = pC_3 + (a-p)K_1$ be any induced row. For any complete row-complement process of word

$$9^h 8^t 7^u 6^\beta 5^u 4^t 3^h$$

with terminal R and degree-j root sets T_j, both $F[A\cup T_9\cup T_8]$ and $F[R\cup T_3\cup T_4]$ are isomorphic to

$$(h-t)K_4 + t(K_4 - K_4) + (p-h-t)C_3 + (a-p)K_1.$$

Here $0 \le t \le h$, $h+t \le p$, and the quartic vertices induce exactly tK_2. Their isolate sets are exactly those of A and R, respectively. Consequently $T_9 \cup T_8$ avoids every A-isolate, and $T_3 \cup T_4$ avoids every R-isolate. These are components of the indicated induced stages only.

(III) *The entire octic excess-seven residual has*

$$(k, a, |V(F)|, \lambda, \delta) = (24, 14, 49, 1, 1), \qquad \epsilon \in \{0, 1\}.$$

Here a row is an induced A with

$$|V(A)| = 23, \quad \Delta(A) \le 2, \quad \mathrm{ps}(A) = 0, \quad \nu(A) = 5, \quad c_1(A) = 5 - \epsilon, \quad c_2(A) = 4.$$

Rows exist, and every such row is an actual source terminal. Every epsilon-zero row is exactly $3C_3 + C_5 + 9K_1$; every epsilon-one row is exactly $4C_3 + P_2 + 9K_1$. All epsilon-zero P_3 rows are excluded.

There is a source-dependent $q_w \in \{0, 1\}$ with

$$(c_0, \ldots, c_8) = (31, 31 - \epsilon, 30, 26 - q_w, 23 + q_w, 23, 18 + \epsilon, 14, 1).$$

Every row-complement process has word 444 if $q_w = 0$, and 543 if $q_w = 1$, with actual reversal between its two rows and three roots.

For every row A and every exterior vertex v, put

$$K = F[V(A) \cup \{v\}], \qquad d_A = |N_F(v) \cap V(A)|.$$

Then

$$3 \le d_A \le 8, \quad \Delta(K) = d_A, \quad \alpha(K) = 14, \quad \nu(K) = 6, \quad \mathrm{ps}(K) = 0,$$

$$c_1(K) = 6 - \epsilon, \quad c_2(K) = 5, \quad c_i(K) = 1 \ (3 \le i < d_A), \quad c_i(K) = 0 \ (i \ge d_A).$$

The interval $3 \le i < d_A$ may be empty. Neither remaining epsilon class nor either word is excluded, and K is not assigned source minimality.

(IV) *Let G be any well-covered graph containing an induced*

$$A = mC_3 + P_3 + zK_1, \qquad m, z \in \mathbb{Z}_{\ge 0}.$$

With $u_{\mathrm{out}} = |V(G) \setminus V(A)|$,

$$u_{\mathrm{out}} \ge 1, \qquad c_2(G) \le u_{\mathrm{out}} + m - 1.$$

In particular, $c_2(G) = u_{\mathrm{out}} + m$ is impossible. No equality for $\alpha(G)$, regularity, or minimality is assumed.

The sextic condition remains separate. For $F \in \mathcal{M}_6$, put $q = \delta - \epsilon$. If some complete source process has positive loss $a - \alpha(R)$, where R is its terminal, then

$$q \ge 1, \qquad k - a \le 85622q - 5766,$$

$$k \le 276200q - 18600, \qquad |V(F)| \le 552400q - 37199.$$

For each positive-loss process, the number of terminal whole triangles and whole single-edge components is at most $2760q - 185$. This is a component count, not a source-order bound.

All native sufficient contact patterns retain their exact deficit, row, and entire-neighborhood hypotheses. Their occurrence in every source is not asserted. The equal-deficit and universal-zero-loss residuals remain separate.

(V) *For a nonempty batch $B \subseteq E(F)$, retain all vertices and define*

$$R_F(B) = \Sigma(F) - \Sigma(F - B), \qquad U_F(B) = R_F(B) - |B|.$$

Call B budgeted if $U_F(B) \le 0$. At every budgeted state of any source with $k \ge 20$, for any matching P in $F - B$ avoiding any forbidden edge set T_e, and any vertex $z_0 \notin V(P)$, the retained residual

$$Q_K = (F - B - T_e) - V(P) - z_0$$

has Tutte deficiency at least

$$2\left\lceil \frac{k-7}{7} \right\rceil.$$

Here

$$\operatorname{def}(Q) = \max_{X \subseteq V(Q)} \big(o(Q - X) - |X|\big),$$

where o counts odd connected components. Every perfect matching of the ambient residual

$$Q_F = (F - T_e) - V(P) - z_0$$

uses at least $\lceil (k-7)/7 \rceil$ actual B-edges. Hence no finite sequence of these budgeted states on the same F reaches a perfectly matchable retained residual.

The four proposed batch tasks in degrees ten and twelve remain distinct: entry into a specified admissible family; an exchange decreasing the full U_F while retaining all active constraints; exclusion of a positive-cost terminal with only whole canonical blocks at its responsible cuts; and legality of roots, forced matchings, and allocation data. The pointwise bound applies whenever any task claims a budgeted state. Failure of a particular batch programme does not imply absence of a source.

The source-realized forced-edge triangle interval, its exact retained deficiency and ambient batch-edge counts, and its physical row-interior ledgers retain their earlier statements. In its ordinary C_7 case, the one-triangle edit can lower full U_F from one to zero while increasing retained deficiency. Its C_5 interval has no budgeted endpoint. For its allowed matching-floor trigger graphs Γ, the C_5 case has $\operatorname{def}_{\mathrm{L}}(\Gamma) = 0$ for every Γ; the C_7 case has

$$\operatorname{def}_{\mathrm{L}}(\Gamma) \le 1 \quad \text{for every allowed } \Gamma, \qquad \max_{\text{allowed } \Gamma} \operatorname{def}_{\mathrm{L}}(\Gamma) = 1.$$

Within a fixed trigger graph,

$$\operatorname{def}_{\mathrm{L}}(\Gamma) = \max_{X} \big(|X| - |N_\Gamma(X)|\big)$$

over its permitted token subsets. This inner maximum and the outer maximum over allowed graphs are distinct. No interval assertion identifies every trigger graph with a native independent-row restoration graph, or supplies arbitrary-family entry or allocation-label transport.

(VI) *Assume exactly the five entire-degree exclusions $N_4, N_6, N_8, N_{10}, N_{12}$. For all positive integers s_f, t_f, and all positive nonincreasing integer lists*

$$n_1 \ge \cdots \ge n_{s_f} \ge 1, \qquad m_1 \ge \cdots \ge m_{t_f} \ge 1,$$

let

$$F_1 = \sum_{i=1}^{s_f} K_{1,n_i}, \qquad F_2 = \sum_{j=1}^{t_f} K_{1,m_j}.$$

Then

$$\widehat{R}(F_1, F_2) = \sum_{l=2}^{s_f + t_f} \max_{\substack{1 \le i \le s_f,\ 1 \le j \le t_f \\ i+j=l}} (n_i + m_j - 1).$$

Here $\widehat{R}$ is the minimum edge count of a finite simple host whose every red/blue edge coloring contains a red F_1 or a blue F_2. These five entire-degree exclusions suffice and remain assumptions.

Proof. We use the earlier inputs in Assumed input 4.15, keeping their stated scope.

[math/blocker] [GAP: Assumed input 4.15 records only the independently supplied floors and three additional statements. The remaining appeals below to source restrictions, stage classifications, process contracts, batch results, and interval statements require the independent prerequisites flagged as missing there. Those appeals remain incomplete proof steps.][addressed-partial: circular assumptions removed; unresolved uses of the earlier residual assembly remain flagged.]

For (I), the earlier residual assembly gives every listed source restriction except the improved quartic floor. If a quartic source had $E = 5$, its connected four-regular factor-critical structure and gap one would satisfy the exclusion in Assumed input 4.15(I). This is impossible. The preceding floor was five, and E is an integer, so the new floor is six. The other four floors and all rank and sector distinctions are unchanged. Retaining earlier implications does not turn descriptions of formerly open cases into existence assertions.

For (II), the octic source, actual rows, process word, and both induced stages have exactly the hypotheses of the earlier four-type stage input. That input excludes every free quartic root, gives the four whole component types with the displayed multiplicities, and proves both phase-choice assertions. A joined pair has eight vertices. Its joining edge is a bridge and lies on no cycle; each K_4 has at most four vertices. The remaining components are triangles and isolates. Hence every ordinary cycle in the stage has length at most four. Equal multiplicities give only the stated unlabeled isomorphism.

The entire supported-child unpaid-cost minimum one is a separate conclusion of the earlier residual assembly, under these unchanged source and stage hypotheses. The stronger classification does not enlarge the supported family. The degree-twelve formula, quartic matching, and literal isolate identifications are its separate degree-twelve assertions. No boundary transport between the degrees is used.

For (III), the earlier assembly gives the excess-seven parameters, row existence, actual reversal, cost vector, word dichotomy, and singleton formulas. The whole-P_3 exclusion of Assumed input 4.15(II) applies under the same proper-subgraph source hypothesis. It removes every epsilon-zero P_3 row, leaving exactly $3C_3{+}C_5{+}9K_1$, while retaining the epsilon-one form $4C_3{+}P_2{+}9K_1$. The singleton conclusion of the earlier assembly applies to both remaining row forms and every exterior vertex, including $d_A = 3$. This is not an application of the equality-sector stage classification.

The generic assertion in (IV) is Assumed input 4.15(III), with its parameters named m, u_{out}. It applies to every well-covered G with the stated induced row and uses no source-row identity. The sextic inequalities are separate conditional conclusions of the earlier residual assembly. Their premise is existence of some positive-loss process. Their terminal estimate concerns only the number of triangles and whole single edges. The generic saving neither removes that existence premise nor changes a terminal component count into a source-order estimate.

For (V), the universal budgeted-state input gives the pointwise Tutte deficiency and actual batch-edge lower bounds with the specified matching, root, and forbidden-edge quantifiers. At $k \geq 20$,

$$2\left\lceil \frac{k-7}{7} \right\rceil \geq 4.$$

Thus every individual retained residual in such a state has positive deficiency. A perfect matching would imply $o(Q{-}X) \leq |X|$ for every X, because each odd component of $Q{-}X$ requires a matching edge to a distinct vertex of X. Hence no state has a perfectly matchable retained residual, and no finite sequence of such states can end with one.

The four batch tasks are separate operational obligations, not additional existence results. Each claimed budgeted state is subject to the same pointwise theorem. The fixed triangle interval and

its costs, ambient matching counts, and physical ledgers are the inherited interval statements, not a new family of operations. Its Hall-type assertion first maximizes over token subsets inside one fixed Γ, and only then permits Γ to vary. It says that every allowed graph has deficiency at most one and that some allowed graph attains one in the C_7 case. It does not say that every allowed graph attains one. The missing full interval data remain identified in Assumed input 4.15.

Finally, the earlier conditional original-list formula, also retained in Theorem 4.20, applies under the same five entire emptiness assumptions. The source classes and the meanings of N_d are unchanged, so it proves (VI). No individual N_d and no converse is established. □

5. Degree six: native rows, contact menus and the sextic cost ledger

5.1. **Degree six.** This chapter develops the native rows of a minimal degree-six obstruction, their actual deletion processes, and the associated contact and component counts. We assume the certificate theory of Companion I and the universal source structure and matching interfaces established earlier in this series. The setup records the precise consequences used below; the subsequent arguments retain the original source graph throughout every row exchange and simultaneous component choice.

Remark 5.1 (Draft status). [frontmatter/blocker] Subject classification and keywords await operator input.

5.2. **Setup.** We fix the native certificate conventions and state explicitly the results assumed from Companion I and the earlier development of this series.

Notation 5.2 (Graphs and integer domains)**.** All graphs are finite and simple. For a graph G, write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for its vertex set, edge set, edge count, matching number, independence number, and maximum degree. Vertex deletion always leaves the induced graph on the remaining vertices. A whole component is an entire connected component.

Every degree capacity, matching budget, least universal total, cost index, cardinality, component count, phase count, and finite optimization variable ranges over the nonnegative integers, unless a different integer range is stated. Generic gaps and explicitly defined differences are integers and may be signed. Paths P_j, cycles C_j, and cliques K_j have j vertices; path indices are positive integers and cycle indices are at least three. Graph sums denote vertex-disjoint unions.

Definition 5.3 (Native certificate costs)**.** For nonnegative integers A_c, B_c, let $c_{(A_c,B_c)}(G)$ be the least nonnegative integer t_c such that every pair of nonnegative integers x_c, y_c with $x_c + y_c = t_c$ permits a partition of $E(G)$ into four parts whose maximum degrees are at most A_c, B_c and whose matching numbers are at most x_c, y_c, respectively. For every nonnegative integer i, put

$$c_i(G) = \max_{\substack{A_c,B_c\in\mathbb{Z}_{\geq 0}\\ A_c+B_c=i}} c_{(A_c,B_c)}(G).$$

We also write $c0 = c_0$, and define

$$\operatorname{ex}(G) = c_0(G) - \nu(G), \qquad \operatorname{ps}(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G), \qquad \operatorname{gap}(G) = \sum_{i\geq 0} c_i(G) - e(G).$$

If $c_0(G) = 0$, put $\operatorname{tr}(G) = 0$. Otherwise, $\operatorname{tr}(G)$ is the greatest nonnegative integer length of a consecutive run of integers $x \in \{0, \dots, c_0(G) - 1\}$ for which $E(G)$ has no partition into two parts of matching numbers at most x and $c_0(G) - 1 - x$. Property T_- is the implication

$$\operatorname{gap}(G) > 0 \quad\Longrightarrow\quad \operatorname{gap}(G) = 1 \quad\text{and}\quad \operatorname{tr}(G) + \operatorname{ex}(G) \leq \max\left\{4, \left\lfloor \frac{\Delta(G)}{2} \right\rfloor + 1\right\}.$$

These are the generic costs used throughout the chapter.

Setup 5.4 (The sextic source)**.** Let F have maximum degree six, violate T_-, and have every proper subgraph satisfy T_-. Proper subgraphs here include arbitrary vertex deletions, arbitrary edge deletions, and their combinations. Put

$$k=\nu(F), \qquad a=\alpha(F), \qquad E=\operatorname{ex}(F),$$
$$\lambda=2E-3-(k-a), \qquad \delta=c_0(F)-c_2(F), \qquad \epsilon=c_0(F)-c_1(F),$$
$$q_{\mathrm{def}}=\delta-\epsilon, \qquad p=E-2-\lambda, \qquad r=k-\lambda-1+\delta,$$
$$s=2\lambda+3-2\delta, \qquad z=a-p-\delta.$$

All these parameters are integers. Their required nonnegativity and the positivity of s are supplied or proved below.

Definition 5.5 (Processes and native rows)**.** A complete source process first deletes any maximal independent set S of F, and then repeatedly deletes a vertex of current maximum degree while that degree is at least three. Its terminal has maximum degree at most two. A complete cubic process on a subcubic graph repeatedly deletes a current degree-three vertex until no such vertex remains.

A native row is any induced subgraph A of F satisfying

$$|V(A)|=r, \quad \Delta(A)\le 2, \quad \mathrm{ps}(A)=0, \quad \nu(A)=p+\delta, \quad c_1(A)=p+\delta-\epsilon=p+q_{\mathrm{def}}, \quad c_2(A)=p.$$

An exposed port of a component C is a vertex v with $\nu(C-v)=\nu(C)$.

For a native row A, an exterior vertex with exactly three A-neighbors is pure if its entire A-neighborhood is one whole triangle of A; otherwise it is nonpure. Equivalently, such exterior vertices are the cubic vertices of $F-V(A)$, once six-regularity is available.

Definition 5.6 (Component constructions)**.** For a graph B, the corona $\mathrm{cor}(B)$ attaches one distinct private pendant leaf to every vertex of B. The graph B_{ij} consists of disjoint cycles of orders i,j, joined by exactly one bridge. Define S_6 on distinct vertices d_1,d_2,d_3,d_4,x,i_0 by the exact edge set

$$\{d_1d_2,d_2d_3,d_3d_4,xd_1,xd_3,xi_0\}.$$

Thus S_6 is a square with private leaves at two adjacent square vertices.

The following setups record assumed statements, rather than additional results proved in this chapter. Their restricted formulations are the ones used in the supplied arguments.

[cite/blocker] Exact locations in Companion I, [I, Theorem number pending], and Chapter 1 were not supplied for the recalled statements below. No bibliography key or theorem number has been invented.

[math/blocker] [GAP: The complete earlier statements and their proofs were not supplied. The arguments below are conditional on the precise consequences recorded in the following setups. In particular, the terminology of covered processes is used only through the explicitly stated cost identities.]

Setup 5.7 (Generic cost properties)**.** We assume the earlier certificate properties:

(I) If $c_0(G)>0$, a missing matching-budget split exists at total $c_0(G)-1$; exchanging the two budgets preserves missingness.

(II) For a vertex set W and every nonnegative integer i,

$$c_i(G)\le |W|+c_i(G-W).$$

(III) Positive costs are subadditive over disjoint components:

$$c_i(G_1\sqcup G_2)\le c_i(G_1)+c_i(G_2) \qquad (i\ge 1).$$

(IV) Costs vanish at and above the maximum-degree threshold. In particular, c_1 vanishes on a matching with isolates. A zero-total $(3,0)$-certificate implies maximum degree at most three.

(V) For every graph under consideration, $\mathrm{ps}(G)\ge 0$.

Setup 5.8 (Universal source structure and actual prefixes)**.** We assume the earlier source theorem and its actual-prefix consequences. The graph F is connected, six-regular, factor-critical, and well-covered, with

$$|V(F)| = 2k+1, \qquad E \geq 5, \qquad 1 \leq \delta \leq \lambda + 1, \qquad 0 \leq \epsilon \leq \delta.$$

Every maximal independent set has cardinality a. The earlier independent-cut and universal-coveredness statements give optimal actual prefixes and unit losses in the active positive cost rows.

More explicitly, if a source process has terminal Q, has t roots after its initial independent set, and has $h = a - \alpha(Q)$, then

$$\begin{aligned} c_0(F) &= a + t + \nu(Q), \\ c_1(F) &= a + t + c_1(Q), \\ c_2(F) &= a + t + c_2(Q). \end{aligned}$$

The terminal is zero-slack and has only

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7$$

as whole components. Its complement $U = F - V(Q)$ is subcubic and zero-slack, and satisfies

$$\alpha(U) = a, \qquad \nu(U) = E + \lambda + 1 - h - \delta,$$

$$\nu(U) - c_1(U) = \epsilon, \qquad \nu(U) - c_2(U) = \delta.$$

Such a complement U admits a complete cubic process preserving $\alpha(U)$.

If C, B are the lengths of the complete degree-five and degree-four phases of any actual source process, the optimal-prefix identities give

$$c_5(F) = a, \qquad c_4(F) = a + C, \qquad c_3(F) = a + C + B.$$

At an actual cubic entry obtained after deleting a vertex set of size v, its matching number and first two costs are respectively $c_0(F) - v, c_1(F) - v, c_2(F) - v$; its slack is zero and its costs with indices at least three vanish.

Setup 5.9 (Subcubic transport)**.** We assume the earlier zero-slack subcubic transport theorem. If C is subcubic and zero-slack, every complete cubic process with root set X_C and terminal R_C satisfies

$$\begin{aligned} \nu(C) &= |X_C| + \nu(R_C), \\ c_1(C) &= |X_C| + c_1(R_C), \\ c_2(C) &= |X_C| + c_2(R_C). \end{aligned}$$

Every prefix is zero-slack, and the terminal has precisely the nine permitted atom types in Setup 5.8. For a degree-at-most-two zero-slack graph R,

$$c_2(R) = o(R), \qquad c_1(R) = \nu(R) - b(R),$$

where $o(R)$ counts whole odd cycles and $b(R)$ counts positive even-order paths. Consequently,

$$\alpha(R) = |V(R)| - \nu(R) - c_2(R).$$

Complete cubic root sets are exactly the maximal independent sets of the graph induced by the original cubic vertices. If every complete cubic process preserves independence at every prefix, this cubic-induced graph is well-covered. We also assume the generic terminal-isolate statement: if C is subcubic, zero-slack, and $c_1(C) = c_2(C)$, every isolate in every cubic terminal has degree at most two in the original C. This statement does not require source minimality.

Setup 5.10 (Missing-budget bounds)**.** We assume the earlier connected odd-order matching theorem. In the source of Setup 5.4,

$$k \geq 3E - 1.$$

For every missing nonnegative integer pair (b_r, b_s) at total $c_0(F) - 1$, put

$$u_6 = \left\lfloor \frac{2E - 3}{4} \right\rfloor.$$

Then $u_6 \geq 1$ and

$$b_r \leq k - E - u_6.$$

The analogous inequality holds after exchanging the budgets.

Setup 5.11 (Earlier source independence preservation)**.** We assume the earlier universal preservation statement: every complete source process preserves independence a at every prefix, and every actual terminal is a native row. We also assume the earlier exclusion of a whole P_6 from every native row. These statements are used only after the independent construction of a row in Theorem 5.17.

The clean unit-difference version has the same prefix conclusion under $\delta - \epsilon = 1$ and $\epsilon \geq 1$, with no diagonal-source premise. The diagonal version has the same conclusion when $\delta = \epsilon \geq 1$. Neither version assigns source minimality to a proper remainder.

Setup 5.12 (Port matching and localization)**.** We assume the following earlier matching interfaces in their native-row setting. Let A have whole components $K_1, P_2, P_4, C_3, C_4, C_5, C_7$, let Z be its isolates, let D be its entire even-component union, and let p count its odd cycles. Put $u = |V(F) \setminus V(A)|$. Assume

$$\alpha(A) = \alpha(F), \qquad c_0(F) = u + \nu(A).$$

For a missing pair (b_r, b_s), put $j = u - b_r$.

(I) Every exterior subset of cardinality at most j can be assigned by one actual matching to exposed ports of distinct odd whole A-components, together with a maximum internal matching in every A-component.

(II) When $j \geq p + 4$, all exterior vertices avoiding Z can be matched to distinct nontrivial odd components.

(III) Under $j \geq p + 4$, a three-contact vertex with three isolate contacts would imply $c_2(F) \leq u + p - 1$. Under $j \geq p + 3$, the same conclusion follows from two isolate contacts and one contact at an arbitrary vertex of D.

(IV) Let Y be an independent exterior set, each member of which has exactly one isolate contact and no odd-cycle contact. If $|Y| \leq j - p - 1$, then

$$D - N_D(Y)$$

is well-covered with independence number $\alpha(D)$. The set $N_D(Y)$ contains at most one endpoint of each edge of every perfect matching of D.

Here D is bipartite with a perfect matching.

We also assume the earlier degree-free singleton localization statement in the form used below. If a three-contact extension $K = A + x$ satisfies

$$\alpha(K) = \alpha(A), \quad c_1(K) = \nu(A) - \epsilon + 1, \quad c_2(K) = p + 1,$$

and x has exactly one odd-cycle contact, its other two contacts are both endpoints of one whole P_2 or P_4.

Setup 5.13 (The earlier path-companion exchange)**.** For a native row whose possible components are $K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7$, let Z_0 be its isolates and let B_0 be the exterior vertices avoiding Z_0. Form the bipartite actual port-incidence graph Γ from B_0 to the nontrivial odd row components.

We assume the earlier statements that Γ has a matching saturating B_0, that every whole P_3 is covered by each such matching used in the exchange, and that prescribed exposed ports extend to maximum independent sets on the relevant components.

A matched whole C_5 or C_7 is marked when its assigned exterior vertex meets both neighbors of some cycle vertex. Among the finitely many B_0-saturating matchings, choose one maximizing the number of marked long cycles. If a whole P_3 is assigned to b, the earlier unmarked-cycle statement supplies a matched unmarked whole C_5 or C_7, say C, with chosen actual port c, such that

$$N_F(b) \cap V(C) = N_C(c).$$

Setup 5.14 (Earlier component and source estimates)**.** We assume the earlier whole-component catalogue: every native complement is subcubic, every whole component has at most fourteen vertices, and its possible types are the eighteen types in Table 1, together with K_4. The earlier local-count statement gives the five columns of that table for every local cubic choice; the additional K_4 row is

$$(s_C, \epsilon_C, q_C, p_C, z_C) = (1, 0, 0, 1, 0).$$

It also gives simultaneous actual realization of all local choices and the global column sums.

We assume the earlier universal exclusion of a K_4 subgraph in a native complement and of a cubic vertex whose entire row-neighborhood is a whole triangle. The weaker earlier estimate, used separately in the packing argument, says that a whole K_4 has at most two pure vertices relative to the row.

Finally, the earlier source matching-budget estimate gives

$$z - \epsilon \geq q_{\mathrm{def}} + 2 + 2(\lambda - \delta + 1) + 2u_6 \geq q_{\mathrm{def}} + 4.$$

This is an estimate in the original source, with no additional numerical premise.

Setup 5.15 (Covering matching and payment)**.** We assume the earlier retained-vertex interface. For any subset X of the cubic vertices of a native complement, retain all vertices of $A \cup X$ and remove, as exceptional edges, all edges with both ends in X. The resulting spanning graph has maximum degree at most three and has one matching covering every degree-three vertex. Moreover, $|X| \leq r$.

We also assume the generic payment statement at degree total three. Let H be the full induced graph on a retained vertex set, let $P \subseteq E(H)$, and let L be the spanning graph with edge set $E(H) \setminus P$. Assume that L is subcubic and has a matching covering every cubic vertex. If $m = \nu(P)$ and the actual omitted-vertex count N satisfies

$$N \geq \max\{0, m - 1\},$$

then

$$c_3(F) \leq N + m.$$

This conclusion accounts for every omitted-vertex incident edge, including edges joining two omitted vertices, every exceptional edge, and every nonnegative integer capacity and matching-budget split.

Setup 5.16 (Diagonal contacts and overlap)**.** We assume the earlier diagonal contact and overlap statements. In a row consisting of triangles, whole edges, and isolates, every three-contact exterior vertex has one of the following entire row-neighborhoods:

(I) one whole triangle;

(II) one whole edge and one triangle vertex;

(III) one isolate and one endpoint of each of two distinct whole edges.

Distinct cubic exterior vertices touch distinct odd row components. For two nonadjacent such vertices, an edge used by a vertex of the second type is used by no other vertex of the second or third type. Two supports of the third type are disjoint unless they use the same two edges with the same endpoint choices.

For a fixed vertex of the third type, there is at most one nonadjacent exterior companion with the same two endpoint contacts and a distinct isolate contact. This is the precise empty-mask companion consequence used below; its application has $\delta \geq 2$.

5.3. **Uniform early rows and replay.** We first construct a native row without assuming that an original source terminal preserves independence.

Theorem 5.17 (Uniform early rows and arbitrary-row replay)**.** *Let F satisfy Setup 5.4. Native rows exist, and*

$$p \geq 0, \qquad 1 \leq \delta \leq \lambda + 1, \qquad 0 \leq \epsilon \leq \delta, \qquad s \in 2\mathbb{Z}_{\geq 0} + 1.$$

Every native row has independence number a and whole components among

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

For every maximal independent set S and every complete source process beginning with S, the degree-five and degree-four phases have lengths

$$p + \epsilon, \qquad p + \delta - \epsilon.$$

Their union with S induces a native row. In particular,

$$c_5(F) = a, \qquad c_4(F) = a + p + \epsilon, \qquad c_3(F) = r.$$

For every induced native row A, there is a maximum independent set contained in A and a legal source prefix deleting exactly A, with those two phase lengths. Put

$$M = k + E - r.$$

Then $J = F - V(A)$ is subcubic and zero-slack, with

$$\nu(J) = M, \qquad c_1(J) = M - \epsilon, \qquad c_2(J) = M - \delta, \qquad c_i(J) = 0 \quad (i \geq 3).$$

For any source process, let R be its terminal, put $h = a - \alpha(R)$, let A be its early row, and let X be the remaining cubic root set. Then

$$0 \leq h \leq s, \qquad |V(R)| = r + h,$$
$$\nu(R) = p + \delta + h, \qquad c_1(R) = p + \delta + h - \epsilon, \qquad c_2(R) = p + h.$$

Its full degree word is

$$5^{p+\epsilon} 4^{p+\delta-\epsilon} 3^{s-h},$$

where exponents denote repetition. The sets $V(A), V(R), X$ partition $V(F)$. The set X is independent, has cardinality $s - h$, and every member has exactly three neighbors in each of A and R. Both $F - V(A)$ and $F - V(R)$ are subcubic and zero-slack.

Proof. We use the generic properties, source structure, transport, and matching-budget statements in Setups 5.7–5.10. We do not use Setup 5.11.

Step 1. We determine every terminal from its independence loss. Fix any source process with terminal Q, root count t, and $h = a - \alpha(Q) \geq 0$. The actual-prefix identities give

$$\nu(Q) - c_2(Q) = \delta, \qquad \nu(Q) - c_1(Q) = \epsilon.$$

For a zero-slack graph of maximum degree at most two, $c_2(Q)$ counts odd cycles. On paths and even cycles, independence number plus matching number equals the order. On odd cycles, the sum is one less than the order. Hence

$$\alpha(Q) = |V(Q)| - \nu(Q) - c_2(Q).$$

Substitute $|V(Q)| = 2k + 1 - a - t$ and $c_0(F) = k + E$. Solving gives

$$t = 2E - 1 - \delta - h, \qquad |V(Q)| = r + h,$$
$$\nu(Q) = p + \delta + h, \qquad c_1(Q) = p + \delta + h - \epsilon, \qquad c_2(Q) = p + h. \tag{8}$$

No zero-loss terminal is required for these identities.

Step 2. We construct an induced row. Put $U = F - V(Q)$. Setup 5.8 gives

$$\Delta(U) \le 3, \quad \mathrm{ps}(U) = 0, \quad \alpha(U) = a, \quad \nu(U) = E + \lambda + 1 - h - \delta,$$

with matching-to-cost differences ϵ, δ. Choose its independence-preserving cubic process, let A be the terminal, and let v be the root count. Transport preserves these two differences and zero slack. Thus

$$\alpha(A) = a, \qquad |V(A)| = a + 2\nu(A) - \delta.$$

Also,

$$|V(A)| = |V(U)| - v, \qquad \nu(A) = \nu(U) - v,$$

and (8) gives

$$|V(U)| = a + 2E - 1 - h - \delta.$$

Solving these equations yields

$$\nu(A) = E - 2 - \lambda + \delta = p + \delta, \qquad |V(A)| = a + 2p + \delta = r.$$

Therefore $c_2(A) = p$ and $c_1(A) = p + \delta - \epsilon$, so A is a native row. Since $c_2(A) \ge 0$, we obtain $p \ge 0$.

Every graph satisfying the row definition has the nine atom types by Setup 5.9. Its independence identity gives

$$\alpha(A) = r - (p + \delta) - p = a.$$

The bounds on δ, ϵ come from Setup 5.8; they imply that $s = 2\lambda + 3 - 2\delta$ is positive and odd.

Step 3. We bound the first two phases in every source process. Choose a missing split (x, y) with

$$x + y = k + E - 1, \qquad x \le y.$$

Such a split exists by Setup 5.7. Setup 5.10 gives $k \ge 3E - 1$. Since $p \ge 0$, we have $\lambda \le E - 2$, and therefore

$$r \ge k - E + 1, \qquad x \le \frac{k + E - 1}{2} \le k - E + 1 \le r.$$

For every vertex set W of cardinality x,

$$\nu(F - W) \ge y + 1.$$

Otherwise, putting all edges incident with W in the first part and all remaining edges in the second would realize the missing split: a matching in the first part has at most $|W| = x$ edges.

Let L be induced and have at least x vertices outside it. Choose W outside L. A matching in $F - W$ has at most $\nu(L)$ edges internal to L. Each other matching edge uses a distinct vertex outside L. Consequently,

$$\nu(L) \ge y + 1 - (2k + 1 - x - |V(L)|) = |V(L)| - k + E - 1. \tag{9}$$

In an arbitrary source process, let C, B count the degree-five and degree-four roots, put $m = C + B$, and let L be induced by S and these roots. Its complement contains the eventual terminal, whose order is at least $r \ge x$ by (8). Thus

$$\nu(L) \ge m - p.$$

The degree-five roots are independent and each has exactly one neighbor in S. Their S-neighbors are distinct: two independent roots with the same sole S-neighbor could replace that neighbor and enlarge S. Hence S together with the five-roots induces a matching with isolates, whose first cost is zero. Inserting the four-roots gives $c_1(L) \le B$. Insertion from F gives

$$c_1(L) \ge m - p - \epsilon, \qquad c_2(L) \ge m - p - \delta.$$

It follows that $C \le p + \epsilon$.

The degree-four roots are independent and each has exactly two neighbors in S and the earlier five-roots. Therefore

$$e(L) = C + 2B = 2m - C.$$

Nonnegative slack now yields

$$2m - C \geq (m - p) + (m - p - \epsilon) + (m - p - \delta),$$

or

$$m + C \leq 3p + \epsilon + \delta. \tag{10}$$

These inequalities hold for every source process.

Step 4. We replay any prescribed induced row. Label paths consecutively and cycles cyclically. In each component choose S_A, T_5 as follows:

component	S_A	T_5
K_1	$\{1\}$	$\varnothing$
P_2	$\{1\}$	$\{2\}$
P_3	$\{1,3\}$	$\varnothing$
P_4	$\{1,3\}$	$\{4\}$
P_6	$\{1,3,5\}$	$\{6\}$
C_3	$\{1\}$	$\{2\}$
C_4	$\{1,3\}$	$\varnothing$
C_5	$\{1,3\}$	$\{4\}$
C_7	$\{1,3,5\}$	$\{6\}$

Unite these choices over all components. The set S_A has cardinality a, and is therefore maximum in F. The set T_5 is independent and has one vertex for each odd cycle and each positive even-order path. Each member has exactly one neighbor in S_A. The odd-cycle count is p, and $\nu(A) - c_1(A) = \epsilon$ counts the positive even-order paths. Hence

$$|T_5| = p + \epsilon.$$

Delete T_5 in any order as degree-five roots of $F - S_A$. If a further degree-five vertex remained, a completion would contradict $C \leq p + \epsilon$. This exhausts that phase.

Put $T_4 = V(A) \setminus (S_A \cup T_5)$. The displayed choices show that T_4 is independent, has cardinality $p+\delta-\epsilon$, and each member has exactly two A-neighbors already removed. Each is therefore a current degree-four vertex. Delete them in any order. Any completion has $C = p + \epsilon$ and $m \geq 2p + \delta$. Inequality (10) gives $m \leq 2p + \delta$, so the four-phase is also complete.

This prefix deletes exactly A. The optimal-prefix identities now give

$$c_5(F) = a, \qquad c_4(F) = a + p + \epsilon, \qquad c_3(F) = a + 2p + \delta = r.$$

The actual cubic-entry statement in Setup 5.8 gives all asserted properties of $J = F - V(A)$.

Step 5. We identify every early pool. The fixed values of $c_4(F)$ and $c_3(F)$ force

$$C = p + \epsilon, \qquad m = 2p + \delta$$

in every source process. For its early pool L, the earlier lower bounds become

$$\nu(L) \geq p + \delta, \qquad c_1(L) \geq p + \delta - \epsilon, \qquad c_2(L) \geq p.$$

Their sum is

$$3p + 2\delta - \epsilon = e(L).$$

Nonnegative slack forces equality in all three bounds, zero slack, and zero costs at every index at least three. The zero-total $(3, 0)$-certificate gives $\Delta(L) \leq 3$. Also $|V(L)| = r$ and $\alpha(L) = a$, since L contains S.

Take a complete cubic process on L, with v roots and terminal L_0. Transport gives

$$\nu(L_0) = p + \delta - v, \qquad c_2(L_0) = p - v, \qquad |V(L_0)| = r - v,$$

and $\mathrm{ps}(L_0) = 0$. Thus

$$\alpha(L_0) = r - v - (p + \delta - v) - (p - v) = a + v.$$

Since L_0 is induced in L, its independence number is at most a. Hence $v = 0$, and L has maximum degree at most two. It is a native row.

Step 6. We recover the complete degree word and the two-row partition. For the original terminal R, (8) gives $2E - 1 - \delta - h$ total roots. The first two phases use $2p + \delta$ roots, leaving

$$2E - 1 - \delta - h - (2p + \delta) = s - h.$$

Thus $0 \le h \le s$ and the stated word follows. Cubic roots in a subcubic graph are pairwise nonadjacent. Each retains exactly three neighbors in the terminal R. Its source degree is six, so its other three neighbors lie in the early row A. The vertex partition and both complement assertions follow from the replay just proved and Setup 5.8. □

Remark 5.18 (Range of the replay theorem). Theorem 5.17 removes the independence-preserving-terminal premise of the earlier conditional canonical-row argument by constructing an induced row independently. Its proof requires no independently assumed covered process and no source terminal with $h = 0$. The theorem alone neither asserts the existence of such a terminal nor asserts that all terminals have independence a. It assigns no source minimality to either row complement and makes no strict-sector or whole-degree exclusion.

5.4. **Well-covered native rows.** We use the earlier path-companion exchange to remove the remaining odd path from every native row.

Theorem 5.19 (Universal path exclusion and well-covered rows)**.** *Let F satisfy Setup 5.4. Every native row is well-covered and has no whole P_3 or P_6 component. More precisely, its whole-component form is*

$$zK_1 + bP_2 + tP_4 + c_3C_3 + c_4C_4 + c_5C_5 + c_7C_7,$$

with nonnegative integer multiplicities satisfying

$$\begin{gathered} c_3 + c_5 + c_7 = p, \qquad b + t = \epsilon, \\ b + 2t + 2c_4 + c_5 + 2c_7 = \delta, \qquad z = a - p - \delta, \\ q_{\mathrm{def}} = t + 2c_4 + c_5 + 2c_7. \end{gathered} \tag{11}$$

Every maximal independent set of the row is a maximum independent set of F. These assertions apply to every terminal of every complete source process.

Proof. Theorem 5.17 gives rows and $\alpha(A) = a$. Setup 5.8 gives well-coveredness of F. Setup 5.11 excludes P_6 and makes every actual terminal native. Fix a native row A. Suppose that it has a positive integer number of whole P_3 components.

Use Z_0, B_0, Γ from Setup 5.13. The finite nonempty set $\mathcal{M}$ of Γ-matchings saturating B_0 contains a matching maximizing the number $\mu(M)$ of marked matched long cycles. Fix such an M, and fix actual edges realizing all its component assignments.

Choose a whole P_3, say Q, and let b be its assigned exterior companion. Setup 5.13 supplies a matched unmarked whole C_5 or C_7, say C, whose chosen actual port c satisfies

$$N_F(b) \cap V(C) = N_C(c).$$

Let w be the exterior vertex assigned to C. The vertices b, w are distinct, since M assigns distinct left vertices to distinct components. The two vertices of $N_C(c)$ are distinct and nonadjacent.

Construct $S \subseteq V(A)$ componentwise. In Q, choose the center. In every other whole P_3, choose both endpoints. Include every row isolate. In each P_2, P_4, C_4, choose a maximum independent

set. In each odd cycle other than C, choose a maximum independent set containing its assigned actual port if the cycle is matched, and choose any maximum independent set otherwise. These port choices are allowed by Setup 5.13.

On C, choose a maximum independent set containing both neighbors of c. For C_5, label $c = 0$ and choose $\{1, 4\}$. For C_7, label $c = 0$ and choose $\{1, 3, 6\}$. The selected sets are independent and attain the cycle bound $\lfloor |V(C)|/2 \rfloor$.

The whole-component partition makes S independent. Every component choice is maximal. Only the choice in Q falls short of a maximum independent set, and it loses exactly one vertex. Hence S dominates A and

$$|S| = a - 1.$$

An exterior vertex outside B_0 is dominated by a selected isolate. A vertex of $B_0 \setminus \{b, w\}$ is dominated by its selected assigned port. This includes assignments to other P_3's, since both endpoints were selected. The vertex b is dominated by the selected pair $N_C(c)$. Thus only w could extend S.

Extend S to a maximal independent set of F. Well-coveredness forces the extension to have cardinality a. Therefore w has no neighbor in S, and $S \cup \{w\}$ is independent.

The vertex w meets an endpoint of Q. To prove this, suppose that it avoids both endpoints. Replace the center of Q in S by both endpoints, obtaining an independent set S' of cardinality a. The vertex w avoids S and, under the supposition, both new vertices. Then $S' \cup \{w\}$ is independent of cardinality $a + 1$, a contradiction.

Replace the assignments bQ, wC in M by wQ, bC. Use the endpoint contact just proved for wQ, and either contact in $N_C(c)$ for bC. The new assignment M' still saturates B_0. Only two distinct left vertices and two distinct right components have changed. Both new contacts are exposed ports: the endpoints of P_3 are exposed, and every vertex of C_5 or C_7 is exposed.

The cycle C is marked under M', because b meets both neighbors of c. It was unmarked under M. No other long cycle changes assignee, and Q is not a long cycle. Marking depends on the assignee's actual cycle contacts, not on the selected single port edge. Therefore

$$\mu(M') = \mu(M) + 1,$$

contradicting maximality. This exchange takes place in one fixed row and its actual incidence graph; the changed matching is not assumed extremal.

The remaining whole components are $K_1, P_2, P_4, C_3, C_4, C_5, C_7$. They are all well-covered. For K_1, P_2, C_3, this follows directly from their edges. For P_4, C_4, C_5, independence number is two and no single closed neighborhood covers the component. Hence every maximal independent set has cardinality two. For C_7, independence number is three, while the union of the closed neighborhoods of one or two vertices has at most six vertices. Thus no smaller independent set dominates C_7. Maximal independent sets of a disjoint union are exactly the unions of component maximal independent sets. Consequently A is well-covered. Its maximal independent sets have cardinality a, so they are maximum in F.

For the exact counts, Setup 5.9 identifies $c_2(A)$ with the odd-cycle count and $\nu(A) - c_1(A)$ with the positive even-path count. This gives the first two equations in (11). The matching numbers on $P_2, P_4, C_3, C_4, C_5, C_7$ are $1, 2, 1, 2, 2, 3$, respectively. Subtracting the odd-cycle count from $\nu(A) = p + \delta$ gives the third equation. Subtracting $b + t = \epsilon$ then gives the formula for q_{def}. Every nonisolated remaining component has independence number equal to matching number, so

$$a = \alpha(A) = \nu(A) + z = p + \delta + z.$$

This proves the formula for z. Every multiplicity is a count of actual whole components. Setup 5.11 extends the assertions to every actual terminal. □

Remark 5.20 (Scope of the row classification). Theorem 5.19 has no bounded-deficit, rank, order, or no-long-cycle premise. It is a row-structure statement and does not exclude the remaining source class or any whole degree, or assert the full Ramsey formula.

5.5. **Three-contact neighborhoods and singleton exchange.** We turn a singleton attachment into the final cubic step of an actual source process.

Theorem 5.21 (Universal three-contact menu)**.** *Let A be any native row of a source F as in Setup 5.4. Let $x \notin V(A)$ have exactly three A-neighbors, and put $K = F[V(A) \cup \{x\}]$.*

(I) *Every complete cubic process on K deletes exactly one vertex, and its terminal is another native row of the same F. The cubic vertices of K form a clique. Moreover,*

$$\alpha(K) = a, \qquad \nu(K) = p + \delta + 1,$$
$$c_1(K) = p + \delta - \epsilon + 1, \qquad c_2(K) = p + 1,$$
$$\mathrm{ps}(K) = 0, \qquad c_i(K) = 0 \quad (i \geq 3).$$

(II) *The entire A-neighborhood of x belongs to one of the following five necessary families:*
   (a) *one whole row triangle;*
   (b) *one vertex of a whole C_3 or C_5, and both endpoints of one whole P_2 or P_4;*
   (c) *one row isolate and one endpoint of each of two distinct whole P_2's;*
   (d) *one row isolate, a P_2 endpoint, and an internal vertex of a different whole P_4;*
   (e) *one row isolate and one of $\{d_1, d_3\}, \{d_2, d_3\}, \{d_2, d_4\}$ on a whole path $d_1d_2d_3d_4$.*

   In the second family, if the contacted cycle is C_{2i+1} and the path is P_{2j}, where $i, j \in \{1, 2\}$, exchanging x for its unique cycle contact is a reversible actual native-row move replacing exactly

$$C_{2i+1} + P_{2j} \quad \text{by} \quad P_{2i} + C_{2j+1}.$$

(III) *Every such x avoids whole row components C_4, C_7. If $\epsilon = 0$, it also avoids every whole C_5. Consequently every whole C_4 or C_7 of every actual source terminal is already a whole component of its actual cubic entry. The same holds for C_5 when $\epsilon = 0$.*

Proof. Theorems 5.17 and 5.19 give the row structure and replay. Setup 5.11 makes every source terminal native and every complete cubic continuation have s roots. Let Z be the row isolates and D its even-component union. Then D is bipartite with a perfect matching. Put

$$u = |V(F) \setminus V(A)| = r + s.$$

Substitution in the source parameters gives

$$c_0(F) = u + \nu(A), \qquad c_1(F) = u + \nu(A) - \epsilon, \qquad c_2(F) = u + p. \tag{12}$$

Step 1. We prove the actual singleton continuation. Six-regularity makes x cubic in $J = F - V(A)$. Extend $\{x\}$ to a maximal independent set X of the graph induced by the cubic vertices of J. Finiteness permits this extension by successive additions. Since selected vertices are nonadjacent, every ordering of X deletes current degree-three vertices. Maximality ensures that each unselected original cubic vertex loses a selected neighbor, so the process is complete.

Prepend the replay of A. Independence preservation and Theorem 5.17 give $|X| = s$ and a native terminal $R = J - X$. The reverse complement is

$$F - V(R) = F[V(A) \cup X],$$

which is subcubic and zero-slack. Every X-vertex has exactly three A-neighbors.

Replay R and delete $X \setminus \{x\}$ in any order. This reaches exactly K. Any continuation is a continuation of the same actual source prefix. It has only one remaining cubic root, since $s - 1$ of the prescribed s roots have already been deleted. Thus every cubic vertex y of K may be chosen first, and $K - y$ is native. If two cubic vertices were nonadjacent, deleting one would leave the other cubic, requiring two steps. Hence the cubic vertices form a clique.

Transport in Setup 5.9, applied to the deletion of x with terminal A, gives all matching and cost identities in part (1). Since A is induced in K and K is induced in F,

$$a = \alpha(A) \le \alpha(K) \le \alpha(F) = a.$$

Step 2. We classify contacts meeting an odd cycle. Suppose that x meets a whole odd cycle C. Its contacts on C have degree three in K, so they are pairwise adjacent. If there are three, they form a whole triangle. Otherwise there are one or two.

Suppose that there are two, say c, d, which are adjacent. The third contact v lies outside C. It has row degree at most one: otherwise it would be cubic in K and would have to be adjacent to c across distinct row components. Thus v is an isolate or an endpoint of an even path. Delete c, which is a native-row exchange by part (1). The path $C - c$ joins at d through x to v and its path or singleton component. This forms one path and no cycle. All other components are unchanged. The resulting row has $p - 1$ odd cycles, contradicting its native value p.

There is therefore exactly one cycle contact c. The degree-free singleton statement in Setup 5.12 applies: the component catalogue comes from Theorem 5.19, and its independence, first-cost, and second-cost hypotheses were proved in part (1). It places the other two contacts at the endpoints of one whole P_2 or P_4.

Deleting c turns C into the even path $C - c$, and turns the supported path together with x into an odd cycle. If $C = C_7$, the new row contains a whole P_6, contrary to Theorem 5.19. Thus $C = C_3$ or C_5, and the replacement is exactly the one stated.

In the new row, the exterior vertex c has exactly three contacts: the two endpoints of $C - c$ and x. The vertex x has row degree two. Part (1), applied again, permits deleting x after adding c. This is the reverse move.

Step 3. We classify contacts avoiding all odd cycles. Choose a missing pair (b_r, b_s) at total $c_0(F)-1$, and put

$$u_6 = \left\lfloor \frac{2E - 3}{4} \right\rfloor, \qquad j_0 = u - b_r.$$

Setup 5.10 gives

$$j_0 - p \ge u - (k - E - u_6) - p = s + \delta + 1 + u_6 \ge 4. \tag{13}$$

Each row component has matching number equal to the floor of half its order. Every odd-cycle port lies in a maximum independent set of that cycle. The isolate-avoiding matching interface in Setup 5.12 therefore matches all exterior vertices avoiding Z to distinct odd row cycles. A vertex with no odd-cycle contact must consequently meet Z.

Three isolate contacts would imply $c_2(F) \le u+p-1$, by the corresponding localization statement and (13). This contradicts (12). If x has two isolate contacts, the third is not on an odd cycle by Step 2. It lies in D. The two-isolate localization statement gives the same contradiction. Thus x has exactly one isolate contact i and two distinct contacts v, w in D.

Apply Setup 5.12 to the singleton $\{x\}$. Its cardinality is at most $j_0 - p - 1$. We obtain that $D - \{v, w\}$ is well-covered with independence number $\alpha(D)$, and that $\{v, w\}$ contains at most one endpoint of every edge of every perfect matching of D.

Suppose first that v, w lie in different components. Deleting each contact must leave that component well-covered with its independence number unchanged. Otherwise either a component would have maximal independent sets of unequal sizes, or the sum of the nonnegative component independence losses would be positive. The possible components of D are P_2, P_4, C_4. Deleting a P_2 endpoint leaves an isolate. Deleting a P_4 internal vertex leaves an isolate and an edge. Both preserve independence and well-coveredness. Deleting a P_4 endpoint or a C_4 vertex leaves P_3, which has maximal independent sets of cardinalities one and two. Thus contacts on different components are P_2 endpoints or P_4 internal vertices. They cannot both be internal vertices of different P_4's, because they would be nonadjacent cubic vertices of K. This gives the third and fourth families.

Suppose next that both contacts lie in one component. Removing both vertices of P_2 lowers independence. On C_4, adjacent deletion lowers independence from two to one, while opposite contacts would be nonadjacent cubic vertices of K. Neither is possible. On $P_4 = d_1d_2d_3d_4$, the perfect matching $\{d_1d_2, d_3d_4\}$ excludes the pairs $\{d_1, d_2\}$ and $\{d_3, d_4\}$. The endpoint pair $\{d_1, d_4\}$ leaves an edge and lowers independence. The remaining pairs are exactly

$$\{d_1, d_3\}, \quad \{d_2, d_3\}, \quad \{d_2, d_4\}.$$

This gives the fifth family.

No surviving family meets C_4 or C_7. When $\epsilon = 0$, the row has no P_2 or P_4, so the cycle/path family cannot meet C_5.

Finally, take any actual cubic entry H, its complete root set T, and terminal row A. The roots are independent and have exactly three neighbors in A, by Theorem 5.17 and Setup 5.11. The proved avoidance excludes edges from T to the specified terminal cycles. Such a cycle has no edge to another row component either. Since $V(H) = V(A) \cup T$, it is already a whole component of H. □

Remark 5.22 (Necessary contact families). The menu is necessary at all deficits, ranks, orders, and processes. It does not assert occurrence of any listed neighborhood, make a whole-source exclusion, or assign source membership to an induced remainder.

5.6. **Simultaneous free odd ports.** We reserve all owned odd components within one matching request.

Theorem 5.23 (Simultaneous free-port extensions)**.** *Let A be any native row of F, put $H = F - V(A)$, and define*

$$C_A = \{v \in V(H) \mid d_H(v) = 3\}, \qquad L_A = V(H) \setminus C_A.$$

An odd whole A-component is owned if a vertex of C_A has a source-neighbor in it, and free otherwise. Every member of C_A has exactly one odd component neighbor, and every owned component has exactly one cubic owner.

Fix any missing nonnegative integer pair (b_r, b_s) at total $c_0(F) - 1$, and put

$$u = |V(H)|, \qquad j = u - b_r, \qquad u_6 = \left\lfloor \frac{2E - 3}{4} \right\rfloor.$$

Then

$$j \geq p + s + \delta + 1 + u_6 = E + \lambda + 2 - \delta + u_6 \geq E + 1 + u_6 \geq 7.$$

For any $Y \subseteq L_A$, let $\mathcal{O}(Y)$ be the family of owned odd row components touched by Y, and put $m(Y) = |\mathcal{O}(Y)|$. If

$$|Y| + m(Y) \leq j,$$

one actual matching assigns every Y-vertex to an exposed port in a distinct free odd row component, assigns the unique owner of each member of $\mathcal{O}(Y)$ to that component, and includes a maximum internal matching in every A-component.

No independence hypothesis on Y is required. In particular, every noncubic native-complement vertex has a source-neighbor in a free odd component.

Proof. Source structure and Theorem 5.17 give

$$E \geq 5, \quad 0 \leq \lambda \leq E - 2, \quad 1 \leq \delta \leq \lambda + 1, \quad u = k + \lambda + 2 - \delta,$$

and $c_0(F) = u + \nu(A)$. Theorem 5.19 gives the seven row atoms. Each has matching number equal to the floor of half its order. The even components have perfect matchings. Removing any vertex from an odd cycle leaves an even path with a perfect matching; for an isolate the empty matching suffices. Thus every vertex of every odd component is exposed.

Setup 5.10 gives $b_r \le k - E - u_6$. Therefore

$$j \ge E + \lambda + 2 - \delta + u_6 = p + s + \delta + 1 + u_6.$$

Since $\delta \le \lambda + 1$, this is at least $E + 1 + u_6 \ge 7$. Also $j \le u$, because $b_r \ge 0$. The simultaneous matching statement of Setup 5.12 now applies to every exterior subset of cardinality at most j.

Every cubic member of H has exactly three A-neighbors by six-regularity. Theorem 5.21 shows that it has exactly one odd component neighbor. The pure family uses a triangle; the cycle/path family uses one odd cycle; each other family uses exactly one isolate. If two distinct cubic vertices had the same odd destination, the matching statement applied to that pair would require distinct destinations when only one is available. Its size condition holds because $j \ge 7$. Thus every owned odd component has a unique owner. This argument does not require the owners to be nonadjacent.

Fix Y as in the statement, and let $U(Y)$ be the set of owners of the members of $\mathcal{O}(Y)$. Different owned components have different owners, since an owner has only one odd component neighbor. Hence

$$|U(Y)| = m(Y), \qquad Y \cap U(Y) = \varnothing.$$

Apply the same matching theorem to $Y \cup U(Y)$. Each owner must be assigned to its unique owned component, so all members of $\mathcal{O}(Y)$ are reserved. If a Y-vertex were assigned to an owned component, that component would lie in $\mathcal{O}(Y)$ and would already be assigned to its owner. Distinctness excludes this. Every Y-vertex is therefore assigned to a distinct free component, in the same matching that contains all required internal maximum matchings.

For a single $v \in L_A$, at most six owned components are touched, because F is six-regular. Thus

$$|\{v\}| + m(\{v\}) \le 7 \le j.$$

The asserted free contact follows. All matching edges lie in the unchanged source. □

Remark 5.24 (Scope of the port interface). Theorem 5.23 applies to every source deficit, rank, excess, order, and native row. It is a matching interface, with no source-cell exclusion, additional coverage claim, cost-saving claim, or Ramsey formula.

5.7. **Even-path packing and cubic-phase length.** We charge nonpure roots to the positive even paths of a row.

Theorem 5.25 (Uniform even-path packing)**.** *Let A be any native row of F.*

(I) *Let X be any complete independent cubic root set in $F - V(A)$, with actual native terminal $R = (F - V(A)) - X$. At most ϵ members of X are nonpure relative to A.*

If a root has one contact on a whole C_3 or C_5 and both endpoints of a whole P_2 or P_4 as its other contacts, those two components and the root form a whole component of $F - V(R)$. Its type is B_{33}, B_{35}, or B_{55}.

(II) *The number of whole K_4 components of $F - V(A)$ is at most ϵ.*

(III) *The source parameters satisfy*

$$s \le 2\epsilon - 1, \qquad \lambda \le \delta + \epsilon - 2.$$

In particular, $\epsilon = 1$ forces

$$\lambda = \delta - 1, \qquad s = 1,$$

without any bound on $\delta - \epsilon$.

Proof. Theorems 5.17 and 5.19 give rows of independence a, exactly ϵ positive even paths, and positive odd s. Every independent cubic set extends to a maximal independent cubic set. Its vertices, in any order, form a complete cubic process: independence preserves their current degrees, and maximality lowers every remaining original cubic degree. Setup 5.11 makes its terminal native and its length s.

Step 1. We record the pair restrictions for any native row. Let Z be its isolates, D its entire P_2, P_4, C_4 union, and $u = r + s$. The remaining components are p odd cycles. The graph D is bipartite with a perfect matching, and $c_0(F) = u + \nu(A)$.

Choose a missing pair and put $j = u - b_r$. As in (13),

$$j - p \geq s + \delta + 1 + u_6 \geq 4.$$

The matching interface applies to any exterior pair. Thus two three-contact vertices with only one odd destination cannot share that destination. This applies both to vertices avoiding isolates with one odd-cycle destination and to vertices with no odd-cycle contact and one isolate destination. Otherwise two distinct assignments would have only one available odd component.

For an independent pair with exactly one isolate contact each and no odd-cycle contacts, Setup 5.12 also says that the union of its D-neighbors contains at most one endpoint of each edge of every perfect matching of D. The pair-size requirement follows from $2 \leq j - p - 1$. These restrictions concern individual pairs and impose no bound on $|X|$.

Step 2. We prove that cycle/path supports are whole and exclusive. Fix X as in part (1), and put

$$G = F[V(A) \cup X] = F - V(R).$$

This graph is subcubic. Let $x \in X$ meet an odd cycle C at c and an even path P at both endpoints. Theorem 5.21 gives an actual native row

$$A' = (A - c) + x.$$

It turns $P + x$ into an odd cycle and $C - c$ into an even path, and leaves all other components, including all isolates, unchanged.

Take $y \in X \setminus \{x\}$. It is nonadjacent to x. It has no contact on C: the odd-destination restriction excludes this for a cycle-contact vertex, while isolate-bearing vertices have no odd-cycle contact. Thus its three row-neighbors are unchanged in A'.

Suppose that y meets P. A pure vertex cannot meet P. A cycle/path vertex meeting P must use P as its whole path support. In A', it would meet the new odd cycle $P + x$ twice and a distinct unchanged odd cycle once, contrary to Theorem 5.21. An isolate-bearing vertex would retain its isolate contact and acquire an odd-cycle contact on $P + x$, also forbidden by that theorem. Hence no other root meets P, and no other root meets C.

The components C, P are whole in A, and X is independent. Their union with x is consequently whole in G. Its edges form the original C, the new cycle $P + x$, and the single bridge cx. The two cycle orders are three or five. This proves the whole-component assertion and the exclusivity of every cycle/path support.

Step 3. We pack the remaining nonpure roots. Call an isolate-bearing root paired on P_4 if both its even-path contacts lie on one whole $P_4 = d_1d_2d_3d_4$. Its contact pair is one of

$$\{d_1, d_3\}, \qquad \{d_2, d_3\}, \qquad \{d_2, d_4\}.$$

We show that no other isolate-bearing root meets that path.

For the pair $\{d_1, d_3\}$, the perfect-matching restriction forbids another root at d_2, d_4. Subcubicness forbids another root at d_3, since d_3 already has two row-neighbors and the first root. A contact at d_1 would, by the menu, require the same pair $\{d_1, d_3\}$, already excluded. For $\{d_2, d_4\}$, the matching edges forbid contacts at d_1, d_3, the degree bound forbids d_2, and a contact at d_4 would require d_2 as well. For $\{d_2, d_3\}$, the matching edges forbid d_1, d_4, and the degree bound forbids d_2, d_3.

A cycle/path root cannot meet this path, since its own support would then fail the exclusivity just proved. A pure root cannot meet an even path. Thus every paired P_4 support is exclusive.

Remove all exclusive paths and their associated cycle/path or paired-P_4 roots. Every remaining nonpure root is isolate-bearing and meets two distinct remaining paths. Its contacts are either endpoints of two P_2's or a P_2 endpoint and a P_4 internal vertex.

Each remaining path supports at most two roots. For $P_2 = vw$, roots cannot touch both endpoints: distinct roots doing so would violate the matching-edge restriction for vw. Only one endpoint is used, and its row degree one permits at most two root-neighbors in the subcubic graph G. For a remaining P_4, only single internal-vertex contacts remain. Each of its two internal vertices has row degree two and permits at most one root-neighbor.

Let b_{ex} count the removed exclusive paths and let h_{rem} count the remaining nonpure roots. Counting root-to-path incidences gives

$$2h_{\mathrm{rem}} \leq 2(\epsilon - b_{\mathrm{ex}}).$$

The total nonpure-root count is therefore

$$b_{\mathrm{ex}} + h_{\mathrm{rem}} \leq \epsilon.$$

Step 4. We bound pure roots by reversing the frame. For a fixed native A, every whole K_4 in $F - V(A)$ has at most two pure vertices by Setup 5.14. Choose a nonpure vertex in each such whole component. The chosen vertices are independent and extend to a complete cubic phase. Part (1) bounds their number by ϵ, proving part (2).

In the frame (A, R, X), a pure root together with its whole row triangle forms a whole K_4 of $F - V(R)$. The triangle vertices have degree three there, the root has exactly that triangle as its neighborhood, and different pure roots have distinct triangle destinations. Since R is native, part (2) bounds the pure roots by ϵ. Part (1) bounds the nonpure roots by ϵ. Hence $s = |X| \leq 2\epsilon$. The integer s is positive and odd, so

$$s \leq 2\epsilon - 1.$$

Substituting its definition gives $\lambda \leq \delta + \epsilon - 2$. At $\epsilon = 1$, this gives $\lambda \leq \delta - 1$, while $\delta \leq \lambda + 1$ gives the reverse inequality. Thus $\lambda = \delta - 1$ and $s = 1$. □

Remark 5.26 (Range of the packing bound). Theorem 5.25 applies at arbitrary deficit, rank, excess, order, and process. It asserts neither source existence nor a whole-degree exclusion or the full Ramsey formula.

5.8. **The whole-component ledger.** We assemble the earlier whole-component catalogue and local counts while retaining all component padding.

Theorem 5.27 (Whole-component ledger and small-phase skeletons)**.** *Let F satisfy Setup 5.4, with maximum degree exactly six. Fix any native row A, and put $H = F - V(A)$. For each graph T, let $n_H(T)$ count whole H-components isomorphic to T.*

(I) *The graph H is subcubic and has no K_4 subgraph. No cubic vertex of H has its entire A-neighborhood equal to a whole row triangle. Every whole component belongs to the eighteen types in Table 1, and has at most fourteen vertices.*

For a whole component C, let X_C be the root set of any complete local cubic process and let $R_C = C - X_C$. Define $s_C = |X_C|$. Define ϵ_C to count the whole P_2, P_4 components of R_C, define

$$q_C = n_{R_C}(P_4) + 2n_{R_C}(C_4) + n_{R_C}(C_5) + 2n_{R_C}(C_7),$$

let p_C count its whole odd cycles, and let z_C count its isolates. The five values are independent of the local choice and are exactly those in Table 1.

All component choices can be made independently and realized simultaneously in one complete source process. This process first replays the same A, then deletes the union of the local root sets in any order. Its terminal is native in the same F, and

$$\sum_C (s_C, \epsilon_C, q_C, p_C, z_C) = (s, \epsilon, q_{\mathrm{def}}, p, z).$$

(II) *The exact nonnegative path ledger is*

$$\begin{aligned}\epsilon - s = {} & n_H(P_2) + n_H(P_4) + n_H(\mathrm{cor}(P_3)) \\ & + n_H(\mathrm{cor}(P_4)) + n_H(\mathrm{cor}(P_6)).\end{aligned} \tag{14}$$

Consequently,

$$1 \le s \le \epsilon, \qquad s \text{ is odd}, \qquad \delta - 1 \le \lambda \le \left\lfloor \frac{2\delta + \epsilon - 3}{2} \right\rfloor.$$

For a fixed native row, $s = \epsilon$ if and only if its complement has none of the five types on the right of (14)*. If equality holds for one row, it holds for every row. At $q_{\mathrm{def}} = 1$, the right side equals $3\epsilon - 2\lambda - 1$.*

Put $w = n_H(K_1)$ and $u_6 = \lfloor (2E-3)/4 \rfloor$. Then

$$w \ge z - \epsilon \ge q_{\mathrm{def}} + 2 + 2(\lambda - \delta + 1) + 2u_6 \ge q_{\mathrm{def}} + 4. \tag{15}$$

These w vertices are whole original H-isolates, and all six source-neighbors of each lie in A.

(III) *If $\epsilon = 2$, then*

$$s = 1, \qquad \lambda = \delta - 1 = q_{\mathrm{def}} + 1, \qquad r = k.$$

The entire H has exactly one of the following forms, with nonnegative integer padding counts u_3, u_4, u_5, u_7, w:

$$H = C + u_3C_3 + u_4C_4 + u_5C_5 + u_7C_7 + wK_1,$$

where $C \in \{\mathrm{cor}(P_3), \mathrm{cor}(P_4)\}$, or

$$H = D + C + u_3C_3 + u_4C_4 + u_5C_5 + u_7C_7 + wK_1,$$

where $D \in \{P_2, P_4\}$ and

$$C \in \{B_{33}, B_{35}, B_{55}, \mathrm{cor}(C_3), S_6\}.$$

In both forms, C is the unique cubic-bearing whole component.

For the first form, the triples (q_C, p_C, z_C) are

$$\mathrm{cor}(P_3) : (0,0,1), \qquad \mathrm{cor}(P_4) : (1,0,1),$$

and put $q_D = 0$. For the second form, they are

C	B_{33}	B_{35}	B_{55}	$\mathrm{cor}(C_3)$	S_6
(q_C, p_C, z_C)	$(0,1,0)$	$(1,1,0)$	$(2,1,0)$	$(1,0,1)$	$(1,0,1)$.

Put $q_D = 0$ for $D = P_2$ and $q_D = 1$ for $D = P_4$. The remaining exact equations are

$$q_{\mathrm{def}} = q_D + q_C + 2u_4 + u_5 + 2u_7, \qquad p = p_C + u_3 + u_5 + u_7, \qquad z = z_C + w.$$

Proof. We use the earlier catalogue, local-count statement, exclusion, and source estimate recorded in Setup 5.14. Their proper-subgraph source assumptions, native certificate meanings, and integer domains are those of Setups 5.4 and 5.2.

Step 1. We identify the complete remaining catalogue and its local counts. The earlier catalogue contains exactly the eighteen displayed types and K_4, with component order at most fourteen. The universal K_4-subgraph exclusion removes precisely that extra type. The same earlier exclusion gives the entire triangle-contact prohibition. Subcubicness and the order bound are unchanged.

The definitions of the constructions agree with the earlier catalogue. In S_6, the square is $d_1d_2d_3xd_1$, with leaves d_4 at d_3 and i_0 at x. These attachment vertices are adjacent.

The earlier local-count statement gives every entry of Table 1 for this same native A and original complement H. It checks every local root choice. In particular, both possible B_{35} terminals, $P_2 + C_5$ and $P_4 + C_3$, have the same five contributions. The omitted K_4 row cannot occur in the source.

Whole component C	s_C	ϵ_C	q_C	p_C	z_C
K_1	0	0	0	0	1
P_2	0	1	0	0	0
P_4	0	1	1	0	0
C_3	0	0	0	1	0
C_4	0	0	2	0	0
C_5	0	0	1	1	0
C_7	0	0	2	1	0
B_{33}	1	1	0	1	0
B_{35}	1	1	1	1	0
B_{55}	1	1	2	1	0
$\mathrm{cor}(P_3)$	1	2	0	0	1
$\mathrm{cor}(P_4)$	1	2	1	0	1
$\mathrm{cor}(P_6)$	2	3	1	0	2
$\mathrm{cor}(C_3)$	1	1	1	0	1
$\mathrm{cor}(C_4)$	2	2	0	0	2
$\mathrm{cor}(C_5)$	2	2	1	0	2
$\mathrm{cor}(C_7)$	3	3	1	0	3
S_6	1	1	1	0	1

TABLE 1. Exact local contributions for the eighteen necessary whole-component types.

For simultaneous realization, a local complete cubic root set is a maximal independent set in its component's original cubic graph. Different whole components have disjoint vertex sets and no joining edges. The union of their local root sets is therefore a maximal independent set of the entire cubic graph. Every ordering deletes current degree-three vertices and lowers every unselected original cubic vertex. Prepend the actual replay of the same A from Theorem 5.17. The earlier local-count statement and source preservation give s roots, a native terminal, and the five global column sums. Thus independently chosen local processes are realized in one actual source process.

Step 2. We subtract the root column and bound original isolates. In Table 1, $\epsilon_C - s_C$ is one exactly for

$$P_2,\ P_4,\ \mathrm{cor}(P_3),\ \mathrm{cor}(P_4),\ \mathrm{cor}(P_6),$$

and zero for each other type. Summation gives (14). Every summand is a nonnegative integer. Since s is positive and odd, this gives $1 \le s \le \epsilon$. Its definition and $\delta \le \lambda + 1$ give

$$\delta - 1 \le \lambda, \qquad 2\lambda + 3 - 2\delta \le \epsilon,$$

which prove the two bounds on λ.

The right side of (14) vanishes exactly when all five counts vanish. The left side depends only on F, so vanishing for one row is equivalent to vanishing for every row. At $q_{\mathrm{def}} = 1$, substitution of $\delta = \epsilon + 1$ gives

$$\epsilon - s = 3\epsilon - 2\lambda - 1.$$

For every non-isolate type in the table, $z_C \le \epsilon_C$. Summing this inequality and retaining the w original isolate components gives $z \le w + \epsilon$. The source estimate in Setup 5.14 then gives (15). These are original whole H-isolates, rather than vertices that become isolated after a later deletion. Six-regularity puts all six of their source-neighbors in A.

Step 3. We determine the entire complement when $\epsilon = 2$. A positive odd integer $s \le 2$ equals one. Hence

$$\lambda = \delta - 1 = q_{\mathrm{def}} + 1, \qquad r = k.$$

The path ledger has total one, so precisely one of its five component types occurs, once. The type $\mathrm{cor}(P_6)$ has local root count two and is impossible when the global root count is one.

If the distinguished component is $\mathrm{cor}(P_3)$ or $\mathrm{cor}(P_4)$, it uses the unique root unit and both even-path units. All remaining components have zero root and zero even-path contributions. The table leaves exactly C_3, C_4, C_5, C_7, K_1 as padding. This gives the first form.

Otherwise, the distinguished path-ledger component is $D = P_2$ or P_4, with root count zero. Exactly one further component has positive root count, necessarily one. The root-one types are

$$B_{33}, B_{35}, B_{55}, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(C_3), S_6.$$

The two path coronas would add a second path-ledger unit. Excluding them leaves the five choices in the second form. Every further component has zero roots and cannot contribute another P_2 or P_4, so again only the stated cycle and isolate padding remains. Root counts two or three cannot occur. Thus C is the unique cubic-bearing whole component in either form.

The listed triples are the last three columns of the table. The padding contributions are

	C_3	C_4	C_5	C_7	K_1
(q_C, p_C, z_C)	$(0,1,0)$	$(2,0,0)$	$(1,1,0)$	$(2,1,0)$	$(0,0,1)$.

The contribution of P_2 is $(0,0,0)$, and that of P_4 is $(1,0,0)$. Summing proves all three remaining equations. □

Remark 5.28 (Necessary whole-component classification). The component bound concerns each whole component, not the order of F or the number of components. The catalogue and the skeletons classify actual whole components necessarily; they do not assert that any listed type occurs or survives further exclusions. No integer solution of the displayed equations is asserted to be realized by a source. No source existence, strict cost saving, entire deficit-cell exclusion, whole-degree exclusion, or star-forest Ramsey formula is concluded.

5.9. **Odd coronas and the cubic-induced graph.** We retain all cubic vertices simultaneously and account for their internal edges.

Theorem 5.29 (Whole odd-corona exclusion)**.** *Let F satisfy Setup 5.4, put $n = |V(F)|$, and fix any native row A. Define*

$$J = F - V(A), \qquad U = \{x \in V(J) \mid d_J(x) = 3\}, \qquad G = J[U].$$

Let b count the whole $\mathrm{cor}(P_3)$ components of J.

(I) *The graph J has no whole $\mathrm{cor}(C_3), \mathrm{cor}(C_5)$, or $\mathrm{cor}(C_7)$ component, with no restriction on the whole P_4 count of A.*

(II) *The graph G is a vertex-disjoint union of K_1, K_2, P_4, C_4, and is therefore bipartite. Its exact invariants are*

$$|U| = 2s - b, \qquad \nu(G) = s - b, \qquad \alpha(G) = s.$$

Its singleton components are precisely the cubic vertices of the b whole $\mathrm{cor}(P_3)$ components.

(III) *Before exclusion, let o count the three whole odd-corona types together. Then*

$$u = |U| = 2s - b + o, \qquad m = \nu(G) = s - b.$$

Retaining all vertices of A and U, with exceptional edge set $E(G)$, omits exactly

$$N = n - r - u = r + s - u$$

vertices. These satisfy $N \geq s \geq m$, and payment of every omitted-vertex incident edge gives

$$c_3(F) \leq N + m = r - o$$

against the exact source value $c_3(F) = r$.

Proof. Theorem 5.17 makes J an actual cubic entry and gives $c_3(F) = r$. The component and local-count statements, assembled in Theorem 5.27, give the full catalogue without K_4 and local root contributions summing to s. Every complete choice has a native terminal of order r. Thus

$$|V(J)| = r + s, \qquad n = 2r + s.$$

Step 1. We compute the entire cubic-induced graph before excluding odd coronas. The seven degree-at-most-two component types contribute no cubic vertices. Each of

$$B_{33}, B_{35}, B_{55}, \mathrm{cor}(P_4), S_6$$

has two adjacent cubic vertices. Its cubic-induced graph is K_2, with order two, matching number one, and local phase contribution one.

A whole $\mathrm{cor}(P_3)$ has only its middle base vertex cubic. It contributes a singleton, with order one, matching number zero, and phase contribution one. In $\mathrm{cor}(P_6)$, the four internal base vertices induce P_4; its order, matching number, and phase contribution are $4, 2, 2$. In $\mathrm{cor}(C_4)$, the four cubic base vertices induce C_4, with the same three contributions.

For each $h \in \{1, 2, 3\}$, the cubic vertices of $\mathrm{cor}(C_{2h+1})$ induce exactly the base cycle. Its matching number is h: alternating edges avoiding one base vertex attain this value, and a larger matching would require at least $2h + 2$ vertices. Its local phase contribution is h, by Table 1.

Different whole J-components have no joining edges. Matching numbers add over their cubic-induced pieces: restricting a matching gives the upper bound, and uniting maximum component matchings attains it. The cubic-vertex count is twice the phase count, except that each $\mathrm{cor}(P_3)$ subtracts one and each odd corona adds one. The matching contribution equals the phase count, except that each $\mathrm{cor}(P_3)$ subtracts one. Therefore

$$u = 2s - b + o, \qquad m = s - b,$$

and $0 \le m \le s$.

Step 2. We pay for the full retained graph in the original source. Retain

$$H = F[V(A) \cup U],$$

including every source edge between retained vertices. Let $P = E(G)$, and let L be the spanning graph on $V(H)$ with edge set $E(H) \setminus P$. Setup 5.15, applied to the arbitrary cubic subset U, gives

$$\Delta(L) \le 3,$$

one matching covering every cubic vertex of L, and $u \le r$. No independence assumption is imposed on U; all its internal edges belong to P. The matching number of P is exactly m.

The actual omitted-vertex count is

$$N = n - |V(H)| = 2r + s - r - u = r + s - u.$$

Since $u \le r$, we have $N \ge s \ge m$, and therefore $N \ge \max\{0, m - 1\}$. The generic payment statement now applies to the full induced remainder H, the actual omitted vertex set, and the exceptional edges P. It accounts for every omitted-vertex incident edge, including edges between omitted vertices, and gives

$$\begin{aligned} c_3(F) &\le N + m \\ &= r + s - (2s - b + o) + (s - b) \\ &= r - o. \end{aligned}$$

Since $c_3(F) = r$ and $o \ge 0$, it follows that $o = 0$. This proves both the exclusion and its full payment ledger.

Step 3. We compute the remaining invariants. The explicit list from Step 1 now leaves only K_1, K_2, P_4, C_4 in G. These are bipartite: use the singleton in one part, opposite ends for K_2, alternating positions for P_4, and opposite pairs for C_4. Unite these bipartitions across components.

Substituting $o = 0$ gives the stated order and matching number. The independence numbers of K_1, K_2, P_4, C_4 are $1, 1, 2, 2$, respectively. The indicated vertex choices attain them; disjoint edges bound the values on K_2, P_4, C_4. They equal the corresponding local phase contributions. Independence number adds over disjoint components by restricting and uniting independent sets. Hence $\alpha(G) = s$. Only $\mathrm{cor}(P_3)$ contributes a cubic singleton, which proves the final assertion. □

Remark 5.30 (Range of the odd-corona exclusion). Theorem 5.29 applies at every sextic deficit and retains the complete original-source payment. It excludes specified whole components, not the entire source class or the Ramsey formula.

5.10. **The quartic-entry support ledger.** We recover the support of each quartic choice from the whole atoms of its early row.

Theorem 5.31 (Exact quartic-entry support)**.** *Let F satisfy Setup 5.4. Take any maximal independent set S and any full quintic phase T_5 of an actual source process. Put*

$$B = F[S \cup T_5], \qquad H = F - V(B),$$

and let K be the graph induced by all vertices of H-degree four. Then

$$B = (p + \epsilon)P_2 + (a - p - \epsilon)K_1.$$

Every maximal independent set Q of K, in any order, is an actual full quartic phase. It has cardinality $p + q_{\mathrm{def}}$, and $A = F[V(B) \cup Q]$ is a native early row.

Every vertex of K has exactly two B-neighbors and belongs to precisely one of the following pools:

(I) *P_K, whose pure vertices meet both endpoints of one whole B-edge;*

(II) *L_K, whose bridge vertices meet one B-edge endpoint and one whole B-isolate;*

(III) *U_K, whose linker vertices meet two distinct whole B-isolates.*

Contacts in two distinct B-edges are impossible.

For any such Q, let t, c_3, c_4, c_5, c_7 count the whole P_4, C_3, C_4, C_5, C_7 components of A. Then

$$|Q \cap P_K| = c_3,$$
$$|Q \cap L_K| = t + 2c_5 + 2c_7,$$
$$|Q \cap U_K| = 2c_4 + c_7.$$

Consequently,

$$|Q \cap (L_K \cup U_K)| = q_{\mathrm{def}} + c_5 + c_7 \in [q_{\mathrm{def}}, 2q_{\mathrm{def}}],$$

and

$$|Q \cap L_K| + 2|Q \cap U_K| = 2q_{\mathrm{def}} - t \le 2q_{\mathrm{def}}.$$

Every independent set I of K satisfies

$$|I \cap L_K| + 2|I \cap U_K| \le 2q_{\mathrm{def}}.$$

The entire pools obey

$$|L_K| + 2|U_K| \le 10q_{\mathrm{def}}, \qquad |L_K \cup U_K| \le 10q_{\mathrm{def}}, \qquad |U_K| \le 5q_{\mathrm{def}}.$$

Pure vertices supported by any fixed whole B-edge form a clique of order at most five.

Proof. Six-regularity and well-coveredness give $|S| = a$. Theorem 5.17 gives $|T_5| = p + \epsilon$, quartic length $p + q_{\mathrm{def}}$, and a native early row after the two phases. Theorem 5.19 gives the seven row atoms and

$$q_{\mathrm{def}} = t + 2c_4 + c_5 + 2c_7, \qquad c_3 + c_5 + c_7 = p.$$

Step 1. We identify the entire quintic prefix. At its deletion, a quintic root has exactly one previously deleted neighbor, because its source degree is six and its current degree is five. Maximality

of S gives every exterior vertex an S-neighbor. Thus each quintic root has exactly one S-neighbor and no earlier quintic neighbor. The set T_5 is independent.

If two quintic roots had the same sole S-neighbor, replacing that neighbor by both roots would enlarge S. Their S-neighbors are therefore distinct. The entire induced B is a matching of $|T_5|$ edges with $a - |T_5|$ isolates, as claimed.

Step 2. We identify every possible quartic phase. The graph H has maximum degree at most four. A maximal independent set Q of its original quartic graph K can be deleted in any order at current degree four. Once it is deleted, every remaining original degree-four vertex has lost a neighbor, so the remainder is subcubic. This is a full legal quartic phase. Complete it arbitrarily to a source process; Theorem 5.17 gives its length and native early row.

Conversely, an actual full quartic phase is independent in H, since a selected degree-four vertex cannot have lost an earlier selected neighbor in a graph initially of maximum degree four. It is maximal among the original quartic vertices, since no such vertex can remain untouched when the phase ends. This describes every allowed phase. Six-regularity gives exactly two B-neighbors for every vertex of K.

Step 3. We determine the inherited phase counts in each row atom. For a whole component C of A, put

$$i_C = |S \cap V(C)|, \quad m_C = |T_5 \cap V(C)|, \quad y_C = |Q \cap V(C)|.$$

The set S is maximum in A, so $i_C = \alpha(C)$. Otherwise enlarging its choice in C would enlarge S. Also

$$|V(C)| = i_C + m_C + y_C, \qquad e(C) = m_C + 2y_C.$$

The second equation counts the S-to-T_5 matching edges and the two B-edges from every Q-vertex; Q is independent.

Alternating independent vertices attain the path and cycle independence numbers, and consecutive adjacent pairs give the corresponding upper bounds. Substitution in the two equations gives

C	K_1	P_2	P_4	C_3	C_4	C_5	C_7
m_C	0	1	1	1	0	1	1
y_C	0	0	1	1	2	2	3.

Thus every whole row component contains at most one B-edge.

Fix $v \in V(K)$ and extend it to a maximal independent set Q. If its two contacts belonged to distinct B-edges, its whole component in the resulting row would contain both edges. This contradicts the table. Since B is a matching with isolates, the pure, bridge, and linker forms are disjoint and exhaustive.

Step 4. We count all support types. The atoms K_1, P_2 contain no quartic vertex. In C_3, the single quartic vertex joins both ends of the inherited B-edge, so it is pure. In P_4, the inherited B-edge lies at one end, and the quartic vertex joins its nearer endpoint to the remaining B-isolate. This is one bridge.

A C_4 has no B-edge and alternates two quartic vertices with two B-isolates, giving two linkers. In a C_5, the path complementary to its B-edge alternates

$$\text{endpoint},\ Q,\ \text{isolate},\ Q,\ \text{endpoint}.$$

Both quartic vertices are bridges. In a C_7, that complementary path alternates

$$\text{endpoint},\ Q,\ \text{isolate},\ Q,\ \text{isolate},\ Q,\ \text{endpoint}.$$

Its end quartic vertices are bridges and its middle quartic vertex is a linker. These descriptions use every edge and vertex of each whole component.

Summing gives the three exact pool counts. Their nonpure sum is

$$t + 2c_4 + 2c_5 + 3c_7 = q_{\mathrm{def}} + c_5 + c_7.$$

Since $0 \le c_5 + c_7 \le q_{\mathrm{def}}$, the stated interval follows. The weighted sum is

$$t + 4c_4 + 2c_5 + 4c_7 = 2q_{\mathrm{def}} - t \le 2q_{\mathrm{def}}.$$

Every independent $I \subseteq V(K)$ extends to a maximal independent Q. The weighted count is nonnegative and monotone under inclusion, so the same bound holds for I. The graph induced by $L_K \cup U_K$ has maximum degree at most four. Color it with five colors by processing vertices in any order and choosing a color absent from already colored neighbors. At most four colors are forbidden at each step. Each color class is independent in K and has weighted count at most $2q_{\mathrm{def}}$. Summing the five classes gives

$$|L_K| + 2|U_K| \le 10q_{\mathrm{def}}.$$

The remaining two bounds follow from nonnegativity.

Step 5. We bound a fixed pure support. Fix a whole B-edge uv. If two of its pure supporting vertices were nonadjacent, extend them to a maximal independent set Q of K. In the resulting native row, u would have its neighbor v and both supporting vertices, contradicting maximum degree at most two. The support pool is therefore a clique. Its members are common neighbors of u, v outside B. The source vertex u has degree six and already has neighbor v, so the pool has at most five vertices. □

Remark 5.32 (Universality of the support ledger). Theorem 5.31 is universal in the initial independent set, quintic choices, quartic choices, source rank, deficits, order, and excess. It makes no source-class exclusion or Ramsey-formula claim.

5.11. **One nonnegative component defect.** We locate the unit difference $c_1 - c_2$ in one fixed whole component and retain all simultaneous cubic choices.

Theorem 5.33 (Component-defect ledger and simultaneous replacement)**.** *Let F satisfy Setup 5.4, and assume*

$$\delta - \epsilon = 1, \qquad \epsilon \ge 1.$$

A native row has $c_1(A) = p + 1$. Such rows exist. Fix any native A, put $J = F - V(A)$, and for each whole connected component C of J define

$$h_C = c_1(C) - c_2(C), \quad \epsilon_C = \nu(C) - c_1(C), \quad \delta_C = \nu(C) - c_2(C).$$

Then $\mathrm{ps}(C) = 0$, all three quantities are nonnegative integers, and

$$\sum_C h_C = 1, \qquad \sum_C \epsilon_C = \epsilon, \qquad \sum_C \delta_C = \delta.$$

Moreover,

$$c_i(J) = \sum_C c_i(C) \qquad (i \in \mathbb{Z}_{\ge 1}).$$

Exactly one component $C_\star$ has $h_{C_\star} = 1$; all others have $h_C = 0$. These labels depend only on the fixed induced graph J.

Independently choose in each C any maximal independent set X_C of its induced cubic graph, including the empty choice when there are no cubic vertices. Put

$$X = \bigcup_C X_C, \qquad R_C = C - X_C, \qquad R = J - X.$$

Every ordering of X is a complete legal cubic continuation after an actual replay of A. It satisfies $|X| = s$, has native terminal R, and preserves independence a at every source prefix.

Each R_C is zero-slack and has only the nine atoms

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

Writing $n_Q(C)$ for the number of its whole components of type Q, one has

$$h_C = n_{P_3}(C) + n_{P_4}(C) + 2n_{P_6}(C) + 2n_{C_4}(C) + n_{C_5}(C) + 2n_{C_7}(C). \tag{16}$$

Also, ϵ_C counts its positive even-order paths. Thus $R_{C_\star}$ has exactly one whole component among P_3, P_4, C_5, and all its other components belong to K_1, P_2, C_3. Every unmarked R_C consists only of K_1, P_2, C_3.

Every complete cubic process on every C preserves $\alpha(C)$ at every prefix, and each cubic-induced graph of C is well-covered. In an unmarked C, every isolate of every R_C has degree at most two in C.

Local choices in any collection of components can be replaced simultaneously while retaining A, the same induced J, every label h_C, and every source edge. The resulting terminal remains an actual native source terminal.

Proof. Theorem 5.17 gives existence and actual replay of native rows. Put $M = k + E - r$. It gives

$$\nu(J) = M, \quad c_1(J) = M - \epsilon, \quad c_2(J) = M - \delta, \quad c_i(J) = 0 \ (i \geq 3), \quad \mathrm{ps}(J) = 0.$$

Step 1. We derive positive-cost additivity. For each positive integer i, define

$$d_i = \sum_C c_i(C) - c_i(J).$$

Component subadditivity makes d_i a nonnegative integer. Edges add over whole components. Matching numbers add as well, by restricting any matching to the components and uniting component maximum matchings. The slack definitions therefore give

$$\mathrm{ps}(J) = \sum_C \mathrm{ps}(C) + \sum_{i \geq 1} d_i.$$

Only finitely many terms are nonzero, since costs vanish above the maximum-degree threshold. Every summand is nonnegative by Setup 5.7. Since $\mathrm{ps}(J) = 0$, each $\mathrm{ps}(C)$ and each d_i is zero.

Step 2. We compute the nonnegative component defects. Each C is subcubic and zero-slack. For every complete local process, transport gives

$$\begin{aligned} \nu(C) &= |X_C| + \nu(R_C), \\ c_1(C) &= |X_C| + c_1(R_C), \\ c_2(C) &= |X_C| + c_2(R_C). \end{aligned}$$

Setup 5.9 gives the nine terminal atoms and

$$c_2(R_C) = o(R_C), \qquad c_1(R_C) = \nu(R_C) - b(R_C).$$

Consequently,

$$\epsilon_C = b(R_C), \quad \delta_C = \nu(R_C) - o(R_C), \quad h_C = \nu(R_C) - o(R_C) - b(R_C).$$

Alternating edges attain matching number equal to the floor of half the order on each path or cycle, and disjoint endpoints bound every matching by that value. Subtracting one for an odd cycle and one for a positive even-order path gives the h_C-weights

$$0, 0, 1, 1, 2, 0, 2, 1, 2$$

on $K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7$, respectively. This proves (16), the path interpretation of ϵ_C, and

$$h_C \geq 0, \qquad \epsilon_C \geq 0, \qquad \delta_C = \epsilon_C + h_C \geq 0.$$

Cost and matching additivity give

$$\sum_C \epsilon_C = \epsilon, \qquad \sum_C \delta_C = \delta, \qquad \sum_C h_C = \delta - \epsilon = 1.$$

Nonnegative integer summands force exactly one unit component. The definitions use only the fixed C, so the labels are independent of process choices. The marked and unmarked terminal alternatives follow from the displayed atom weights.

Step 3. We realize all local choices simultaneously. Choose the sets X_C independently. Their union is independent because different whole components have no edges between them. Every selected vertex starts cubic in J and loses no earlier selected neighbor. After all selected vertices are removed, every unselected original cubic vertex has lost a neighbor by local maximality. Every other vertex started with degree at most two. Thus every ordering of X is a complete cubic process on the same J.

Prepend the native replay of A. The clean unit-difference preservation statement in Setup 5.11 gives independence a at every prefix. Hence the loss h in Theorem 5.17 is zero. Its terminal identities yield

$$|X| = s, \quad |V(R)| = r, \quad \nu(R) = p + \delta, \quad c_1(R) = p + 1, \quad c_2(R) = p,$$

and $\mathrm{ps}(R) = 0$. Thus R is native.

Step 4. We retain preservation and labels under replacement. Prescribe any complete process on one component C, and choose arbitrary complete processes elsewhere. Perform the prescribed component roots first, or interleave them. This is one of the actual source continuations just constructed.

At each prefix J', independence numbers add over its remaining component pieces C'. Each satisfies $\alpha(C') \le \alpha(C)$, while source preservation gives

$$\alpha(J') = \alpha(J) = a.$$

Thus the sum of the nonnegative losses $\alpha(C) - \alpha(C')$ is zero. Each individual loss is zero. Performing the prescribed component first verifies every prefix of that component process. Since it was arbitrary, Setup 5.9 gives well-coveredness of its cubic-induced graph.

For an unmarked C, we have $\mathrm{ps}(C) = 0$, $\Delta(C) \le 3$, and $c_1(C) = c_2(C)$. The generic terminal-isolate statement in Setup 5.9 gives degree at most two in C for every terminal isolate. It requires no source minimality of C.

Replacing any collection of local sets merely changes maximal independent cubic choices in the same whole components. Step 3 realizes the replacements simultaneously. The vertex sets A, J, C and all source edges, including the edges between A and J, stay fixed. Terminal membership may change, but the values h_C cannot. □

Remark 5.34 (Native replacement without inherited minimality). Theorem 5.33 uses no diagonal-source premise, extremal-frame assertion, or source membership of a proper remainder. Its nine-atom accounting is retained as stated; the universal row theorem can impose further restrictions on actual native terminals.

5.12. **Whole complements on the positive diagonal.** We use the diagonal overlap restrictions to identify whole components and all their simultaneous cubic choices.

Theorem 5.35 (Whole-complement normal form on the positive diagonal)**.** *Let F satisfy Setup 5.4, and assume*

$$\delta = c_0(F) - c_1(F) = c_0(F) - c_2(F) \ge 1.$$

Thus $\epsilon = \delta$, and every native row consists of p triangles, δ whole edges, and $z = a - p - \delta$ isolates.

Let $B = B_{33}$ be the barbell formed by two triangles joined by one bridge. Put

$$P^+ = \mathrm{cor}(P_3), \qquad C^+ = \mathrm{cor}(C_4).$$

For every native row A, the entire complement $J = F - V(A)$ is a disjoint union of whole components from

$$K_1,\ K_2,\ C_3,\ K_4,\ B,\ P^+,\ C^+.$$

If their nonnegative integer multiplicities are $w, m, t, q_4, q_B, q_P, q_C$, respectively, then

$$\begin{aligned}\delta &= m + q_B + 2q_P + 2q_C,\\ s &= q_4 + q_B + q_P + 2q_C,\\ p &= t + q_4 + q_B,\\ z &= w + q_P + 2q_C.\end{aligned}$$

Every complete cubic process on J independently chooses one vertex of each K_4, one bridge endpoint of each B, the unique cubic vertex of each P^+, and one of the two opposite pairs of cycle vertices of each C^+. Its terminal is again a native row. The cubic-induced graph of J is a disjoint union of K_4, K_2, K_1, C_4, with these respective whole-component origins.

Every row and process in this statement is actual in the original source. No source-minimality assumption is inherited by J.

Proof. The row count equations in Theorem 5.19, with $q_{\mathrm{def}} = 0$, give the stated triangle-edge-isolate form. We use arbitrary-row replay, diagonal preservation from Setup 5.11, and the contact, overlap, and companion statements in Setup 5.16.

Step 1. We classify the reverse complement of an arbitrary actual frame. Fix a native row Q and a complete cubic process on $F - V(Q)$, with root set X and terminal R. Theorem 5.17 makes this process actual. Diagonal preservation gives another native row R with the same p, δ, z, r, together with

$$|X| = s, \qquad X \text{ independent.}$$

Both complements are subcubic, and every X-vertex has exactly three neighbors in each row. We classify

$$G = F - V(R) = F[V(Q) \cup X].$$

The three possible entire Q-neighborhoods of a root are: a whole triangle; a whole edge and one triangle vertex; or an isolate and one endpoint of each of two different whole edges. Call these types I, II, and III. Every triangle or isolate meets at most one root by odd-component uniqueness.

A type I root and its triangle induce K_4. The triangle has no further row-neighbors, no other root touches it, and X is independent. This is a whole K_4 of G.

For a type II root v, write its contacts as the endpoints b, c of a whole row edge and a vertex t_0 of a whole triangle T. The vertices v, b, c form one triangle, joined by vt_0 to T. The overlap restriction prevents any other type II or III root from using bc; type I roots use only triangles. Odd-component uniqueness prevents another root from touching T. No row edge leaves this six-vertex set. Thus it is a whole barbell component.

Group type III roots by their identical two endpoint contacts. Distinct groups use disjoint whole row edges by the overlap statement. Every group has cardinality at most two. Fix a member x. If there were two other members, both would be nonadjacent to x, and their whole row-neighborhoods would use the same two endpoint contacts and their distinct isolates. They would give two of the companions prohibited by Setup 5.16. Its hypotheses hold, and type III ensures $\delta \geq 2$.

For a group of one root x, write its contacts as an isolate i and endpoints a_1, a_2 of whole edges a_1b_1, a_2b_2. The exact edges on these six vertices are

$$xi,\ xa_1,\ xa_2,\ a_1b_1,\ a_2b_2.$$

This is P^+, with path vertices a_1, x, a_2 and private leaves b_1, i, b_2.

For a group of two roots x, y, let their distinct isolate contacts be i, j. Its exact edges are

$$xa_1,\ xa_2,\ ya_1,\ ya_2,\ xi,\ yj,\ a_1b_1,\ a_2b_2.$$

This is C^+, with cycle order x, a_1, y, a_2 and private leaves i, b_1, j, b_2.

In either case, no other root uses either row edge, and no other root touches the relevant isolates. The roots are independent and the row components are whole. Hence each constructed graph is a

whole component of G. Every unused triangle, edge, or isolate of Q is untouched by X, and remains a whole C_3, K_2, or K_1. These cases account for every root and every row vertex.

Step 2. We classify the originally specified complement. Take $Q = A$ and any complete cubic process with root set X and terminal R. The same X is a complete cubic root set in $F - V(R)$: it is independent, each member has three neighbors in A, and deleting it leaves the degree-at-most-two row A. Since R is native, its arbitrary-row replay makes the reverse process actual. Apply Step 1 with row R and terminal A. This classifies the entire $J = F - V(A)$.

Step 3. We determine all choices and multiplicities. In a whole K_4, every vertex is cubic and any first deletion leaves a triangle, so exactly one vertex is chosen. In a barbell, only the bridge endpoints are cubic. Deleting either lowers the other to degree two and leaves one triangle and one edge. In P^+, only the middle path vertex is cubic; deleting it leaves two edges and one isolate.

In C^+, precisely the four cycle vertices are cubic. Deleting one lowers its two cycle-neighbors to degree two, while its opposite vertex remains cubic and must also be deleted. The terminal has two edges and two isolates. The components K_1, K_2, C_3 have no cubic vertices.

Different whole components have no joining edges. Thus all listed choices are independent, and every combination is realized by a complete cubic process in any interleaving. The terminal is native by replay and diagonal preservation. Counting its whole edges, roots, triangles, and isolates gives the four stated multiplicity equations.

The cubic-induced pieces are K_4 from a whole K_4, K_2 from a barbell, K_1 from P^+, and C_4 from C^+. There are no other cubic vertices. All multiplicities count actual whole components and are nonnegative integers. □

Remark 5.36 (Necessary diagonal normalization). Theorem 5.35 is a source-wide whole-component and simultaneous-choice normalization. Its permitted list is necessary and does not assert occurrence; in particular, the universal K_4 exclusion in Theorem 5.27 remains applicable. The normal form does not exclude the remaining positive diagonal sector or prove the full Ramsey formula.

6. Degree eight: rows, the cubic phase graph and exterior capacities

We study the degree-eight sources arising in the earlier development of this series. The arguments compare their native rows with actual cubic entries, retaining the original exterior edges throughout the localization, exchange, and certificate constructions.

6.1. **Setup.** We fix the certificate notation and state explicitly the results assumed from Companion I and the earlier chapter on universal structure. All graphs are finite and simple. Every capacity, matching budget, cost index, least universal total, degree, cardinality, component count, and phase count belongs to $\mathbb{Z}_{\geq 0}$. Differences of such quantities belong to $\mathbb{Z}$ unless a nonnegative range is stated.

Definition 6.1 (Certificates and the source property)**.** Let G be a finite simple graph and let $A_c, B_c \in \mathbb{Z}_{\geq 0}$. Define $c_{(A_c,B_c)}(G)$ to be the least $t \in \mathbb{Z}_{\geq 0}$ such that, for every $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = t$, the entire edge set of G has a partition into four parts whose respective maximum degrees and matching numbers satisfy

$$\Delta(G_1) \leq A_c, \qquad \Delta(G_2) \leq B_c, \qquad \nu(G_3) \leq x, \qquad \nu(G_4) \leq y.$$

For $i \in \mathbb{Z}_{\geq 0}$, put

$$c_i(G) = \max_{\substack{A_c, B_c \in \mathbb{Z}_{\geq 0} \\ A_c + B_c = i}} c_{(A_c,B_c)}(G), \qquad c_0(G) = c_{(0,0)}(G).$$

Write $\nu(G), \alpha(G), e(G), \Delta(G)$ for the matching number, independence number, edge count, and maximum degree. Define

$$\Sigma(G) = \sum_{i \geq 0} c_i(G), \qquad \mathrm{ex}(G) = c_0(G) - \nu(G),$$

$$\mathrm{ps}(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G), \qquad \mathrm{gap}(G) = \Sigma(G) - e(G).$$

If $c_0(G) = 0$, set $\mathrm{tr}(G) = 0$. Otherwise, $\mathrm{tr}(G)$ is the greatest length of a consecutive run of integers $x \in \{0, \dots, c_0(G) - 1\}$ for which no partition into two edge parts has matching bounds $x, c_0(G) - 1 - x$. The property T_- is

$$\mathrm{gap}(G) > 0 \quad \Longrightarrow \quad \mathrm{gap}(G) = 1 \ \text{ and } \ \mathrm{tr}(G) + \mathrm{ex}(G) \leq \max\left\{4, \left\lfloor \frac{\Delta(G)}{2} \right\rfloor + 1\right\}.$$

A pair that fails the relevant partition condition is called a *missing allocation*.

Setup 6.2 (The original source)**.** Throughout the chapter, F has maximum degree eight, violates T_-, and every proper subgraph of F satisfies T_-. Proper subgraphs include arbitrary edge deletions, vertex deletions, and combinations of the two. Put

$$k = \nu(F), \quad a = \alpha(F), \quad E = \mathrm{ex}(F), \quad \delta = c_0(F) - c_2(F), \quad \epsilon = c_0(F) - c_1(F),$$
$$\lambda = 2E - 3 - (k - a), \qquad p = E - 2 - \lambda, \qquad r = k - \lambda - 1 + \delta,$$
$$s = 2\lambda + 3 - 2\delta, \qquad M = E + \lambda + 1 - \delta, \qquad q = q_8 = c_4(F) - r.$$

These expressions are integers; their ranges are established below or specified in the statements where they are used. The two sectors are $\delta = 0$ and $\delta > 0$. The positive-cost ledger consists of $c_1(F), \dots, c_6(F)$; the full ledger will be determined in Theorem 6.15.

Definition 6.3 (Processes, rows, and good graphs)**.** A *complete source process* first deletes a maximal independent set S of F, then repeatedly deletes an arbitrary vertex of current maximum degree while that degree is at least three. Its *terminal* is the remaining induced graph. A *cubic entry* is the actual induced remainder at the start of its degree-three phase. A word $5^q4^b3^z$ records q deletions of degree five, followed by b deletions of degree four and z deletions of degree three.

A *native row*, or simply a *row*, is an induced graph A satisfying

$$|V(A)| = r, \qquad \Delta(A) \leq 2, \qquad \mathrm{ps}(A) = 0,$$
$$\nu(A) = p + \delta, \qquad c_1(A) = p + \delta - \epsilon, \qquad c_2(A) = p.$$

A graph is d-good, for $d \in \mathbb{Z}_{\geq 0}$, if its maximum degree is at most d and a matching covers all its degree-d vertices.

Notation 6.4 (Components and neighborhoods)**.** Paths P_j, cycles C_j, and cliques K_j are indexed by their vertex orders, with $j \geq 1$ for paths and cliques and $j \geq 3$ for cycles. A *whole* subgraph is an entire connected component. The notation $n_L(H)$ counts whole components of L isomorphic to H. For vertex sets U, V, the notation $e_F(U, V)$ counts edges between them when the sets are disjoint. We write $N_F(Q)$ for the union of open neighborhoods of vertices in Q, and $N_F[Q]$ for $Q \cup N_F(Q)$.

Definition 6.5 (The component constructions)**.** For a finite simple graph B, let $\mathrm{cor}(B)$ be obtained by attaching one distinct private pendant leaf to every vertex of B. For $i, j \in \{3, 5\}$, let B_{ij} consist of disjoint cycles C_i, C_j joined by one bridge. Let $L_\square$ be a four-cycle with one distinct private pendant leaf at each of two adjacent cycle vertices.

We assume, established earlier in this series, the statements in Setups 6.6–6.13. They are prerequisites, not additional conclusions proved in this chapter.

[cite/blocker] The supplied material does not identify the theorem numbers or the precise allocation of these prerequisites between Companion I and the earlier chapters. The requested reference [I, Theorem X] therefore remains unresolved. The labeled setups below state the assumptions explicitly; no bibliographic entry is invented. [deferred: to reference_verify]

Setup 6.6 (General costs and universal source accounting)**.** The following facts are assumed.

(I) For every finite simple graph G, $\mathrm{ps}(G) \geq 0$. Inserting one vertex increases each $c_{(A_c,B_c)}$, and hence each c_i, by at most one.

(II) The source F is connected, eight-regular, factor-critical, well-covered, and has order $2k+1$. Its excess satisfies $E \geq 6$. Every prefix of every source process, including its terminal, has independence number a.

(III) At a source-process parent H after the initial independent-set deletion, let $D = \Delta(H) \geq 3$, and let v be a current maximum-degree vertex. Then

$$c_i(H) = 0 \quad (i \geq D), \qquad c_i(H) - c_i(H - v) = 1 \quad (1 \leq i < D).$$

Also $c_8(F) = 1$.

(IV) If L is an actual cubic entry, its full cubic phases are exactly the maximal independent sets in the graph induced by its degree-three vertices. Every ordering of such a set is legal.

(V) For the source, $c_2(F)$ equals the minimum number of vertices whose deletion leaves a 2-good graph. If such an optimal deletion leaves A_0, and a capacity pair (A_c, B_c) attains $c_2(F)$, then

$$c_{(A_c,B_c)}\big(F[V(A_0) \cup T]\big) = |T| \quad \text{for every } T \subseteq V(F) \setminus V(A_0).$$

(VI) The parameter bounds $p \geq 1$ and $\delta \leq \lambda + 1$ hold.

Setup 6.7 (Conditional row transfer)**.** The earlier degree-eight row-transfer statement applies whenever at least one source terminal preserves independence a. Under that hypothesis,

$$0 \leq \epsilon \leq \delta, \qquad s > 0, \qquad s \text{ is odd.}$$

Rows have independence a, edge count $3p + 2\delta - \epsilon$, and components in

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

They have exactly p odd-cycle components and ϵ positive even-order path components. Every source process has initial phase lengths $p + \epsilon$ at degree seven and $p + \delta - \epsilon$ at degree six. The initial independent set together with those phases induces a row, and

$$c_7(F) = a, \qquad c_6(F) = a + p + \epsilon, \qquad c_5(F) = r.$$

For every prescribed induced row A, the graph $J = F - V(A)$ satisfies

$$|V(J)| = r + s, \quad \nu(J) = M, \quad \mathrm{ps}(J) = 0, \quad e(J) = 3p + 2\delta - \epsilon + 4s,$$

is 5-good, and satisfies

$$c_i(J) = c_i(F) - r \quad (1 \leq i \leq 5), \qquad c_1(J) = M - \epsilon, \quad c_2(J) = M - \delta, \quad c_5(J) = 0.$$

The transfer construction deletes any prescribed row as an initial maximum independent set followed by the degree-seven and degree-six phases. Componentwise, it chooses a maximum independent set in the row, then one vertex with one remaining neighbor for each odd cycle and positive even-order path, and finally the remaining independent vertices of row degree two.

Setup 6.8 (Missing allocations and native localization)**.** The following earlier statements are used in their native source domain.

(I) Put

$$u_8 = \max\left\{0, \left\lfloor \frac{2E - 3}{6} \right\rfloor\right\}.$$

For every missing pair $(x, y) \in \mathbb{Z}^2_{\geq 0}$ at total $c_0(F) - 1$,

$$x \leq k - E - u_8.$$

(II) Let A be an induced graph with independence a, the nine-component list of Setup 6.7, and

$$c_0(F) = u + \nu(A), \qquad u = |V(F) \setminus V(A)|.$$

Let ℓ count its nontrivial odd-order components, and put $j = u - x$ for a missing pair. If $j \geq \ell + 1$, and every exposed port of each nontrivial odd component extends to a maximum independent set of that component, there is one independent set I_{ev} that is maximum on the union of the odd components and satisfies

$$F - N_F[I_{\mathrm{ev}}] = F[D_{\mathrm{ev}}],$$

where D_{ev} is the union of the even-order components. Every maximal independent set of $F[D_{\mathrm{ev}}]$, united with this same I_{ev}, is maximum in F. The even remainder is well-covered, and P_6 is excluded.

(III) For the resulting component list $K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7$, retain the same native identity and missing-pair margin. Let $R = V(F) \setminus V(A)$, let Z_A be the isolates of A, and put

$$B_0 = \{v \in R : N_F(v) \cap Z_A = \varnothing\}.$$

Let $\mathcal{O}$ be the nontrivial odd components. Join $v \in B_0$ to $C \in \mathcal{O}$ if v has an actual neighbor at a vertex whose deletion leaves a perfect matching of C. This incidence graph has a matching saturating B_0, and every such matching covers every P_3.

For each fixed such matching, the alternating closure of the P_3 components gives sets $B \subseteq B_0$ and $\mathcal{H} \subseteq \mathcal{O}$ such that all P_3's lie in $\mathcal{H}$, the incidence neighborhood of B is exactly $\mathcal{H}$, and the matching pairs B bijectively with $\mathcal{H}$. Put

$$Y = D_{\mathrm{ev}} \cup \bigcup_{C \in \mathcal{H}} V(C), \qquad W = Y \cup B.$$

There is an independent set $I \subseteq V(A) \setminus Y$, maximum on every discarded component, with

$$F - N_F[I] = F[W].$$

The graph $F[W]$ is well-covered and has a perfect matching. If h counts the P_3's, $p_{\mathcal{H}}$ counts the cycles in $\mathcal{H}$, $b = |B|$, and $t = \nu(A[Y])$, then

$$b = |\mathcal{H}| = h + p_{\mathcal{H}}, \qquad \nu(F[W]) = t + b, \qquad \alpha(F[W]) = t + h.$$

For every $T \subseteq R \setminus B$ with $|T| \leq j - b$, one matching takes $B \cup T$ to exposed ports in distinct odd components, takes B into $\mathcal{H}$ and T outside $\mathcal{H}$, and is compatible with maximum internal matchings of all components of A. It need not extend the fixed incidence matching.

(IV) When $\alpha(A) = \alpha(F) = a$, every set Q independent in the actual exterior graph satisfies

$$\alpha(A) - \alpha(A - N_F(Q)) \geq |Q|.$$

Setup 6.9 (The earlier path-root restrictions)**.** In the native localization setting of Setup 6.8, fix any incidence matching saturating all of B_0. Let B_P be the roots assigned to P_3 components, and let P_b be the path assigned to $b \in B_P$. Then B_P is independent in F. Each $b \in B_P$ has a nonempty neighborhood on the endpoints of P_b and avoids every other P_3, every P_3 center, D_{ev}, and Z_A. If a cycle D is assigned to w_D, then

$$N_F(B_P) \cap V(D) \subseteq \bigcap_{z \in N_F(w_D) \cap V(D)} N_D(z).$$

No vertex of B_P meets an unmatched cycle. Every vertex of B_P has two contacts on some C_5 or C_7. The family of permitted incidence matchings is finite, so an integer-valued score on its nonempty family has a maximum.

Setup 6.10 (Singleton exchanges and three-contact restrictions)**.** For a native row A and an exterior vertex v with exactly three A-neighbors, put $K = F[V(A) \cup \{v\}]$. Then

$$c_2(K) = p + 1.$$

The cubic vertices of K, namely v and its neighbors of A-degree two, form a clique. Deleting any of these cubic vertices leaves another actual native row of the same source.

The earlier degree-free odd-cycle statement has the following specialization. Assume the row has components in $K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7$, independence a, exactly p odd cycles and ϵ positive even-order paths, and

$$c_1(F) = u + \nu(A) - \epsilon, \qquad u = |V(F) \setminus V(A)|.$$

If v has exactly one neighbor on an odd-cycle component and $c_2(K) \geq p+1$, its other two neighbors are the endpoints of one whole P_2 or P_4.

For an exterior vertex with one isolate neighbor $i(v)$ and two even-component contacts, the earlier internal-contact restrictions exclude C_4 contacts. If a degree-two contact lies on $P_4 = u_1u_2u_3u_4$, its entire mask, after reflection if needed, is one of

$$\{u_2, u_3, i(v)\}, \qquad \{u_2, u_4, i(v)\}, \qquad \{u_2, i(v), w\},$$

where w in the third case is an endpoint of a distinct whole P_2. The first mask is excluded at every admitted ϵ by the earlier full first-cost payment.

Setup 6.11 (Full independent-phase incidence)**.** Let A be a native row with components in $K_1, P_2, P_4, C_3, C_4, C_5, C_7$ Let X be an actual independent full quintic phase in the reverse row-complement process, with $F[V(A) \cup X]$ the corresponding actual cubic entry. A root is *pure* if its entire row mask is a whole triangle. Every nonpure root has exactly two even-path contacts, and each whole P_2 or P_4 receives at most two contacts from nonpure roots.

Each root has one odd-component owner: a whole triangle, a C_3 or C_5 contacted once together with both endpoints of an even path, or a whole isolate together with two even-component contacts. The owners of distinct roots are distinct. Among isolate-bearing roots, all contacts to a fixed whole P_2 lie at one endpoint. An isolate-bearing root meeting u_1 of a whole $P_4 = u_1u_2u_3u_4$ has its other path contact at u_3; the reflected assertion holds at u_4. These conclusions use protection of perfect-matching edges, well-coveredness after singleton projection, and the preserved independence number.

A root with an odd-cycle owner contacted once, together with its supporting even path, forms a whole B_{33}, B_{35}, or B_{55} in the cubic entry. These packets contain no other root.

Setup 6.12 (Earlier payments on the even remainder)**.** Let A be a native row with components in $K_1, P_2, P_4, C_3, C_4, C_5, C_7$, let D be its entire even part, let Z be its isolates, and let

$$U = V(F) \setminus V(A), \quad u = |U|, \quad m = \nu(D).$$

Assume $\alpha(A) = a$, $c_0(F) = u + \nu(A)$, and that a missing pair defines $j = u - x$. The part D is bipartite and has a perfect matching. The following native statements are assumed.

(I) If an exterior vertex has three isolate neighbors and no other row neighbors, then $j \geq p + 4$ implies $c_2(F) \leq u + p - 1$.

(II) If it has two isolate neighbors and one D-neighbor, then $m \geq 1$ and $j \geq p + 3$ imply the same inequality.

(III) If it has exactly one isolate neighbor and two D-neighbors, then the one-isolate-port statement applies to an independent singleton whenever $1 \leq j - p - 1$. It gives

$$D - N_F(v) \text{ well-covered}, \qquad \alpha(D - N_F(v)) = m,$$

and the two contacts cannot contain both endpoints of any edge in any perfect matching of D.

Setup 6.13 (Earlier phase closure and row partition)**.** At $\delta = \lambda$, the earlier complete boundary closure gives

$$q_8 = 1, \qquad \text{common row-complement word } 543.$$

Under

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E - 3, \qquad \epsilon \geq 2, \qquad q \geq 2,$$

the earlier higher-deficit statement excludes a whole K_4 from every actual cubic entry and gives $2 \leq q \leq \epsilon$.

The earlier explicit row partition writes all edges of a native row with components in $K_1, P_2, P_4, C_3, C_4, C_5, C_7$ as a matching together with exactly

$$c_1(A) = \nu(A) - \epsilon$$

pieces of matching number at most one. The componentwise choices are restated in the proof of Theorem 6.25 to specify every edge of the source certificate used there.

Remark 6.14 (Automated assistance). The main results of this chapter were obtained with the assistance of the Ansatz system, an automated proof-search and verification system. Because automated systems have limitations, it is possible that related references have been missed, and comments from experts are welcome.

[acknowledged: A verification statement remains pending; no independent re-verification claim has been added.] [addressed-partial: existing placeholders preserved | deferred: author details and subject classification require information from the operator.] [acknowledged: Compilation and page-range checks remain pending external verification.]

6.2. **Rows and legal reversal.** In this section, we prove that universal independence preservation removes the terminal-preservation hypothesis from the row transfer and determines both orientations of every row-complement process.

Theorem 6.15 (Rows, the common word, and legal reversal)**.** *Let F and its parameters be as in Setup 6.2.*

(I)

$$p \geq 1, \qquad 0 \leq \epsilon \leq \delta \leq \lambda + 1, \qquad s > 0, \quad s \text{ odd}.$$

Rows exist. Every source process has first two phase lengths $p+\epsilon$ at degree seven and $p+\delta-\epsilon$ at degree six, and its initial maximal independent set together with these phases induces a row.

Every row has independence a, edge count $3p + 2\delta - \epsilon$, and components in

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

It has exactly p odd-cycle components and ϵ positive even-order path components. Every induced row is an actual terminal of a complete source process.

(II) *For every row A, put $J = F - V(A)$. Then*

$$\alpha(J) = a, \quad |V(J)| = r + s, \quad \nu(J) = M, \quad \mathrm{ps}(J) = 0,$$

$$e(J) = 3p + 2\delta - \epsilon + 4s,$$

and J is 5-good. Moreover,

$$c_i(J) = c_i(F) - r \quad (1 \leq i \leq 5),$$

$$c_1(J) = M - \epsilon, \qquad c_2(J) = M - \delta, \qquad c_5(J) = 0.$$

Every complete current-maximum-degree process on J has exactly s roots, ends in a row, and preserves independence a at every prefix. Every root deletion loses one matching unit, and every prefix has zero slack.

(III) *The integer $q = q_8$ depends only on F, satisfies*

$$0 \le q \le \lambda + 1 - \delta,$$

and every process in the preceding item has word

$$5^q 4^{s-2q} 3^q.$$

Every complete source process, from any maximal independent complement, has word

$$7^{p+\epsilon} 6^{p+\delta-\epsilon} 5^q 4^{s-2q} 3^q.$$

The source cost profile is

$$\big(c_0(F), \dots, c_8(F)\big) = \big(k+E,\ k+E-\epsilon,\ k+E-\delta,\ r+s-q,\ r+q,\ r,\ a+p+\epsilon,\ a,\ 1\big). \tag{17}$$

Let A be any row, and let a continuation have terminal row R and ordered roots $x_1, \dots, x_s$, with forward deletion degrees $d_1, \dots, d_s$. Then $x_s, \dots, x_1$ is a legal complete current-maximum-degree process of $F - V(R)$, ends at A, and has the same word. In particular,

$$d_i + d_{s+1-i} = 8 \qquad (1 \le i \le s).$$

(IV) *Put $X = \{x_1, \dots, x_s\}$. Then $V(A), V(R), X$ partition $V(F)$, both rows have order r, and*

$$\Delta(F[X]) \le 2, \qquad e_F(X, V(A)) = e_F(X, V(R)) = 4s - e(F[X]).$$

No rank, sector, excess ceiling, order bound, or preserving-terminal premise is imposed.

[out-of-scope: thm-body-math; env=proof; label=thm:rows] [addressed: The proof now establishes matching loss and zero slack before identifying the terminal. The degree-two second-cost identity and independence preservation determine the root count, terminal order, matching number, and costs before the terminal edge count is used.]

Proof. We first discharge the hypothesis of the earlier row transfer, then use its accounting to identify the common word. The reversal will be checked on the actual vertices.

Step 1. The earlier row transfer applies to every source in the statement. Setup 6.6 says that every prefix of every source process preserves independence a, including the terminal. Thus at least one preserving terminal exists, as required by Setup 6.7. The two deficit parameters in that transfer are the present δ, ϵ; its parameters p, r, s, M are unchanged. We reserve q for the number of degree-five roots.

Setups 6.6 and 6.7 give $p \ge 1$, $0 \le \epsilon \le \delta \le \lambda + 1$, and positive odd s. They also give the row list, independence, edge count, component counts, early-row construction, and the values of $c_5(F), c_6(F), c_7(F)$. For $J = F - V(A)$, they give all the displayed identities in the second item except $\alpha(J) = a$ and the subsequent process conclusions.

The canonical construction in Setup 6.7 applies to any prescribed induced row A. It chooses componentwise a maximum independent set in A, then the one-neighbor vertices for the degree-seven phase, and finally the remaining independent degree-two row vertices for the degree-six phase. Thus $F - V(A)$ is an actual source prefix. This use of the construction occurs after its preserving-terminal hypothesis has been discharged.

Fix any complete maximum-degree continuation from J. Prepend the canonical deletion of A. The resulting process is a complete source process, so every prefix, including J and its terminal R, has independence a by Setup 6.6. Let $t \in \mathbb{Z}_{\ge 0}$ be the number of continuation roots. The stopping condition gives $\Delta(R) \le 2$.

Each continuation root loses one matching unit and preserves zero slack. Initially $\mathrm{ps}(J) = 0$. At a parent of maximum degree $D \ge 3$, let $\mu \in \{0, 1\}$ be the matching loss under the root deletion. The edge loss is D. The positive-cost loss is $D - 1$ by Setup 6.6. Consequently,

$$\mathrm{ps}(\text{parent}) - \mathrm{ps}(\text{child}) = D - \mu - (D-1) = 1 - \mu.$$

If the parent has zero slack, nonnegativity of the child's slack forces $\mu = 1$ and zero child slack. Induction proves the claim at every root.

Every continuation deletion has degree at least three, so Setup 6.6 also gives loss one in each of c_1, c_2. Starting from the known values on J, we obtain

$$|V(R)| = r + s - t, \qquad \nu(R) = M - t, \qquad \mathrm{ps}(R) = 0,$$
$$c_1(R) = M - \epsilon - t, \qquad c_2(R) = M - \delta - t.$$

These identities do not assume that R is a row.

Step 2. The degree-two component identities determine the terminal and the number of roots. Let H be any finite simple graph of maximum degree at most two, and let $o(H)$ count its odd-cycle components. Its components are paths, including isolated vertices, and cycles. All its edges fit in one capacity-two part, so

$$c_{(2,0)}(H) = c_{(0,2)}(H) = 0.$$

For every $i \geq 3$, each capacity pair totaling i has at least one capacity at least two. Thus $c_i(H) = 0$ for $i \geq 3$.

We compute the remaining capacity pair totaling two. Remove one edge from each odd-cycle component. The remaining graph consists of paths and even cycles, whose edges alternate between two matchings. The removed edges themselves form a matching because they belong to distinct components. For any $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = o(H)$, put x removed edges in the first budget part and the other y in the second. Together with the two alternating matchings, this is a certificate at capacity $(1, 1)$. Hence $c_{(1,1)}(H) \leq o(H)$.

Conversely, in any certificate at capacity $(1, 1)$, the two parts controlled by degrees are matchings. They cannot cover an entire odd cycle: their colors would have to alternate around a cycle of odd length. Each odd-cycle component therefore contributes an edge to one of the two parts controlled by matching budgets. Choose one such edge from each odd-cycle component. The chosen edges are pairwise vertex-disjoint. Their numbers in the two budget parts are bounded by the respective budgets, so every feasible budget total is at least $o(H)$. Consequently,

$$c_2(H) = c_{(1,1)}(H) = o(H).$$

For a path on $m \geq 1$ vertices and a cycle on $m \geq 3$ vertices, respectively, the component values are

$$\nu(P_m) = \left\lfloor \frac{m}{2} \right\rfloor, \qquad \alpha(P_m) = \left\lceil \frac{m}{2} \right\rceil, \qquad \nu(C_m) = \alpha(C_m) = \left\lfloor \frac{m}{2} \right\rfloor.$$

Alternating edges and vertices attain these values. A matching uses two vertices per edge. Consecutive vertex pairs, with a final singleton when needed, bound the size of an independent set on a path. On a cycle, pairing each independent vertex with its clockwise successor gives disjoint pairs and the stated upper bound. Adding the component values gives

$$\alpha(H) + \nu(H) = |V(H)| - o(H) = |V(H)| - c_2(H).$$

Apply this identity to the terminal R, using $\alpha(R) = a$ and the accounting from Step 1:

$$\begin{aligned} a &= |V(R)| - \nu(R) - c_2(R) \\ &= (r + s - t) - (M - t) - (M - \delta - t) \\ &= r + s - 2M + \delta + t. \end{aligned}$$

The parameter definitions give

$$M - s = E - 2 - \lambda + \delta = p + \delta, \qquad r - 2p - \delta = k - 2E + \lambda + 3 = a.$$

Substitution yields $a = a + t - s$, so $t = s$. The terminal therefore satisfies

$$|V(R)| = r, \qquad \Delta(R) \leq 2, \qquad \mathrm{ps}(R) = 0,$$
$$\nu(R) = p + \delta, \qquad c_1(R) = p + \delta - \epsilon, \qquad c_2(R) = p.$$

It is an induced graph satisfying every native-row condition in Definition 6.3. Finally, the vanishing of $c_i(R)$ for $i \geq 3$, together with zero slack, gives

$$e(R) = \nu(R) + c_1(R) + c_2(R) = 3p + 2\delta - \epsilon = e(A).$$

Thus every continuation has exactly s roots and ends in a row, with its matching losses, independence preservation, and zero slack established before the common-word calculation.

Step 3. The edge and cost ledgers determine one word for every continuation. Let $q_A, b_A, z_A \in \mathbb{Z}_{\geq 0}$ count the roots of degrees five, four, and three. Since R and A have the same edge count,

$$q_A + b_A + z_A = s, \qquad 5q_A + 4b_A + 3z_A = e(J) - e(R) = 4s.$$

Subtracting four times the first identity from the second gives

$$q_A = z_A, \qquad b_A = s - 2q_A.$$

Positive-cost accounting identifies the degree-five count as $c_4(J)$, because $c_5(J) = 0$ and row four loses one exactly at degree-five deletions. It identifies the degree-four count as $c_3(J) - c_4(J)$. Hence

$$q_A = c_4(F) - r = q, \qquad b_A = c_3(F) - c_4(F).$$

Both quantities depend only on F. Vertex deletion cannot increase degrees, so the word is $5^q 4^{s-2q} 3^q$. The middle count is nonnegative and odd, because s is odd. It is therefore at least one, giving

$$0 \leq q \leq \frac{s-1}{2} = \lambda + 1 - \delta.$$

Every source process has the early phases and row from Setup 6.7, so this same word follows its degree-seven and degree-six phases.

The identities just proved give

$$c_4(F) = r + q, \qquad c_3(F) = r + s - q.$$

The definitions of c_0, c_1, c_2, the known values of c_5, c_6, c_7, and $c_8(F) = 1$ now give (17).

Step 4. Complementary degrees give a legal reversal. Fix $A, R, x_1, \ldots, x_s$ as in the statement. Just before x_i is deleted forward, the other remaining vertices are

$$V(R) \cup \{x_{i+1}, \ldots, x_s\}.$$

Just before its proposed reverse deletion, the other remaining vertices are

$$V(A) \cup \{x_1, \ldots, x_{i-1}\}.$$

These sets partition every other vertex of F. Eight-regularity therefore makes the proposed reverse degree exactly $8 - d_i$. Reversing the forward word and subtracting its degrees from eight gives the same word $5^q 4^{s-2q} 3^q$.

It remains to verify current maximality. Since R is a row, $F - V(R)$ is 5-good and canonically realizes a source prefix. Every complete continuation from it has the phase lengths already proved. If $q > 0$, the first q proposed reverse roots have current degree five in a graph of maximum degree at most five. They are legal. After them no degree-five vertex can remain: otherwise that vertex could be deleted and the process completed, giving more than q degree-five roots. If $q = 0$, this same continuation argument shows that no degree-five vertex is present initially.

The graph now has maximum degree at most four. The next $s-2q$ proposed roots have degree four and are legal. If a degree-four vertex remained afterward, completing a process through that vertex would contradict the fixed degree-four count. The remaining q proposed roots have degree three in a graph of maximum degree at most three, so they are legal. Their deletion leaves exactly A, of maximum degree at most two. This proves complete legal reversal and the identity $d_i + d_{s+1-i} = 8$.

Prepend the canonical initial deletion of R to this reversed continuation. The resulting full source process has terminal A. Since A was arbitrary, every induced row is an actual terminal.

Step 5. The root graph and both boundary counts retain all original contacts. The partition and row orders have been established. Each $x \in X$ belongs to both row complements, each of maximum

degree at most five. Its total degree eight forces at least three neighbors in A and at least three in R. Thus $d_{F[X]}(x) \le 2$.

Finally,

$$e(J) = e(R) + e(F[X]) + e_F(X, V(R)).$$

Since $e(J) - e(R) = 4s$, this gives $e_F(X, V(R)) = 4s - e(F[X])$. Applying the same identity to the opposite row complement gives the formula toward A. □

Remark 6.16 (Scope of the row accounting). Theorem 6.15 does not identify $c_0(J)$ with $\nu(J)$. No remainder is assumed to inherit source minimality. The theorem does not exclude the source class and makes no unrestricted Ramsey formula assertion.

6.3. **Native localization and exclusion of three-vertex paths.** The goal of this section is to retain every exterior conflict in the exposed-port localization and then use the resulting incidence matching to exclude whole P_3 components.

Theorem 6.17 (Native exposed-port localization)**.** *Let F satisfy Setup 6.2, and let A be any terminal of any complete source process. Put*

$$R = V(F) \setminus V(A), \qquad u_R = |R|.$$

Let $\ell \in \mathbb{Z}_{\ge 0}$ count the nontrivial odd-order components of A. For every missing $(x, y) \in \mathbb{Z}^2_{\ge 0}$ with $x + y = c_0(F) - 1$, set

$$j = u_R - x \in \mathbb{Z}, \qquad u_8 = \max\left\{0, \left\lfloor \frac{2E - 3}{6} \right\rfloor\right\}.$$

Then the following statements hold.

(I)

$$j - \ell \ge 2\lambda + 4 - 2\delta + u_8 \ge 2 + u_8.$$

In particular $j > 0$. Let D_{ev} be the vertex union of all even-order components of A. There is an independent set $I_{\mathrm{ev}} \subseteq V(A) \setminus D_{\mathrm{ev}}$ such that

$$|I_{\mathrm{ev}}| = a - \frac{|D_{\mathrm{ev}}|}{2}, \qquad F - N_F[I_{\mathrm{ev}}] = F[D_{\mathrm{ev}}].$$

Every maximal independent set of $F[D_{\mathrm{ev}}]$ unites with this same I_{ev} to form a maximum independent set of F. Every even-order terminal component is well-covered. Every terminal has components only in

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

Thus P_6 is impossible, including in terminals containing C_5 or C_7.

(II) *Let Z_A be the terminal isolates, put*

$$B_0 = \{v \in R : N_F(v) \cap Z_A = \varnothing\},$$

and let $\mathcal{O}$ be the nontrivial odd-order components. For $C \in \mathcal{O}$, define its exposed ports by

$$P(C) = \{v \in V(C) : C - v \text{ has a perfect matching}\}.$$

Let Π be the bipartite graph on $B_0, \mathcal{O}$, with vC an edge exactly when v has an actual F-neighbor in $P(C)$. All root conflicts are retained in the separate induced graph $F[R]$.

The graph Π has a matching saturating B_0, and every such matching covers every P_3. For every fixed such matching $\mathcal{M}$, its alternating closure from the P_3 components gives $B \subseteq B_0$ and $\mathcal{H} \subseteq \mathcal{O}$ such that

$$\{P_3 \text{ components}\} \subseteq \mathcal{H}, \qquad N_\Pi(B) = \mathcal{H},$$

and $\mathcal{M}$ pairs B bijectively with $\mathcal{H}$. Put

$$Y = D_{\mathrm{ev}} \cup \bigcup_{C \in \mathcal{H}} V(C), \qquad W = Y \cup B.$$

There is an independent set $I \subseteq V(A) \setminus Y$ with

$$|I| = a - \alpha(A[Y]), \qquad F - N_F[I] = F[W].$$

The graph $F[W]$ is well-covered and has a perfect matching. For the integers

$$h = n_A(P_3), \quad p_{\mathcal{H}} = |\{C \in \mathcal{H} : C \text{ is a cycle}\}|, \quad b = |B|, \quad t = \nu(A[Y]),$$

the exact identities are

$$b = |\mathcal{H}| = h + p_{\mathcal{H}}, \qquad \nu(F[W]) = t + b, \qquad \alpha(F[W]) = t + h,$$

$$\nu(F[W]) - \alpha(F[W]) = p_{\mathcal{H}}.$$

For every $T \subseteq R \setminus B$ with $|T| \le j - b$, one matching takes $B \cup T$ to exposed ports in distinct odd A-components, takes B into $\mathcal{H}$ and T outside $\mathcal{H}$, and is simultaneously compatible with maximum internal matchings of all A-components. It need not extend $\mathcal{M}$.

(III) *For every $Q \subseteq B$, define the nonnegative integer losses*

$$\ell_A(Q) = \alpha(A) - \alpha(A - N_F(Q)),$$

$$\ell_Y(Q) = \alpha(A[Y]) - \alpha(A[Y] - N_F(Q)).$$

Then $\ell_A(Q) = \ell_Y(Q)$. If Q is independent in the actual graph $F[B]$, then $\ell_Y(Q) \ge |Q|$. Conversely, an independent $Q \subseteq B$ with $|Q| - \ell_Y(Q) \ge 1$ would give an independent set of the same F of size at least $a + 1$.

These assertions hold in both sectors, at every source rank and excess, and in particular throughout $\delta > 0, E \ge 8$, without a bound on λ or $|B|$.

Proof. We check the native hypotheses of the earlier localization and then prove the exact equality of losses.

Step 1. Every missing pair has the required margin. Theorem 6.15 gives

$$|V(A)| = k - \lambda - 1 + \delta, \qquad \nu(A) = p + \delta = E - 2 - \lambda + \delta,$$

as well as $\alpha(A) = a$, zero slack, and the initial nine-component list. Setup 6.6 gives connectedness, eight-regularity, factor-criticality, well-coveredness, and order $2k + 1$. Hence

$$u_R = k + \lambda + 2 - \delta, \qquad c_0(F) = u_R + \nu(A).$$

These are properties of the original F. No minimality is transferred to a remainder. Also $E \ge 6$, so the positive-excess hypothesis of the missing-allocation bound is satisfied.

For every missing pair, Setup 6.8 gives $x \le k - E - u_8$. Such a pair exists because $c_0(F) > 0$ is the least integer total at which all pairs are feasible. Therefore

$$j = u_R - x \ge E + \lambda + 2 - \delta + u_8.$$

Every nontrivial odd component on the initial list has matching number at least one. Matching-number additivity gives

$$\ell \le \nu(A) = E - 2 - \lambda + \delta.$$

Subtracting yields

$$j - \ell \ge 2\lambda + 4 - 2\delta + u_8.$$

The bound $\delta \le \lambda + 1$ from Theorem 6.15 gives the further lower bound $2 + u_8$. The calculation applies separately to every missing pair and does not select a restricted rank range.

Step 2. The even part has one fixed independent exterior. Every component C on the initial list satisfies $\nu(C) = \lfloor |V(C)|/2 \rfloor$. Every exposed port of an odd cycle C_3, C_5, C_7 extends to a maximum

independent set by cyclic alternating choices. The exposed ports of P_3 are its endpoints, and both lie in its maximum independent endpoint pair.

Thus the even-localization statement in Setup 6.8 applies: F is well-covered, the native c_0-identity holds, and $j \geq \ell + 1$. It gives the exact closed-neighborhood remainder, well-coveredness, and exclusion of P_6. Before that exclusion, each even component is bipartite with a perfect matching. Its independence number is half its order, using a bipartition for the lower bound and its perfect matching for the upper bound. The independent set supplied by localization is maximum on the odd components. Since $\alpha(A) = a$, its size is $a - |D_{\mathrm{ev}}|/2$. The same earlier statement gives the assertion for every maximal independent set of the even remainder.

Step 3. The alternating localization applies with actual endpoints. After Step 2 the row has the exact component domain required by the port statement in Setup 6.8. The native identity, well-coveredness, and missing-pair margin have already been checked. Its exterior is the present R; its isolates and exposed-port graph are precisely Z_A, Π.

That statement gives the saturating matching and its universal coverage of P_3's. For every fixed $\mathcal{M}$, it gives $B, \mathcal{H}$, their exact neighborhood equality, and the witness I. Since I is maximum on every discarded component,

$$|I| = a - \alpha(A[Y]).$$

It also gives the perfect matching, well-coveredness, and the identities involving $t, h, p_{\mathcal{H}}, b$. Subtracting the matching and independence identities gives $p_{\mathcal{H}}$. Its joint matching conclusion applies to every prescribed T of size at most $j - b$, retaining the actual matching endpoints. No assertion that this matching extends $\mathcal{M}$ is needed.

Step 4. Discarded components have exactly zero loss. The equality $F - N_F[I] = F[W]$ implies that no edge joins B to I. The discarded row components are the isolates and the odd cycles outside $\mathcal{H}$, because all even components and all P_3's lie in Y. On each discarded component C, the set $I \cap V(C)$ is maximum independent.

Fix any $Q \subseteq B$, whether independent or not. The set $I \cap V(C)$ survives deletion of $N_F(Q)$. Hence

$$\alpha(C - N_F(Q)) \geq |I \cap V(C)| = \alpha(C).$$

Vertex deletion gives the reverse inequality. Thus every discarded component has zero independence loss. Since Y consists of whole row components, additivity gives $\ell_A(Q) = \ell_Y(Q)$.

If Q is independent in $F[B]$, it is independent in the unchanged exterior graph $F[R]$. The native loss inequality in Setup 6.8, with the independence discrepancy zero because $\alpha(A) = a$, gives $\ell_A(Q) \geq |Q|$. Substitution proves the stated inequality.

Finally, let Q be independent in $F[B]$, and choose a maximum independent set J in $A[Y] - N_F(Q)$. The sets I, Q, J are disjoint. Exact localization excludes edges from I to Q or J, and the definition of J excludes edges from Q to J. Thus their union is independent in F, of size

$$|I| + |Q| + \alpha(A[Y]) - \ell_Y(Q) = a + |Q| - \ell_Y(Q).$$

A surplus of at least one would give the asserted contradiction. Every conflict inside Q has been tested in $F[B]$. □

Remark 6.18 (The remaining cost question). Theorem 6.17 does not assert that an augmenting Q exists. It does not prove $c_2(F) \geq c_0(F)$, exclude every remaining strict terminal, or establish a whole-degree or Ramsey formula conclusion. The remaining issue is a native cost contradiction within or through the localized configuration.

Theorem 6.19 (Exclusion of three-vertex paths)**.** *Every native row of the source in Setup 6.2 has no whole P_3 component. Rows are exactly the terminals of the complete source processes of Definition 6.3. Every row is well-covered, has independence a, and has components only in*

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7.$$

The assertion holds at every source rank in both sectors, with no upper excess or order bound, including throughout the strict range $E \geq 8$.

Proof. Fix any row A. Theorem 6.15 makes it an actual terminal and gives $\alpha(A) = a$. The original F is well-covered by Setup 6.6. We use a saturating port matching that maximizes the number of assigned cycles with multiple contacts.

Step 1. Choose an extremal matching in the full permitted family. Use $R, Z_A, D_{\mathrm{ev}}, B_0, \mathcal{O}, \Pi$ from Theorem 6.17, and let $\mathcal{C} \subseteq \mathcal{O}$ be its odd-cycle components. Put $h = n_A(P_3)$. For an incidence bC, with $C \in \mathcal{C}$, define

$$d(b, C) = |N_F(b) \cap V(C)|.$$

For a matching $\mathcal{M}$ saturating all of B_0, put

$$\Phi(\mathcal{M}) = |\{bC \in \mathcal{M} : C \in \mathcal{C},\ d(b, C) \geq 2\}|.$$

The permitted family is nonempty by Theorem 6.17. Its finiteness, recorded in Setup 6.9, permits a choice maximizing Φ over the entire family.

Suppose that $h > 0$. Let B_P be the h roots assigned to P_3's, let P_b be assigned to b, and let w_D be assigned to a matched odd cycle D. Setup 6.9 applies to this particular matching. Thus B_P is independent, each of its roots meets the endpoints of its assigned path, and it avoids the other P_3's, all their centers, D_{ev}, and Z_A. For a matched cycle,

$$N_F(B_P) \cap V(D) \subseteq \bigcap_{z \in N_F(w_D) \cap V(D)} N_D(z). \tag{18}$$

There are no contacts to unmatched cycles, and every B_P root has two contacts on a C_5 or C_7.

Choose $b \in B_P$ and such a cycle C. It is matched; write $w = w_C$. Choose $z \in N_F(w) \cap V(C)$. The two contacts of b exhaust the two-element set $N_C(z)$ by (18). Let $m = |V(C)| \in \{5, 7\}$, and label the cycle $0, 1, \ldots, m-1$ with $z = 0$. The contacts are $1, m-1$. Since $2 \neq m-2$,

$$N_C(1) \cap N_C(m-1) = \{0\}.$$

Applying (18) to every contact of w therefore gives

$$N_F(w) \cap V(C) = \{0\}, \qquad N_F(b) \cap V(C) = \{1, m-1\}, \tag{19}$$

and $N_F(B_P) \cap V(C) \subseteq \{1, m-1\}$.

Step 2. A changed independent set on C forces the old cycle root to replace b. For each matched cycle other than C, choose an actual contact of its assigned root and a maximum independent set containing that contact. For each unmatched cycle, choose any maximum independent set. Cyclic alternating choices give the required sets on C_3, C_5, C_7. Let their union outside C be τ_{other}.

Every B_P root avoids τ_{other}. For a matched cycle, (18) places its contacts inside the open neighborhood of the chosen port, which an independent set containing that port avoids. Unmatched cycles have no B_P contacts.

On C, choose

$$J_C = \begin{cases} \{1, 4\}, & m = 5, \\ \{1, 3, 6\}, & m = 7. \end{cases}$$

These sets are independent, contain both contacts in (19), and have maximum size $(m-1)/2$. Put $\tau' = \tau_{\mathrm{other}} \cup J_C$. Let C_P be the set of all P_3 centers. The even part is a disjoint union of P_2, P_4, C_4. Choose its bipartition D_0, D_1 componentwise. Each shore is independent, dominates the even part, and has size $|D_{\mathrm{ev}}|/2$. Both are empty if the even part is empty.

For $i \in \{0, 1\}$, put

$$S_i = Z_A \cup \tau' \cup C_P \cup D_i.$$

Each S_i is independent and dominates A. Its size is $a - h$: every component contributes its independence number except that each P_3 contributes its center instead of its endpoint pair.

Let t_{hit} count the roots of B_P meeting J_C. Since b is one, $t_{\mathrm{hit}} \geq 1$. Exterior vertices outside B_0 are dominated by Z_A. A B_0-vertex assigned to a cycle other than C is dominated by its chosen port in τ_{other}. Consequently, any vertex that could be added to S_i lies among the surviving B_P roots or is w. There are at most $h - t_{\mathrm{hit}} + 1$ such vertices. No row vertex can be added because S_i dominates A.

Extend S_i to a maximal independent set of F. Well-coveredness forces the extension to add h vertices. Thus

$$h \leq h - t_{\mathrm{hit}} + 1,$$

and $t_{\mathrm{hit}} = 1$. Its unique hit root is b. The entire possible h-vertex pool is

$$B'_P = (B_P \setminus \{b\}) \cup \{w\}.$$

For both choices of i, a maximal extension must use every vertex of B'_P. Hence B'_P is independent in the actual F and avoids both S_0, S_1. In particular, w avoids all P_3 centers and the entire even part.

Step 3. Exact localization supplies the missing endpoint incidence. Put

$$I = Z_A \cup \tau' \cup D_0.$$

This is S_0 with the P_3 centers removed. Its closed neighborhood removes every row component except the P_3's: maximum independent sets dominate the cycles, and D_0 dominates the even part. All P_3 vertices avoid I, because distinct row components are anticomplete.

The exterior outside B'_P is also dominated. Vertices outside B_0 meet Z_A, cycle-assigned roots other than w meet their chosen ports, and b meets J_C. Conversely, B'_P avoids I. Therefore

$$F - N_F[I] = F\big[(\text{all } P_3 \text{ vertices}) \cup B'_P\big] =: H.$$

This equality retains all actual edges. The graph H is well-covered: any maximal independent set K of H gives a maximal independent set $I \cup K$ of F. Every vertex outside H lies in I or has an I-neighbor, and every vertex of $H \setminus K$ has a K-neighbor. Thus every such K has size $a - |I|$.

Suppose that w has no neighbor on the endpoints of P_b. It already avoids the center. Every other vertex of B'_P avoids P_b by Setup 6.9. There are no edges between distinct row components. Hence P_b is a whole component of H. Take any maximal independent set K_0 of $H - V(P_b)$. Uniting K_0 with the center of P_b, or with its two endpoints, gives maximal independent sets of different sizes in H, a contradiction. Thus w has an actual endpoint neighbor in P_b.

Step 4. Swapping two assignments increases the extremal score. Replace bP_b, wC in $\mathcal{M}$ by bC, wP_b. Both incidences are legal: every cycle vertex is an exposed port, and the endpoint neighbor just proved is an exposed port of P_b. The four incidence endpoints are distinct. The new matching $\mathcal{M}'$ still saturates the same full B_0.

The cycle C was not counted in $\Phi(\mathcal{M})$, since w had exactly one contact there. It is counted in $\Phi(\mathcal{M}')$, since b has exactly two contacts. No other cycle assignment changes. Therefore

$$\Phi(\mathcal{M}') = \Phi(\mathcal{M}) + 1,$$

contrary to maximality. This excludes $h > 0$.

Theorem 6.17 already excludes P_6, so the remaining list is the one stated. Each permitted component is well-covered. This follows directly for K_1, P_2, C_3. For P_4, C_4, C_5, the independence number is two, and a singleton cannot dominate because its closed neighborhood has size at most three, smaller than the component order. For C_7, the independence number is three, whereas the closed neighborhoods of two vertices cover at most six vertices. Thus no independent set of size at most two is maximal. A maximal independent set of a disjoint union restricts to a maximal independent set in each component. Hence A is well-covered. Its independence a and the equivalence between rows and terminals were supplied by Theorem 6.15. □

Remark 6.20. Theorem 6.19 is a restriction on every row of a source. It does not assert that the remaining source class is empty.

6.4. **Fixed exterior capacities and odd-cycle transport.** In this section, we prove that one capacity pair remains sharp under every exterior insertion and use singleton exchanges to describe all three-contact interactions with odd row cycles.

Theorem 6.21 (Fixed native capacity and odd-cycle transport)**.** *Let F satisfy Setup 6.2, let A be any native row, and put*

$$R = V(F) \setminus V(A), \qquad u = |R|.$$

(I) *For every $B \subseteq R$, with $b = |B|$, put $H_B = F[V(A) \cup B]$. Then*

$$c_{(1,1)}(H_B) = c_2(H_B) = p + b,$$

$$c_{(2,0)}(H_B) = c_{(0,2)}(H_B) \le b, \qquad c_1(H_B) = p + \delta - \epsilon + b.$$

Thus $(1,1)$ uniquely attains the maximum defining $c_2(H_B)$. At that capacity, at least one budget pair of total $p+b-1$ is missing, and every pair of total $p+b$ is feasible. No independence assumption is imposed on B.

Choose any set Z containing exactly one vertex from each odd-cycle component of A, and no other vertices. Put $A_0 = A - Z$. For every $T \subseteq R \cup Z$,

$$c_{(1,1)}\big(F[V(A_0) \cup T]\big) = |T|.$$

The same capacity pair works for every choice of Z, T.

(II) *Let $v \in R$ have exactly three A-neighbors, and let C be a whole odd-cycle component meeting v. Exactly one of the following holds:*

(a) *C is a triangle and $N_F(v) \cap V(A) = V(C)$;*

(b) *C is a triangle or pentagon, its sole v-neighbor is c, and the other two A-neighbors of v are the endpoints of one whole P_2 or P_4, denoted by P.*

In the second case, write $C = C_{2i+1}$ and $P = P_{2j}$, where $i, j \in \{1, 2\}$. Then

$$A' = F[(V(A) \setminus \{c\}) \cup \{v\}]$$

is another actual native row of the same source. Its only component changes are

$$C_{2i+1} + P_{2j} \quad \longmapsto \quad P_{2i} + C_{2j+1}.$$

The move is reversible. Every exterior vertex with exactly three row neighbors avoids every whole row C_7. If $\epsilon = 0$, it also avoids every whole row C_5.

(III) *Let J_3 be an actual cubic entry and A its terminal. Every whole C_7 of A is already whole in J_3. If $\epsilon = 0$, the same holds for every whole C_5.*

In every two-row frame $A, \widehat{A}, X$ of Theorem 6.15, each vertex of $F[X]$-degree two avoids all whole C_7's of both rows. At $\epsilon = 0$, it also avoids all whole C_5's of both rows.

All assertions hold at arbitrary admitted $\lambda, \delta, \epsilon$.

Proof. Theorem 6.15 gives actual terminal realizability, $\alpha(A) = a$, $p \ge 1$, and the component counts p, ϵ. Its source profile gives

$$c_2(F) = u + p, \qquad c_1(F) = u + p + \delta - \epsilon.$$

Theorem 6.17 excludes P_6 from every row. We retain its component domain when applying the earlier degree-free exchange statement.

Step 1. Insertion inequalities force the split capacity. Since $\Delta(A) \le 2$, all its edges fit in a capacity-two part, giving

$$c_{(2,0)}(A) = c_{(0,2)}(A) = 0.$$

Repeated insertion from Setup 6.6 gives

$$c_{(2,0)}(F) = c_{(0,2)}(F) \le u < u + p = c_2(F).$$

The only capacity pairs totaling two are $(2,0), (1,1), (0,2)$. Hence

$$c_{(1,1)}(F) = c_2(F) = u + p.$$

For $B \subseteq R$, insertion from H_B to F gives

$$c_{(1,1)}(H_B) \ge c_{(1,1)}(F) - (u - b) = p + b.$$

Insertion from A to H_B gives $c_2(H_B) \le c_2(A) + b = p + b$. Since $c_{(1,1)} \le c_2$, equality follows. The same insertion bound gives $c_{(2,0)}(H_B) = c_{(0,2)}(H_B) \le b$. For the first cost,

$$c_1(F) \le u - b + c_1(H_B) \le u - b + c_1(A) + b = c_1(F).$$

Thus $c_1(H_B) = c_1(A) + b$. Since $p \ge 1$, the attaining capacity pair is unique. Minimality of its integer universal total gives a missing pair at $p + b - 1$, and every pair at $p + b$ is feasible. The missing pair need not be the same for different B.

Delete the specified Z from A. Every component of A_0 is a path or an even cycle. Alternating path edges cover all degree-two vertices of a path, leaving at most the opposite endpoint uncovered. Alternating edges give a perfect matching on each even cycle. Thus A_0 is 2-good. Moreover,

$$|V(A_0)| = r - p, \qquad |V(F) \setminus V(A_0)| = u + p = c_2(F).$$

This is an optimal threshold-two deletion by Setup 6.6. Its sharp insertion statement, using the attaining pair $(1,1)$, proves

$$c_{(1,1)}\big(F[V(A_0) \cup T]\big) = |T| \quad (T \subseteq R \cup Z).$$

Step 2. Three-contact exchanges determine the odd-cycle interaction. Fix v, C, and put $t_C = |N_F(v) \cap V(C)| > 0$. In $K = F[V(A) \cup \{v\}]$, the cubic vertices are v and its neighbors of row degree two. Setup 6.10 makes them a clique and allows deletion of any of them to give another actual row with p odd cycles. The contacts on C are therefore pairwise adjacent.

If $t_C = 3$, the three contacts form a triangle component, and they exhaust the row neighborhood of v. This is the first alternative.

Suppose that $t_C = 2$, with adjacent contacts c, d. The third neighbor z lies outside C. It has row degree at most one, since otherwise it would be cubic in K and would have to be adjacent to c, contrary to the distinct row components being anticomplete. Thus z is an isolate or an endpoint of a path. Deleting c is a legal row exchange. The path $C - c$, with endpoint d, joins through v to z and its path or singleton component. Their union is one path. All other components are unchanged, so the new row would have $p - 1$ odd cycles, contradicting Theorem 6.15.

It remains that $t_C = 1$. Write c for the sole contact and x, y for the other row neighbors. The degree-free statement in Setup 6.10 applies: its component domain follows from Theorem 6.17, its independence hypothesis is $\alpha(A) = a$, and its first-cost identity is

$$c_1(F) = u + \nu(A) - \epsilon.$$

Its component counts are p, ϵ, and its singleton second-cost bound is $c_2(K) = p + 1$. It follows that x, y are the endpoints of one whole P_2 or P_4, denoted by P. This application uses no restriction on ordinary seven-cycles elsewhere in F.

The exchanged graph $A' = K - c$ is an actual row. The old cycle becomes $C - c$, and $P + v$ becomes an odd cycle. Every other component remains unchanged. If $C = C_7$, the path $C - c$ is a whole P_6, contrary to Theorem 6.17. Hence $C = C_3$ or C_5, and the displayed replacement follows.

In A', the exterior vertex c has exactly three row neighbors: its two old cycle neighbors, now the endpoints of $C - c$, and v. The vertex v has degree two on the new odd cycle. Setup 6.10 allows exchange of v back for c, proving reversibility. If $\epsilon = 0$, no P_2 or P_4 occurs in A, so the second alternative is impossible. The asserted cycle avoidances follow.

Step 3. The avoidances pass to cubic entries and both frame boundaries. Let T_3 be the final cubic phase in J_3. Setup 6.6 makes T_3 independent and gives $J_3 - T_3 = A$. Each $v \in T_3$ has degree three in J_3, with all three neighbors in A; these are exactly its F-neighbors in A. Step 2 excludes contact with any whole C_7, and at $\epsilon = 0$ with any whole C_5. Such a row cycle also has no edge to another row component. It is therefore already whole in J_3.

For a frame $A, \widehat{A}, X$, Theorem 6.15 gives eight-regularity, the vertex partition, and maximum degree at most five in both row complements. A vertex $v \in X$ with $d_{F[X]}(v) = 2$ has at least three neighbors in each row. Its total number of row neighbors is $8 - 2 = 6$. Thus it has exactly three in each row, and Step 2 applies separately to both. □

Remark 6.22 (Scope of transport). All edges on B and between B and A remain in the insertion graphs. Theorem 6.21 gives no implication $c_2(F) \geq c_0(F)$, no guaranteed augmenting exterior set, and no whole-sector or Ramsey formula exclusion. Its cycle restrictions concern whole row components, not arbitrary cycles elsewhere in F.

6.5. **The cubic-entry catalogue and path surplus.** The goal of this section is to reconstruct every whole component of an actual cubic entry from the independent full-phase contacts and then count its exact contribution to the even-path ledger.

Theorem 6.23 (Cubic-entry components and exact path surplus)**.** *Let F satisfy Setup 6.2, and assume*

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E - 3, \qquad q \geq 1.$$

Retain every admitted value of ϵ. Let L be any actual cubic entry of any complete source process.

(I) *Every whole component of L belongs to the eighteen-type list*

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7, K_4,$$
$$B_{33}, B_{35}, B_{55}, L_{\square}, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(P_6), \mathrm{cor}(C_4), \mathrm{cor}(C_5), \mathrm{cor}(C_7).$$

Every such component has at most fourteen vertices. This is a necessary list; occurrence of a listed type is not asserted.

(II) *Every actual cubic entry satisfies*

$$\epsilon - q + n_L(K_4) = n_L(P_2) + n_L(P_4) + n_L(\mathrm{cor}(P_3)) + n_L(\mathrm{cor}(P_4)) + n_L(\mathrm{cor}(P_6)). \tag{20}$$

The left side is nonnegative and is zero exactly when all five types on the right are absent. Every choice of maximal independent sets in the cubic graphs of the separate whole components is simultaneously realizable as one actual full cubic phase of L.

No optimizing choice of L or its phase is required.

Proof. Fix L and any full cubic phase X. Setup 6.6 identifies X as a maximal independent set in the graph induced by the degree-three vertices of L, and every ordering is legal. Theorem 6.15 gives $|X| = q$ and makes $A = L - X$ a native row with exactly ϵ positive even-order paths.

Complete the original source process with this chosen X, then reverse its row-complement portion using Theorem 6.15. The reversed process begins from $F - V(A)$, and X is its full quintic phase. Also $L = F[V(A) \cup X]$. Thus Setup 6.11 applies with the original vertices and edges unchanged. The row catalogue is $K_1, P_2, P_4, C_3, C_4, C_5, C_7$ by Theorem 6.19.

Step 1. Separate the whole cycle and skip components. Every root in X has exactly three A-neighbors because X is independent and L is subcubic. Setup 6.11 gives each root a distinct odd-component owner. Its possibilities are a whole triangle, a C_3 or C_5 contacted once with both endpoints of an even path, or a whole isolate with two even contacts. Setup 6.10 excludes C_4 contacts and gives the internal-P_4 menu. Theorem 6.21 excludes contacts to whole C_7's.

An isolate owner is a private leaf in L: it had no row neighbor, and owner injectivity prevents another root from using it. Roots with odd-cycle owners avoid row isolates. A pure root and its triangle form a whole K_4. Each of these four vertices has all three L-neighbors inside that set. An odd-cycle-transfer root and its two supporting row components form a whole B_{33}, B_{35}, or B_{55} by Setup 6.11. These whole components are pairwise disjoint and contain no other root. Set them aside.

Every row cycle outside those components is untouched by X. Odd-cycle contacts have been exhausted by the owner alternatives, and even-cycle contacts are excluded. Such cycles remain whole and belong to C_3, C_4, C_5, C_7.

Consider a remaining root whose two even contacts lie on one whole row path. They cannot both lie on P_2, by protection of its perfect-matching edge. For $P_4 = u_1u_2u_3u_4$, the pair must be

$$\{u_1, u_3\}, \qquad \{u_2, u_3\}, \qquad \text{or} \qquad \{u_2, u_4\}.$$

When an endpoint is contacted, this is the endpoint restriction from Setup 6.11; otherwise both contacts are internal. That endpoint restriction uses singleton and independent-pair projection: a lone endpoint contact would leave a non-well-covered P_3, while the pairs $\{u_1, u_2\}$ and $\{u_1, u_4\}$ violate perfect-matching-edge protection and the preserved independence number, respectively. Among roots meeting one whole P_2, the same protection places all contacts at only one endpoint.

The internal pair $\{u_2, u_3\}$, with the isolate owner, is excluded for every admitted ϵ by Setup 6.10. Thus a skip pair remains. It uses both allowed nonpure incidences on this P_4, so no other root meets the path. Its isolate owner is private. The path, root, and isolate therefore form a whole six-vertex component. Its edges are the three path edges, the two skip edges, and the isolate edge. They form a four-cycle with private leaves at two adjacent vertices, namely $L_\square$. Set aside all these whole components.

Step 2. Every remaining component is a path or cycle corona. Each remaining root has a private isolate leaf and two contacts on distinct even row paths. It cannot meet a P_4 endpoint, since that would force both contacts onto the same path. If it has an internal P_4 contact, the precise menu in Setup 6.10 forces its other contact to be an endpoint of a distinct whole P_2. Otherwise its contacts are endpoints of two distinct whole P_2's.

A contacted P_2 has only one contacted endpoint. Its other endpoint is a private leaf in L. The contacted endpoint has at most two root neighbors because its row degree is one and L is subcubic. An uncontacted P_2 remains whole. In every remaining P_4, neither endpoint has a root neighbor. Its two endpoints are private leaves of the two internal vertices. Each internal vertex has its internal path edge, its private leaf, and at most one root neighbor. An unowned row isolate remains a whole K_1. Set aside these whole isolates and uncontacted P_2's.

The remaining graph is a corona with base vertices consisting of the remaining roots, the contacted endpoints of contacted P_2's, and both internal vertices of every remaining P_4. Their respective private leaves are the isolate owners, the other P_2-endpoints, and the two P_4-endpoints. These vertex sets are disjoint and cover the remaining graph. Its base edges are exactly the root-to-path contacts and the internal edges of the P_4's. There are no root-to-root edges or edges between distinct row components.

Every root has base degree two. Every contacted P_2-endpoint and every internal P_4-vertex has base degree one or two. Thus the finite simple base graph has maximum degree at most two. A connected component with a degree-one vertex is a path: following its unused continuing edge cannot branch and exhausts the component. If every vertex has degree two, following edges returns to a cycle, and connectedness leaves no additional vertex. An isolated base vertex would give P_2. Hence every remaining whole component is $\mathrm{cor}(P_t)$ for an integer $t \geq 1$, or $\mathrm{cor}(C_t)$ for an integer $t \geq 3$.

Step 3. The common phase length bounds the base orders. Let

$$K = L[\{v \in V(L) : d_L(v) = 3\}].$$

Every maximal independent set of K is a full legal cubic phase, and every such phase has size q, by Setup 6.6 and Theorem 6.15. Thus K is well-covered. Each whole component of K is well-covered: two maximal independent sets of different sizes in one component could be united with fixed maximal independent sets in the others.

For a whole $\mathrm{cor}(P_t)$ with $t \geq 3$, its cubic graph is P_{t-2} on the internal base vertices. For a whole $\mathrm{cor}(C_t)$, its cubic graph is C_t on all base vertices. These are whole components of K.

We verify the required degree-two classification. For an integer $m \geq 1$,

$$\alpha(P_m) = \left\lceil \frac{m}{2} \right\rceil .$$

Alternating vertices attain the bound, and consecutive pairs, with a possible final singleton, give the upper bound. A maximal independent set dominates the path. Its closed neighborhoods have size at most three, so its size is at least $\lceil m/3 \rceil$. This lower bound is attained, with vertices labeled $1, \ldots, m$, by

m	independent dominating set
$3t$	$2, 5, \ldots, 3t-1$
$3t+1$	$1, 4, \ldots, 3t+1$
$3t+2$	$2, 5, \ldots, 3t+2$

where $t \in \mathbb{Z}_{\geq 0}$, and $t \geq 1$ in the first row. The gaps are three and the end gaps dominate the endpoints. Thus P_m is well-covered exactly when

$$\left\lceil \frac{m}{2} \right\rceil = \left\lceil \frac{m}{3} \right\rceil ,$$

namely for $m \in \{1, 2, 4\}$. For exhaustion, write $m = 6t + j$, where $t \in \mathbb{Z}_{\geq 0}$ and $0 \leq j \leq 5$. The difference of the ceilings is

$$t + \left\lceil \frac{j}{2} \right\rceil - \left\lceil \frac{j}{3} \right\rceil .$$

It is positive for $t \geq 1$; among the remaining positive orders it vanishes exactly at $1, 2, 4$. Therefore $\mathrm{cor}(P_t)$ with $t \geq 3$ requires $t \in \{3, 4, 6\}$. The cases $t = 1, 2$ give P_2, P_4.

For an integer $m \geq 3$,

$$\alpha(C_m) = \left\lfloor \frac{m}{2} \right\rfloor .$$

A selected vertex and its clockwise successor form disjoint pairs for an independent set, giving the upper bound; alternating choices attain it. A maximal independent set again has size at least $\lceil m/3 \rceil$. Put $t = \lceil m/3 \rceil$, so $2t \leq m \leq 3t$. Arrange t selected cycle vertices with $3t - m$ clockwise gaps of length two and $m - 2t$ gaps of length three. The nonnegative integer counts sum to t, and the gap lengths sum to m. The selected vertices are independent and dominate the cycle. Hence C_m is well-covered exactly when

$$\left\lfloor \frac{m}{2} \right\rfloor = \left\lceil \frac{m}{3} \right\rceil ,$$

which gives $m \in \{3, 4, 5, 7\}$. To check exhaustion, write $m = 6t + j$ with $0 \leq j \leq 5$. The difference is

$$t + \left\lfloor \frac{j}{2} \right\rfloor - \left\lceil \frac{j}{3} \right\rceil .$$

It is positive for $t \geq 2$, positive at $m = 6, 8, 9, 10, 11$, and zero at $m = 7$. The cases $3, 4, 5$ also give zero.

The remaining possibility $\mathrm{cor}(C_3)$ is excluded by an actual phase choice. Select one of its three cubic base vertices and select maximal independent sets in the cubic graphs of all other whole components. Their union is a legal full phase. Its native row contains, inside this corona, a whole P_4 and the selected base's private leaf as a whole isolate. The selected base has exactly the two internal P_4-vertices and that isolate as its row mask. This is the excluded mask of Setup 6.10. Thus only $\mathrm{cor}(C_4), \mathrm{cor}(C_5), \mathrm{cor}(C_7)$ remain among cycle coronas.

The reconstruction yields all eighteen listed types. The largest is $\mathrm{cor}(C_7)$, of order fourteen. All other listed orders are at most twelve. Every separated subgraph was a whole component, and the corona reconstruction covered every remaining vertex and edge.

Step 4. Each path corona contributes exactly one surplus unit. Because whole components have no edges between them, arbitrary maximal independent choices in their cubic graphs unite to a maximal independent set of K. They therefore form one actual full cubic phase.

For a whole component C, let

$$b_C = |X \cap V(C)|,$$

and let ϵ_C count the positive even-order whole paths in $C - X$. Such paths are also whole in $A = L - X$. Consequently,

$$\sum_C b_C = q, \qquad \sum_C \epsilon_C = \epsilon.$$

In a whole K_4, exactly one vertex is selected and the terminal is C_3, so $\epsilon_C - b_C = -1$. In B_{ij}, the cubic graph is the edge between its bridge endpoints. Exactly one endpoint is selected, leaving the opposite odd cycle and an even path of order two or four. Thus $\epsilon_C - b_C = 0$. The two cubic vertices of $L_\square$ are adjacent. Deleting either leaves one whole P_4 and one isolate, again giving zero. An ordinary whole P_2 or P_4 has no selected root and contributes one. The ordinary cycles and K_1 contribute zero.

Let $C = \mathrm{cor}(P_t)$, with $t \in \{3, 4, 6\}$, and put $b = b_C$. The selected roots form a maximal independent set of the internal base path. Every unselected internal base vertex has a selected neighbor. Each endpoint had at most one base neighbor initially. Hence the remaining base graph has maximum degree at most one. It has

$$t - b \text{ vertices}, \qquad t - 1 - 2b \text{ edges},$$

because selected internal vertices each have base degree two and no edge joins two selected vertices. Its number of components is

$$(t - b) - (t - 1 - 2b) = b + 1.$$

Each component is an isolate or an edge, whose corona is P_2 or P_4. Private leaves of selected bases become isolates. Therefore $\epsilon_C = b + 1$, giving surplus one.

For $C = \mathrm{cor}(C_t)$, with $t \in \{4, 5, 7\}$, the selected roots form a maximal independent set of the entire base cycle. Every unselected base has a selected neighbor, so the remaining base graph has maximum degree at most one. It has $t - b$ vertices and $t - 2b$ edges, hence

$$(t - b) - (t - 2b) = b$$

components. Their coronas are P_2's or P_4's, and selected bases again leave isolated private leaves. Thus $\epsilon_C = b$, giving surplus zero.

Summing over the complete catalogue gives

$$\epsilon - q = -n_L(K_4) + n_L(P_2) + n_L(P_4) + n_L(\mathrm{cor}(P_3)) + n_L(\mathrm{cor}(P_4)) + n_L(\mathrm{cor}(P_6)).$$

This is (20). Nonnegativity and its zero criterion follow from the nonnegative integer component counts. □

Remark 6.24 (Scope of the catalogue). The bounds concern whole components of actual cubic entries and their native terminal rows. They give no bound on the number of components or on the source order. No simultaneous restoration-cost claim, source-class exclusion, or Ramsey formula conclusion is made.

6.6. **Odd-corona exclusion and the cubic phase graph.** In this section, we exclude the two odd cycle coronas by a certificate on every edge of the original source, then determine the entire graph on the cubic vertices.

Theorem 6.25 (Odd-corona exclusion and the exact cubic phase graph)**.** *Let F be an eight-regular source as in Setup 6.2, and assume*

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E-3, \qquad q \geq 1.$$

Retain every admitted ϵ, and let L be any actual cubic entry.

(I) *No whole component of L is $\mathrm{cor}(C_5)$ or $\mathrm{cor}(C_7)$. Its complete necessary component list is*

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7, K_4,$$
$$B_{33}, B_{35}, B_{55}, L_{\square}, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(P_6), \mathrm{cor}(C_4).$$

Every component has at most twelve vertices. For either excluded type, the contradiction is the original-source inequality

$$c_1(F) \leq c_1(F) - 1,$$

with every source edge and every nonnegative integer budget split included.

(II) *Put*

$$U_3 = \{v \in V(L) : d_L(v) = 3\}, \qquad G_3 = L[U_3],$$
$$b = n_L(\mathrm{cor}(P_3)), \qquad t = n_L(K_4).$$

The entire G_3 is a vertex-disjoint union of K_1, K_2, P_4, C_4, K_4, and

$$|U_3| = 2q - b + 2t, \qquad \nu(G_3) = q - b + t, \qquad \alpha(G_3) = q.$$

Its isolates are exactly the cubic middle vertices of the b whole $\mathrm{cor}(P_3)$'s. Every maximum matching of G_3 covers all nonisolated vertices.

(III) *If also $\epsilon \geq 2$ and $q \geq 2$, then $t = 0$. Hence G_3 is bipartite, its entire component list is K_1, K_2, P_4, C_4, and*

$$|U_3| = 2q - b, \qquad \nu(G_3) = q - b, \qquad 2 \leq q \leq \epsilon.$$

The list in the first item loses K_4 and has fifteen types. Moreover,

$$\epsilon - q = n_L(P_2) + n_L(P_4) + n_L(\mathrm{cor}(P_3)) + n_L(\mathrm{cor}(P_4)) + n_L(\mathrm{cor}(P_6)).$$

Proof. All constructions remain in the same original source F. No induced remainder is assumed source-minimal.

Step 1. An odd corona admits a simultaneous actual phase choice. Suppose that a whole component of L is

$$D = \mathrm{cor}(C_{2h+1}), \qquad h \in \{2, 3\}.$$

Label its base cycle $v_0, v_1, \ldots, v_{2h}, v_0$, and its private leaves $w_0, \ldots, w_{2h}$, with pendant edges $v_j w_j$. These cycle and pendant edges are all the edges of D. Let

$$S = \{v_0, v_2, \ldots, v_{2h-2}\}.$$

This is an independent dominating set of the base cycle, of size h. The intervening odd-index vertices lie between selected vertices; v_{2h-1} meets v_{2h-2}, and v_{2h} meets v_0. Thus S is maximal independent in the cubic graph of D.

Choose maximal independent sets in the cubic graphs of all other whole components of L. By Theorem 6.23, their union with S is one actual full cubic phase X. Theorem 6.15 gives $|X| = q \geq h$ and makes $A = L - X$ a native row.

The unselected bases in D have exactly one remaining base edge, $v_{2h-1}v_{2h}$, and $h-1$ isolated base vertices. With their private leaves, the local row consists of the whole path

$$w_{2h-1} - v_{2h-1} - v_{2h} - w_{2h},$$

$h-1$ whole P_2's, and the h private leaves of selected bases as whole isolates. These are whole row components because D was whole in L.

Put

$$C = c_1(A) = \nu(A) - \epsilon, \qquad u = |V(F) \setminus V(A)|.$$

Theorem 6.15 gives

$$u = r + s, \qquad s \geq 2q+1, \qquad c_1(F) = u + C.$$

For the last identity, substituting the definitions gives

$$u + C = (r+s) + (p + \delta - \epsilon) = k + E - \epsilon.$$

Step 2. The row partition specifies all edges that will remain unpaid. Restate the partition from Setup 6.13 as follows. On each whole P_2, retain its edge in a matching. On each whole P_4, retain its two outer edges and take its middle edge as one unit piece. On each whole C_4, retain a perfect matching and take each other edge as a separate unit piece.

On a whole C_{2j+1}, with $j \in \{1,2,3\}$, retain an alternating j-edge matching leaving one vertex unmatched. The unretained edges consist of a two-edge star at that vertex and $j-1$ isolated intervening edges. Take the star as one unit piece and each intervening edge as another. Isolates contribute nothing.

The retained graph is a matching, all pieces are edge-disjoint, and the number of unit pieces is $\nu(A) - \epsilon = C$. Each piece has matching number at most one. Every odd-cycle component contributes at least one piece, so $C \geq p \geq 1$. The designated local P_4 has exactly one unit piece, the edge $v_{2h-1}v_{2h}$.

Step 3. The entire source has a first-cost certificate saving one unit. Retain the full induced graph

$$H = F[V(A) \cup S].$$

Since $V(A) \cup S \subseteq V(L)$, D is whole in L, and no selected root outside D is retained, H is the disjoint union of D and the row components outside D. No edge of this retained induced graph is omitted.

On D, put all base-cycle edges into one exceptional piece. Its matching number is exactly h: alternating edges give a matching of size h, and at most h disjoint edges fit on $2h+1$ vertices. Retain all $2h+1$ pendant edges in the free matching. Outside D, use the row partition of Step 2. The local P_4 accounts for exactly one of its C unit pieces; the local P_2's and isolates account for none. Thus outside D there are exactly $C-1$ unit pieces. The unpaid graph is a matching, consisting of the pendant matching of D and the row matching outside D, on disjoint vertex sets. This partitions all edges of H.

There are $N = u - h$ omitted source vertices. Order them arbitrarily and assign each source edge having an omitted endpoint to its first omitted endpoint. The edges assigned to each vertex form a star, possibly empty, of matching number at most one. This partitions every edge outside H into exactly N unit pieces. Edges between two omitted vertices are assigned once, not discarded.

The entire $E(F)$ has therefore been partitioned into a free matching, one piece of matching number h, and $C - 1 + N$ unit pieces. Put

$$T = h + (C - 1 + N) = u + C - 1.$$

Since

$$u = r + s \geq s \geq 2q + 1 \geq 2h + 1, \qquad C \geq 1,$$

we have $T \geq 2h+1$, and hence $T \geq 2h-1$.

Take any $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = T$. At least one is at least h; otherwise $x + y \leq 2h - 2 < T$. Place the whole cycle piece in a part with budget at least h, and reserve h units of that budget. The remaining budgets are nonnegative integers summing to $C - 1 + N$. Assign that many unit pieces to each part.

A matching in a union of the pieces restricts to a matching in each piece, so its size is at most the sum of their matching-number bounds. Thus both exceptional parts respect their budgets. For either capacity split totaling one, the free matching fits in the capacity-one part, and the capacity-zero part is empty. This supplies a certificate for every capacity split totaling one and every exceptional split totaling T, on all source edges. Therefore

$$c_1(F) \leq T = u + C - 1 = c_1(F) - 1,$$

a contradiction.

This excludes both odd coronas at every admitted ϵ, q. No universal local first-cost inequality for H or for the corona alone has been used. Removing these two types from Theorem 6.23 leaves the stated sixteen types. The largest remaining type is $\mathrm{cor}(P_6)$, of order twelve.

Step 4. The whole-component catalogue determines G_3 exactly. The degree-at-most-two types $K_1, P_2, P_4, C_3, C_4, C_5, C_7$ contribute no cubic vertices and no cubic phase root. Each of $B_{33}, B_{35}, B_{55}, L_\square, \mathrm{cor}(P_4)$ has two adjacent cubic vertices. Its cubic graph is K_2, and each full phase selects one vertex. The middle base of $\mathrm{cor}(P_3)$ is its unique cubic vertex, contributing K_1 and one phase root. The four internal bases of $\mathrm{cor}(P_6)$ induce P_4, and a full phase selects two. The four bases of $\mathrm{cor}(C_4)$ induce C_4, and a full phase selects two. A whole K_4 contributes K_4 and one phase root.

There are no edges between whole components of L. Thus these cubic graphs form the entire disjoint union G_3. The total phase contribution is q by Theorem 6.15. The order, matching number, and independence number of each piece are

piece	$\lvert V\rvert$	ν	α
K_1	1	0	1
K_2	2	1	1
P_4	4	2	2
C_4	4	2	2
K_4	4	2	1

Alternating edges and vertices give the entries for paths and the four-cycle. Two disjoint edges give a perfect matching in K_4, and pairwise adjacency bounds its independence by one.

Relative to twice the phase contribution, K_1 subtracts one and K_4 adds two. Relative to the phase contribution in matching number, K_1 subtracts one and K_4 adds one. Matching and independence numbers are additive across components: restriction gives the upper bounds, and unions of component maximizers attain them. Therefore

$$|U_3| = 2q - b + 2t, \qquad \nu(G_3) = q - b + t, \qquad \alpha(G_3) = q.$$

The only cubic singleton comes from $\mathrm{cor}(P_3)$, proving the isolate assertion. Every other cubic piece has a perfect matching. Every maximum matching of their disjoint union is maximum on each piece, so covers all nonisolated vertices.

Step 5. The additional higher-deficit hypotheses remove K_4. Under $\epsilon \geq 2, q \geq 2$, Setup 6.13 excludes every whole K_4 and gives $2 \leq q \leq \epsilon$. Hence $t = 0$. Each remaining cubic piece K_1, K_2, P_4, C_4 is bipartite, and their bipartitions unite to one for G_3. Substituting $t = 0$ gives the stated exact invariants. The component list loses K_4 and has fifteen types. Finally, substituting $n_L(K_4) = 0$ into (20) gives the last identity. □

Remark 6.26 (Scope of the phase graph). These are native component and matching restrictions. They do not exclude an entire source sector, establish a strict fifth-cost payment, or give a Ramsey formula conclusion.

6.7. **The seven components at the boundary.** The goal of this section is to describe the unique cubic interaction at $\delta = \lambda$ and calculate its exact one-pair contributions for every admitted ϵ.

Theorem 6.27 (Seven whole cubic components and their contributions)**.** *Let F satisfy Setup 6.2, and assume $\delta = \lambda$.*

(I) *For every native row A and every exterior vertex v with exactly three A-neighbors, the whole component C containing v in $F[V(A) \cup \{v\}]$ is one of*

$$K_4,\ B_{33},\ B_{35},\ B_{55},\ L_{\square},\ \mathrm{cor}(P_3),\ \mathrm{cor}(P_4).$$

Every other component is an unchanged whole A-component. The component C contains all cubic vertices, and those vertices induce a nonempty clique. Deleting any one of them leaves another actual native row of the same source.

(II) *Every such source has $q_8 = 1$. Every row-complement process has word 543. Its cubic entry has exactly one whole component C from the seven-item list, with every other component in*

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7.$$

In particular, $|V(C)| \le 10$. For every actual frame $(A, \{x\}, \{y\}, \{z\}, R)$, both cubic entries $F[V(A) \cup \{x\}]$ and $F[V(R) \cup \{z\}]$ have this description.

(III) *For any cubic vertex w of a listed C, let $o_C(w)$ be the number of odd-cycle components of $C - w$, let $b_C(w)$ be the number of positive even-order path components of $C - w$, and put*

$$d_C(w) = \nu(C - w) - o_C(w).$$

These nonnegative integer values are independent of w, and are

C	d_C	b_C	o_C
K_4	0	0	1
B_{33}	1	1	1
B_{35}	2	1	1
B_{55}	3	1	1
$L_{\square}$	2	1	0
$\mathrm{cor}(P_3)$	2	2	0
$\mathrm{cor}(P_4)$	3	2	0

Thus the entire cubic interaction uses at most two positive even-order terminal paths and contributes at most three units to the second deficit. Every other row component and its contribution remain unchanged under every cubic choice. In particular, $\mathrm{cor}(P_3)$ and $\mathrm{cor}(P_4)$ require $\epsilon \ge 2$, and only K_4 is possible at $\epsilon = 0$.

Proof. We first classify an arbitrary three-contact singleton at the boundary. The earlier boundary closure will then identify every actual cubic entry with such a singleton insertion.

Step 1. The native missing allocation supplies the payment margins. Fix any row A. The source is well-covered by Setup 6.6. Theorem 6.15 gives $\alpha(A) = a$, terminal realizability, and exactly p odd cycles. Theorem 6.19 gives its seven-type row catalogue.

Let D be the union of its whole even-order components, let Z be its isolates, and put

$$U = V(F) \setminus V(A), \qquad u = |U|, \qquad m = \nu(D).$$

The graph D, possibly empty, is bipartite, has maximum degree at most two, and has a perfect matching. The exact source identities are

$$c_0(F) = u + \nu(A), \qquad c_2(F) = u + p.$$

Since $c_0(F) > 0$ is the least integer universal total, choose a missing pair $(b_r, b_s) \in \mathbb{Z}^2_{\ge 0}$ of total $c_0(F) - 1$, and put $j = u - b_r$. Every nontrivial odd row component is a cycle, so their count is p.

At $\delta = \lambda$, Theorem 6.17 gives

$$j - p \geq 2\lambda + 4 - 2\delta + u_8 = 4 + u_8 \geq 4, \qquad u_8 = \max\left\{0, \left\lfloor \frac{2E-3}{6} \right\rfloor\right\}.$$

This verifies the missing-allocation margins for Setup 6.12 without assuming $E \geq 8$.

Step 2. Classify the three-contact mask. Let $v \in U$ have exactly three row neighbors. If all three are isolates, the first payment in Setup 6.12 applies with $j \geq p + 4$. It gives $c_2(F) \leq u + p - 1$, contradicting $c_2(F) = u + p$.

If exactly two neighbors are isolates, the third cannot lie on an odd cycle by Theorem 6.21: its odd-cycle alternatives use either a whole triangle or one cycle vertex and both endpoints of an even path. Thus the third contact lies in D, so $m \geq 1$. The second payment in Setup 6.12, with $j \geq p+3$, gives the same contradiction. Therefore v has at most one isolate neighbor.

If v meets an odd cycle, Theorem 6.21 is exhaustive. Its whole-triangle case gives a whole K_4, since the triangle is the entire row neighborhood. Otherwise the neighborhood consists of one vertex of a C_3 or C_5 and both endpoints of a whole P_2 or P_4. The path together with v is a C_3 or C_5, joined to the original odd cycle by the sole edge from v to that cycle. This gives a whole B_{33}, B_{35}, or B_{55}. Whole-component status follows from the whole row components and the absence of further row neighbors of v.

Now assume that v avoids every odd cycle. If it also avoided Z, it would lie in B_0 of Theorem 6.17, but would have no adjacent exposed port in a nontrivial odd row component. That contradicts the B_0-saturating matching. Hence v has exactly one isolate neighbor $i(v)$; its other two neighbors lie in D.

The one-isolate-port statement of Setup 6.12 applies to the independent singleton $\{v\}$. Its size satisfies $1 \leq j - p - 1$, since $j \geq p + 4$. It gives

$$D - N_F(v) \text{ well-covered}, \qquad \alpha(D - N_F(v)) = m, \tag{21}$$

and forbids both endpoints of any perfect-matching edge among the two contacts.

If one contact has row degree two, Setup 6.10 excludes C_4 and gives, on a reflected and labeled $P_4 = u_1u_2u_3u_4$, the masks

$$\{u_2, u_3, i(v)\}, \quad \{u_2, u_4, i(v)\}, \quad \{u_2, i(v), w\},$$

where w is an endpoint of a distinct whole P_2. The first is excluded by the earlier full first-cost payment. The second gives $L_\square$. For the third, write w' for the other endpoint of its P_2. The base path

$$u_3 - u_2 - v - w$$

has respective private leaves $u_4, u_1, i(v), w'$. It is therefore $\mathrm{cor}(P_4)$. These are whole components, because the listed row components and the full neighborhood of v exhaust the connected part.

It remains that neither even contact has row degree two. Both are endpoints of P_2's or P_4's. If they lie in one P_2, they are the endpoints of a perfect-matching edge, which is forbidden. If they lie in one P_4, they are its two endpoints. Deleting them leaves one P_2, of independence one instead of two. Every other even component is unchanged, giving $\alpha(D - N_F(v)) = m - 1$, contrary to (21).

Thus the contacts lie on distinct paths. If one path were P_4, deleting its contacted endpoint would leave a whole P_3 in $D - N_F(v)$. Its center and endpoint pair, united with the same maximal independent set of the other components, would give maximal independent sets of different sizes. This contradicts well-coveredness in (21). Both contacted paths are therefore distinct whole P_2's. Their contacted endpoints, joined through v, form the base of $\mathrm{cor}(P_3)$; their other endpoints and $i(v)$ are the private leaves. This exhausts the possible masks.

Every listed component is subcubic. In K_4, all four vertices are cubic and form a clique. In B_{ij}, precisely the two bridge endpoints are cubic and adjacent. In $L_\square$, precisely the adjacent leaf-bearing cycle vertices are cubic. In $\mathrm{cor}(P_3)$, only the middle base is cubic. In $\mathrm{cor}(P_4)$, the two internal bases are cubic and adjacent. All other components are unchanged row components of maximum degree

at most two. Thus the cubic set is a nonempty clique lying in C. Setup 6.10 permits deletion of any of its vertices and gives another actual native row of the same F. This proves the first item.

Step 3. Boundary closure gives one cubic component on both sides. Setup 6.13 gives $q_8 = 1$ and common word 543 at $\delta = \lambda$. Thus any actual cubic entry is $F[V(R) \cup \{z\}]$, where R is its native terminal and z has exactly three R-neighbors. Step 2 identifies its unique component containing z and all cubic vertices. Every other component is a row component. The largest listed order is ten, for B_{55}.

For a frame $(A, \{x\}, \{y\}, \{z\}, R)$, the forward cubic entry is $F[V(R) \cup \{z\}]$. Legal reversal from Theorem 6.15 deletes z, y, x from $F - V(R)$, so its cubic entry is $F[V(A) \cup \{x\}]$. Applying the first item to each actual row gives the two-sided description.

Step 4. Direct deletions give the contribution table. Deleting any vertex from K_4 leaves C_3, with matching number one, one odd cycle, and no positive even path. For B_{ij}, deleting the bridge endpoint on C_i leaves

$$P_{i-1} \dot\cup C_j.$$

Its matching number is

$$\frac{i-1}{2} + \frac{j-1}{2},$$

its odd-cycle count is one, and its positive even-path count is one. Therefore

$$d_C = \frac{i+j-4}{2},$$

which is $1, 2, 3$ for B_{33}, B_{35}, B_{55}. The expression is symmetric in i, j, so the other cubic choice has the same values.

For $L_\square$, label the cycle $a - b - c - d - a$, with private leaves at a, b. Deleting a leaves its leaf isolated and the path from the leaf at b through b, c, d, a P_4. Deleting b gives the reflected graph. Thus the matching number is two, with no odd cycle and one positive even path.

Deleting the middle base of $\mathrm{cor}(P_3)$ leaves two whole P_2's and its private leaf as an isolate. The matching number is two, the odd-cycle count zero, and the positive even-path count two. Deleting either internal base of $\mathrm{cor}(P_4)$ leaves

$$P_4 \dot\cup P_2 \dot\cup K_1.$$

The corresponding values are matching number three, no odd cycle, and two positive even paths. These deletions verify every table entry and its independence of the cubic choice.

For a native terminal row, Theorem 6.15 gives ϵ positive even-order paths, p odd cycles, and matching number $p + \delta$. Hence δ is the sum of component matching numbers minus the odd-cycle count. Every unchanged component contributes a nonnegative integer. The changed component contributes precisely d_C and b_C, with $d_C \le 3$ and $b_C \le 2$. This proves the stated bounds and parameter requirements. All other whole components remain literally unchanged under every cubic choice. □

Remark 6.28 (Boundary scope). The seven components describe necessary native interfaces. Their occurrence is not asserted. The result is not a complete boundary or cost-saving theorem.

7. Degree eight: matching transfer, port saturation and the deficit-two class

The matching arguments in this chapter take place in induced subgraphs of one fixed source. We keep every original edge on a retained vertex set, account for every edge incident with an omitted vertex, and require changes to the initial deletion phases to return to the same induced complement.

7.1. **Setup.** We assume, as established earlier in this series, the prerequisites stated below from Companion I, Chapter 1 on universal structure and the generic prefix lemma, and Chapter 3 on octic rows and the cubic phase graph. We state the precise portions used here and do not reproduce their proofs.
[cite/blocker] Exact theorem locators in Companion I and the earlier chapters were not supplied. The placeholder [I, Theorem X] and the earlier-series attributions below require those locators; no bibliography entry for the series has been invented.

Notation 7.1. All graphs are finite and simple. For a graph G, write $e(G)$, $\nu(G)$, $\alpha(G)$, $\Delta(G)$, and $N_G(v)$ for its edge count, matching number, independence number, maximum degree, and open vertex neighborhood. For $P \subseteq E(G)$, degree and matching notation applied to P refers to the spanning graph $(V(G), P)$.

The graphs P_j, C_j, and K_j are the path, cycle, and complete graph on j vertices, with integer orders in their usual ranges. A whole component is an entire connected component of the indicated graph. Vertex deletions are induced deletions and retain every source edge on the remaining vertices. Edge deletions retain the entire vertex set.

Definition 7.2 (Certificate costs and the source property)**.** Let $A_c, B_c \in \mathbb{Z}_{\geq 0}$. Define $c_{(A_c,B_c)}(G)$ to be the least $T_c \in \mathbb{Z}_{\geq 0}$ such that, for every $x_c, y_c \in \mathbb{Z}_{\geq 0}$ with $x_c + y_c = T_c$, there is a partition

$$E(G) = Q_R \sqcup Q_B \sqcup P_R \sqcup P_B$$

satisfying

$$\Delta(Q_R) \leq A_c, \qquad \Delta(Q_B) \leq B_c, \qquad \nu(P_R) \leq x_c, \qquad \nu(P_B) \leq y_c.$$

For $i \in \mathbb{Z}_{\geq 0}$, put

$$c_i(G) = \max_{\substack{A_c, B_c \in \mathbb{Z}_{\geq 0} \\ A_c + B_c = i}} c_{(A_c,B_c)}(G), \qquad c_0 = c_{(0,0)}.$$

Define

$$\mathrm{ex}(G) = c_0(G) - \nu(G), \qquad \mathrm{ps}(G) = e(G) - \nu(G) - \sum_{i \geq 1} c_i(G), \qquad \mathrm{gap}(G) = \sum_{i \geq 0} c_i(G) - e(G).$$

These sums are finite. The quantity $\mathrm{gap}(G)$ is a signed integer.

Set $\mathrm{tr}(G) = 0$ when $c_0(G) = 0$. Otherwise, $\mathrm{tr}(G)$ is the greatest nonnegative integer length of a consecutive run of integers

$$x \in \{0, \ldots, c_0(G) - 1\}$$

for which $E(G)$ has no partition into two parts with matching numbers at most x and $c_0(G) - 1 - x$. Property T_- is the implication

$$\mathrm{gap}(G) > 0 \quad \Longrightarrow \quad \left[\mathrm{gap}(G) = 1 \text{ and } \mathrm{tr}(G) + \mathrm{ex}(G) \leq \max\left\{4, \left\lfloor \frac{\Delta(G)}{2} \right\rfloor + 1\right\}\right].$$

A *missing pair* for G is a pair $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = c_0(G) - 1$ for which the corresponding two-part matching-bounded partition does not exist.

Setup 7.3 (Source parameters)**.** Let $\mathcal{M}_8$ be the class of finite simple graphs of maximum degree exactly eight that violate T_-, while every proper subgraph satisfies T_-. Proper subgraphs include arbitrary vertex deletions, arbitrary edge deletions, and combinations of the two.

For a source F, write

$$k = \nu(F), \qquad a = \alpha(F), \qquad E = \mathrm{ex}(F), \qquad \lambda = 2E - 3 - (k - a),$$
$$\delta = c_0(F) - c_2(F), \qquad \epsilon = c_0(F) - c_1(F), \qquad p = E - 2 - \lambda, \qquad r = k - \lambda - 1 + \delta,$$
$$s = 2\lambda + 3 - 2\delta, \qquad q = q_8 = c_4(F) - r, \qquad \zeta = q_{\mathrm{def}} = \delta - \epsilon, \qquad \beta = p + \zeta.$$

Also put

$$u_{8} = \max\left\{0, \left\lfloor \frac{2E-3}{6} \right\rfloor\right\}.$$

Every capacity, matching budget, least universal total, cost index, degree, order, cardinality, matching size, omission count, component count, phase length, and finite choice domain is a nonnegative integer. Parameter differences and generic gaps may be signed integers unless an inequality establishes their nonnegativity.

Definition 7.4 (Native rows and source processes)**.** With the parameters of Setup 7.3, a *native row* is an induced subgraph A of F satisfying

$$|V(A)| = r, \qquad \Delta(A) \le 2, \qquad \mathrm{ps}(A) = 0, \qquad \nu(A) = p+\delta, \qquad c_{1}(A) = p+\delta-\epsilon = \beta, \qquad c_{2}(A) = p.$$

An exposed port of a whole component C is a vertex $v \in V(C)$ such that $\nu(C-v) = \nu(C)$. A graph is well-covered when every maximal independent set is maximum.

A complete source process first deletes any maximal independent set of F, and then repeatedly deletes any current-maximum-degree vertex while that degree is at least three. A full degree-D phase consists of all consecutive deletions at current degree D. A complete continuation on an induced graph repeats current-maximum-degree deletions until maximum degree at most two. An actual cubic or quartic entry is the induced graph at the start of the corresponding full phase in such a process.

Setup 7.5 (Universal octic source data)**.** The earlier universal and octic developments give the following conclusions for every $F \in \mathcal{M}_{8}$:

$$F \text{ is connected and eight-regular}, \qquad |V(F)| = 2k+1, \qquad c_{0}(F) = k+E,$$

and

$$E \ge 10, \qquad 2 \le \epsilon \le \delta \le \lambda \le \left\lfloor \frac{E-4}{2} \right\rfloor, \qquad \lambda \ge 3, \qquad p \ge \lambda+2, \qquad q \ge 1.$$

At $E = 10$, the exact additional conclusions are

$$\lambda = \delta = 3, \qquad q = 1, \qquad \epsilon \in \{2,3\}.$$

The following three alternatives are exhaustive and disjoint:

$$\begin{array}{l} \delta = \lambda \ge 3, \quad q = 1; \\ \delta = \epsilon = 2 < \lambda, \quad q = 1; \\ 3 \le \delta < \lambda. \end{array}$$

For every native row A, the complement $J = F - V(A)$ has the phase word

$$5^{q}\,4^{\,s-2q}\,3^{q}.$$

Here exponents are nonnegative integer phase counts. In particular, its cubic phase is nonempty.

Setup 7.6 (Native row structure and source costs)**.** The earlier octic row results give native row existence and $p \ge 1$. Every native row A is well-covered, has $\alpha(A) = a$, and has only whole components

$$\mathsf{K}_{1},\ \mathsf{P}_{2},\ \mathsf{P}_{4},\ \mathsf{C}_{3},\ \mathsf{C}_{4},\ \mathsf{C}_{5},\ \mathsf{C}_{7}.$$

It has exactly p odd-cycle components and exactly ϵ positive even-order path components. In particular, whole P_{3} and P_{6} components do not occur.

For $u = |V(F) \setminus V(A)|$, the actual source costs satisfy

$$c_{1}(F) = u+\beta, \qquad c_{2}(F) = u+p.$$

Setup 7.7 (Controlled early realization)**.** The earlier octic realization result applies at every deficit. Fix a native row A and any maximum independent set S of A. For each whole component T, put

$$B_7(T) = \{v \in V(T) \setminus S : \ |N_F(v) \cap S| = 1\}.$$

On an odd cycle, $B_7(T)$ is the pair of endpoints of the unique edge of $T-S$. On a positive even-order path, it is either one endpoint of T, with $T-S$ independent, or the pair of endpoints of the unique edge of $T-S$, with all other vertices of $T-S$ isolated. It is empty on K_1 and C_4; the earlier clause also states emptiness on P_3, which Assumed result 7.6 excludes here.

Choose independently one vertex from $B_7(T)$ on every odd cycle and every positive even-order path. Let U_7 be their union and $U_6 = V(A) \setminus (S \cup U_7)$. Then

$$|S| = a, \qquad |U_7| = p + \epsilon, \qquad |U_6| = p + \delta - \epsilon.$$

Both U_7 and U_6 are independent. Their vertices have degrees seven in $F-S$ and six in $F-(S\cup U_7)$, respectively. The latter graph has maximum degree six if $U_6 \neq \varnothing$, and at most five otherwise.

Deleting S, then U_7 in any order, then U_6 in any order gives the complete degree-seven and degree-six phases and leaves exactly $F - V(A)$. Any prescribed complete continuation on that same complement may then be appended, with every later induced prefix, root, and terminal row unchanged. Component choices can be changed independently.

Setup 7.8 (Missing-budget bounds)**.** The degree-eight specialization of the earlier missing-budget result applies to a connected graph of order $2k+1$, matching number k, maximum degree at most eight, and excess $E \geq 1$. With

$$u_8 = \max\left\{0, \left\lfloor \frac{2E-3}{6} \right\rfloor\right\},$$

it gives

$$k \geq 3E - 1 + 2u_8, \qquad x \leq k - E - u_8, \qquad y \leq k - E - u_8$$

for both coordinates of every missing pair.

For an actual full cubic entry K of an octic source, the earlier cubic-entry result gives

$$|V(K)| = r + q, \qquad \nu(K) = p + \delta + q,$$
$$|V(F) \setminus V(K)| = r + s - q, \qquad c_0(F) = |V(F) \setminus V(K)| + \nu(K).$$

For the same missing pair (x, y), it also gives

$$|V(F) \setminus V(K)| - x \geq p + 2\lambda + 4 - \delta - q + u_8.$$

Setup 7.9 (Saturation over an original row)**.** The earlier saturation result has the following form. Let R be an induced subgraph of a finite simple graph F, and assume that every whole component C of R satisfies $\nu(C) = \lfloor |V(C)|/2 \rfloor$. Put $U = V(F) \setminus V(R)$. Assume

$$c_0(F) = |U| + \nu(R),$$

and let (x, y) be a missing pair with $j = |U| - x > 0$. For every $W \subseteq U$ with $|W| \leq j$, there is a matching on the full induced graph $F[V(R) \cup W]$ consisting of maximum internal matchings in every whole R-component and an edge from every $w \in W$ to an exposed port of a distinct odd whole R-component.

Setup 7.10 (Original quintic owners)**.** In the octic source range

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E - 3, \qquad q \geq 1,$$

fix any native row A, put $J = F - V(A)$, and define

$$Q_5 = \{c \in V(J) : d_J(c) = 5\}.$$

The earlier retained-quintic result gives a unique odd whole A-component $O_A(c)$ having an original neighbor of c, for each $c \in Q_5$. These components are pairwise distinct. Its prescribed-owner interface permits simultaneous prescriptions of actual edges into these owners and compatible maximum internal matchings leaving their endpoints uncovered.

Setup 7.11 (Generic first-residual equality)**.** We use the following specialization of the generic prefix lemma from Chapter 1. Let R be a retained induced row, let W be disjoint from $V(R)$, and put

$$H = F[V(R) \cup W], \quad h = |W|, \quad u = |V(F) \setminus V(R)|, \quad N = u - h.$$

Let M be a maximum matching of H whose internal restriction is maximum in R and whose other h edges saturate W into R. Put

$$m = \nu\big((V(R), E(R) \setminus M)\big).$$

If

$$c_1(F) = u + m, \qquad N \geq \max\{0, m + h - 1\},$$

then

$$\nu\big((V(H), E(H) \setminus M)\big) = m + h, \qquad N + \nu\big((V(H), E(H) \setminus M)\big) = c_1(F).$$

The two row arguments in the generic result may both equal R. Its generic statement imposes no six-regular source hypothesis.

Setup 7.12 (Covering-matching splitting)**.** The certificate splitting result of Companion I, with reference placeholder [I, Theorem X], states the following. Let D be a positive integer. If a finite simple graph K has maximum degree at most D and one matching covering every degree-D vertex, then, for every $A_c, B_c \in \mathbb{Z}_{\geq 0}$ with $A_c + B_c = D$, its edges can be partitioned into parts of maximum degrees at most A_c and B_c. No exceptional edge part is required.

Setup 7.13 (Unit-deficit source profile)**.** In the earlier unit-deficit source profile, assume

$$c_0(F) - c_2(F) = 1, \qquad c_0(F) - c_1(F) = 0.$$

Then $r = k - \lambda$. If q_8 is the common quintic and cubic count and

$$b = 2\lambda + 1 - 2q_8 > 0$$

is the positive odd quartic count, the total root count and the third source cost satisfy

$$2\lambda + 1 = 2q_8 + b, \qquad c_3(F) = r + (2\lambda + 1) - q_8 = r + q_8 + b.$$

Definition 7.14 (Cubic-bearing graphs)**.** For $i, j \in \{3, 5\}$, let B_{ij} consist of disjoint cycles of orders i, j joined by one bridge. Let $L_\square$ be a square with one distinct private leaf at each of two adjacent square vertices. For a positive integer j, let $\mathrm{cor}(\mathsf{P}_j)$ be obtained by attaching one distinct private leaf to every vertex of P_j.

For an integer $\ell \geq 3$, let B_ℓ consist of ℓ vertex-disjoint copies of K_4, one distinguished vertex in each, and exactly the edges of an ℓ-cycle among the distinguished vertices. The notation B_ℓ is separate from the two-cycle notation B_{ij}.

Setup 7.15 (One-pair entries and exterior obstructions)**.** The following earlier-series statements are used in the source range

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E - 3, \qquad q_8 = 1,$$

with every allowed value of ϵ.

First, every actual cubic entry L is a singleton cubic insertion over a native row. Deleting any cubic vertex of L gives an actual native row of the same source. Exactly one whole component bears cubic vertices, and it belongs to

$$\mathsf{K}_4,\ B_{33},\ B_{35},\ B_{55},\ L_\square,\ \mathrm{cor}(\mathsf{P}_3),\ \mathrm{cor}(\mathsf{P}_4).$$

Every other whole component belongs to

$$\mathsf{K}_1,\ \mathsf{P}_2,\ \mathsf{P}_4,\ \mathsf{C}_3,\ \mathsf{C}_4,\ \mathsf{C}_5,\ \mathsf{C}_7.$$

Second, if L has a whole K_4 component C, then $H = F - V(L)$ is an actual quartic entry. There is a cubic vertex v of H whose entire neighborhood in L consists of three vertices of C, a whole L-isolate i, and an internal vertex t of a whole L-component P_4.

Third, the earlier exterior-pair obstruction forbids the following configuration over a native row A: whole components consisting of a triangle T, an isolate i, and a path $P = \mathsf{P}_4$, together with distinct exterior vertices u, v such that

$$uv \in E(F), \qquad N_F(u) \cap V(A) = V(T),$$

and $N_F(v) \cap V(A)$ consists exactly of two vertices of T, the vertex i, and an internal vertex of P. The obstruction retains every allowed first deficit and phase count; only its $q_8 = 1$ specialization is used here.

Setup 7.16 (Equal-deficit source exclusion)**.** The earlier source exclusion states that there is no finite simple eight-regular, proper-subgraph-minimal violation of T_- satisfying

$$E \geq 8, \qquad \delta = \epsilon = 2 < \lambda \leq E - 3, \qquad q_8 = 1.$$

It excludes the entire source class under these hypotheses, with no additional pivot, row, process, or marking premise.

7.2. **Prescribed starts and unchanged continuation.** In this section, we extract the prescribed-start statement from the earlier row structure and realization results, keeping the later induced graphs fixed.

Lemma 7.17 (Prescribed starts and unchanged continuation)**.** *Let F be any finite simple eight-regular graph violating T_- such that every proper subgraph satisfies T_-, including arbitrary vertex and edge deletions. Use the integer parameters of Setup 7.3 and the native rows of Definition 7.4.*

Fix any native row A, any maximum independent set S of A, and any prescribed complete continuation on $J = F - V(A)$. The row A is well-covered, has independence number a, and has only whole components

$$\mathsf{K}_1, \mathsf{P}_2, \mathsf{P}_4, \mathsf{C}_3, \mathsf{C}_4, \mathsf{C}_5, \mathsf{C}_7,$$

with exactly p odd-cycle components and ϵ positive even-order path components. For each whole component T, put

$$B_7(T) = \{v \in V(T) \setminus S :\ |N_F(v) \cap S| = 1\}.$$

Then the following conclusions hold.

(I) *If T is an odd cycle, $B_7(T)$ consists of the two endpoints of the unique edge in $T - S$. If T is a positive even-order path, $B_7(T)$ is either one endpoint of T, with $T - S$ independent, or the two endpoints of the unique edge in $T - S$, with all other vertices of $T - S$ isolated. For $T = \mathsf{K}_1$ or C_4, it is empty.*

(II) *Choose independently one vertex from $B_7(T)$ for every odd cycle and every positive even-order path, and let U_7 be the union of these choices. Put*

$$U_6 = V(A) \setminus (S \cup U_7).$$

Then

$$|S| = a, \qquad |U_7| = p + \epsilon, \qquad |U_6| = p + \delta - \epsilon.$$

Both U_7 and U_6 are independent. Every vertex of U_7 has degree seven in $F - S$. Every vertex of U_6 has degree six in $F - (S \cup U_7)$. The latter graph has maximum degree six if $U_6 \neq \varnothing$, and at most five otherwise.

(III) *The set S is a maximum independent set of the entire source F. Starting from F, delete S, then U_7 in any order, then U_6 in any order. These are the complete degree-seven and degree-six phases. The remaining induced graph is exactly J.*

Appending the prescribed continuation on this same J gives a legal complete source process. Every later induced prefix, chosen root, and terminal row is identical to the prescribed continuation. This holds for every chosen maximum independent set S of A and every collection of component choices in part (II)*. Choices on different whole components may therefore be changed independently and in any order while preserving the same row A and the same later continuation.*

The statement applies in both sectors $\delta = 0$ and $\delta > 0$, without an additional excess, rank, order, first-deficit, or phase-count hypothesis.

Proof. The eight-regular source F belongs to the source class in Setup 7.3. Its proper-subgraph minimality is exactly the one required by the earlier octic results. The native-row condition there is

$$c_1(A) = p + q_{\mathrm{def}}, \qquad q_{\mathrm{def}} = \delta - \epsilon,$$

which is the condition $c_1(A) = p + \delta - \epsilon$ used here. The other four native-row conditions agree as well. Thus the source parameters and the physical rows are unchanged.

Assumed result 7.6 gives $\alpha(A) = a$, well-coveredness, the seven whole-component types, and the two exact component counts. It also gives row existence, although a row has already been fixed. The absence of P_3 and P_6 is part of that component restriction.

Apply Assumed result 7.7 to this A and the arbitrary maximum independent set S. Its definition of $B_7(T)$ is the displayed definition in the statement. Its component assertions give part (I); the additional P_3 case in that earlier clause is irrelevant because P_3 is absent.

The same result permits independent choices on every odd cycle and every positive even-order path. It gives the three cardinalities, independence of U_7, U_6, the exact degrees, and the maximum-degree assertion in part (II). In every deletion, all original edges on the remaining vertices are retained.

By Assumed result 7.6,

$$|S| = \alpha(A) = a = \alpha(F).$$

Since A is induced, S is independent in F. Hence S is maximum in F, and is a permitted initial set for a complete source process.

Finally, Assumed result 7.7 gives the complete degree-seven and degree-six phases in any respective order and leaves exactly $F - V(A)$. It permits the initially prescribed continuation on that very graph, with every later induced prefix, root, and terminal row unchanged. This proves part (III), including independent changes on distinct row components. No parameter specialization was made in applying the realization clause. □

Remark 7.18. Lemma 7.17 prescribes neither certificate colors nor arbitrary initial-owner labels. It does not prescribe a common matching through changed early phases. It asserts no strict cost saving, source-class exclusion, whole-degree exclusion, or Ramsey formula, and incorporates no further unstated conclusions.

7.3. **Original-source ranges and port saturation.** The goal of this section is to obtain the scalar range and the original-row matching assignment separately, so that both can later be applied on one retained induced graph.

Proposition 7.19 (Uniform assignment range and absorption margin)**.** *Let $F \in \mathcal{M}_8$, with the parameters of Setup 7.3. Native rows exist. Fix any native row A, and put*

$$J = F - V(A), \qquad u = |V(J)|,$$

and let z_A be the number of isolated vertices of A. Here

$$u_8 = \left\lfloor \frac{2E-3}{6} \right\rfloor.$$

Then the following conclusions hold.

(I) *The source and row orders satisfy*

$$u = r + s, \qquad r = 3p + 2\delta + z_A, \qquad z_A = a - p - \delta, \qquad u_8 \geq 2,$$

and

$$z_A \geq s + \delta + 1 + 2u_8.$$

At least one missing pair exists. For every missing pair (x, y), put

$$j_x = u - x, \qquad j_y = u - y.$$

Both coordinates of that same pair satisfy

$$j_x \geq p + s + \delta + 1 + u_8, \qquad j_y \geq p + s + \delta + 1 + u_8.$$

(II) *Fix integers g, e_{sel} with*

$$0 \leq g \leq \min\{q_8, \epsilon\}, \qquad 1 \leq e_{\mathrm{sel}} \leq 4.$$

Put

$$h = s + \beta + g + e_{\mathrm{sel}}, \qquad N = u - h, \qquad \rho = \beta + h.$$

These are nonnegative integers. For every missing pair, simultaneously,

$$h \leq j_x, \qquad h \leq j_y.$$

Moreover,

$$N \geq \rho - 1 = \beta + h - 1.$$

Consequently every $W \subseteq V(J)$ of size h gives a full induced graph $F[V(A) \cup W]$ with exactly N omitted original vertices and with this omission threshold. Such h-element subsets exist because $h \leq u$.

Proof. Assumed result 7.5 gives connectedness, eight-regularity,

$$|V(F)| = 2k + 1, \qquad c_0(F) = k + E,$$

and

$$E \geq 10, \quad 2 \leq \epsilon \leq \delta \leq \lambda \leq \left\lfloor \frac{E-4}{2} \right\rfloor, \quad \lambda \geq 3, \quad p \geq \lambda + 2, \quad q_8 \geq 1.$$

At $E = 10$, it also gives $\lambda = \delta = 3$, $q_8 = 1$, and $\epsilon \in \{2, 3\}$. Thus $u_8 \geq 2$, and the floor defining u_8 here equals the maximum with zero in Assumed result 7.8.

At zero degree capacities, both degree-bounded edge parts are empty. Hence Definition 7.2 agrees with the two-part split definition used in that missing-budget result. Its hypotheses hold: F is connected, has order $2k + 1$, has matching number k, has maximum degree at most eight, and has $E \geq 1$. Therefore

$$k \geq 3E - 1 + 2u_8, \qquad x, y \leq k - E - u_8 \tag{22}$$

for every missing pair. Such a pair exists because $c_0(F) = k + E > 0$ is the least universal total. If every split at total $c_0(F) - 1$ were feasible, that smaller total would contradict its minimality.

Assumed result 7.6 gives native row existence, $\alpha(A) = a$, and the seven whole-component types. On P_2, P_4, and C_4, independence and matching numbers are respectively $1, 2, 2$. On C_3, C_5, and C_7, they are respectively $1, 2, 3$. An isolate contributes one to independence number and zero to matching number. Adding over all whole components gives

$$z_A = \alpha(A) - \nu(A) = a - p - \delta. \tag{23}$$

The parameter definitions yield

$$k - a = 2E - 3 - \lambda = 2p + \lambda + 1.$$

Consequently

$$r = k - \lambda - 1 + \delta = a + 2p + \delta = 3p + 2\delta + z_A.$$

Using $|V(F)| = 2k + 1$, we obtain

$$u = 2k + 1 - r = k + \lambda + 2 - \delta = r + s. \tag{24}$$

Here u denotes the exterior cardinality.

Since $E = p + \lambda + 2$, the bound on either coordinate in (22) gives

$$\begin{aligned} j_x &\geq k + \lambda + 2 - \delta - (k - E - u_8) \\ &= E + \lambda + 2 - \delta + u_8 = p + s + \delta + 1 + u_8. \end{aligned}$$

The same calculation applies to y in the same pair.

Substituting $a = k - 2E + 3 + \lambda$ and $p = E - 2 - \lambda$ into (23) gives

$$z_A = k - 3E + 5 + 2\lambda - \delta.$$

Thus (22) implies

$$z_A \geq 4 + 2\lambda - \delta + 2u_8 = s + \delta + 1 + 2u_8.$$

This proves part (I).

Fix the integer parameters of part (II). The source inequalities give

$$\beta = p + \delta - \epsilon \geq p \geq 5, \qquad s = 2(\lambda - \delta) + 3 \geq 3.$$

Hence h and ρ are positive integers. For either coordinate of a missing pair, part (I) gives

$$\begin{aligned} j_x - h &\geq p + s + \delta + 1 + u_8 - (s + p + \delta - \epsilon + g + e_{\mathrm{sel}}) \\ &= \epsilon + 1 + u_8 - g - e_{\mathrm{sel}}. \end{aligned}$$

If $E \geq 11$, then $u_8 \geq 3$, and this is at least

$$\epsilon + 1 + 3 - \epsilon - 4 = 0.$$

If $E = 10$, the exact source signature gives $q_8 = 1$, so $g \leq 1$. Together with $\epsilon \geq 2$ and $u_8 = 2$, the bound is at least

$$2 + 1 + 2 - 1 - 4 = 0.$$

These cases exhaust the allowed integer values of E. Thus $h \leq j_x, j_y$ for every missing pair. Since $x \geq 0$, $j_x = u - x \leq u$, so $h \leq u$ and $N \geq 0$. An exterior set of cardinality u therefore has an h-element subset.

The exact absorption margin is

$$\begin{aligned} N - \rho + 1 &= u - \beta - 2h + 1 \\ &= r - s - 3\beta - 2g - 2e_{\mathrm{sel}} + 1 \\ &= z_A - s - \delta + 3\epsilon - 2g - 2e_{\mathrm{sel}} + 1. \end{aligned} \tag{25}$$

The isolate bound in part (I) implies

$$N - \rho + 1 \geq 3\epsilon + 2 + 2u_8 - 2g - 2e_{\mathrm{sel}}.$$

Since $g \leq \epsilon$, $\epsilon \geq 2$, $u_8 \geq 2$, and $e_{\mathrm{sel}} \leq 4$, the right side is at least

$$\epsilon + 2 + 2u_8 - 2e_{\mathrm{sel}} \geq 2 + 2 + 4 - 8 = 0.$$

Therefore $N \geq \rho - 1$. For every exterior W of size h, the number of omitted original vertices is

$$|V(F)| - |V(A)| - |W| = u - h = N.$$

This proves part (II). □

Remark 7.20. Proposition 7.19 asserts a scalar assignment range and an omission margin for every source and every native row. It does not choose a selector, port matching, sequential matching prefix, or edge certificate, and it does not exclude a source.

Theorem 7.21 (Original-row Hall saturation with prescribed owners)**.** *Let $F \in \mathcal{M}_8$, and use Setup 7.3. Fix any native row A, and put*

$$J = F - V(A), \qquad u = |V(J)|, \qquad Q_5 = \{c \in V(J) : d_J(c) = 5\}.$$

For $c \in Q_5$, let $O_A(c)$ be its unique odd whole A-component having an original neighbor of c. These owners are pairwise distinct.

(I) *Missing pairs exist. For every missing pair (x, y), put $j_A = u - x$. Then*

$$c_0(F) = u + \nu(A), \qquad j_A \geq p + s + \delta + 1 + u_8.$$

The row A, any maximum independent start inside A, and any prescribed complete current-maximum-degree continuation on J can be realized in one complete source process. Every later physical graph of the prescribed continuation remains unchanged. The bound uses the same pair (x, y) and the same original row A.

(II) *For every $W \subseteq V(J)$ with $h = |W| \leq j_A$, put*

$$Q_W = W \cap Q_5, \qquad O_W = \{O_A(c) : c \in Q_W\}.$$

Let $\mathcal{O}_A$ be the family of all odd whole A-components. Define the bipartite graph Π_W between W and $\mathcal{O}_A$ by joining w to C precisely when an original F-edge joins w to an exposed port of C. For every $X \subseteq Q_W$ and $Y \subseteq W \setminus Q_5$,

$$|N_{\Pi_W}(X \cup Y)| \geq |X| + |Y|, \qquad |N_{\Pi_W}(Y) \setminus O_W| \geq |Y|.$$

Prescribe one actual source edge ct_c into $O_A(c)$ for every $c \in Q_W$, and prescribe a maximum internal matching of each such component leaving t_c uncovered. There is one maximum matching M_0 of the full induced graph

$$H = F[V(A) \cup W]$$

that contains these prescriptions, assigns every $w \in W$ to an exposed port of a distinct odd whole original A-component, and contains a maximum internal matching in every whole A-component. Its size is

$$\nu(H) = |M_0| = p + \delta + h.$$

Any prescribed perfect internal matching on every even whole component can also be retained. Any prescribed maximum internal matching on an odd whole component with no W-neighbor can be retained. All prescriptions may be fixed before the remaining destinations and matchings are chosen.

Every vertex of $W \setminus Q_5$ is assigned outside O_W. All induced edges of H, including edges within W, remain present. There is no independence hypothesis on W, and no cardinality restriction on its separate phase parts. In particular, the conclusion applies whenever

$$h \leq p + s + \delta + 1 + u_8.$$

(III) *If instead*

$$|W \cup Q_5| \leq j_A,$$

all nonquintic vertices of W can simultaneously be assigned into distinct odd whole A-components having no original neighbor in the entire set Q_5. Arbitrary prescribed owners of the quintics in W and the internal prescriptions in part (II) *are retained. The matching remains on exactly H, with size $p + \delta + h$.*

Without this stronger cardinality hypothesis, part (II) *reserves exactly O_W, not the owners of omitted quintics.*

Proof. Assumed result 7.5 gives eight-regularity,

$$E \geq 10, \quad 2 \leq \epsilon \leq \delta \leq \lambda \leq \left\lfloor \frac{E-4}{2} \right\rfloor, \quad \lambda \geq 3, \quad p \geq 5, \quad q \geq 1.$$

Hence all numerical hypotheses $E \geq 8$, $1 \leq \delta \leq \lambda \leq E-3$, and $q \geq 1$ of the earlier octic inputs hold.

Lemma 7.17 gives the row catalog and realizes any prescribed continuation on the same J. Such a continuation exists because J is finite: successively delete a current-maximum-degree vertex until the maximum degree is at most two. The phase word in Assumed result 7.5 has a nonempty cubic phase. Let K be its actual full cubic entry in the prescribed replay. We use K only for the numerical range; the matching assignments below remain in the original A.

Assumed result 7.8 gives

$$|V(K)| = r + q, \quad \nu(K) = p + \delta + q, \quad |V(F) \setminus V(K)| = r + s - q,$$

and

$$c_0(F) = |V(F) \setminus V(K)| + \nu(K).$$

Since $|V(A)| = r$,

$$u = |V(F) \setminus V(K)| + q.$$

Consequently

$$c_0(F) = u + p + \delta = u + \nu(A) > 0. \tag{26}$$

Minimality of the nonnegative integer universal total $c_0(F)$ gives a missing pair at total $c_0(F) - 1$.

Fix any such pair (x, y). The cubic-entry bound for this same pair gives

$$j_K := |V(F) \setminus V(K)| - x \geq p + 2\lambda + 4 - \delta - q + u_8.$$

Adding q yields

$$j_A = j_K + q \geq p + 2\lambda + 4 - \delta + u_8 = p + s + \delta + 1 + u_8. \tag{27}$$

In particular $j_A > 0$. The replay assertions are precisely Lemma 7.17(III). No replay or assignment changes the already fixed A. This proves part (I).

Every whole component C of A satisfies

$$\nu(C) = \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

For an isolate this follows from the absence of edges. On the even paths and the even cycle in the catalog, an alternating matching is perfect. On an odd cycle, deleting one vertex leaves an even path, whose alternating matching attains the vertex-count upper bound. Thus Assumed result 7.9 applies with

$$R = A, \qquad U = V(J),$$

using (26), the same missing pair, and $j_A > 0$.

For any W of size $h \leq j_A$, it gives one matching assigning every $w \in W$ along an original edge to an exposed port of a distinct odd whole A-component, together with maximum internal matchings in all whole components. Denote the resulting injective component assignment by ϕ.

Assumed result 7.10 gives the unique odd owner for each retained quintic and its simultaneous prescribed-owner interface. For $c \in Q_W$, the component $\phi(c)$ is an odd whole component contacted by c; therefore

$$\phi(c) = O_A(c).$$

Distinct retained quintics have distinct owners. Injectivity of ϕ then places every $\phi(w)$, for $w \in W \setminus Q_5$, outside O_W. Hence

$$\phi(X \cup Y) \subseteq N_{\Pi_W}(X \cup Y)$$

is a set of $|X| + |Y|$ distinct components. Also,

$$\phi(Y) \subseteq N_{\Pi_W}(Y) \setminus O_W$$

has cardinality $|Y|$. These inclusions prove both Hall inequalities on the original exposed-port incidence graph.

Impose all prescriptions in part (II). On $O_A(c)$, replace the supplied assignment edge by the prescribed edge ct_c, and replace its internal matching by the prescribed maximum matching leaving t_c uncovered. Distinct owners do not meet another assigned port, because ϕ is injective. Every prescribed neighbor can be exposed: an odd component is an isolate or an odd cycle, and deleting any vertex leaves, respectively, the empty graph or an even path with a perfect matching.

On each even whole component, replace its internal matching by the prescribed perfect matching. No exterior matching edge enters such a component. On each odd whole component with no W-neighbor, replace its internal matching by the prescribed maximum matching. Such a component is not a destination of ϕ.

These replacements do not conflict. An owned component has a neighbor in W, so it does not belong to the last replacement class. Parity separates the even-component replacements. Keep the assignment edges and internal matchings on all remaining components. Their union is a matching of the full induced H, contains every prescription, and has

$$\nu(A) + h = p + \delta + h$$

edges.

For any matching M in H, at most $\nu(A)$ edges lie entirely in A. Every other edge meets W, and disjoint matching edges meet distinct vertices of W. There are at most h such edges. Thus

$$|M| \leq \nu(A) + h.$$

The constructed matching attains this bound. Unused edges of $H[W]$ remain in the graph. This argument includes $W = \varnothing$, $Q_W = \varnothing$, and empty component families. The final numerical range in part (II) follows from (27).

For part (III), put $W^+ = W \cup Q_5$. Apply the already proved part (II) to W^+. Prescribe the required owner edges for Q_W, and choose owner edges and compatible internal matchings for $Q_5 \setminus W$.

We must preserve the original internal prescriptions on odd components having no W-neighbor. If such a component is owned by a quintic outside W, its prescribed maximum matching has a unique uncovered vertex, which need not be adjacent to that owner. Do not impose that internal matching at the intermediate stage. Keep instead the compatible intermediate internal matching, and restore the original prescription after removing the added owner's assignment edge. If a no-W-neighbor component is unowned, it has no W^+-neighbor, so its matching may be prescribed from the start. Every even-component prescription remains valid.

The intermediate matching assigns every nonquintic of W outside all Q_5-owners, since all quintics participate simultaneously. Delete exactly the assignment edges of the quintics in $Q_5 \setminus W$, and delete those exterior vertices. Keep the internal row matchings, except that on the owned no-W-neighbor components just described, restore the originally prescribed maximum matchings. Those components now have no matching edge to W and are disjoint from all remaining destinations.

The resulting matching lies on exactly H, retains every required prescription, and has $\nu(A)+|W|$ edges. Every nonquintic of W is assigned into a distinct odd component having no original neighbor in Q_5. The upper bound from part (II) proves maximal cardinality. The stronger hypothesis $|W \cup Q_5| \leq j_A$ has been used explicitly. □

Remark 7.22. Theorem 7.21 concerns original-source Hall inequalities and existence of the first matching. It includes every retained nonquintic vertex, including quartic-phase vertices. It asserts no subsequent matching ranks, strict integer payment, whole-source exclusion, or Ramsey formula.

7.4. **Full-boundary absorption.** In this section, we pay every omitted-incident edge by stars and keep one arbitrary matching-bounded edge set intact.

Theorem 7.23 (Full-boundary absorption and an odd-cycle obstruction)**.** *Use the certificate costs of Definition 7.2, with all capacities, budgets, and totals in* $\mathbb{Z}_{\geq 0}$.

(I) *Let* F *be any finite simple graph, let* $S \subseteq V(F)$*, and put*

$$N = |S|, \qquad J = F - S.$$

Let $P \subseteq E(J)$*, and put*

$$m = \nu\big((V(J), P)\big).$$

Let D *be a positive integer. Assume that*

$$K = (V(J), E(J) \setminus P)$$

has maximum degree at most D *and has one matching covering every degree-*D *vertex. If*

$$N \geq \max\{0, m-1\},$$

then

$$c_D(F) \leq N + m.$$

There is no independence assumption on S*, component condition on* P*, or restriction on a source boundary edge.*

(II) *Let* F *be a finite simple graph of maximum degree eight violating* T_-*, while every proper subgraph satisfies* T_-*, including arbitrary edge and vertex deletions. Assume*

$$c_0(F) - c_2(F) = 1, \qquad c_0(F) - c_1(F) = 0.$$

Put

$$k = \nu(F), \quad a = \alpha(F), \quad E = \mathrm{ex}(F), \quad \lambda = 2E - 3 - (k-a), \quad p = E - 2 - \lambda, \quad r = k - \lambda.$$

Let q_8 *be the common quintic and cubic count, and let*

$$b = 2\lambda + 1 - 2q_8 > 0$$

be the positive odd quartic count.

Let A *be any native row of order* r *with*

$$\Delta(A) \leq 2, \quad \mathrm{ps}(A) = 0, \quad \nu(A) = c_1(A) = p + 1, \quad c_2(A) = p.$$

Let $X_0 \subseteq V(F) \setminus V(A)$ *have cardinality* $t \in \mathbb{Z}_{\geq 0}$*. Assume that the entire induced graph*

$$F - (V(A) \cup X_0)$$

is a vertex-disjoint union of blocks B_{2m_j+1} *from Definition 7.14, where every* m_j *is a positive integer, together with any nonnegative integer number of copies of* K_4 *and a graph of maximum degree at most two. Put*

$$M = \sum_j m_j,$$

with $M = 0$ *if there are no blocks. If*

$$r + t \geq \max\{0, M-1\},$$

then

$$M + t \geq q_8 + b.$$

Consequently no such actual decomposition can have $M + t \leq q_8 + b - 1$*, and in particular none can have* $M + t = q_8$.

The decomposition in part (II) *is an additional hypothesis, not a normalization assertion about surviving sources.*

Proof. We first prove part (I). Fix any nonnegative integer capacity split $A_c + B_c = D$. Assumed result 7.12 partitions $E(K)$ into Q_R, Q_B with maximum degrees at most A_c, B_c, without exceptional edges. Regard every vertex of S as isolated in these two parts.

Order S as $s_1, \dots, s_N$. Assign every original edge having an endpoint in S to its smallest-index endpoint in S. Let S_j be the set assigned to s_j. Each S_j is a possibly empty star, and

$$\nu(S_j) \le 1.$$

The sets S_j are pairwise edge-disjoint and cover every edge incident with S, including edges with both endpoints in S. They are disjoint from P and $E(K)$. Thus

$$Q_R,\ Q_B,\ P,\ S_1, \dots, S_N$$

account for all source edges. Their vertex sets need not be disjoint.

Fix any nonnegative integer matching budgets $x + y = N + m$. If $m = 0$, then $P = \varnothing$, since a nonempty simple edge set has matching number at least one. Put x stars in P_R and the other y stars in P_B. Their matching numbers are at most x, y.

Assume $m \ge 1$. Since $N \ge m - 1$,

$$x + y = N + m \ge 2m - 1.$$

Both budgets cannot be at most $m - 1$, since their sum would then be at most $2m - 2$. Suppose first that $x \ge m$. Put all of P, together with exactly $x - m$ stars, in P_R, and put the remaining stars in P_B. The chosen number is an integer satisfying

$$0 \le x - m \le N.$$

The second part receives

$$N - (x - m) = y$$

stars. A matching in P_R uses at most m edges of P and at most one edge from each selected star, so it has size at most x. A matching in P_B has size at most y. These restrictions remain valid when pieces share vertices: the pieces are edge-disjoint, and every matching edge belongs to exactly one piece. If instead $y \ge m$, interchange P_R and P_B.

The four parts form a certificate for every capacity split and every matching-budget split under consideration. Hence $c_D(F) \le N + m$. This includes $N = 0$ and all zero-budget cases.

For part (II), set

$$S = V(A) \cup X_0, \qquad N = r + t, \qquad J = F - S.$$

Let P consist of all distinguished cycle edges in all blocks B_{2m_j+1}. These cycles are vertex-disjoint. A cycle on $2m_j + 1$ vertices has matching number m_j: the vertex count gives the upper bound, and alternate edges leaving one vertex uncovered attain it. Matching number adds over vertex-disjoint components, so $\nu(P) = M$.

After removing P, each block becomes $2m_j + 1$ disjoint copies of K_4. The other stated copies of K_4 and the maximum-degree-two graph are unchanged. Therefore $K = J - P$ has maximum degree at most three. Choose a perfect matching in every copy of K_4. Their union is one matching covering every degree-three vertex of K; the remaining part has no degree-three vertex.

Apply part (I) with

$$D = 3, \qquad N = r + t, \qquad m = M.$$

Its numerical hypothesis is exactly the one in part (II). We obtain

$$c_3(F) \le r + t + M.$$

Assumed result 7.13 gives

$$c_3(F) = r + (2\lambda + 1) - q_8 = r + q_8 + b.$$

Comparison yields

$$q_8 + b \le t + M.$$

The forbidden ranges follow by integer comparison and $b > 0$.

Part (I) paid every edge incident with A or X_0, including edges between deleted vertices. No edge-free boundary or source-matching extension was assumed. □

Remark 7.24. Theorem 7.23 gives a strengthened payment range and a conditional source obstruction. It asserts neither a whole-source exclusion nor a Ramsey formula.

7.5. **Three matchings on one retained graph.** The goal of this section is to use the exact source costs to force three successive matchings to respect every native row contained in the same retained graph.

Theorem 7.25 (Simultaneous three-matching transfer)**.** *Let F be any finite simple eight-regular graph violating T_-, while every proper subgraph satisfies T_-, including arbitrary vertex and edge deletions. Use Setup 7.3, in particular*

$$\zeta = q_{\mathrm{def}} = \delta - \epsilon, \qquad \beta = p + \zeta.$$

Native rows exist. Fix any native row A and any $W \subseteq V(F) \setminus V(A)$. Put

$$H = F[V(A) \cup W], \quad h = |W|, \quad u = |V(F) \setminus V(A)|, \quad N = |V(F) \setminus V(H)| = u - h.$$

Assume the following two additional conditions:

$$\begin{aligned} &H \text{ has a matching of size } p + \delta + h; \\ &N \geq \max\{0, \beta + h - 1\}. \end{aligned} \tag{28}$$

The first condition is equivalent to one assignment of every vertex of W to an actual adjacent vertex in a distinct odd whole A-component. Every assigned vertex is an exposed port. The assignment edges and maximum internal matchings avoiding these ports realize the condition.

Choose any maximum matching M_0 of H, and put

$$P_0 = H, \qquad P_1 = (V(H), E(H) \setminus M_0).$$

Choose any maximum matching M_1 of P_1, and put

$$P_2 = (V(H), E(H) \setminus (M_0 \cup M_1)).$$

Choose any maximum matching M_2 of P_2, and put

$$P_3 = (V(H), E(H) \setminus (M_0 \cup M_1 \cup M_2)).$$

These are edge deletions only. Then the following conclusions hold.

(I) *The exact ranks are*

$$\nu(P_0) = p + \delta + h, \qquad \nu(P_1) = \beta + h, \qquad \nu(P_2) = p + h.$$

For every native row R contained in the same H, put $W_R = V(H) \setminus V(R)$, so $|W_R| = h$. The same M_0 is maximum internally in every whole R-component. Its other h edges saturate W_R into distinct odd whole R-components. The entire first residual row is

$$R_1 = (V(R), E(R) \setminus M_0) = p\mathsf{P}_3 + \zeta\mathsf{P}_2 + \textit{isolates},$$

where the right side denotes disjoint union.

The same M_1 has β internal R_1-edges forming a maximum R_1-matching, and h edges saturating W_R into R. The entire second residual row

$$R_2 = (V(R), E(R) \setminus (M_0 \cup M_1))$$

consists of exactly p pairwise vertex-disjoint edges and isolates. The same M_2 contains all these edges and h edges saturating W_R into R.

None of M_0, M_1, M_2 has an edge inside W_R. Their union contains every edge of every contained native row. All these assertions hold simultaneously for the one chosen sequence.

(II) *For $i \in \{1,2\}$, put*

$$K_i = \left(V(H), \bigcup_{j=0}^{i-1} M_j \right), \qquad \widehat{D}_i = N + \nu(P_i).$$

For every nonnegative integer capacity split $A_c + B_c = i$ and every nonnegative integer matching-budget split $x_c + y_c = \widehat{D}_i$, there is a complete F-certificate whose degree-bounded parts, restricted to H, have union exactly $E(K_i)$.

Every original omitted-incident edge is included, including edges whose two endpoints are omitted. Every fixed M_0-edge remains unpaid by the matching-bounded parts. The full debits are neutral:

$$\widehat{D}_1 = u + \beta = c_1(F), \qquad \widehat{D}_2 = u + p = c_2(F).$$

(III) *For any contained native row R, put*

$$Q_5(R) = \{v \in V(F) \setminus V(R) : d_{F-V(R)}(v) = 5\}.$$

At every $v \in W_R \cap Q_5(R)$, the first three matchings use all three original edges into R. Thus P_3 has no edge inside R, and no edge between R and $W_R \cap Q_5(R)$. Every original edge of $H[W_R]$ survives in P_3.

For any nonempty finite family $(R_j)_{j\in I}$ of contained native rows, put

$$C_{\mathrm{common}} = \bigcap_{j\in I} V(R_j), \qquad X_{\mathrm{common}} = V(H) \setminus \bigcup_{j\in I} V(R_j).$$

Each of M_0, M_1, M_2 saturates X_{common} into C_{common}. Their restrictions are three pairwise edge-disjoint matchings. Every vertex of X_{common} has at least three distinct original neighbors in C_{common}. No disjointness of the three sets of mates is asserted.

Proof. We use the octic row and source-cost results, the generic first-residual equality, and part (I) of Theorem 7.23. The latter two inputs impose no six-regular source condition.

Step 1. Row data and source costs. We first record the component counts governing the residual rows.

Assumed result 7.6 gives row existence and $p \geq 1$. Lemma 7.17 gives, for every native R, the seven whole-component types, exactly p odd cycles, and exactly ϵ positive even-order paths. Let

$$b, t, n_3, n_4, n_5, n_7 \in \mathbb{Z}_{\geq 0}$$

count its whole components $\mathsf{P}_2, \mathsf{P}_4, \mathsf{C}_3, \mathsf{C}_4, \mathsf{C}_5, \mathsf{C}_7$, respectively. Then

$$b + t = \epsilon, \qquad n_3 + n_5 + n_7 = p.$$

The component matching counts give

$$\begin{aligned} p + \delta = \nu(R) &= b + 2t + n_3 + 2n_4 + 2n_5 + 3n_7 \\ &= p + \epsilon + t + 2n_4 + n_5 + 2n_7. \end{aligned}$$

Therefore

$$\zeta = \delta - \epsilon = t + 2n_4 + n_5 + 2n_7 \geq 0, \qquad \beta \geq p. \tag{29}$$

Applied to the entire exterior of A, Assumed result 7.6 gives the actual source identities

$$c_1(F) = u + \beta, \qquad c_2(F) = u + p. \tag{30}$$

Every contained native row has order r, so the same h, u, N apply to all of them.

Step 2. The initial rank respects every contained row. We prove both the simultaneous assertion for M_0 and the assignment equivalence.

Relative to any contained native row R, a matching of H has at most $\nu(R) = p+\delta$ internal edges and at most h other edges, because each other edge meets W_R. Hence

$$\nu(H) \le p + \delta + h.$$

The first condition in (28) forces equality.

For any maximum M_0, both bounds are attained for every R. Its internal restriction is maximum in R, and is maximum in every whole component: otherwise the sum of component matching numbers would not be attained. Its h other edges use exactly one W_R-vertex each and saturate all of W_R into R.

An even component in the catalog has a perfect internal matching, so it supplies no crossing endpoint. An odd component is an isolate or an odd cycle; a maximum internal matching leaves exactly one vertex uncovered. Thus crossing endpoints are exposed ports in distinct odd whole components. This proves the simultaneous M_0 assertion and one direction of the assignment equivalence.

Conversely, assume that the stated assignment relative to A is given. The vertex of an isolate is exposed by its empty matching. Deleting any vertex of an odd cycle leaves an even path, whose alternating perfect matching is maximum in the cycle and avoids that vertex. Choose these internal matchings in the distinct assigned components, and any maximum internal matching in each unassigned component. Their union with the assignment edges has size $\nu(A) + h = p + \delta + h$. This proves the equivalence without a Hall hypothesis and without changing H.

Step 3. The entire first residual row has a fixed form. We determine the edges left by the actual internal restrictions of M_0.

Deleting a maximum matching from P_2 leaves no edge. Deleting one from P_4 leaves its middle edge. Deleting one from C_4 leaves two disjoint edges.

On C_{2v+1}, where $v \in \{1,2,3\}$, a maximum matching exposes one vertex and is a perfect matching of the even path obtained by deleting it. That path has a unique perfect matching: its endpoint forces its incident edge, and repetition on the remaining shorter path forces every edge. The residual cycle edges therefore form one P_3, centered at the exposed vertex, and $v-1$ further disjoint edges. Isolates remain isolated.

Applying these descriptions component by component gives exactly $n_3 + n_5 + n_7 = p$ copies of P_3, and $t + 2n_4 + n_5 + 2n_7 = \zeta$ copies of P_2. Hence the entire R_1 has the asserted form and

$$\nu(R_1) = p + \zeta = \beta$$

for every contained R.

Step 4. The first residual rank is exact on H. We apply the generic prefix equality on the same retained graph.

In Assumed result 7.11, take both row arguments equal to any one contained R, the fixed matching equal to this M_0, and the residual row rank equal to β. The result does not require distinct rows. Its omission hypothesis is the second condition in (28), and its source equality is (30). Therefore

$$\nu(P_1) = \beta + h, \qquad N + \nu(P_1) = c_1(F).$$

Fix any maximum M_1 of this same P_1. For each contained R, its internal and noninternal edge counts are bounded by β and h. Their sum is $|M_1| = \beta + h$, so both bounds are attained. The h noninternal edges each use exactly one W_R-vertex and saturate W_R into R. Its internal restriction is maximum in R_1.

Such a restriction uses one edge in each P_3 and the sole edge of each P_2. Its deletion leaves exactly p disjoint edges and isolates. This proves the entire R_2 description for the same M_1, simultaneously for all R.

Step 5. The second residual rank is exact on H. We force equality by a full-boundary certificate.

A matching of P_2 has at most p internal row edges and at most h other edges, so

$$\nu(P_2) \le p + h.$$

The unpaid graph $K_2 = M_0 \cup M_1$ has maximum degree at most two. Every degree-two vertex has one incident edge from each matching and is covered by the same M_0. By (29) and the omission condition,

$$N \ge \max\{0, \beta + h - 1\} \ge \max\{0, \nu(P_2) - 1\}.$$

Apply Theorem 7.23(I) with degree total two, omitted set $V(F)\setminus V(H)$, exceptional edge set $E(P_2)$, and complementary graph K_2. Using (30),

$$u + p = c_2(F) \le N + \nu(P_2) \le u - h + p + h = u + p.$$

The u in the first expression is the exterior cardinality. All inequalities are equalities. Thus

$$\nu(P_2) = p + h, \qquad N + \nu(P_2) = c_2(F).$$

Fix any maximum M_2 of this same P_2. Relative to every contained R, its internal and noninternal counts attain p, h. Its internal edges are all the edges of R_2. Its other h edges saturate W_R into R, with none inside W_R. All original row edges therefore belong to $M_0 \cup M_1 \cup M_2$. This proves part (I).

Step 6. The certificates retain the exact unpaid graphs. We give every capacity split and every integer budget endpoint.

For $i = 1, 2$, the graph K_i is a union of i edge-disjoint matchings, and M_0 covers every degree-i vertex. The ranks just proved and the omission hypothesis give

$$N \ge \max\{0, \nu(P_i) - 1\}.$$

Thus Theorem 7.23(I) applies.

To identify the exact degree-bounded part, use its construction. At any capacities totaling i, partition all K_i-edges into the two degree-bounded parts. Order the N omitted vertices and assign every original edge meeting them to its first omitted endpoint. This gives N pairwise edge-disjoint, possibly empty stars covering all omitted-incident edges exactly once.

Put $\rho = \nu(P_i)$, and fix budgets $x_c + y_c = N + \rho$. If $\rho = 0$, then P_i is empty; assign x_c stars to the first matching-bounded part and y_c to the other. If $\rho \ge 1$, then

$$N + \rho \ge 2\rho - 1,$$

so at least one budget is at least ρ. If $x_c \ge \rho$, put all of P_i and exactly $x_c - \rho$ stars in the first matching-bounded part, and all remaining stars in the second. The selected number lies between zero and N. Restricting any matching to the edge-disjoint pieces bounds the two matching numbers by x_c, y_c. If $y_c \ge \rho$, use the reversed allocation.

Every source edge is assigned, including every edge with two omitted endpoints. Inside H, the degree-bounded parts use exactly K_i. No fixed M_0-edge is charged to a matching-bounded part or changed. The established equalities give

$$\widehat{D}_1 = N + \beta + h = u + \beta = c_1(F), \qquad \widehat{D}_2 = N + p + h = u + p = c_2(F).$$

This proves part (II).

Step 7. The three matchings clear row edges and agree on common exteriors. We now verify part (III) for the same chosen sequence.

None of the three matchings has an edge inside W_R, so every original edge of $H[W_R]$ survives. Part (I) clears every row edge. If $v \in W_R \cap Q_5(R)$, eight-regularity gives exactly three original neighbors of v in R. Each matching uses a v-to-R edge, and their edge sets are disjoint. They therefore use precisely these three edges.

For a stated family of contained rows, a vertex of X_{common} has its unique mate in each M_i inside every row R_j. That mate lies in C_{common}. Restricting each matching to edges incident with

X_{common} gives three pairwise edge-disjoint matchings, each saturating X_{common} into C_{common}. At a fixed exterior vertex, their three different edges have distinct other endpoints because F is simple. No assertion about disjointness of the three sets of mates is needed. □

Remark 7.26. Theorem 7.25 is conditional on an assignment covering the entire W, equivalently the first condition in (28), and on the numerical omission condition there. It does not establish a selector for a cycle of length 26, an unrestricted original-row Hall condition, a strict payment, a source-class exclusion, or a Ramsey formula.

7.6. **The one-pair cubic-bearing inventory.** In this section, we combine the earlier whole-clique contact pattern with the exterior-pair obstruction to remove the whole-K_4 alternative for every allowed first deficit.

Theorem 7.27 (Whole-clique exclusion and the six-type inventory)**.** *Let F be any finite simple graph of maximum degree eight violating T_-, while every proper subgraph satisfies T_-, including arbitrary edge and vertex deletions. Use the integer parameters of Setup 7.3, and assume*

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E-3, \qquad q_8 = 1.$$

Retain every allowed value of ϵ.

Start a complete source process by deleting any maximal independent set, and then delete arbitrary current-maximum-degree vertices until the maximum degree is at most two. Every actual cubic entry L of every such process has no whole K_4 component. Its unique whole component bearing cubic vertices is exactly one of

$$B_{33}, \quad B_{35}, \quad B_{55}, \quad L_{\square}, \quad \mathrm{cor}(\mathsf{P}_3), \quad \mathrm{cor}(\mathsf{P}_4),$$

as defined in Definition 7.14. This is an exhaustive necessary list, with no occurrence assertion. Every other whole L-component belongs to

$$\mathsf{K}_1, \mathsf{P}_2, \mathsf{P}_4, \mathsf{C}_3, \mathsf{C}_4, \mathsf{C}_5, \mathsf{C}_7.$$

The entire cubic set

$$D_3 = \{v \in V(L) : d_L(v) = 3\}$$

has one or two vertices and is a clique. It is a singleton exactly for $\mathrm{cor}(\mathsf{P}_3)$*, where it is the middle base vertex. Otherwise it consists of two adjacent vertices: the bridge endpoints in a barbell, the leaf-bearing square vertices in $L_{\square}$, or the internal base vertices in* $\mathrm{cor}(\mathsf{P}_4)$*.*

Deleting any member of D_3 is a full legal cubic phase and gives an actual native row of the same source.

Proof. Fix any actual cubic entry L, and suppose it contains a whole K_4 component C. The source range is exactly the range in Assumed result 7.15. Its whole-clique contact statement gives an actual quartic entry

$$H = F - V(L)$$

and a cubic vertex v of H whose entire L-neighborhood consists of three vertices of C, a whole isolate i, and an internal vertex t of a whole path $P = \mathsf{P}_4$.

Choose any one of the three contacted vertices of C, and call it u. Since C is a whole K_4, u is cubic in L. By the singleton insertion and arbitrary cubic deletion statements in Assumed result 7.15,

$$A = L - u$$

is an actual native row of the same source.

The row A contains the whole triangle $T = C - u$, the same whole isolate i, and the same whole path P. Since C was whole,

$$N_F(u) \cap V(A) = V(T).$$

Since u was one of the three contacted vertices of C, $N_F(v) \cap V(A)$ consists exactly of two vertices of T, the vertex i, and the internal vertex t of P. Also $uv \in E(F)$. The vertices u, v are distinct and both lie outside A. Reversing the ordering of P, if needed, makes t its second vertex.

These are precisely the hypotheses of the exterior-pair obstruction in Assumed result 7.15. Its source range has already been checked, and it imposes no additional restriction on ϵ. The contradiction proves that L has no whole K_4. Since L was arbitrary, this holds at every actual cubic entry of every complete source process.

The complete one-pair inventory in Assumed result 7.15 gives exactly one whole cubic-bearing component from its seven types. Removing K_4 leaves exactly the six displayed alternatives. The other whole components remain in the stated seven-item degree-two catalog. This proves the necessary inventory, without asserting realization of any alternative.

The entire cubic sets follow from the component edges. In B_{ij}, only the two bridge endpoints have their cycle degree increased from two to three, and they are adjacent by the bridge. In $L_\square$, only the two leaf-bearing square vertices have degree three, and they are adjacent by definition. In $\mathrm{cor}(\mathsf{P}_3)$, only the middle base vertex has two base neighbors and a leaf. In $\mathrm{cor}(\mathsf{P}_4)$, exactly the two adjacent internal base vertices have these three neighbors. Every remaining vertex in these components has degree at most two. Every vertex in every other whole L-component also has degree at most two.

Finally, Assumed result 7.15 gives an actual native row after deleting any cubic vertex. That row has maximum degree at most two. Hence one such deletion is the full cubic phase, and it is legal because the chosen vertex has degree three in the actual cubic entry. This proves all process quantifiers. □

Remark 7.28. Theorem 7.27 excludes the entire whole-K_4 component branch for all first deficits. It imposes neither $\epsilon = 1$ nor a diagonal restriction. It does not exclude all one-pair sources, an entire degree class, or the Ramsey formula.

7.7. **Excluding the second-deficit-two class.** The goal of this section is to combine the exhaustive octic source alternatives with the earlier equal-deficit source exclusion.

Theorem 7.29 (The second deficit is at least three)**.** *Every $F \in \mathcal{M}_8$ satisfies*

$$c_0(F) - c_2(F) \geq 3.$$

In particular, there is no such source with second deficit two, at any order, excess, rank, first deficit, source phase count, native row, or actual process. The source class uses proper-subgraph minimality under arbitrary vertex and edge deletions, and all costs, capacities, budgets, indices, counts, and finite optimization domains have the integer domains of Definitions 7.2 and 7.4 and Setup 7.3.

Proof. Let F be an arbitrary member of $\mathcal{M}_8$. Define

$$k = \nu(F), \quad a = \alpha(F), \quad E = \mathrm{ex}(F), \quad \lambda = 2E - 3 - (k - a),$$

and

$$\delta = c_0(F) - c_2(F), \quad \epsilon = c_0(F) - c_1(F), \quad r = k - \lambda - 1 + \delta, \quad q_8 = c_4(F) - r.$$

Assumed result 7.5 gives eight-regularity,

$$E \geq 10, \qquad 2 \leq \epsilon \leq \delta \leq \lambda \leq \left\lfloor \frac{E-4}{2} \right\rfloor, \qquad \lambda \geq 3.$$

It also gives the three exhaustive and disjoint alternatives

$$\delta = \lambda \geq 3,\ q_8 = 1; \qquad \delta = \epsilon = 2 < \lambda,\ q_8 = 1; \qquad 3 \leq \delta < \lambda.$$

No further source conclusions are needed.

Suppose $\delta = 2$. The first and third alternatives require $\delta \geq 3$. Hence the second applies to this same original F, giving

$$\delta = \epsilon = 2 < \lambda, \qquad q_8 = 1.$$

Also $E \geq 10$ implies $E \geq 8$, and

$$\left\lfloor \frac{E-4}{2} \right\rfloor \leq E-3.$$

For the latter inequality, $(E-4)/2 \leq E-3$ is equivalent to $E \geq 2$. Thus $\lambda \leq E-3$.

We have checked every hypothesis of Assumed result 7.16 on the same original source: finite simplicity, eight-regularity, proper-subgraph-minimal violation of T_-,

$$E \geq 8, \qquad \delta = \epsilon = 2 < \lambda \leq E-3, \qquad q_8 = 1.$$

That result excludes the entire source class, without an additional pivot, row, process, or marking hypothesis. This is a contradiction.

The original integer bound is $\delta \geq 2$. Excluding equality therefore gives $\delta \geq 3$. Since F was arbitrary, the conclusion holds throughout $\mathcal{M}_8$. □

Remark 7.30. Theorem 7.29 excludes the entire second-deficit-two source class. It is not a whole-degree exclusion and does not assert the Ramsey formula.

8. Degree ten: native rows, port localization and margins

The degree-ten arguments below keep the source graph, its induced rows, and every deletion phase on the same physical vertex set. We assume the earlier certificate and structural statements listed in Section 8.1; the resulting interfaces supply the degree-ten inputs for the strict-sector selector and prefix argument and for the equality-sector prefix argument.

8.1. **Setup.** We first specify the certificate conventions and list explicitly the results assumed from the earlier development of the series.

Notation 8.1. All graphs are finite and simple. For a graph G, write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for its vertex set, edge set, edge count, matching number, independence number, and maximum degree. Write $N_G(v)$ for the open neighborhood of v, and $N_G[I]$ for the union of I with all vertices adjacent to I.

The symbols P_j, C_j, and K_j denote paths, cycles, and complete graphs on j vertices, respectively. Paths are indexed consecutively and cycles cyclically. A whole component means an entire connected component.

For an induced subgraph A of a source F, put

$$N_A(v) = N_F(v) \cap V(A).$$

For $v \in V(A)$, put $d_A(v) = |N_A(v)|$. For disjoint vertex sets U, W, let $e_F(U,W)$ count the actual F-edges between them. Deleting a vertex set always means induced deletion, retaining every source edge between remaining vertices. We also write $F - A$ for $F - V(A)$.

Every capacity, budget, cost index, least universal total, degree, order, cardinality, component count, matching size, phase length, debit, and finite choice domain is a nonnegative integer unless another integer range is stated. Generic gaps and parameter differences may be signed integers. A deletion word such as $6^t5^b4^t$ records t deletions at degree six, followed by b at degree five and t at degree four.

Definition 8.2 (Certificate costs)**.** For $A_c, B_c \in \mathbb{Z}_{\geq 0}$, let $c_{A_c,B_c}(G)$ be the least $T \in \mathbb{Z}_{\geq 0}$ such that every pair $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = T$ admits a partition

$$E(G) = Q_R \,\dot\cup\, Q_B \,\dot\cup\, P_R \,\dot\cup\, P_B$$

satisfying

$$\Delta(Q_R) \leq A_c, \qquad \Delta(Q_B) \leq B_c, \qquad \nu(P_R) \leq x, \qquad \nu(P_B) \leq y.$$

Quantities associated with an edge set refer to its spanning graph. For $i \in \mathbb{Z}_{\geq 0}$, put

$$c_i(G) = \max_{\substack{A_c, B_c \in \mathbb{Z}_{\geq 0} \\ A_c + B_c = i}} c_{A_c, B_c}(G).$$

These are the generic certificate costs, also denoted c_i^* in the earlier development. Put

$$\operatorname{ex}(G) = c_0(G) - \nu(G), \qquad \operatorname{ps}(G) = e(G) - \nu(G) - \sum_{i \geq 1} c_i(G),$$

and

$$\operatorname{gap}(G) = \sum_{i \geq 0} c_i(G) - e(G) = \operatorname{ex}(G) - \operatorname{ps}(G).$$

The sums are finite.

If $c_0(G) = 0$, put $\operatorname{tr}(G) = 0$. Otherwise, $\operatorname{tr}(G)$ is the greatest length of a consecutive integer run of values

$$x \in \{0, \ldots, c_0(G) - 1\}$$

for which $E(G)$ has no partition into two parts with matching numbers at most x and $c_0(G) - 1 - x$. Such a pair of budgets is called a *missing pair*.

Definition 8.3 (Sources, goodness, and processes)**.** A graph satisfies T_- if

$$\operatorname{gap}(G) > 0 \quad \Longrightarrow \quad \operatorname{gap}(G) = 1 \quad \text{and} \quad \operatorname{tr}(G) + \operatorname{ex}(G) \leq \max\left(4, \left\lfloor \frac{\Delta(G)}{2} \right\rfloor + 1\right).$$

Let $\mathcal{M}_{10}$ consist of all graphs of maximum degree exactly ten that violate T_-, while every proper subgraph satisfies T_-. Proper subgraphs here allow arbitrary vertex deletions, edge deletions, or both.

A graph is well-covered if every maximal independent set is maximum. It is factor-critical if deleting any one vertex leaves a perfect matching. For $j \in \mathbb{Z}_{\geq 1}$, a graph is j-good if its maximum degree is at most j and one matching covers every vertex of degree j.

A complete source process deletes any maximal independent set of F, then repeatedly deletes any vertex of current maximum degree while that degree is at least three. A full degree-j phase consists of all deletions at current degree j. A complete continuation on a row complement follows the same rule until the maximum degree is at most two.

Setup 8.4 (Source parameters)**.** For $F \in \mathcal{M}_{10}$, put

$$n = |V(F)|, \qquad k = \nu(F), \qquad a = \alpha(F), \qquad E = \operatorname{ex}(F),$$
$$\lambda = 2E - 3 - (k - a), \qquad \delta = c_0(F) - c_2(F), \qquad \epsilon = c_0(F) - c_1(F),$$
$$p = E - 2 - \lambda, \qquad r = k - \lambda - 1 + \delta, \qquad s = 2\lambda + 3 - 2\delta,$$
$$u = r + s, \qquad M = k + E - r.$$

These parameters are integers. Their nonnegative and positive ranges are conclusions below.

Definition 8.5 (Native rows)**.** For the parameters of Setup 8.4, a native row is an induced subgraph A of F satisfying

$$|V(A)| = r, \qquad \Delta(A) \leq 2, \qquad \operatorname{ps}(A) = 0,$$

and

$$\nu(A) = p + \delta, \qquad c_1(A) = p + \delta - \epsilon, \qquad c_2(A) = p.$$

In the equality sector $c_2(F) = c_0(F)$, the earlier saturated-row statement identifies these rows with

$$pC_3 \,\dot\cup\, zK_1, \qquad z = a - p.$$

For a whole odd component C, an exposed port is a vertex v for which $C - v$ has a perfect matching. Thus

$$\operatorname{Port}(C) = \{v \in V(C) \mid C - v \text{ has a perfect matching}\}.$$

The unique vertex of an isolated component is its exposed port.

We assume, as established earlier in this series, the following statements. The certificate inputs belong to Companion I, and the universal source and process inputs belong to the earlier universal development. The octic row and matching technology from Chapters 3–4 is used only through the degree-ten specializations expressly recorded below.

[cite/blocker] Numbered references to Companion I and to the earlier chapters were not supplied. The intended placeholders are [I, Theorem X] and the corresponding earlier-chapter theorem numbers. They must be resolved before this chapter is treated as independently referenced.

[math/blocker] [GAP: the complete original hypotheses and earlier-series locations of the unreproduced degree-ten structure, localization, and exchange results were not supplied. The precise specializations used below are recorded as assumptions; their availability before this chapter remains to be checked.]

Setup 8.6 (Earlier certificate and matching statements)**.** We assume the following certificate statements from Companion I.

(I) A one-good graph has first cost zero, and a two-good graph has second cost zero. If H is induced in G, then
$$c_i(G) \le |V(G) \setminus V(H)| + c_i(H)$$
at every nonnegative integer index used below. The insertion certificate restores every original edge incident with an omitted vertex, including edges whose two endpoints were omitted.

(II) For the connected ten-regular near-perfect sources used in the saturated matching argument, put
$$u_{\mathrm{own}} = \max\left(0, \left\lfloor \frac{2E-3}{8} \right\rfloor\right).$$
The regular missing-allocation bound gives
$$k \ge 3E - 1 + 2u_{\mathrm{own}}.$$

(III) Suppose that $V(F) = A \dot\cup J$, every whole component of $F[A]$ is odd and has a near-perfect matching, every vertex of each component is an exposed port, and
$$c_0(F) = |J| + \nu(F[A]).$$
For a missing pair (x, y), put $j = |J| - x$. Every prescribed subset of J of size at most j can be matched by actual source edges into distinct whole A-components, together with a maximum internal matching in every whole A-component. We use this statement for rows consisting of triangles and isolates.

Setup 8.7 (Earlier source and conditional row statements)**.** The following are the earlier source and process inputs, always with the proper-subgraph convention of Definition 8.3.

(I) Every degree-ten source is connected, ten-regular, factor-critical, and well-covered, with
$$n = 2k + 1, \qquad E \ge 12, \qquad \lambda \ge 2.$$
Every actual prefix and terminal of every complete source process has independence number a.

(II) The conditional row bookkeeping theorem assumes the existence of a complete source process whose terminal has independence number a. Under that premise it gives the row existence, component list, component counts, phase counts, source costs, and complement formulas stated in Theorem 8.13(I)–(IV). Its root-partition conclusion gives the bound and boundary identities in Theorem 8.13(V) for an actual full process. The componentwise replay used in the proof below is part of this earlier row development.

(III) Under the same terminal-existence premise, the top-two-defect theorem gives exactly the inequalities and the rank-two and rank-three restrictions stated in Theorem 8.13(VI).

(IV) The stronger degree-ten source theorem gives, in both sectors,

$$\operatorname{gap}(F) = 1, \qquad E \geq 12, \qquad \lambda \geq 3, \qquad p \geq 5.$$

It also gives the three low source-cost identities

$$c_0(F) = k + E, \qquad c_1(F) = k + E - \epsilon, \qquad c_2(F) = k + E - \delta.$$

Setup 8.8 (Earlier even-component and port-localization statements)**.** For the same source and every native row, the earlier even-component localization theorem gives exactly the fixed independent witness, closed-neighborhood complement, well-coveredness, and extension statement in Theorem 8.15(III). It excludes a whole P_6 component.

The earlier complete port-localization theorem gives the entire statement of Theorem 8.15(IV): existence of missing pairs; its bound for every ordered missing pair; saturation of the entire incidence side; coverage of every whole P_3 by every saturating matching; the alternating closure and entire-neighborhood identity; the localized independent witness and matching and independence formulas; and the simultaneous physical reservation for the whole prescribed set. It does not require the reservation matching to extend the incidence matching defining the alternating closure.

The same earlier development gives

$$p = \tau(F) - c_0(F),$$

where $\tau(F)$ is the minimum vertex-cover cardinality. It also gives the conditional assertion that $p = 0$ forces every native row to have components only from K_1, P_2, P_4, C_4.

Setup 8.9 (Earlier selector and shortest-cycle reserve statements)**.** We use the following specializations of the earlier generic statements.

(I) For a maximum-degree-two terminal R, let $m_R = \nu(R)$, let ϵ_R count its positive even-order path components, and put

$$\beta_R = m_R - \epsilon_R.$$

For $D_2 = \{v \in V(R) \mid d_R(v) = 2\}$, a maximum independent set T_0 of $R[D_2]$, and a whole odd cycle C, put

$$e_C = \frac{|V(C)| + 1}{2}, \qquad Y = T_0 \cup V(C).$$

The physical selector statement gives

$$\alpha(R[D_2]) = |T_0| = \beta_R, \qquad |Y| = \beta_R + e_C,$$

and $R[Y]$ is C together with isolates, with matching number $e_C - 1$. In an actual partition

$$V(F) = V(A) \dot\cup X \dot\cup V(R),$$

all original neighbors of Y outside R lie in $V(A) \cup X$. If F is ten-regular, each vertex of Y has exactly eight such neighbors.

(II) For the native terminals used here, with p whole odd cycles, $m_R = p + \delta$, and $\beta = m_R - \epsilon$, the generic shortest-cycle reserve applies with parameters

$$B_{\mathrm{res}} = E, \qquad \ell_{\mathrm{res}} = 8,$$

once

$$E \geq m_R + 1, \qquad 8 < 2p.$$

For a shortest whole odd cycle it gives

$$\beta \geq p, \qquad e_C \leq 1 + \left\lfloor \frac{\beta}{p} \right\rfloor, \qquad e_C \leq 1 + \left\lfloor \frac{2E - 3}{8} \right\rfloor.$$

The last inequality is the strict-divisor case of that result.

(III) Put $v_{10} = \lfloor (2E-3)/8 \rfloor$. For the same actual partition and selector, the additional premise

$$r \geq s + 3m_R + 1 + 2v_{10}$$

gives

$$N - (\beta + h - 1) \geq 3\epsilon + 2(v_{10} + 1 - e_C),$$

where $h = s + \beta + e_C$ and $N = r - \beta - e_C$. Its numerical-range conclusion says that

$$j \geq s + m_R + v_{10} + 1 \quad \Longrightarrow \quad j - h \geq \epsilon + v_{10} + 1 - e_C.$$

Setup 8.10 (Earlier saturated-row statement)**.** In the equality sector, the earlier saturated-row theorem gives

$$\lambda \geq 2, \qquad p, z \geq 0,$$

and identifies the terminals as $pC_3 \,\dot\cup\, zK_1$. For every such induced row A, with $J = F - A$, it gives

$$|V(J)| = k + \lambda + 2, \qquad \Delta(J) \leq 7, \qquad c_0(F) = k + E = |V(J)| + p.$$

Every vertex of J either meets a whole A-triangle or has at least three neighbors among the A-isolates.

Setup 8.11 (Three-neighbor dichotomy for saturated rows)**.** Let $F \in \mathcal{M}_{10}$ satisfy $c_2(F) = c_0(F)$, and let A be any induced row from Setup 8.10. We assume the following additional attachment statement. Every vertex $v \in V(F) \setminus V(A)$ with $|N_A(v)| = 3$ satisfies one of these alternatives:

(I) All three vertices of $N_A(v)$ are whole A-isolates.

(II) The set $N_A(v)$ is the vertex set of one whole A-triangle.

Setup 8.12 (Earlier equality-frame and exchange statements)**.** The following statements concern the original degree-ten source with $c_2(F) = c_0(F)$, every equality-native row, and actual complete processes.

(I) The equality process theorem removes the entire degree-seven phase, bounds every row complement by degree six, and gives the common word, common parameter t, terminal rows, and complete source-cost profile in Theorem 8.22(I).

(II) The reversible-frame theorem gives the partition, block independence, reversal, all block reorderings, five-neighbor boundaries, actual quartic outers, all outer costs, and five-good complements in Theorem 8.22(II). It also gives the simultaneous phase choices in part (IV).

(III) The triangle-contact and outer-insertion statement gives contact with at most one whole row triangle, exclusion of an exact triangle neighborhood, and the complete third-cost insertion identity in Theorem 8.22(IV). Its two hypothetical outer certificates have bounds $t-2$ and $t-1$ for, respectively, a whole $K_{4,4}$ and two free quartic roots sharing a terminal isolate.

(IV) The pair-exclusion and physical-exchange statement gives the pair exclusion and its paid-rank-p+1 certificate in Theorem 8.22(III). It gives $\Delta(F[X]) \leq 2$, the entire exchanged quartic set, the literal fixed-row identity in part (VI), and iteration under finite compositions of those legal exchanges.

(V) The triangle-plus-two exclusion gives the three independent exterior vertices with the same hypothetical mask, the paid-rank-$p+2$ certificate, and its complete bipartite residual in part (III). It also gives every isolate lower bound and row-neighbor mask in part (V).

(VI) The complete outer-component statement excludes the four-isolate mask and gives the five component types, the unique full quartic phase, disjoint entire row-neighborhoods, and the extension of every independent subset of the entire degree-six pool in part (IV).

(VII) The boundary matching and swap statement gives the entire Y-to-U matching, the unit matching increment for every middle vertex, and the legal full-phase exchange on either fixed outer in part (VI).

The precise conclusions referred to here are displayed in full in Theorem 8.22; no additional clauses of the earlier statements are assumed.

8.2. **Universal rows and the four-degree core.** In this section, we remove the terminal-existence premise from the earlier row bookkeeping and retain the quantifiers over every row and every continuation.

Theorem 8.13 (Universal rows and top-two-defect restrictions)**.** *Let $F \in \mathcal{M}_{10}$, with the parameters of Setup 8.4.*

(I) *The source is connected, ten-regular, factor-critical, and well-covered, with*

$$n = 2k + 1, \qquad E \geq 12, \qquad \lambda \geq 2.$$

Every actual prefix and terminal of every source process has independence number a. Moreover,

$$p \geq 0, \qquad 0 \leq \epsilon \leq \delta \leq \lambda + 1,$$

s is positive and odd, and

$$n = 2r + s, \qquad M = p + \delta + s.$$

(II) *Native rows exist, and every terminal is a native row. Every native row A has*

$$\alpha(A) = a, \qquad e(A) = 3p + 2\delta - \epsilon,$$

exactly p odd-cycle components, and exactly ϵ positive even-order path components. Its whole components belong to

$$K_1,\ P_2,\ P_3,\ P_4,\ P_6,\ C_3,\ C_4,\ C_5,\ C_7.$$

(III) *Every source process has exactly $p+\epsilon$ degree-nine roots and $p+\delta-\epsilon$ degree-eight roots. Their induced union with the initial independent set is a native row.*

Conversely, every induced native row is the entire vertex set removed by the initial independent set and the first two degree phases of some actual source process. The source costs satisfy

$$c_9(F) = a, \qquad c_8(F) = a + p + \epsilon, \qquad c_7(F) = r.$$

(IV) *For every induced native row A, its complement $J = F - A$ is seven-good. It satisfies*

$$|V(J)| = r + s, \qquad \alpha(J) = a, \qquad \nu(J) = M, \qquad \mathrm{ps}(J) = 0,$$
$$e(J) = 3p + 2\delta - \epsilon + 5s,$$

and

$$c_i(J) = c_i(F) - r \qquad (1 \leq i \leq 7).$$

In particular,

$$c_1(J) = M - \epsilon, \qquad c_2(J) = M - \delta, \qquad \sum_{i \geq 3} c_i(J) = 2s.$$

(V) *For every induced native row A, every complete maximum-degree process on $J = F - A$ is an actual continuation of a source process. It has exactly s roots, its deletion degrees sum to $5s$, and its terminal R is another native row.*

If X is its entire root set, then

$$V(F) = V(A) \,\dot\cup\, V(R) \,\dot\cup\, X, \qquad \Delta(F[X]) \leq 4,$$

and

$$e_F(X, V(A)) = e_F(X, V(R)) = 5s - e(F[X]).$$

These assertions prescribe no matching exposure, independent cut, positive-row allocation, or source-minimality inheritance for J.

(VI) *Uniformly in both source sectors, if* $\delta = \lambda$, *then*

$$k \geq 8E - 25 - 8\lambda, \qquad 32E \leq 63\lambda + 199.$$

If $\delta = \lambda + 1$, *then*

$$k \geq 8E - 24 - 8\lambda, \qquad 32E \leq 63\lambda + 190.$$

At $\lambda = 2$, *one has* $\delta \leq 1$. *At* $\lambda = 3$, *one has* $\delta \leq 3$, *and* $\delta = 3$ *forces*

$$(E, k, a, n) = (12, 47, 29, 95).$$

None of these numerical profiles is asserted realizable.

No terminal-existence premise remains in these conclusions. They concern the full class $\mathcal{M}_{10}$, *with both sectors and the same proper-subgraph hypotheses.*

Proof. We use the source, universal independence, conditional row bookkeeping, and top-two-defect statements recorded in Setup 8.7. Every application permits both vertex and edge deletion in the original minimality hypothesis. No remainder is assumed minimal.

The source statement gives the structural assertions, $E \geq 12$, $\lambda \geq 2$, and well-coveredness. Therefore every maximal independent initial set has size a. A complete source process exists: whenever the current maximum degree is at least three, choose a vertex of that degree and delete it. Each step decreases the finite number of vertices.

The universal independence statement applies to every such process and gives independence number a at each prefix and terminal. Hence the existence premise of the conditional row bookkeeping theorem is satisfied. Apply that theorem at degree ten, with its two deficits equal to δ and ϵ. It gives the row and initial-phase assertions, positive odd s, and the complement formulas, apart from the strengthened continuation quantifiers verified below. Since

$$s = 2\lambda + 3 - 2\delta > 0,$$

integrality gives $\delta \leq \lambda + 1$. Substitution in the definitions gives

$$M = p + \delta + s.$$

Every terminal preserves independence, so the terminal formulas identify every terminal as a row.

We next verify the converse row quantifier and the actual replay needed for arbitrary continuations. In any induced row, label paths consecutively and cycles cyclically. The componentwise choices of initial independent vertices and first-phase vertices are

component	initial vertices	first-phase vertices
K_1	$\{1\}$	$\varnothing$
P_2	$\{1\}$	$\{2\}$
P_3	$\{1,3\}$	$\varnothing$
P_4	$\{1,3\}$	$\{4\}$
P_6	$\{1,3,5\}$	$\{6\}$
C_3	$\{1\}$	$\{2\}$
C_4	$\{1,3\}$	$\varnothing$
C_5	$\{1,3\}$	$\{4\}$
C_7	$\{1,3,5\}$	$\{6\}$

Let S be the union of the initial sets. The row independence identity gives $|S| = a$, so S is a maximum independent set of F.

Let T_1 be the union of the first-phase sets. It is independent, has size $p + \epsilon$, and each of its vertices has exactly one neighbor in S. Ten-regularity therefore permits each vertex of T_1 to be deleted at degree nine. The universal first-phase count from the conditional row theorem says that these vertices exhaust that phase.

The remaining row vertices form an independent set T_2 of size $p+\delta-\epsilon$. Each has exactly two row-neighbors, both already deleted. They can therefore be deleted at degree eight. The universal second-phase count says that these exhaust that phase. Thus the removed set is exactly $V(A)$, and $J = F - A$ is an actual source prefix.

Fix any complete maximum-degree process on this J. Append it to the displayed replay of A. The result is a full source process, so universal independence gives

$$\alpha(J) = a$$

and terminal independence a. Its terminal R is consequently a row of order r. Since $|V(J)| = r+s$, the appended process has exactly s roots.

Each deleted edge is counted once by its deletion-time degree. Hence the sum of those degrees is

$$e(J) - e(R) = (3p + 2\delta - \epsilon + 5s) - (3p + 2\delta - \epsilon) = 5s.$$

The actual-process conclusion of the conditional row theorem now gives the partition into A, R, X, the bound

$$\Delta(F[X]) \le 10 - 6 = 4,$$

and both boundary identities $5s - e(F[X])$. This establishes the every-row and every-continuation assertions without selecting a special terminal.

For the remaining cost identity, $\mathrm{ps}(J) = 0$ gives

$$\sum_{i\ge 3} c_i(J) = e(J) - \nu(J) - c_1(J) - c_2(J).$$

Substituting the displayed formulas yields

$$\begin{aligned}\sum_{i\ge 3} c_i(J) &= 3p + 2\delta - \epsilon + 5s - M - (M-\epsilon) - (M-\delta)\\ &= 3p + 3\delta + 5s - 3M\\ &= 2s,\end{aligned}$$

because $M = p + \delta + s$.

Finally, the top-two-defect theorem has the same source-minimality premise and the same existence-of-an-independence-preserving-terminal premise already discharged above. Its degree-ten specialization gives every assertion in part (VI), including the rank-three deficit-three tuple. This application is uniform in λ and ϵ, and introduces no equality-sector premise. □

Remark 8.14. Theorem 8.13 strengthens the earlier conditional row and defect statements by discharging their terminal premise throughout the original degree-ten source class. It leaves all other necessary restrictions on that class in force. It proves neither whole-degree emptiness nor an unconditional star-forest formula.

8.3. **Native rows and complete port localization.** The goal of this section is to combine Theorem 8.13 with the stronger source bounds and the earlier localization statements, retaining the choices of rows, budgets, and matchings.

Theorem 8.15 (Native rows and complete port localization)**.** *Let $F \in \mathcal{M}_{10}$, with the parameters and native rows of Section 8.1.*

(I) *The source is connected, ten-regular, factor-critical, and well-covered, with*

$$\mathrm{gap}(F) = 1, \qquad n = 2k+1 = 2r+s,$$

$$E \ge 12, \qquad \lambda \ge 3, \qquad p \ge 5, \qquad 0 \le \epsilon \le \delta \le \lambda + 1.$$

The integer s is positive and odd, and $M = p + \delta + s$. These assertions apply in both sectors $\delta = 0$ and $\delta > 0$, without asserting either sector empty.

Every actual prefix remainder and every terminal of every complete source process has independence number a. Native rows exist, and every terminal is a native row. Every native row A satisfies

$$\alpha(A) = a, \qquad e(A) = 3p + 2\delta - \epsilon,$$

has exactly p odd-cycle components and exactly ϵ positive even-order path components, and has only whole components from

$$K_1,\ P_2,\ P_3,\ P_4,\ C_3,\ C_4,\ C_5,\ C_7.$$

This is a necessary list; no occurrence of P_3, or of any other listed type, is asserted.

Every complete source process has exactly $p+\epsilon$ degree-nine roots and $p+\delta-\epsilon$ degree-eight roots. The initial independent set and these two full phases induce a native row. Conversely, every induced native row is the entire early deleted pool of some complete source process.

The exact source costs include

$$c_0(F) = k + E,$$
$$c_1(F) = k + E - \epsilon = u + p + \delta - \epsilon,$$
$$c_2(F) = k + E - \delta = u + p,$$
$$c_9(F) = a, \qquad c_8(F) = a + p + \epsilon, \qquad c_7(F) = r.$$

(II) *For any native row A, put $J = F - A$. Then J is seven-good, and*

$$|V(J)| = u = r + s, \qquad \alpha(J) = a, \qquad \nu(J) = M, \qquad \mathrm{ps}(J) = 0,$$
$$e(J) = 3p + 2\delta - \epsilon + 5s.$$

Moreover,

$$c_i(J) = c_i(F) - r \qquad (1 \le i \le 7),$$

and

$$c_1(J) = M - \epsilon, \qquad c_2(J) = M - \delta, \qquad \sum_{i \ge 3} c_i(J) = 2s.$$

Every complete continuation on this same J is an actual continuation of a source process. It has exactly s roots, deletion-degree sum $5s$, and a native terminal R. If X_{root} is its entire root set, then

$$V(F) = V(A) \,\dot\cup\, V(R) \,\dot\cup\, X_{\mathrm{root}},$$
$$\Delta(F[X_{\mathrm{root}}]) \le 4,$$

and

$$e_F(X_{\mathrm{root}}, V(A)) = e_F(X_{\mathrm{root}}, V(R)) = 5s - e(F[X_{\mathrm{root}}]).$$

(III) *Let Z be the union of the vertex sets of all even-order whole A-components. There is an independent set*

$$I_{\mathrm{even}} \subseteq V(A) \setminus Z$$

such that

$$|I_{\mathrm{even}}| = a - \frac{|Z|}{2}, \qquad V(F) \setminus N_F[I_{\mathrm{even}}] = Z.$$

The graph $F[Z]$ and each of its whole components are well-covered. Every maximal independent set K_{even} of $F[Z]$ extends with the same I_{even} to a maximum independent set

$$I_{\mathrm{even}} \cup K_{\mathrm{even}}$$

of F. In particular, no native row has a whole P_6 component.

(IV) *Put* $U = V(F) \setminus V(A)$. *Let* h_3 *count the whole* P_3 *components of* A, *let* I_0 *be the set of all whole* A*-isolates, and put*

$$B_0 = \{v \in U \mid N_F(v) \cap I_0 = \varnothing\}.$$

Let $\mathcal{O}$ *be the family of whole* A*-components of odd order greater than one. Define the bipartite incidence graph* Γ, *with vertex classes* $B_0, \mathcal{O}$, *by joining* v *to* C *exactly when* v *has an actual source neighbor in* $\mathrm{Port}(C)$.

Choose any missing pair $b_r, b_s \in \mathbb{Z}_{\geq 0}$ *with*

$$b_r + b_s = c_0(F) - 1.$$

Such pairs exist, and no ordering of their coordinates is assumed. Put

$$j_{\mathrm{range}} = |U| - b_r.$$

Then

$$j_{\mathrm{range}} \geq s + p + \delta + \left\lfloor \frac{2E - 3}{8} \right\rfloor + 1.$$

The graph Γ *has a matching saturating the entire* B_0, *and every such matching covers every whole* P_3 *on the* $\mathcal{O}$*-side. Fix any such matching* M_{inc}. *Starting from all* P_3 *component vertices, traverse* M_{inc}*-edges from* $\mathcal{O}$ *to* B_0 *and non-*M_{inc}*-edges from* B_0 *to* $\mathcal{O}$. *Let* B *and* $\mathcal{H}$ *be the reached vertices on the respective sides.*

Then

$$N_\Gamma(B) = \mathcal{H},$$

every whole P_3 *belongs to* $\mathcal{H}$, *and* M_{inc} *pairs* B *bijectively with* $\mathcal{H}$. *If* $p_{\mathcal{H}}$ *counts the whole odd cycles in* $\mathcal{H}$, *then*

$$|B| = |\mathcal{H}| = h_3 + p_{\mathcal{H}}, \qquad 0 \leq p_{\mathcal{H}} \leq p.$$

Here $N_\Gamma(B)$ *is the entire incidence-neighbor set of* B.

With Z *as in part* (III), *put*

$$V_{\mathcal{H}} = \bigcup_{C \in \mathcal{H}} V(C), \qquad W_{\mathrm{loc}} = Z \cup B \cup V_{\mathcal{H}}, \qquad t_{\mathrm{loc}} = \nu(F[Z \cup V_{\mathcal{H}}]).$$

There is an independent set

$$I_{\mathrm{loc}} \subseteq V(A) \setminus (Z \cup V_{\mathcal{H}})$$

such that

$$V(F) \setminus N_F[I_{\mathrm{loc}}] = W_{\mathrm{loc}}.$$

The graph $F[W_{\mathrm{loc}}]$ *is well-covered, has a perfect matching, and satisfies*

$$\nu(F[W_{\mathrm{loc}}]) = t_{\mathrm{loc}} + |\mathcal{H}|,$$
$$\alpha(F[W_{\mathrm{loc}}]) = t_{\mathrm{loc}} + h_3,$$
$$\nu(F[W_{\mathrm{loc}}]) - \alpha(F[W_{\mathrm{loc}}]) = p_{\mathcal{H}}.$$

For every set $T_{\mathrm{extra}} \subseteq U \setminus B$ *with*

$$|T_{\mathrm{extra}}| \leq j_{\mathrm{range}} - |B|,$$

there is one matching in F *that sends every vertex of* $B \cup T_{\mathrm{extra}}$ *to an exposed port in a distinct odd whole* A*-component. It sends* B *into* $\mathcal{H}$ *and* T_{extra} *outside* $\mathcal{H}$, *and includes a maximum internal matching in every whole* A*-component. For a singleton component the port is its unique vertex.*

This physical matching need not extend M_{inc}. *No independence hypothesis on* T_{extra} *is imposed.*

(V) *Let $\tau(F)$ be the least cardinality of a vertex set meeting every edge of F. Then*

$$p = \tau(F) - c_0(F).$$

If $\delta = \lambda$, then

$$k \geq 8E - 25 - 8\lambda, \qquad 32E \leq 63\lambda + 199.$$

If $\delta = \lambda + 1$, then

$$k \geq 8E - 24 - 8\lambda, \qquad 32E \leq 63\lambda + 190.$$

At $\lambda = 3$, one has $\delta \leq 3$, and $\delta = 3$ forces

$$(E, k, a, n) = (12, 47, 29, 95).$$

The conditional consequences

$$\lambda = 2 \Longrightarrow \delta \leq 1$$

and

$$p = 0 \Longrightarrow \textit{every native row has only } K_1, P_2, P_4, C_4$$

also hold. Their antecedents are empty under the unconditional bounds in part (I)*, so they give no additional occurrence or payment coverage.*

All assertions concern the original proper-subgraph source, every indicated row and process, and every indicated budget or matching choice.

Proof. We apply Theorem 8.13 together with the stronger source statement and the two localization statements recorded in Setups 8.7 and 8.8. Their certificate conventions, proper-subgraph source hypothesis, and native-row definition agree with those used here.

The stronger source statement gives connectedness, ten-regularity, factor-criticality, well-coveredness, gap one, order $2k + 1$, and the three low source-cost identities. It gives

$$E \geq 12, \qquad \lambda \geq 3, \qquad p \geq 5$$

in both sectors. These are consequences for arbitrary sources, rather than additional hypotheses selecting a subfamily.

Theorem 8.13(I) gives

$$0 \leq \epsilon \leq \delta \leq \lambda + 1, \qquad n = 2r + s, \qquad M = p + \delta + s,$$

positive odd s, and independence number a at every actual prefix and terminal. Its part (II) gives row existence, every terminal's row status, every row's independence and edge count, and both component counts.

The even-component localization statement applies to this same arbitrary row and removes P_6 from the initial nine-item list. The remaining necessary list is exactly the one in part (I). That statement does not remove P_3.

The two full early phases, their counts, realization of every induced row, and the source coefficients c_9, c_8, c_7 follow from Theorem 8.13(III). The parameter identities give

$$u + p = k + E - \delta, \qquad u + p + \delta - \epsilon = k + E - \epsilon.$$

Together with the low source-cost identities, these give the alternative formulas for $c_1(F)$ and $c_2(F)$.

For part (II), Theorem 8.13(IV) supplies every complement identity and seven-goodness for every induced row. Its part (V) says that every complete process on this same complement can be appended to an actual early realization. It supplies the exact number s of roots, deletion-degree sum $5s$, native terminal, full partition, root-degree bound, and both cross-edge identities. No independently chosen terminal or minimality assumption on J is introduced.

For part (III), apply the even-component localization statement with its even vertex set equal to the entire Z defined here. Denote its independent witness by I_{even}. It gives the exact closed-neighborhood complement, the size of I_{even}, well-coveredness of $F[Z]$ and its components, and extension of every maximal independent K_{even} with this same witness. The source and row hypotheses have already been matched.

For part (IV), apply the complete port-localization statement to the same native row A. Its count of whole P_3's is h_3; its missing budgets are b_r, b_s; its range is j_{range}; its arbitrary saturating incidence matching is M_{inc}; and its localized vertex set and retained-row matching number are W_{loc} and t_{loc}. The actual-edge incidence through exposed ports is unchanged.

That statement gives a missing nonnegative integer pair and applies to every ordered missing pair without restricting the first coordinate to be smaller. Hence it gives the displayed range inequality. It saturates the entire B_0 and covers every P_3 with every saturating incidence matching.

For every chosen M_{inc}, it gives exactly the specified alternating closure, including the entire-neighborhood equality, the bijection, and the cardinality $h_3 + p_{\mathcal{H}}$. Its independent witness lies in the discarded row components, namely

$$V(A) \setminus (Z \cup V_{\mathcal{H}}).$$

It gives well-coveredness, a perfect matching, and all three matching and independence formulas for $F[W_{\mathrm{loc}}]$.

Its final reservation statement applies to the entire set $B \cup T_{\mathrm{extra}}$ at once. Each incidence edge is realized by an actual source edge to an exposed port. Distinct assigned whole components make these source edges disjoint and permit maximum internal matchings avoiding the assigned ports. This gives the stated physical matching, including maximum internal matchings in every whole row component. The statement imposes no independence condition on T_{extra}, and does not require the physical matching to extend M_{inc}. On a singleton, deletion of the port leaves the empty perfect matching.

For part (V), Setup 8.8 supplies the vertex-cover identity and the conditional $p = 0$ consequence. Theorem 8.13(VI) supplies the two top-deficit inequalities, the rank-three restrictions and exact tuple, and the conditional rank-two consequence. Their source hypotheses are unchanged and no equality-sector condition is added. Since $\lambda \geq 3$ and $p \geq 5$, the two retained conditionals have empty antecedents and give no fresh exclusion. □

Remark 8.16. No remainder in Theorem 8.15 is asserted source-minimal. No arbitrary ambient maximum matching is asserted to lift, and separately chosen matching witnesses are not identified. The theorem asserts no supported-triangle packet reachability, positive correction margin, full process reversal, degree-ten sector exclusion, whole-degree exclusion, or Ramsey formula. Only its displayed clauses are used below.

8.4. **Shortest terminal cycles and both missing-budget margins.** In this section, we obtain the row-order reserve from both coordinates of an original missing pair and apply it to every permitted shortest-cycle selector.

Theorem 8.17 (Shortest-cycle reserve and original-budget margins)**.** *Let $F \in \mathcal{M}_{10}$, and use the parameters of Setup 8.4. Put*

$$m_R = p + \delta, \qquad \beta = p + \delta - \epsilon, \qquad v_{10} = \left\lfloor \frac{2E - 3}{8} \right\rfloor.$$

Fix any native row A and any complete continuation on $F - A$. Let X_{root} be its entire root set and R its terminal. Put

$$D_2 = \{v \in V(R) \mid d_R(v) = 2\}.$$

Choose any maximum independent set T_0 of $R[D_2]$, and any shortest whole odd-cycle component C of R. Such cycles exist. Define

$$e_C = \frac{|V(C)|+1}{2}, \qquad Y = T_0 \cup V(C), \qquad W = X_{\mathrm{root}} \cup Y,$$
$$h = |W|, \qquad H = F[V(A) \cup W], \qquad N = |V(F) \setminus V(H)|.$$

All these are the same original physical vertices, and H retains all induced source edges.

(I) *The terminal R is a native row with exactly $p \geq 5$ whole odd cycles and exactly ϵ positive even-order whole path components. Moreover,*

$$|X_{\mathrm{root}}| = s, \qquad |V(R)| = r, \qquad \nu(R) = m_R, \qquad \alpha(R[D_2]) = \beta \geq p,$$

and

$$2 \leq e_C \leq \min\left(4,\ v_{10}+1,\ 1+\left\lfloor \frac{\beta}{p} \right\rfloor\right).$$

The selector satisfies

$$|T_0| = \beta, \qquad |Y| = \beta + e_C, \qquad h = s + \beta + e_C,$$
$$N = u - h = r - \beta - e_C.$$

The entire $R[Y] = H[Y]$ consists of C and isolated vertices, so

$$\nu(R[Y]) = e_C - 1.$$

Every $y \in Y$ has exactly eight original neighbors outside $V(R)$, all retained in $V(A) \cup X_{\mathrm{root}}$.

(II) *The source itself supplies the row-order reserve*

$$r \geq s + 3m_R + 1 + 2v_{10}.$$

Consequently this same retained graph satisfies

$$N - (\beta + h - 1) \geq 3\epsilon + 2(v_{10} + 1 - e_C) \geq 3\epsilon,$$
$$N \geq \max(0, \beta + h - 1), \qquad N \geq e_C - 2.$$

No additional source-order, positive-phase, triangle, or selector-margin hypothesis is imposed.

(III) *Missing pairs exist. For every ordered missing pair (b_R, b_B), put*

$$j_R = u - b_R, \qquad j_B = u - b_B, \qquad j_* = u - \max(b_R, b_B).$$

Then

$$j_R - h \geq \epsilon + v_{10} + 1 - e_C \geq \epsilon,$$
$$j_B - h \geq \epsilon + v_{10} + 1 - e_C \geq \epsilon.$$

In particular, $h \leq j_*$, and the exact identity

$$N - (\beta + h - 1) = |b_R - b_B| + 2(j_* - h) + \epsilon$$

holds on the same retained H, for every missing pair.

The assertions quantify over every source in both sectors, every native row, every actual continuation, every permitted T_0, and every shortest whole odd terminal cycle.

Proof. We use Theorem 8.15 and the generic selector and reserve statements recorded in Setup 8.9. We keep one source, one row, and one actual continuation throughout.

Step 1. Identify the actual terminal and its odd cycles. The source, certificate definitions, and native-row definition agree with those of Theorem 8.15. Its part (I) gives ten-regularity,

$$|V(F)| = 2r + s, \quad E \geq 12, \quad \lambda \geq 3, \quad p \geq 5, \quad 0 \leq \epsilon \leq \delta \leq \lambda + 1,$$

positive odd s, native-row existence, and $c_0(F) = k + E$.

Its part (II) gives the actual partition

$$V(F) = V(A) \,\dot\cup\, X_{\mathrm{root}} \,\dot\cup\, V(R), \qquad |X_{\mathrm{root}}| = s,$$

and says that R is a native row for every indicated continuation. Thus

$$|V(R)| = r, \qquad \nu(R) = p + \delta = m_R.$$

Applied to this same R, part (I) gives exactly p odd-cycle components, exactly ϵ positive even-order path components, and the component list

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

Whole odd cycles therefore exist, and any shortest one satisfies $2 \le e_C \le 4$. Finiteness provides all independent-set and minimum-cycle choices.

Step 2. Derive the reserve from both original missing coordinates. The definitions give

$$u + m_R = (r + s) + (p + \delta) = k + E = c_0(F).$$

Let (b_R, b_B) be any missing pair. Existence follows from Theorem 8.15(IV).

That part applies to an arbitrary ordered missing pair without a restriction on its first coordinate. Apply it first with first coordinate b_R, and then with first coordinate b_B. The reversed pair is missing: exchanging the two parts of a partition for the reversed pair would give a partition for the original pair. Consequently, for this original pair simultaneously,

$$j_R \ge s + m_R + v_{10} + 1, \qquad j_B \ge s + m_R + v_{10} + 1. \tag{31}$$

Both inequalities concern the same F and A, with complement order u; no new selector or matching has been chosen.

Their sum is

$$\begin{aligned} j_R + j_B &= 2u - (b_R + b_B) \\ &= 2u - (c_0(F) - 1) \\ &= u - m_R + 1 \\ &= r + s - m_R + 1. \end{aligned}$$

Adding the two lower bounds in (31) and rearranging gives

$$r \ge s + 3m_R + 2v_{10} + 1.$$

Since at least one missing pair exists, this is an unconditional source consequence. The coordinate bounds themselves hold for every missing pair.

Step 3. Check the generic shortest-cycle hypotheses. Use the same terminal R, with its p whole odd cycles, matching number $m_R = p + \delta$, and even-path count ϵ. In the generic reserve statement take

$$B_{\mathrm{res}} = E, \qquad \ell_{\mathrm{res}} = 8.$$

Its required bound holds because

$$m_R = p + \delta = E - 2 - \lambda + \delta \le E - 1,$$

using $\delta \le \lambda + 1$. Thus $B_{\mathrm{res}} \ge m_R + 1$. Also,

$$\ell_{\mathrm{res}} = 8 \ge 1, \qquad 8 < 2p,$$

because $p \ge 5$. The corresponding reserve integer is exactly v_{10}.

The generic result gives

$$\beta = m_R - \epsilon \ge p, \qquad e_C \le 1 + \left\lfloor \frac{\beta}{p} \right\rfloor.$$

Its strict-divisor case gives $e_C \le v_{10} + 1$. Together with Step 1, these are all the cycle bounds in part (I). No result for a different degree is substituted here.

Step 4. Identify the selector and its physical boundary. Apply the physical selector statement to the same induced R, C, T_0, ambient graph F, and partition with retained row A and root set X_{root}. The maximum-degree-two, whole-cycle, maximum-independent-set, and actual-partition hypotheses have all been checked. Its parameter is

$$\beta_R = \nu(R) - \epsilon = \beta.$$

It yields

$$\alpha(R[D_2]) = |T_0| = \beta, \qquad |Y| = \beta + e_C,$$
$$h = s + \beta + e_C, \qquad N = r - \beta - e_C \geq 0.$$

It also gives that $R[Y] = H[Y]$ is C with isolates, with matching number $e_C - 1$. Since $u = r + s$, one has $N = u - h$.

Every vertex of Y has degree two in R. Ten-regularity therefore gives exactly $10 - 2 = 8$ original neighbors outside R at each such vertex. The actual partition puts all these neighbors in the retained $V(A) \cup X_{\mathrm{root}}$.

Step 5. Obtain the omission and coordinate margins. The additional row-order premise of the generic actual-graph reserve is precisely the inequality proved in Step 2. It gives

$$N - (\beta + h - 1) \geq 3\epsilon + 2(v_{10} + 1 - e_C).$$

The second term is nonnegative by Step 3. This proves the stronger omission inequality in part (II).

In particular, $N \geq \beta + h - 1$. Since $\beta \geq p \geq 5$ and $s \geq 1$,

$$\beta + h - 1 = s + 2\beta + e_C - 1 \geq e_C - 2$$

and $\beta + h - 1 \geq 0$. Hence both remaining bounds in part (II) follow.

Apply the numerical-range conclusion of the generic reserve separately to the two inequalities in (31). It gives

$$j_R - h \geq \epsilon + v_{10} + 1 - e_C \geq \epsilon,$$
$$j_B - h \geq \epsilon + v_{10} + 1 - e_C \geq \epsilon.$$

Since $j_* = \min(j_R, j_B)$, it follows that $h \leq j_*$. No balancing assumption on the missing budgets is used.

Step 6. Verify the exact missing-budget identity. For nonnegative integers b_R, b_B,

$$2\max(b_R, b_B) - |b_R - b_B| = b_R + b_B,$$

as follows by considering which coordinate is larger. Therefore

$$\begin{aligned} |b_R - b_B| + 2(j_* - h) + \epsilon &= 2u - (b_R + b_B) - 2h + \epsilon \\ &= 2u - (u + m_R - 1) - 2h + \epsilon \\ &= u - \beta - 2h + 1 \\ &= (u - h) - (\beta + h - 1) \\ &= N - (\beta + h - 1). \end{aligned}$$

Every equality uses this one pair's original total and this one retained graph H.

All choices at the start were arbitrary within the stated domains. No extra phase count, matching existence, source-order premise, or cross-degree bound has been added. □

Remark 8.18. Theorem 8.17 supplies the reserve and both original missing-budget margins. It asserts no unmentioned matching-prefix or strict cost-certificate conclusion and excludes no whole source class.

8.5. **Attachment rules for an odd-component unit-deficit row.** The next theorem isolates the three-contact and four-contact arguments from all degree and source assumptions.

Theorem 8.19 (Three- and four-neighbor attachment rules)**.** *Use the generic costs of Definition 8.2. Let F be any finite simple graph of order n, and let A be an induced subgraph of order r isomorphic to either*

$$pC_3 \dot\cup P_3 \dot\cup zK_1 \qquad or \qquad (p-1)C_3 \dot\cup C_5 \dot\cup zK_1,$$

where $p \in \mathbb{Z}_{\geq 1}$ and $z \in \mathbb{Z}_{\geq 0}$. Assume

$$\alpha(F) = \alpha(A), \qquad c_1(F) = n - r + p + 1, \qquad c_2(F) = n - r + p.$$

Call the p odd-cycle components of A its cycles.

(I) *If $x \in V(F) \setminus V(A)$ has exactly three neighbors in A and meets at least one cycle, then its entire A-neighborhood is one whole triangle of A.*

(II) *If $x \in V(F) \setminus V(A)$ has exactly four neighbors in A and meets at least two different cycles, then its entire A-neighborhood is one whole triangle together with one vertex of another cycle.*

No regularity, degree bound, minimality, zero-slack condition on F, or source-process provenance is assumed.

Proof. The zero-cost and vertex-insertion statements of Setup 8.6 are the only earlier inputs. A one-good graph has maximum degree at most one. A two-good graph has maximum degree at most two and a matching covering its degree-two vertices. Thus an induced θ-good subgraph of F, for $\theta \in \{1, 2\}$, has order at most

$$n - c_\theta(F).$$

Step 1. Construct maximum-order induced good subgraphs. Let I consist of all isolates of A. Construct D_1 by keeping I, two arbitrary vertices in each triangle, and either both P_3 endpoints or three C_5 vertices obtained by deleting a nonadjacent pair. In the C_5 case, the three retained vertices induce one edge and an isolate. Hence $A[D_1]$ is one-good and

$$|D_1| = r - p - 1 = n - c_1(F).$$

Construct D_2 by keeping I, keeping all vertices of P_3 if it is present, and deleting one arbitrary vertex from each odd cycle. Its nontrivial components are paths. Alternating edges in each such path leave at most an endpoint exposed, so they cover every degree-two vertex. Thus $A[D_2]$ is two-good and

$$|D_2| = r - p = n - c_2(F).$$

Adding an exterior vertex to any such D_1 or D_2, while preserving the corresponding goodness, contradicts the order bound. We choose the omissions separately in each component.

Step 2. Record the independence restriction. Every maximum independent set of A has size $\alpha(F)$. Therefore every exterior vertex has a neighbor in every such set.

Suppose an exterior vertex avoids I and has no whole triangle in its neighborhood. In the P_3 form it must meet a P_3 endpoint. Otherwise, choose a nonneighbor in each triangle and both endpoints of P_3, together with I, to obtain a maximum independent set avoiding the exterior vertex.

In the C_5 form, its nonneighbors in C_5 must induce a clique. Otherwise an independent pair of those nonneighbors, one nonneighbor in every triangle, and I give a maximum independent set avoiding the exterior vertex. In particular, three C_5 contacts that block every maximum independent set of C_5 must be three consecutive cycle vertices, since the complementary two vertices must be adjacent. We use these restrictions only when there is no isolate contact and no whole triangle.

Step 3. Exclude every forbidden three-contact pattern. Fix an exterior vertex x with three A-neighbors and at least one cycle contact. If the three contacts form a triangle, the conclusion holds. Assume they do not.

First suppose that x meets I. If x has only one cycle contact, omit it in D_2. When the other two contacts are isolates, adding x gives a path of two edges on those isolates.

If, besides the isolate, the remaining contact lies in P_3, consider its position. For an endpoint contact, use D_2; adding x joins the isolate to a P_3 endpoint and leaves a path. For a center contact, use D_1, omitting that center and the cycle contact; adding x gives only one edge to its isolate neighbor. These exhaust the possibilities with exactly one cycle contact.

If x has two cycle contacts and one isolate contact, omit both cycle contacts in D_2 when they lie in different cycles. When they lie in the same cycle and are adjacent, omit one in D_2. The remaining contact is an endpoint of the resulting path, and adding x joins it to the isolate. If the two contacts are nonadjacent in the same cycle, that cycle is C_5. Omit both in D_1, leaving x adjacent only to its isolate. Each case contradicts the corresponding maximum order.

Now suppose that x avoids I. In the C_5 form, Step 2 and the absence of a whole triangle force exactly three consecutive C_5 contacts. Omit their two outer vertices in D_1. The middle contact is then isolated, and adding x preserves maximum degree at most one.

In the P_3 form, Step 2 forces an endpoint contact. If there is exactly one P_3 contact, the other two contacts lie on triangles. In D_2, omit one contact in each contacted triangle. If both contacts were in the same triangle, only one remains, and it is an endpoint of the resulting edge. If they were in different triangles, neither remains. Adding x at its P_3 endpoint and at the possible triangle endpoint leaves paths.

If there are two P_3 contacts and one triangle contact, omit the triangle contact. When the P_3 contacts are its two endpoints, use D_2; adding x creates a four-cycle, which has a perfect matching. When the contacts are the center and an endpoint, use D_1. It omits the center and leaves the contacted endpoint isolated, so adding x creates one edge. Three P_3 contacts are impossible because a cycle contact is required. This proves part (I).

Step 4. Exclude every forbidden four-contact pattern. Fix x with four A-neighbors and contacts in at least two different cycles. If x meets a whole triangle, its three triangle neighbors leave exactly one further neighbor. That neighbor lies in another cycle, as required. Assume there is no whole triangle.

Consider first the C_5 form. If x avoids I, Step 2 forces at least three C_5 contacts. Since another cycle is contacted and there are only four neighbors, there are exactly three consecutive C_5 contacts and one triangle contact. Omit the two outer C_5 contacts and the triangle contact in D_1. Only the middle, isolated C_5 contact remains adjacent to x, so adding x preserves one-goodness.

If x meets I, there are either two cycle contacts and two isolate contacts, or three cycle contacts and one isolate contact. With two cycle contacts, they lie in different cycles. Omit both in D_2; adding x gives a two-edge path on the two isolates.

With three cycle contacts, omit all three in D_2 if they lie in three different cycles, leaving only an isolate contact. Otherwise they are distributed two plus one over two cycles. If the double contacts are adjacent, omit one of them and the singleton in D_2. The remaining double contact is a path endpoint, and adding x joins it to the isolate. If the double contacts are nonadjacent, they lie in C_5. Omit both in D_1 and omit the singleton in its triangle. Adding x leaves only one isolate edge. This exhausts the C_5 form.

Now consider the P_3 form. If x has isolate contacts but no P_3 contact, the two-cycle-contact/two-isolate and three-cycle-contact/one-isolate constructions just described apply, with all contacted cycles triangles.

If x has both an isolate contact and a P_3 contact, the assumption of two distinct contacted cycles forces one neighbor in each of two different triangles, one isolate, and one P_3 vertex. Omit both triangle contacts. For a P_3 endpoint, use D_2; then x joins that endpoint to its isolate neighbor and leaves a path. For the P_3 center, use D_1, which omits the center, so only the isolate edge remains.

It remains that x avoids I. Step 2 gives at least one P_3 endpoint contact. If x has one P_3 contact, its three cycle contacts lie in at least two triangles. Omit one contact in every contacted triangle in D_2. At most one triangle contact remains, and it is a path endpoint. Adding x at this possible endpoint and the P_3 endpoint joins paths.

If x has two P_3 contacts, its two cycle contacts lie in different triangles; omit both. When the P_3 contacts are its endpoints, use D_2 and obtain a four-cycle. When they are its center and an endpoint, use D_1 and obtain one edge to an isolated endpoint. There cannot be three P_3 contacts while two distinct cycles are contacted. This exhausts the P_3 form.

In every forbidden case, the augmented graph is a union of paths, isolates, and at most one even four-cycle when D_2 is used, or a union of edges and isolates when D_1 is used. The required matching or degree condition therefore holds on every component. Each augmentation contradicts its maximum induced-good order. The remaining configurations are exactly those in parts (I) and (II). □

8.6. **Saturated owner matching and the swap interface.** The goal of this section is to reserve all triangle owners simultaneously and count every new owner created by a row swap.

Theorem 8.20 (Uniform owner reservation and exact swaps)**.** *Let $F \in \mathcal{M}_{10}$, and assume $c_2(F) = c_0(F)$. Put*

$$k = \nu(F), \qquad a = \alpha(F), \qquad E = \mathrm{ex}(F),$$

$$\lambda = 2E - 3 - (k - a), \qquad p = E - 2 - \lambda, \qquad z = a - p, \qquad s = 2\lambda + 3.$$

Let $\mathcal{R}$ be the family of all vertex sets inducing p disjoint triangles and z isolates. In this theorem a row $A \in \mathcal{R}$ denotes its vertex set. Put

$$J = F - A, \qquad Q = Q(A) = \{b \in V(J) \mid d_J(b) = 7 \text{ and } b \text{ meets an } A\text{-triangle}\},$$

and $o = o(A) = |Q|$.

(I) *The family $\mathcal{R}$ is finite and nonempty. Hence a row minimizing $o(A)$ exists. Every $b \in Q$ has entire A-neighborhood equal to one whole triangle T_b. These owned triangles are distinct, so $o \le p$.*

(II) *Put*

$$u_{\mathrm{own}} = \max\left(0, \left\lfloor \frac{2E - 3}{8} \right\rfloor\right).$$

Choose any missing pair (x, y) with

$$x + y = c_0(F) - 1, \qquad x \le y,$$

and put $j = |V(J)| - x$. Then

$$j \ge p + s + u_{\mathrm{own}} + 1.$$

Every set of at most j vertices of J has an actual matching into distinct A-components, compatible with an internal edge in every A-triangle.

After reserving all owners into their respective triangles, every prescribed

$$W \subseteq V(J) \setminus Q, \qquad |W| \le j - o,$$

can simultaneously be matched into distinct unowned A-components, retaining all those internal triangle edges. Moreover,

$$j - o \ge p - o + s + u_{\mathrm{own}} + 1 \ge s + 1.$$

In particular, every nonowner has a neighbor in an unowned A-component, and no two distinct nonowners have all such neighbors confined to one component.

(III) *For $b \in Q$ and $v \in V(T_b)$, put*

$$A_{bv} = (A \setminus \{v\}) \cup \{b\}.$$

Then $A_{bv} \in \mathcal{R}$.

Let Z_{bv} consist of all $w \in V(J)$ satisfying

$$d_J(w) = 6, \qquad wb \notin E(F),$$

and

$$N_F(w) \cap A = V(D_w) \cup \{v\},$$

where D_w is an unowned A-triangle. Then

$$o(A_{bv}) = o(A) - d_{J[Q]}(b) + |Z_{bv}|.$$

The sets Z_{bv} are pairwise disjoint. Their union Z has an injective map

$$w \longmapsto D_w$$

into the unowned A-triangles. Therefore

$$\sum_{b \in Q} \sum_{v \in V(T_b)} |Z_{bv}| = |Z| \le p - o.$$

These assertions hold at every allowed rank $\lambda \ge 2$ and for every induced row, apart from the explicitly existential minimizing-row assertion.

Proof. The source structure and the saturated-row statement are recorded in Setups 8.7 and 8.10. We also use the regular missing-allocation and distinct-component matching statements of Setup 8.6.

The source is connected, ten-regular, and factor-critical, with order $2k+1$ and $E \ge 1$. The saturated-row statement gives $\lambda \ge 2$, $p, z \ge 0$, and the row conclusions used here. A maximum independent set exists because the source is finite. Successive maximum-degree deletions terminate after finitely many vertices, and their terminal is a row. Thus $\mathcal{R}$ is nonempty. It is a subfamily of the finite power set of $V(F)$, so it is finite. The integer-valued function $o(A)$ attains a minimum. This establishes the existence assertions before any extremal choice is used.

Fix any $A \in \mathcal{R}$, and put $J = F - A$. The saturated-row theorem gives

$$|V(J)| = k + \lambda + 2, \qquad \Delta(J) \le 7, \qquad c_0(F) = k + E = |V(J)| + p.$$

It also gives the attachment alternative: every vertex of J meets a whole A-triangle or at least three A-isolates.

A missing pair at total $c_0(F) - 1$ exists by the least-total definition. Color symmetry permits its coordinates to be ordered as $x \le y$. The regular missing-allocation bound gives

$$k \ge 3E - 1 + 2u_{\mathrm{own}}.$$

Consequently,

$$\begin{aligned} j &= k + \lambda + 2 - x \\ &\ge \left\lceil \frac{k - E + 2\lambda + 5}{2} \right\rceil \\ &\ge E + \lambda + u_{\mathrm{own}} + 2 \\ &= p + s + u_{\mathrm{own}} + 1. \end{aligned}$$

In particular, $j \ge s + 1 \ge 8$.

Each component of $F[A]$ is odd and has a near-perfect matching. Its matching number is p. Every vertex of every component is an exposed port: deleting a vertex from a triangle leaves an edge, and deleting an isolate leaves the empty graph. Since

$$c_0(F) = |V(J)| + p,$$

the distinct-component matching statement applies to the actual partition $A, V(J)$. It gives the prescribed matching, compatible with an internal edge in every triangle, for every set of size at most j.

For $b \in Q$, ten-regularity and $d_J(b) = 7$ give exactly three A-neighbors. Apply the additional three-neighbor dichotomy of Setup 8.11 to this row. Since b meets a triangle, the all-isolate alternative is excluded. Hence its entire A-neighborhood is one whole triangle T_b. If two owners had the same triangle, the prescribed matching for that pair would have to send them into distinct components, although each has only that one possible component. Hence their owned triangles are distinct, and $o \le p$.

For a prescribed W in part (II), apply the matching statement to $Q \cup W$, whose size is at most j. Every owner is forced into its own T_b, so the vertices of W go into distinct other components. Retaining these edges and the internal triangle edges proves simultaneous reservation. Subtracting o from the lower bound for j gives

$$j - o \ge p - o + s + u_{\text{own}} + 1 \ge s + 1.$$

Since this is at least eight, the singleton and two-nonowner consequences are both within the permitted range. This proves parts (I) and (II).

Fix $b \in Q$ and $v \in V(T_b)$. The owner b has no A-neighbors outside T_b, and is adjacent to its other two vertices. Replacing v by b therefore preserves that triangle and every other component, without adding cross-component edges. Thus A_{bv} is a row. Its complement is

$$J' = J - b + v.$$

The new exterior vertex v has exactly three neighbors in A_{bv}: the other two vertices of T_b and b. It therefore has degree seven in J' and owns the changed triangle. The old owner b has left the complement.

Every other old owner b' has its assigned triangle disjoint from T_b, and hence is not adjacent to v. Its new exterior degree is

$$7 - \mathbf{1}_{\{b'b \in E(F)\}}.$$

Exactly the old owners adjacent to b lose owner status; the others retain their unchanged triangles.

We classify every additional new owner. For an old exterior vertex $w \ne b$,

$$d_{J'}(w) = d_J(w) + \mathbf{1}_{\{wv \in E(F)\}} - \mathbf{1}_{\{wb \in E(F)\}}.$$

An old vertex of degree at most five cannot reach degree seven.

An old degree-seven nonowner has three A-neighbors, all isolates by the attachment alternative. It is not adjacent to v. If it is adjacent to b, its exterior degree decreases. Otherwise it retains the same three isolate neighbors. Neither case makes it a triangle owner.

An old degree-six vertex reaches degree seven only when it is adjacent to v and nonadjacent to b. Its four old A-neighbors then lose v under the swap. By Setup 8.11 applied to A_{bv}, becoming an owner requires the other three neighbors to form a whole A_{bv}-triangle. It cannot own the changed triangle, which contains the nonneighbor b. Thus the other three neighbors form a whole unchanged A-triangle D_w.

If D_w were owned, all old A-neighbors of w would lie in D_w and T_b. But w was a nonowner, so part (II) requires a neighbor in an unowned A-component. This is a contradiction. Thus D_w is unowned.

The additional new owners are therefore exactly Z_{bv}. Conversely, each vertex meeting the defining degree and neighborhood conditions of Z_{bv} becomes an owner after the swap. Accounting for the exchanged pair b, v, the old owners adjacent to b, and these new owners gives

$$o(A_{bv}) = o(A) - d_{J[Q]}(b) + |Z_{bv}|.$$

For $w \in Z$, its four A-neighbors consist of a whole unowned triangle and one vertex of an owned triangle. The unowned triangle, the extra vertex v, and the owner b of the latter's triangle are uniquely determined. Hence the sets Z_{bv} are pairwise disjoint.

If distinct $w, w' \in Z$ had $D_w = D_{w'}$, both would be nonowners whose neighbors in unowned A-components are confined to this one triangle. That contradicts the two-nonowner consequence of part (II). Thus $w \mapsto D_w$ is injective. There are $p - o$ unowned triangles, giving the asserted sum bound. □

Remark 8.21. Every swap in Theorem 8.20 occurs inside the original source. No source minimality is transferred to a complement or row. The theorem gives the owner-reservation and exact swap interface with the degree-ten thresholds; it asserts no extremal edge inequality, source deletion lift, sector exclusion, or whole-degree closure.

8.7. **The equality frame and physical row-preserving exchanges.** In this section, we assemble the reversible equality frame, all row-contact exclusions, and the exchanges that preserve both original rows.

Theorem 8.22 (Equality frame, contact exclusions, and fixed-row exchanges)**.** *Let $F \in \mathcal{M}_{10}$, and assume $c_2(F) = c_0(F)$. Put*

$$k = \nu(F), \qquad a = \alpha(F), \qquad E = \operatorname{ex}(F),$$
$$\lambda = 2E - 3 - (k - a), \qquad p = E - 2 - \lambda, \qquad z = a - p,$$
$$r = k - \lambda - 1, \qquad s = 2\lambda + 3.$$

A native row is any induced $pC_3 \dot\cup zK_1$. An exterior vertex completes a whole row triangle T when it is adjacent to every vertex of T.

(I) *The source is connected, ten-regular, factor-critical, and well-covered. Thus every single-vertex deletion leaves a perfect matching, and every maximal independent set has size a. Moreover,*

$$E \geq 12, \qquad \lambda \geq 3, \qquad p \geq 5, \qquad z \geq 0,$$
$$|V(F)| = 2k + 1 = 2r + s, \qquad r = 3p + z, \qquad a = p + z, \qquad \operatorname{gap}(F) = 1.$$

Native rows exist, and every native row complement has maximum degree at most six.

There is one common $t \in \mathbb{Z}_{\geq 0}$, depending only on F, such that every complete process from every native row complement has word

$$6^t 5^b 4^t, \qquad b = s - 2t.$$

Here b is positive and odd, and $0 \leq t \leq \lambda + 1$. The terminal is another native row. The exact source costs are

$$(c_0(F), \dots, c_{10}(F)) = (k + E,\ k + E,\ k + E,\ r + s,\ r + s - t,\ r + t,\ r,\ r,\ a + p,\ a,\ 1), \tag{32}$$

and $c_i(F) = 0$ for $i \geq 11$.

(II) *Choose any native row A and any complete process from $F - A$, with terminal R and actual degree blocks T_6, T_5, T_4. Put*

$$H_A = F[V(A) \cup T_6], \qquad H_R = F[V(R) \cup T_4], \qquad Y = T_5,$$
$$X = V(F) \setminus (V(A) \cup V(R)).$$

Then

$$V(F) = V(H_A) \dot\cup Y \dot\cup V(H_R),$$
$$|V(H_A)| = |V(H_R)| = r + t, \qquad |Y| = b.$$

Each of T_6, Y, T_4 is independent. The reverse order of all roots is a legal complete process from $F-R$ to A, with the same word. Every ordering within each degree block is legal. Every $y \in Y$ has exactly five neighbors in each of H_A, H_R.

Each outer $H \in \{H_A, H_R\}$ is an actual quartic entry, allowing an empty quartic phase when $t = 0$. It satisfies

$$\Delta(H) \le 4, \qquad \alpha(H) = a,$$
$$\nu(H) = c_0(H) = c_1(H) = c_2(H) = p + t,$$
$$c_3(H) = t, \qquad c_i(H) = 0 \quad (i \ge 4),$$
$$e(H) = 3p + 4t, \qquad \mathrm{ps}(H) = \mathrm{ex}(H) = \mathrm{gap}(H) = 0.$$

Every complete process on H has word 4^t and a native terminal. Both full complements $F - H_A$ and $F - H_R$ are five-good and have maximum degree exactly five.

(III) *At every native row D, each exterior vertex contacts at most one whole D-triangle. No exterior vertex has entire D-neighborhood exactly one whole triangle.*

There do not exist distinct exterior vertices u, v completing the same whole D-triangle T with

$$|N_D(u) \setminus V(T)| \le 1, \qquad |N_D(v) \setminus V(T)| \le 1.$$

This exclusion holds whether or not uv is an edge.

No exterior vertex completing a whole triangle has at most two other D-neighbors. Thus the entire triangle-plus-one-isolate and triangle-plus-two-isolate masks are excluded. No exterior vertex has entire D-neighborhood consisting of four whole D-isolates.

The pair exclusion uses the retained graph

$$F[V(D) \cup \{u, v\}],$$

with $N = r + s - 2$ omitted original vertices, a paid graph of matching number at most $p+1$, and a two-good spanning residual. Its complete original-source bound is

$$c_2(F) \le N + p + 1 = r + s + p - 1,$$

against the actual value $c_2(F) = r + s + p$.

In the forbidden triangle-plus-two-isolate case, there are three independent exterior vertices v, u, w with the same entire D-neighborhood

$$V(T) \cup \{i, j\}.$$

Retain $F[V(D) \cup \{v, u, w\}]$. Pay all edges induced on $V(T) \cup \{v, u, w\}$, the edges vi, vj, and all triangle edges of D outside T. The paid matching number is at most $p + 2$. The entire residual is the complete bipartite graph with shores $\{u, w\}$, $\{i, j\}$, together with isolates. With $N = r + s - 3$, this gives

$$c_2(F) \le N + p + 2 = r + s + p - 1.$$

Both certificates restore every original edge incident with an omitted vertex, including edges with both endpoints omitted. These are contradiction certificates for the specified masks, not packets asserted to occur in a surviving source.

(IV) *Every whole component of either outer is either an isolated vertex or the following graph, for some $\ell \in \{0, 1, 2, 3\}$: start with one triangle; choose ℓ distinct triangle vertices; attach to each a distinct new root; and attach three distinct private leaves to each root. All added vertices are distinct, and there are no other edges.*

This is a necessary list with no occurrence assertion. In particular, every cycle in an outer is a triangle.

For either outer H, the entire set

$$U_H = \{v \in V(H) \mid d_H(v) = 4\}$$

is independent, has size t, and is the unique full quartic phase. Every ordering is legal, and

$$D = H - U_H$$

is the same physical terminal row. Every vertex of U_H has entire D-neighborhood consisting of three private isolates and one triangle vertex. Distinct vertices of U_H have disjoint entire D-neighborhoods.

For any native row D, every degree-six vertex of $F - D$ has entire D-neighborhood consisting of three isolates and one triangle vertex. Every independent subset of the entire degree-six set extends to an actual full degree-six phase. Its vertices therefore have pairwise disjoint entire D-neighborhoods.

The phases on both fixed outers are simultaneous actual choices. For any full quartic phases U_A, U_R of H_A, H_R, the induced rows

$$A' = H_A - U_A, \qquad R' = H_R - U_R$$

are native. Deleting U_A, then Y, then U_R is a legal complete process from $F - A'$ to R', with every ordering in every block permitted. All original source edges and both fixed outer vertex sets remain the same.

For either outer H, put $O = V(F) \setminus V(H)$. Then

$$|O| = r + s - t, \qquad c_3(H) = t, \qquad c_3(F) = r + s.$$

For every $D_{\text{outer}} \in \mathbb{Z}_{\geq 0}$ with $c_3(H) \leq D_{\text{outer}}$, restoring all original incidences gives

$$c_3(F) \leq D_{\text{source}} := |O| + D_{\text{outer}},$$

and

$$D_{\text{source}} - c_3(F) = D_{\text{outer}} - c_3(H).$$

The previously excluded hypothetical whole $K_{4,4}$ component has $D_{\text{outer}} = t - 2$. The previously excluded hypothetical pair of distinct free quartic roots sharing a terminal isolate has $D_{\text{outer}} = t - 1$. Here free means having no neighbor in the entire outer degree-four set. Their complete source debits are $r + s - 2$ and $r + s - 1$, respectively. Both configurations are absent under the component list, and these conditional debits give no further source coverage.

(V) *The entire induced root graph satisfies*

$$\Delta(F[X]) \leq 2.$$

Every root $x \in X$ has at least three A-isolate neighbors and at least three R-isolate neighbors. Put

$$d_X(x) = |N_F(x) \cap X|.$$

Its ordered row-neighbor counts satisfy

$d_X(x)$	$\big(\lvert N_A(x)\rvert, \lvert N_R(x)\rvert\big)$
2	$(4,4)$
1	$(4,5)$ *or* $(5,4)$
0	$(4,6)$, $(5,5)$, *or* $(6,4)$.

Every root of positive $F[X]$-degree completes no triangle in either row. A triangle completion is possible only on a six-neighbor side of an isolated root, where the entire row-neighborhood is that triangle and three isolates.

Every middle vertex $y \in Y$, including an isolated vertex of $F[X]$, completes no triangle in either row. These are necessary restrictions; the listed masks are not asserted realized.

(VI) *Fix either outer H, and let H_{op} be the other. Put*

$$G = F[V(H) \cup Y], \qquad U = U_H, \qquad D = H - U.$$

The entire bipartite graph of actual edges between Y and U is a matching. For every $y \in Y$,

$$\nu(H + y) = \nu(H) + 1 = p + t + 1.$$

In particular, its entire H-neighborhood is not contained in one even-order whole H-component with a perfect matching.

Let V_0 be any independent subset of U whose vertices have Y-neighbors. For $v \in V_0$, let $f(v)$ be its unique Y-neighbor. The map f is injective. Put

$$Y' = (Y \setminus f(V_0)) \cup V_0,$$

$$H' = F[(V(H) \setminus V_0) \cup f(V_0)], \qquad U' = (U \setminus V_0) \cup f(V_0).$$

Every ordering of Y' is a legal full degree-five phase in the same actual G, and H' is its actual quartic remainder. The set U' is its entire degree-four set and unique full quartic phase. Its terminal is the same physical row:

$$H' - U' = H - U = D.$$

The opposite outer H_{op} and its terminal row stay physically fixed.

For any full quartic phase U_{op} of H_{op}, in the appropriate forward or reverse orientation, the blocks

$$U_{\mathrm{op}},\ Y',\ U'$$

form an actual complete source process. Both outers, their middle, and all their costs satisfy the conditions in parts (II) *and* (IV).

Every finite composition of these legal exchanges on either side preserves both original rows A, R and the entire induced graph $F[X]$, including every original edge.

No prescribed matching is asserted to survive an exchange, and no arbitrary chosen sequence of moves is asserted legal. All statements concern the original equality-sector source; none is transferred to $c_2(F) < c_0(F)$.

Proof. We use Theorem 8.15 and the seven earlier equality statements recorded in Setup 8.12. Their certificate definitions and proper-subgraph hypotheses agree with those used here.

The equality assumption gives $\delta = 0$. The inequality

$$0 \le \epsilon \le \delta$$

from Theorem 8.15 then gives $\epsilon = 0$. Hence its parameters r, s become

$$r = k - \lambda - 1, \qquad s = 2\lambda + 3.$$

The earlier equality process theorem uses exactly these parameters and defines a native row as p disjoint triangles and z isolates. No application changes the source or sector.

Step 1. Establish the source and common process data. Theorem 8.15(I) gives connectedness, ten-regularity, factor-criticality, well-coveredness,

$$E \ge 12, \qquad \lambda \ge 3, \qquad p \ge 5,$$

order $2k + 1 = 2r + s$, and gap one.

The earlier equality process statement supplies equality-native rows, the bound six on every row complement after removal of the entire degree-seven phase, and the common word and parameter t. It gives

$$0 \le t \le \lambda + 1, \qquad b = s - 2t > 0,$$

with b odd, and the complete eleven-entry source-cost profile (32).

Its native rows satisfy

$$r = 3p + z, \qquad a = p + z.$$

Thus $z \geq 0$ counts their isolates. Native-row and terminal existence follow from Theorem 8.15 and the equality process statement. These conclusions do not select a smaller row class.

Step 2. Keep the actual reversible frame and all outer costs. Apply the earlier reversible-frame theorem to the same arbitrary native row A and the same arbitrary complete row-complement process. Its degree blocks and the definitions of H_A, H_R, Y agree exactly with those in part (II).

It gives the disjoint partition, both outer orders, the cardinality of Y, five neighbors in each outer at every middle vertex, reverse legality, and block reordering. It also gives all outer cost and matching values, actual quartic entry status, and five-goodness of both entire complements.

For the independence of a degree block, an earlier deletion of a neighbor in that block would lower a later selected vertex below the block's degree. The block vertices are therefore independent. Reordering them preserves the degrees of all remaining vertices in that block, so every such ordering is legal. These statements include the case $t = 0$.

The simultaneous-phase conclusion of the reversible-frame theorem gives the choices in part (IV) on the same two outer vertex sets. Every middle ordering is permitted because Y is independent and every middle vertex retains its five neighbors in the opposite outer.

Step 3. Retain all contact exclusions and their original-source certificates. The earlier triangle-contact statement gives contact with at most one whole row triangle and excludes every exact triangle neighborhood, at every native row.

The pair-exclusion statement forbids two distinct exterior vertices completing the same triangle with at most one extra neighbor each, without an adjacency restriction. Its retained graph is

$$F[V(D) \cup \{u, v\}].$$

It pays all edges induced on the five-vertex triangle-and-root set, together with all other row triangles. The paid matching number is at most

$$2 + (p - 1) = p + 1.$$

The unpaid graph consists only of the possible two extra contacts and is two-good by the earlier certificate statement. There are exactly $r+s-2$ omitted vertices. Restoring every original incidence gives

$$c_2(F) \leq (r + s - 2) + (p + 1) = r + s + p - 1,$$

contradicting

$$c_2(F) = k + E = r + s + p.$$

This certificate is conditional on the forbidden pair and asserts no occurrence.

The earlier triangle-plus-two exclusion forbids a full triangle with at most two extra neighbors. In its two-isolate case, extending an independent set of size $a - 2$ gives two additional independent exterior vertices u, w with the same entire mask as v. Thus the three exterior vertices are independent and have neighborhood

$$V(T) \cup \{i, j\}$$

inside D.

That certificate retains D with v, u, w, pays all edges of their induced triangle-and-root set, the edges vi, vj, and the other row triangles. Its paid matching number is at most $p + 2$. Its entire residual is exactly the complete bipartite graph between $\{u, w\}$ and $\{i, j\}$, together with isolates. Since the omitted count is $r + s - 3$, full original-source restoration gives

$$c_2(F) \leq (r + s - 3) + (p + 2) = r + s + p - 1.$$

Every original boundary edge is restored at every budget split, including edges with two omitted endpoints.

The zero-extra and one-extra cases are included in the same exclusion. No separate case involving another triangle is needed, since contact with two row triangles has already been excluded. Finally, the earlier complete outer-component statement excludes every exact four-isolate neighborhood for every exterior vertex at every native row, without a phase restriction. This proves all clauses of part (III).

Step 4. Identify the whole outer components and simultaneous phases. The earlier complete outer-component theorem gives the five types in part (IV) for both actual outers, including distinctness of all added vertices and the absence of any other edges. It gives the entire unique quartic phase, its independent vertex set of size t, and the full disjoint row-neighborhoods of its vertices. The reversible-frame theorem gives legality of every ordering and the actual terminal.

In each listed component, every edge outside the one triangle lies on an attached tree and is a bridge. A cycle cannot use a bridge: the rest of that cycle would still join its endpoints after the edge was deleted. Hence every outer cycle is a triangle.

The degree-six conclusion of the same component theorem applies to every native row complement. It gives the complete neighborhood description and extends every independent subset of the entire degree-six pool to a full actual degree-six phase. In such a phase the entire row-neighborhoods are disjoint. The simultaneous choices on the two fixed outers were supplied in Step 2.

For the third-cost statement, the earlier outer-insertion theorem gives the exact omitted count, the true outer and source third costs, and restoration for every certified nonnegative outer bound. Substituting

$$|O| = r + s - t, \qquad c_3(H) = t, \qquad c_3(F) = r + s$$

gives

$$\begin{aligned} D_{\text{source}} - c_3(F) &= (r + s - t + D_{\text{outer}}) - (r + s) \\ &= D_{\text{outer}} - t \\ &= D_{\text{outer}} - c_3(H). \end{aligned}$$

The two earlier hypothetical certificates have bounds $t-2$ and $t-1$, for a whole $K_{4,4}$ and for a pair of distinct free quartic roots sharing a terminal isolate, respectively. Their source debits are therefore $r + s - 2$ and $r + s - 1$. These configurations are already absent from the complete component list, so the conditional debits add no source coverage.

Step 5. Record the entire induced root graph and both row masks. The earlier pair-exclusion and physical-exchange statement gives

$$\Delta(F[X]) \le 2$$

for the entire induced root graph, with X the complement of both physical rows.

The earlier triangle-plus-two exclusion gives at least three isolate neighbors on each row side, the three ordered count cases in part (V), the prohibition of triangle completion at positive root degree, and the restriction on a six-neighbor side of an isolated root. It also forbids a triangle completion at every middle vertex.

Those statements use the entire row-neighborhoods, so no extra root contact is discarded. The middle set here is exactly the original degree-five block $Y = T_5$. This proves part (V) with the physical rows unchanged.

Step 6. Verify the legal exchanges and fixed-row iteration. Apply the earlier boundary matching and swap statement to this same complete frame and either chosen outer. It gives that the entire actual Y-to-U graph is a matching. It also gives, for every $y \in Y$,

$$\nu(H + y) = \nu(H) + 1 = p + t + 1,$$

and excludes confinement of the entire H-neighborhood to one even-order whole component with a perfect matching.

Its swap conclusion gives the injective attachment map f, every-order legality of the full phase Y', the physical quartic remainder H', and simultaneous realization after any full phase on the fixed opposite outer.

The earlier physical-exchange statement identifies the entire new quartic set as

$$U' = (U \setminus V_0) \cup f(V_0).$$

It gives the literal identity

$$H' - U' = H - U$$

and fixes the opposite terminal row. The new frame is therefore an actual frame of the reversible-frame theorem. All costs from part (II) and the universally quantified component list from part (IV) apply to it. This uses the same original source, rather than a minimality assertion about H'.

Finally, the earlier physical-exchange statement permits induction over every finite composition of these legal exchanges on either side. At each step both physical rows remain fixed. Their complement X in the fixed source remains fixed as well, and every induced $F[X]$-edge remains present. This gives exactly the claimed iteration. It neither prescribes other moves nor asserts survival of a particular matching. □

Remark 8.23. Theorem 8.22 gives necessary equality-sector contracts with complete source, row, process, whole-component, and integer-domain conditions. It implies neither equality-sector exclusion nor whole-degree exclusion. No unmentioned earlier clause, strict-sector conclusion, source occurrence, or sector closure is imported. No claim of literature priority is made for the strengthened ranges and interfaces presented here.

9. Degree twelve: the numerical frame, native rows, selectors and reserves

5.1. **Degree twelve.** We develop the degree-twelve numerical, row, and matching interfaces needed for the shortest-cycle reserve and the degree-twelve matching prefix. The arguments retain the original source, its induced rows, and its actual deletion processes throughout.

5.1.1. *Setup.* We assume, established earlier in this series, the results recorded in Setups 5.5–5.10 below. They comprise the required inputs from Companion I and the earlier universal-structure and matching development.

[addressed-partial: the supplied placeholders are retained | deferred: author, affiliation, and email to the operator]

[metadata/blocker] The Mathematics Subject Classification remains to be supplied. [addressed-partial: the classification placeholder is retained | deferred: classification to the operator]

[cite/blocker] [GAP: verified bibliographic entries and theorem numbers for Companion I and the earlier chapters were not supplied. The earlier results are identified below by their precise mathematical content; no citation key or theorem number has been invented.] [deferred: to reference_verify]

[math/blocker] [GAP: confirm that every unreproduced prerequisite recorded below is available in Companion I or Chapters 1–5. The supplied material does not give their chapter locations.] [addressed-partial: the owner and exposed-port inputs are explicit numbered assumptions | deferred: earlier-series proofs and locations to the operator]

[acknowledged: the chapter format and both boundary comparisons are retained]

[acknowledged: compilation and pagination remain unverified; no verification result is claimed]

Notation 5.1. All graphs are finite and simple. We write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for the vertex set, edge set, edge count, matching number, independence number, and maximum degree. The open neighborhood of v is $N_G(v)$, and its degree is $d_G(v)$. For an induced subgraph A of a graph F, put

$$N_A(v) = N_F(v) \cap V(A).$$

For disjoint vertex sets U, W, let $e_F(U, W)$ count the edges between them. We use $G-Z$ for induced vertex deletion and $G - P$ for deletion of an explicitly specified edge set P. A whole component is an entire connected component. The graphs P_j, C_j, K_j have j vertices.

Every capacity, budget, cost index, least universal total, degree, order, cardinality, component count, matching size, phase length, debit, and finite optimization index is an element of $\mathbb{Z}_{\geq 0}$, unless another integer range is stated. Gaps, differences, and slacks may lie in $\mathbb{Z}$.

Definition 5.2 (Certificate costs)**.** Let $A_c, B_c, x_c, y_c \in \mathbb{Z}_{\geq 0}$. An $(A_c, B_c; x_c, y_c)$-certificate for G is a partition of $E(G)$ into four parts whose spanning subgraphs have maximum degrees at most A_c, B_c and matching numbers at most x_c, y_c, respectively. Define $c_{(A_c,B_c)}(G)$ to be the least $t \in \mathbb{Z}_{\geq 0}$ for which every $x_c, y_c \in \mathbb{Z}_{\geq 0}$ with $x_c + y_c = t$ admits such a certificate. For $i \in \mathbb{Z}_{\geq 0}$, put

$$c_i(G) = \max_{\substack{A_c, B_c \in \mathbb{Z}_{\geq 0} \\ A_c + B_c = i}} c_{(A_c,B_c)}(G), \qquad c_0 = c_{(0,0)}.$$

These are the generic costs also denoted c_i^* and c_{00} in the earlier development. Set

$$\operatorname{ex}(G) = c_0(G) - \nu(G), \qquad \operatorname{ps}(G) = e(G) - \nu(G) - \sum_{i \geq 1} c_i(G),$$

and

$$\operatorname{gap}(G) = \sum_{i \geq 0} c_i(G) - e(G) = \operatorname{ex}(G) - \operatorname{ps}(G).$$

The sums are finite: for $i \geq 2\Delta(G)$, every capacity split has a part of capacity at least $\Delta(G)$, so all edges can be placed in that part and $c_i(G) = 0$.

At zero degree capacities the first two parts are empty. A pair $x, y \in \mathbb{Z}_{\geq 0}$ is therefore *missing* if $E(G)$ has no partition into two parts of matching numbers at most x, y. If $c_0(G) = 0$, set $\operatorname{tr}(G) = 0$. Otherwise $\operatorname{tr}(G)$ is the greatest length of a consecutive run of missing first budgets

$$x \in \{0, \ldots, c_0(G) - 1\}, \qquad y = c_0(G) - 1 - x.$$

The property T_- is

$$\operatorname{gap}(G) > 0 \quad \Longrightarrow \quad \operatorname{gap}(G) = 1 \ \text{ and } \ \operatorname{tr}(G) + \operatorname{ex}(G) \leq \max\left(4, \left\lfloor \frac{\Delta(G)}{2} \right\rfloor + 1\right).$$

Setup 5.3 (Sources and parameters)**.** Let $\mathcal{M}_{12}$ be the entire class of graphs F of maximum degree exactly twelve that violate T_-, while every proper subgraph satisfies T_-. Proper subgraphs here allow arbitrary vertex deletions, edge deletions, or both.

For $F \in \mathcal{M}_{12}$, define

$$\begin{gathered} n = |V(F)|, \qquad k = \nu(F), \qquad a = \alpha(F), \qquad E = \operatorname{ex}(F), \\ \lambda = 2E - 3 - (k - a), \qquad \delta = c_0(F) - c_2(F), \qquad \epsilon = c_0(F) - c_1(F), \\ p = E - 2 - \lambda, \qquad r = k - \lambda - 1 + \delta, \qquad s = 2\lambda + 3 - 2\delta, \\ u = r + s, \qquad M = k + E - r, \qquad \beta = p + \delta - \epsilon. \end{gathered}$$

These parameters are integers; their nonnegativity and stronger ranges will be asserted where established. The notation q, b in the earlier row statements means δ, ϵ, respectively.

Definition 5.4 (Rows and processes)**.** For a source in Setup 5.3, a *native row*, or simply a *row*, is an induced subgraph A satisfying

$$|V(A)| = r, \quad \Delta(A) \leq 2, \quad \operatorname{ps}(A) = 0, \quad \nu(A) = p + \delta, \quad c_1(A) = p + \delta - \epsilon, \quad c_2(A) = p.$$

A complete source process first deletes any maximal independent set of F. It then repeatedly deletes any current maximum-degree vertex while the maximum degree is at least three, and stops at maximum degree at most two. A full degree-j phase consists of all deletions made at current

degree j. A continuation on a row complement follows the same maximum-degree rule. All vertex deletions are induced and retain every original edge between the remaining vertices.

A graph is well-covered if every maximal independent set is maximum. It is factor-critical if deleting any one vertex leaves a perfect matching. It is j-good if its maximum degree is at most j and one matching covers every vertex of degree j. A vertex cover meets every edge; its minimum cardinality is denoted by $\tau(G)$. For an odd component C, define

$$\mathrm{Port}(C) = \{v \in V(C) : \nu(C - v) = \nu(C)\}.$$

When $\nu(C) = \lfloor |V(C)|/2 \rfloor$, these are exactly the vertices whose deletion leaves a perfect matching.

Setup 5.5 (Earlier source and process results)**.** The following results are assumed from the earlier development.

(I) Every $F \in \mathcal{M}_{12}$ is connected, twelve-regular, factor-critical, and well-covered, with

$$|V(F)| = 2k + 1, \qquad \mathrm{gap}(F) = 1, \qquad E \geq 10.$$

The degree-twelve rigid source class used in the earlier finiteness statements is exactly $\mathcal{M}_{12}$; its proper-subgraph condition is unchanged.

(II) Every terminal of every complete source process has independence number a.

(III) *Covered-process applicability assumption.* We assume, established earlier in this series, that if $E \geq 10$ and $\lambda \geq 2$, every complete maximum-degree process from every maximum-independent complement is covered throughout in the sense of the earlier development. The properties of coveredness used here are recorded after this list.

(IV) The earlier strengthened source restriction gives, for every $F \in \mathcal{M}_{12}$,

$$\delta > 0, \qquad \lambda \geq 3, \qquad p \geq 5,$$

and

$$c_0(F) = k + E, \qquad c_1(F) = k + E - \epsilon, \qquad c_2(F) = k + E - \delta.$$

(V) For every actual continuation on a row complement, its full phases $B_9, \ldots, B_3$, including empty phases, partition its entire root set.

Covered-process terminology. The full earlier definition of a covered process is not recoverable from the supplied text. Under $E \geq 10$ and $\lambda \geq 2$, the properties used here are precisely the following assumed consequences from Setup 5.9. Put

$$B_\lambda = \left\lfloor \frac{14\lambda + 21}{13} \right\rfloor, \qquad T_\lambda = 16\lambda + 24.$$

For every terminal vertex set R, there is an actual set $X \subseteq V(F) \setminus R$, of size $t \leq T_\lambda$, such that, for

$$L = V(F) \setminus (R \cup X),$$

both $F[R]$ and $F[L]$ have maximum degree at most two, and

$$|R| \leq k - \lambda - 1 + B_\lambda, \qquad |L| \leq k + \lambda + 2.$$

The number P of whole triangles in $F[R]$ satisfies

$$P \geq E - 2 - \lambda - B_\lambda.$$

The source-cost and rank estimates used with this decomposition are

$$c_9(F) \geq k - 7\lambda - 10, \qquad 23k < 100E - 100 - 50\lambda.$$

Item (iii) is the earlier-series applicability assumption that makes these inputs available for every allowed complete source process. These consequences specify every use of coveredness in this chapter; they do not supply its missing earlier definition. The full-allocation certificate in Setup 5.9 is applied using its explicitly stated partition hypotheses.

[math/blocker] [GAP: the earlier definition of a covered process was not supplied. Its use here is restricted to the explicit applicability assertion in (iii) and the conclusions in Setup 5.9.] [addressed-partial: every covered-process property used here is stated explicitly | deferred: the full earlier definition and applicability proof to the operator]

Setup 5.6 (Earlier conditional row theorem)**.** Assume that a source has at least one complete process whose terminal has independence number a. The earlier conditional row theorem at degree twelve gives the following statements.

(I) Rows exist,

$$p \geq 0, \qquad 0 \leq \epsilon \leq \delta, \qquad s > 0 \text{ odd}, \qquad |V(F)| = 2r + s.$$

Every row has independence number a, edge count $3p + 2\delta - \epsilon$, and components from

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

It has exactly p odd-cycle components and exactly ϵ positive even-order path components. A terminal preserving independence is a row.

(II) Every source process has exactly $p + \epsilon$ degree-eleven roots and $p + \delta - \epsilon$ degree-ten roots. Their union with the initial independent set induces a row. The source costs are

$$c_{11}(F) = a, \qquad c_{10}(F) = a + p + \epsilon, \qquad c_9(F) = r.$$

(III) For every row A, the complement $J = F - V(A)$ is nine-good and satisfies

$$|V(J)| = r + s, \quad \nu(J) = M, \quad \mathrm{ps}(J) = 0,$$
$$e(J) = 3p + 2\delta - \epsilon + 6s, \qquad c_i(J) = c_i(F) - r \quad (1 \leq i \leq 9),$$
$$c_1(J) = M - \epsilon, \qquad c_2(J) = M - \delta.$$

(IV) For an actual source process with early row A, root set X after that row, and terminal row R preserving independence, the sets $V(A), V(R), X$ partition $V(F)$, and

$$\Delta(F[X]) \leq 6, \qquad e_F(X, V(A)) = e_F(X, V(R)) = 6s - e(F[X]).$$

Setup 5.7 (Earlier missing-budget and port interfaces)**.** We assume the following two matching interfaces.

(I) Let G be connected, of maximum degree at most twelve, order $2k + 1$, matching number k, and excess $E \geq 1$. Put

$$v = \max\left(0, \left\lfloor \frac{2E - 3}{10} \right\rfloor\right).$$

Every missing pair $x, y \in \mathbb{Z}_{\geq 0}$ at total $c_0(G) - 1$ satisfies

$$2E - 1 + v \leq x \leq k - E - v, \qquad k \geq 3E - 1 + 2v.$$

The same coordinate bounds apply to y.

(II) *Exposed-port extension and expansion assumption.* We assume, established earlier in this series, the following statement. Let A be induced in F, put $U = V(F) \setminus V(A)$, and assume

$$\nu(C) = \left\lfloor \frac{|V(C)|}{2} \right\rfloor \quad \text{for every component } C \text{ of } A, \qquad c_0(F) = |U| + \nu(A).$$

Fix a missing pair $x, y \in \mathbb{Z}_{\geq 0}$ at total $c_0(F) - 1$, and put $j = |U| - x$. Every $T \subseteq U$ with $|T| \leq j$ can be matched to exposed ports in distinct odd whole components of A, compatibly with maximum internal matchings of all components.

If o is the number of odd components and $\mathcal{C}$ is a family of ℓ odd components with

$$o - j + 1 \leq \ell \leq o,$$

then at least $|U| - o + \ell$ vertices of U have a neighbor in the exposed ports of members of $\mathcal{C}$.

Setup 5.8 (Earlier essential-port localization)**.** Let F be well-covered, and let A be induced with components from

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

Write $U = V(F) \setminus V(A)$, let h count the P_3 components, and let p_A count the odd cycles. Assume

$$c_0(F) = |U| + \nu(A).$$

Fix a missing pair $x, y \in \mathbb{Z}_{\geq 0}$ at total $c_0(F) - 1$, put $j = |U| - x$, and assume $j \geq p_A + h + 1$.

Let I_0 be the isolates of A, put

$$B_0 = \{v \in U : N_F(v) \cap I_0 = \varnothing\},$$

and join $v \in B_0$ to a nontrivial odd component C when v has an actual neighbor in $\mathrm{Port}(C)$. The earlier essential-port theorem asserts that this bipartite graph has a B_0-saturating matching and that every such matching covers every P_3 component.

For any chosen matching, take its alternating closure from all P_3 component vertices. It gives $B \subseteq B_0$ and a family $\mathcal{H}$ of nontrivial odd components such that

$$N_\Gamma(B) = \mathcal{H}, \qquad |B| = |\mathcal{H}| = h + p_H,$$

where $\mathcal{H}$ contains all P_3 components and p_H odd cycles. If Z is the union of the even components and

$$W = Z \cup B \cup \bigcup_{C \in \mathcal{H}} V(C),$$

there is an independent set I outside W with

$$V(F) \setminus N_F[I] = W.$$

The graph $F[W]$ is well-covered, has a perfect matching, and satisfies

$$\nu(F[W]) - \alpha(F[W]) = p_H.$$

For every $T \subseteq U \setminus B$ of size at most $j - |B|$, one simultaneous port matching reserves B into $\mathcal{H}$ and assigns T to distinct odd components outside $\mathcal{H}$, with compatible maximum internal component matchings.

The earlier zero-odd-cycle refinement asserts that if $p_A = 0$, the stated missing-budget hypotheses hold, and every exterior vertex has at least three neighbors in A, then A has no P_3 component.

Setup 5.9 (Earlier covered-process estimates and allocation certificate)**.** For a covered degree-twelve source with $E \geq 10$ and $\lambda \geq 2$, put

$$B_\lambda = \left\lfloor \frac{14\lambda + 21}{13} \right\rfloor, \qquad T_\lambda = 16\lambda + 24.$$

For every terminal vertex set R, the earlier two-row decomposition gives an actual set $X \subseteq V(F) \setminus R$, of size $t \leq T_\lambda$, such that, for

$$L = V(F) \setminus (R \cup X),$$

both $F[R]$ and $F[L]$ have maximum degree at most two, and

$$|R| \leq k - \lambda - 1 + B_\lambda, \qquad |L| \leq k + \lambda + 2.$$

The number P of whole triangles in $F[R]$ satisfies

$$P \geq E - 2 - \lambda - B_\lambda.$$

The earlier source-cost and rank estimates are

$$c_9(F) \geq k - 7\lambda - 10, \qquad 23k < 100E - 100 - 50\lambda.$$

The full-allocation certificate used below requires only a twelve-regular graph and an actual partition R, L, X, with $\Delta(F[L]) \le 2$ and P whole triangles in $F[R]$. For $t = |X|$, $m = P - 4t$, and an integer $\ell \ge 1$, it gives

$$10\big(m - 60(\ell - 1)\big) > |L| \quad \Longrightarrow \quad c_9(F) \le |R| + t - \ell.$$

The certificate covers every capacity split and every matching-budget allocation in the definition of c_9.

Setup 5.10 (Earlier native refinements and owner results)**.** The following statements are assumed from the earlier native-row development.

(I) The first row refinement removes P_6 from the conditional component list. The sharper row theorem says that every native row is well-covered and has components only from

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7.$$

(II) For every row A and $v \notin V(A)$, put $K_v = F[V(A) \cup \{v\}]$ and $d = |N_A(v)|$. The earlier singleton theorem gives

$$3 \le d \le 12, \quad \Delta(K_v) = d, \quad \alpha(K_v) = a, \quad \nu(K_v) = p + \delta + 1, \quad \mathrm{ps}(K_v) = 0,$$
$$c_1(K_v) = p + \delta - \epsilon + 1, \qquad c_2(K_v) = p + 1,$$
$$c_i(K_v) = 1 \ (3 \le i < d), \qquad c_i(K_v) = 0 \ (i \ge d).$$

If $d = 3$, its cubic vertices form a nonempty clique, and deleting any one gives a native row realized as an early source pool. Every complete maximum-degree process on K_v has one deletion.

(III) *Complete-owner assumption.* We assume, established earlier in this series, the following statement. Let $F \in \mathcal{M}_{12}$, let A be any native row, and put

$$J = F - V(A), \qquad Q_9 = \{v \in V(J) : d_J(v) = 9\}, \qquad \Gamma_9 = F[Q_9].$$

The set Q_9 is the entire degree-nine set. No independence or size restriction is imposed on it. Every $v \in Q_9$ has exactly one of the following entire row neighborhoods, allowing reflection of a displayed P_4:

(A) $\{c, x, y\}$, where c lies on a whole C_3 or C_5, and x, y are both endpoints of a distinct whole P_2 or P_4.

(B) $\{u_2, u_4, z\}$, where $u_1 - u_2 - u_3 - u_4$ is a whole P_4 and $\{z\}$ is a whole isolate.

(C) $\{u_2, w, z\}$, where $u_1 - u_2 - u_3 - u_4$ is a whole P_4, w is an endpoint of a different whole edge ww', and $\{z\}$ is a whole isolate.

(D) $\{z, w_1, w_2\}$, where $\{z\}$ is a whole isolate and w_1, w_2 are endpoints of two distinct whole P_2 components.

In type (A), let O_v be the contacted odd cycle and put $o_v = c$. In types (B)–(D), put $O_v = \{z\}$ and $o_v = z$. For all distinct $v \in Q_9$, the whole odd owners O_v are pairwise distinct, and

$$N_A(v) \cap V(O_v) = \{o_v\}.$$

For every $X \subseteq Q_9$, including the empty set and the entire set, put

$$h = |X|, \qquad G_X = F[V(A) \cup X].$$

Then

$$\nu(G_X) = \nu(A) + h.$$

Every maximum matching contains all physical owner edges vo_v, a maximum internal matching in every whole component of A, and no other edge incident with X. Conversely, any compatible choice of maximum internal matchings exposing the prescribed owner ports, together with all owner edges, is maximum.

Every maximum matching of G_X covers every vertex of G_X-degree at least three. For every actual edge set $P_{\mathrm{paid}} \subseteq E(G_X)$ disjoint from $E(A)$ and all owner edges, put

$$K_X = G_X - P_{\mathrm{paid}}.$$

Then

$$\nu(K_X) = \nu(G_X),$$

and the entire families of maximum matchings of K_X and G_X coincide as physical edge sets. Every maximum matching of K_X covers every vertex of K_X-degree at least three, and

$$K_X \text{ is three-good} \quad \Longleftrightarrow \quad \Delta(K_X) \le 3.$$

Arbitrary internal X-edges and nonowner contacts may be deleted, while every row edge and owner edge survives. No paid edge is reused in a surviving matching.

The exact costs satisfy

$$c_1(G_X) = c_1(A) + h, \qquad c_2(G_X) = p + h, \qquad \mathrm{ps}(G_X) + \sum_{i \ge 3} c_i(G_X) = e(F[X]).$$

If X is independent, then $\mathrm{ps}(G_X) = 0$, all $c_i(G_X)$ with $i \ge 3$ vanish, and G_X is three-good.

For every row vertex w,

$$\alpha\big(\Gamma_9[N_F(w) \cap Q_9]\big) \le 3 - d_A(w).$$

The Q_9-neighbors of a row vertex of degree two form a clique. Every whole isolate has at most one Q_9-neighbor. Every whole odd cycle has at most one Q_9-neighbor, and a whole C_7 has none.

For every complete continuation on J, its full degree-nine and degree-three phase lengths are

$$q_9 = \alpha(\Gamma_9) = c_8(F) - r, \qquad n_3 = r + s - c_3(F).$$

All the statements in this item are assumed earlier-series inputs.

(IV) *Physical exchange and triangle-contact assumptions.* We assume, established earlier in this series, the following statements for the source, row, and masks in (iii).

In type (A), write the cycle as C_{2i+1} and the path as P_{2j}, with $i, j \in \{1, 2\}$. The native row

$$A' = F[(V(A) \setminus \{c\}) \cup \{v\}]$$

replaces exactly

$$C_{2i+1} + P_{2j} \quad \text{by} \quad P_{2i} + C_{2j+1}.$$

All other components are physically unchanged. Deleting v and reinserting c returns exactly A.

In type (B), exchanging u_2 for v replaces the displayed P_4 and isolate by the whole path $u_3 - u_4 - v - z$ and isolate $\{u_1\}$. In type (C), that exchange gives the whole edge u_3u_4, whole path $z - v - w - w'$, and isolate $\{u_1\}$. These are actual native rows of the same source, and all other components are physically unchanged.

For every whole triangle T of A, no exterior vertex has entire row neighborhood $V(T)$.

(V) The one-unit-per-restored-vertex bound gives a source certificate of debit

$$|V(F) \setminus V(K)| + c_i(K)$$

when the omitted original vertices are restored. Applied to K_v at index nine, its debit is

$$u - 1 + \mathbf{1}_{\{d \ge 10\}}.$$

Items (iii)–(iv) state the earlier-series assumptions in full. Neither item is defined by a reference to a later result in this chapter. Their earlier-series proofs and locations were not supplied; the citation and prerequisite flags above remain open.

5.1.2. *Rows and complementary cores.* We first remove the terminal-existence premise from the conditional row theorem.

Theorem 5.11 (The uniform two-row and core interface)**.** *Let $F \in \mathcal{M}_{12}$, with the notation of Setups 5.3 and 5.4.*

(I) *Every actual prefix remainder and every terminal of every source process has independence number a. Rows exist, every terminal is a row, and*

$$p \geq 0, \qquad 0 \leq \epsilon \leq \delta \leq \lambda + 1, \qquad s > 0 \text{ odd}, \qquad |V(F)| = 2r + s.$$

Every row has independence number a, edge count $3p + 2\delta - \epsilon$, and components from

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

It has exactly p odd-cycle components and exactly ϵ positive even-order path components.

(II) *Every source process has exactly $p+\epsilon$ degree-eleven roots and $p+\delta-\epsilon$ degree-ten roots. Their union with the initial independent set induces a row, and*

$$c_{11}(F) = a, \qquad c_{10}(F) = a + p + \epsilon, \qquad c_9(F) = r.$$

Conversely, every induced row can be realized as this entire early prefix of an actual source process.

(III) *For every induced row A, the complement $J = F - V(A)$ is nine-good and satisfies*

$$\alpha(J) = a, \quad \nu(J) = M, \quad \mathrm{ps}(J) = 0, \quad e(J) = 3p + 2\delta - \epsilon + 6s,$$
$$c_i(J) = c_i(F) - r \quad (1 \leq i \leq 9),$$
$$c_1(J) = M - \epsilon, \qquad c_2(J) = M - \delta, \qquad \sum_{i \geq 3} c_i(J) = 3s.$$

Every complete maximum-degree process on this J is a legal continuation of an actual source prefix. It has exactly s roots, deletion-degree sum $6s$, independence number a at every remaining prefix, and a terminal row R. For its root set X,

$$V(F) = V(A) \,\dot\cup\, V(R) \,\dot\cup\, X,$$

$$\Delta(F[X]) \leq 6, \qquad e_F(X, V(A)) = e_F(X, V(R)) = 6s - e(F[X]).$$

These statements hold uniformly in rank, excess, order, both cost sectors, and every indicated choice. They assume no source minimality for row complements and assert no higher-rank exclusion or unrestricted star-forest formula.

Proof. Setup 5.5(i) gives connectedness, twelve-regularity, factor-criticality, well-coveredness, and order $2k + 1$. Thus every maximal independent start has size a. By Setup 5.5(ii), the terminal R of every complete process has independence number a. Each prefix remainder contains its own terminal and is induced in F. Its independence number is therefore at least $\alpha(R) = a$ and at most $\alpha(F) = a$.

A complete process exists because a finite graph has a maximal independent set and successive vertex deletions terminate. Consequently the existence premise of Setup 5.6 holds. Its first two parts give row existence, the component list, component counts, row independence and edge count, phase lengths, and source costs. Every terminal preserves independence, so every terminal is a row. The positive odd integer

$$s = 2\lambda + 3 - 2\delta$$

satisfies $s \geq 1$, whence $\delta \leq \lambda + 1$. This proves (i) and the forward part of (ii).

We give an actual replay for an arbitrary induced row A. Label paths consecutively and cycles cyclically. Choose sets S, T_1 in each component as follows:

component	S	T_1
K_1	$\{1\}$	$\varnothing$
P_2	$\{1\}$	$\{2\}$
P_3	$\{1,3\}$	$\varnothing$
P_4	$\{1,3\}$	$\{4\}$
P_6	$\{1,3,5\}$	$\{6\}$
C_3	$\{1\}$	$\{2\}$
C_4	$\{1,3\}$	$\varnothing$
C_5	$\{1,3\}$	$\{4\}$
C_7	$\{1,3,5\}$	$\{6\}$

Take their unions over the components. The set S is maximum independent in A, so $|S| = a$. Since A is induced, S is independent in F, and hence maximum there.

The set T_1 is independent and has one vertex per odd cycle or positive even-order path. Thus $|T_1| = p + \epsilon$. Each of its vertices has exactly one neighbor in S, so twelve-regularity gives current degree eleven after deleting S. Since T_1 is independent, its vertices may be deleted in any order at that degree. All remaining degrees are at most eleven because S is maximal independent. After deleting T_1, no degree-eleven vertex remains: deleting such a vertex and completing the process would contradict the universal first-phase length $p + \epsilon$.

Put $T_2 = V(A) \setminus (S \cup T_1)$. The displayed choices show that T_2 is independent and that every vertex in T_2 has exactly two neighbors in A, both already deleted. Thus each has current degree ten. From the parameter definitions,

$$r = a + 2p + \delta,$$

and therefore

$$|T_2| = r - a - (p + \epsilon) = p + \delta - \epsilon.$$

No current degree exceeds ten, so all vertices of T_2 may be deleted in any order. The universal second-phase length then implies that no degree-ten vertex remains. The entire early deleted pool is exactly $V(A)$. This proves the converse in (ii), and any continuation on $J = F - V(A)$ can be appended to this replay.

Setup 5.6(iii) gives nine-goodness, the matching number, potential, edge count, and cost differences of J. Append any complete continuation to the replay. Its terminal has independence a, and the first paragraph gives independence a for J and every intermediate remainder. The terminal is a row, so it has order r and edge count $3p + 2\delta - \epsilon$. Hence the continuation has

$$|V(J)| - r = s$$

roots and deletion-degree sum

$$e(J) - e(R) = 6s.$$

Setup 5.6(iv) applies to this actual process and gives the partition and cross-edge formulas.

Finally, the parameter definitions give $M = p + \delta + s$. Using $\mathrm{ps}(J) = 0$,

$$\begin{aligned} \sum_{i \geq 3} c_i(J) &= e(J) - \nu(J) - c_1(J) - c_2(J) \\ &= 3p + 2\delta - \epsilon + 6s - 3M + \epsilon + \delta \\ &= 3s. \end{aligned}$$

Every row and every continuation were arbitrary. □

5.1.3. *Essential Hall blocks.* The missing-budget range makes the earlier essential-port localization apply to every row.

Theorem 5.12 (Essential Hall blocks in every row)**.** *Let $F \in \mathcal{M}_{12}$, and let A be any row. Put*

$$U = V(F) \setminus V(A),$$

let h count the P_3 components of A, and let I_0 be its isolates. Define

$$B_0 = \{v \in U : N_F(v) \cap I_0 = \varnothing\}.$$

Let Γ be the bipartite graph from B_0 to the nontrivial odd components of A, with vC an edge exactly when v has an actual neighbor in $\mathrm{Port}(C)$.

Choose any missing pair $x, y \in \mathbb{Z}_{\geq 0}$ at total $c_0(F) - 1$, and put $j = |U| - x$. Such pairs exist, and

$$j \geq s + p + \delta + \left\lfloor \frac{2E - 3}{10} \right\rfloor + 1.$$

The graph Γ has a B_0-saturating matching, and every such matching covers every P_3 component.

For every chosen matching, its alternating closure from all P_3 component vertices gives $B \subseteq B_0$ and a family $\mathcal{H}$ of nontrivial odd components such that

$$N_\Gamma(B) = \mathcal{H}, \qquad |B| = |\mathcal{H}| = h + p_H, \qquad 0 \leq p_H \leq p.$$

Here $\mathcal{H}$ contains all P_3 components and p_H odd cycles. The closure follows a matching edge from a component to B_0, then an unchosen edge back to a component, and repeats.

Let Z be the union of all even components of A, and put

$$W = Z \cup B \cup \bigcup_{C \in \mathcal{H}} V(C).$$

There is one independent set I outside W for which

$$V(F) \setminus N_F[I] = W.$$

The graph $F[W]$ is well-covered, has a perfect matching, and satisfies

$$\nu(F[W]) - \alpha(F[W]) = p_H.$$

For every $T \subseteq U \setminus B$ with $|T| \leq j - |B|$, one simultaneous exposed-port matching reserves B into $\mathcal{H}$ and sends T to distinct odd components outside $\mathcal{H}$, compatibly with maximum internal matchings of all components of A.

Moreover,

$$p = \tau(F) - c_0(F).$$

If $p = 0$, every row has components only from K_1, P_2, P_4, C_4. These assertions are uniform in rank, excess, order, both sectors, and every row. This theorem itself excludes neither the $p = 0$ source class, nor the positive-p strict sector, nor the entire degree.

Proof. Setup 5.5(i) gives well-coveredness, twelve-regularity, order $2k+1$, and $E \geq 10$. Theorem 5.11 gives $p, \delta \geq 0$, positive s, order $2r + s$, and exactly p odd cycles in A. The first refinement in Setup 5.10(i) gives the list

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

Each listed component C satisfies $\nu(C) = \lfloor |V(C)|/2 \rfloor$: alternating consecutive edges attain the bound, and disjoint endpoints give the upper bound.

Put

$$v = \left\lfloor \frac{2E - 3}{10} \right\rfloor \geq 1.$$

Since $c_0(F) = k + E \geq 10$, a missing pair exists at total $c_0(F) - 1$; otherwise that smaller total would be universal. Fix any such pair. Setup 5.7(i) applies to the connected graph of order $2k + 1$ and matching number k, and gives $x \leq k - E - v$. Consequently

$$j = r + s - x \geq r + s - k + E + v = s + p + \delta + v + 1. \tag{33}$$

The parameter identities also give

$$|U| + \nu(A) = r + s + p + \delta = k + E = c_0(F). \tag{34}$$

There are $p + h$ nontrivial odd components. Each contributes at least one matching edge, so

$$p + h \leq \nu(A) = p + \delta.$$

Thus (33) implies $j \geq p + h + 1$.

All hypotheses of Setup 5.8 now hold for this actual F, A: well-coveredness, the eight-component list, (34), integer missing budgets, and the required range. Its localization theorem gives the matching, the essential P_3 components, the alternating block, the independent-set witness, the perfect matching, and the simultaneous reservation statement. Since the block contains every P_3 and otherwise only odd cycles, its odd-cycle count satisfies $0 \leq p_H \leq p$. This also gives $|B| = |\mathcal{H}| = h + p_H$ and the stated matching-minus-independence identity.

Suppose $p = 0$. The component list reduces to K_1, P_2, P_3, P_4, C_4. By Theorem 5.11(iii), $\Delta(F - V(A)) \leq 9$. Twelve-regularity therefore gives at least three row neighbors for every vertex of U. Together with (33) and (34), these are the hypotheses of the zero-odd-cycle refinement in Setup 5.8. It excludes P_3.

The complement of an independent set is a vertex cover, and conversely. Hence

$$\tau(F) = |V(F)| - \alpha(F) = 2k + 1 - a.$$

Subtracting $c_0(F) = k + E$ gives

$$\tau(F) - c_0(F) = k + 1 - a - E = (2E - 3 - \lambda) + 1 - E = E - 2 - \lambda = p.$$

No localized graph has been assumed to inherit source minimality. □

5.1.4. *Finiteness at fixed rank defect.* We apply the full-allocation certificate to the actual two-row decomposition.

Theorem 5.13 (Unconditional finiteness throughout the lower ranks)**.** *Let F belong to the unchanged degree-twelve rigid source class, equivalently $\mathcal{M}_{12}$ by Setup 5.5(i). Retain every proper-subgraph T_- condition, allowing both vertex and edge deletions. Assume*

$$E \geq 10, \qquad \lambda \geq 2,$$

and define

$$B_\lambda = \left\lfloor \frac{14\lambda + 21}{13} \right\rfloor, \qquad T_\lambda = 16\lambda + 24, \qquad K_\lambda = 1385\lambda + 2078 + 61B_\lambda.$$

Then

$$E \leq \left\lfloor \frac{23K_\lambda - 1}{13} \right\rfloor \leq 2567\lambda + 3851, \qquad 23k < 100E - 100 - 50\lambda.$$

At $\lambda = 2$, these bounds give

$$E \leq 8900, \qquad k \leq 38686, \qquad |V(F)| \leq 77373.$$

No covered-process premise is imposed. Every complete maximum-degree process from every maximum-independent complement is covered. For every such process with terminal vertex set R, there exists $X \subseteq V(F) \setminus R$, with $|X| \leq T_\lambda$, such that

$$L = V(F) \setminus (R \cup X), \qquad \Delta(F[R]), \Delta(F[L]) \leq 2,$$
$$|R| \leq k - \lambda - 1 + B_\lambda, \qquad |L| \leq k + \lambda + 2,$$

and $F[R]$ has at least $E-2-\lambda-B_\lambda$ whole triangles.

Thus each fixed λ in the stated source class has bounded excess and order. No source existence, finite exhaustion, degree-twelve deletion inequality, lower-rank exclusion, whole-degree closure, or unrestricted Ramsey formula is asserted.

Proof. The class identification in Setup 5.5(i) preserves the entire proper-subgraph condition. We assume, established earlier in this series, the applicability assertion in Setup 5.5(iii): under $E \geq 10$ and $\lambda \geq 2$, every allowed complete process is covered in the earlier-series sense. Such processes exist by finite successive deletion. The assumed conditional estimates in Setup 5.9 therefore apply without adding a hypothesis on F. The properties of coveredness used in this application are stated explicitly after Setup 5.5(v); its full earlier definition remains subject to the accompanying blocker.

Write $B = B_\lambda$, $T = T_\lambda$. For every terminal R, that setup gives an actual set X, of size $t \leq T$, and the asserted degree and order bounds. If P is the number of whole triangles in $F[R]$, then

$$P \geq E - 2 - \lambda - B.$$

It also gives

$$c_9(F) \geq k - 7\lambda - 10, \qquad 23k < 100E - 100 - 50\lambda. \tag{35}$$

In particular,

$$|L| \leq k + \lambda + 2 < \frac{100E - 54 - 27\lambda}{23} < \frac{100E}{23}.$$

Apply the allocation certificate in Setup 5.9 to the actual partition R, L, X. Its hypotheses require twelve-regularity, maximum degree at most two on L, and whole triangles on R, all of which hold. Put

$$\ell = t + 6\lambda + 10 + B \geq 1, \qquad m = P - 4t.$$

Then

$$\begin{aligned} m - 60(\ell - 1) &\geq E - \big[2 + \lambda + B + 4T + 60(T + 6\lambda + 9 + B)\big] \\ &= E - K_\lambda, \end{aligned}$$

because $T = 16\lambda + 24$.

Suppose $13E \geq 23K_\lambda$. It follows that

$$10\big(m - 60(\ell - 1)\big) \geq 10(E - K_\lambda) \geq \frac{100E}{23} > |L|.$$

The certificate gives

$$\begin{aligned} c_9(F) &\leq |R| + t - \ell \\ &\leq k - \lambda - 1 + B - (6\lambda + 10 + B) \\ &= k - 7\lambda - 11, \end{aligned}$$

contrary to (35). Thus $13E < 23K_\lambda$. Since E, K_λ are integers,

$$E \leq \left\lfloor \frac{23K_\lambda - 1}{13} \right\rfloor.$$

The estimate $B \leq (14\lambda + 21)/13$ gives

$$K_\lambda \leq \frac{18859\lambda + 28295}{13},$$

and hence

$$\frac{23K_\lambda - 1}{13} \leq \frac{433757\lambda + 650772}{169}.$$

The exact products

$$169 \cdot 2567 = 433823, \qquad 169 \cdot 3851 = 650819$$

show that

$$433757\lambda + 650772 < 169(2567\lambda + 3851)$$

for $\lambda \geq 2$. Taking floors proves the simpler bound. The rank inequality in (35), together with $|V(F)| = 2k+1$, proves fixed-λ order finiteness.

At $\lambda = 2$,

$$B_2 = 3, \qquad T_2 = 56, \qquad K_2 = 5031.$$

Therefore

$$23K_2 - 1 = 115712, \qquad \left\lfloor \frac{115712}{13} \right\rfloor = 8900.$$

Equation (35) gives

$$23k < 100 \cdot 8900 - 100 - 100 = 889800.$$

Since $23 \cdot 38687 = 889801$, we obtain $k \leq 38686$ and $|V(F)| \leq 77373$. These are necessary bounds; neither existence nor exhaustion is claimed. The certificate used above concerns all capacity and matching-budget allocations. □

5.1.5. *Native rows and degree-nine owners.* We assemble the complete native contract while keeping every matching and exchange on its original physical vertices.

Theorem 5.14 (Native rows, complete owners, and singleton restoration)**.** *Let $F \in \mathcal{M}_{12}$, with the parameters and definitions in Setups 5.3 and 5.4.*

(I) *The source is connected, twelve-regular, factor-critical, and well-covered, with*

$$n = 2k+1 = 2r+s, \qquad \operatorname{gap}(F) = 1,$$

$$E \geq 10, \quad \lambda \geq 3, \quad p \geq 5, \quad 0 \leq \epsilon \leq \delta \leq \lambda + 1, \quad \delta > 0.$$

Thus s is positive and odd, and the entire equality sector $\delta = 0$ is empty. The exact source costs are

$$c_0(F) = k + E, \qquad c_1(F) = k + E - \epsilon = u + p + \delta - \epsilon,$$

$$c_2(F) = k + E - \delta = u + p, \qquad c_{11}(F) = a, \qquad c_{10}(F) = a + p + \epsilon, \qquad c_9(F) = r.$$

Every actual prefix remainder and terminal of every complete source process has independence number a. Native rows exist, and every terminal is native. Every native row A is well-covered, has

$$\alpha(A) = a, \qquad e(A) = 3p + 2\delta - \epsilon,$$

and has components only from

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7.$$

It has exactly p odd-cycle components and exactly ϵ positive even-order path components.

(II) *Every source process has exactly $p+\epsilon$ degree-eleven roots and $p+\delta-\epsilon$ degree-ten roots. Their union with the initial independent set induces a native row. Conversely, every native row is realizable as that entire early deleted pool.*

For any native A, put $J = F - V(A)$. Then

$$|V(J)| = u, \quad \alpha(J) = a, \quad \nu(J) = M, \quad \operatorname{ps}(J) = 0,$$

$$e(J) = 3p + 2\delta - \epsilon + 6s, \qquad c_i(J) = c_i(F) - r \quad (1 \leq i \leq 9),$$

$$c_1(J) = M - \epsilon, \quad c_2(J) = M - \delta, \quad \sum_{i \geq 3} c_i(J) = 3s.$$

The graph J is nine-good. Every complete continuation on this same J is legal after an actual early realization of A. It has s roots, deletion-degree sum $6s$, independence a at every prefix, and a terminal native row R. For its entire root set X_{root},

$$V(F) = V(A) \,\dot\cup\, V(R) \,\dot\cup\, X_{\mathrm{root}},$$

$$\Delta(F[X_{\mathrm{root}}]) \leq 6, \qquad e_F(X_{\mathrm{root}}, V(A)) = e_F(X_{\mathrm{root}}, V(R)) = 6s - e(F[X_{\mathrm{root}}]).$$

(III) *For any native A and any $v \notin V(A)$, put*

$$K_v = F[V(A) \cup \{v\}], \qquad d = |N_A(v)|.$$

Then

$$3 \le d \le 12, \quad \Delta(K_v) = d, \quad \alpha(K_v) = a, \quad \nu(K_v) = p + \delta + 1, \quad \mathrm{ps}(K_v) = 0,$$
$$c_1(K_v) = p + \delta - \epsilon + 1, \qquad c_2(K_v) = p + 1,$$
$$c_i(K_v) = 1 \quad (3 \le i < d), \qquad c_i(K_v) = 0 \quad (i \ge d).$$

If $d = 3$, the cubic vertices of K_v form a nonempty clique. Deleting any one gives another native row that is an actual early pool of the same F. Every complete maximum-degree process on K_v to degree at most two consists of exactly one deletion.

(IV) *Put*

$$Q_9 = \{v \in V(J) : d_J(v) = 9\}, \qquad \Gamma_9 = F[Q_9].$$

This is the entire degree-nine set; no independence or size restriction is assumed. It is exactly the set of exterior vertices with three row neighbors. Every $v \in Q_9$ has exactly one of the following entire masks, allowing reflection of a displayed P_4:

(A) *$\{c, x, y\}$, where c lies on a whole C_3 or C_5, and x, y are both endpoints of a distinct whole P_2 or P_4.*

(B) *$\{u_2, u_4, z\}$, where $u_1 - u_2 - u_3 - u_4$ is a whole P_4 and $\{z\}$ is a whole isolate.*

(C) *$\{u_2, w, z\}$, where $u_1 - u_2 - u_3 - u_4$ is a whole P_4, w is an endpoint of a different whole edge ww', and $\{z\}$ is a whole isolate.*

(D) *$\{z, w_1, w_2\}$, where $\{z\}$ is a whole isolate and w_1, w_2 are endpoints of two distinct whole P_2 components.*

In type (A), *write the cycle as C_{2i+1} and the path as P_{2j}, with $i, j \in \{1, 2\}$. The exchange*

$$A' = F[(V(A) \setminus \{c\}) \cup \{v\}]$$

replaces exactly

$$C_{2i+1} + P_{2j} \quad \text{by} \quad P_{2i} + C_{2j+1}.$$

All other components are physically unchanged. Deleting v and reinserting c returns exactly A.

In type (B), *exchanging u_2 for v replaces the displayed P_4 and isolate by the whole path $u_3 - u_4 - v - z$ and isolate $\{u_1\}$. In type* (C), *that exchange gives the whole edge u_3u_4, whole path $z - v - w - w'$, and isolate $\{u_1\}$. These are actual native rows of the same source.*

For every whole triangle T of A, no exterior vertex has entire row neighborhood $V(T)$. Thus a vertex meeting all of T has an additional row neighbor, at least four row contacts, and J-degree at most eight. The additional-contact case is not excluded.

(V) *In type* (A), *let O_v be the contacted odd cycle and $o_v = c$. In types* (B)–(D), *put $O_v = \{z\}$ and $o_v = z$. The whole odd owners O_v, for all distinct $v \in Q_9$, are pairwise distinct, and*

$$N_A(v) \cap V(O_v) = \{o_v\}.$$

For every $X \subseteq Q_9$, including the empty set and the entire set, put

$$h = |X|, \qquad G_X = F[V(A) \cup X].$$

Then

$$\nu(G_X) = \nu(A) + h.$$

Every maximum matching contains all physical owner edges vo_v, a maximum internal matching in every whole component of A, and no other edge incident with X. Conversely, any compatible choice of maximum internal matchings exposing the prescribed owner ports, together with all owner edges, is maximum.

Every maximum matching of G_X covers every vertex of G_X-degree at least three. Let $P_{\mathrm{paid}} \subseteq E(G_X)$ be any actual edge set disjoint from $E(A)$ and all owner edges, and put $K_X = G_X - P_{\mathrm{paid}}$. Then

$$\nu(K_X) = \nu(G_X),$$

and their entire families of maximum matchings coincide as physical edge sets. Every maximum matching of K_X covers every vertex of K_X-degree at least three. Consequently

$$K_X \text{ is three-good} \quad \Longleftrightarrow \quad \Delta(K_X) \le 3.$$

This permits arbitrary deletion of internal X-edges and nonowner contacts, while every row edge and owner edge survives. No paid edge is reused in a surviving matching.

The exact costs satisfy

$$c_1(G_X) = c_1(A) + h, \qquad c_2(G_X) = p + h, \qquad \mathrm{ps}(G_X) + \sum_{i \ge 3} c_i(G_X) = e(F[X]).$$

If X is independent, then $\mathrm{ps}(G_X) = 0$, all $c_i(G_X)$ with $i \ge 3$ vanish, and G_X is three-good.

For every row vertex w,

$$\alpha\big(\Gamma_9[N_F(w) \cap Q_9]\big) \le 3 - d_A(w).$$

Thus the Q_9-neighbors of a row vertex of degree two form a clique. Every whole isolate has at most one Q_9-neighbor. Every whole odd cycle has at most one Q_9-neighbor, and a whole C_7 has none.

(VI) *Every complete continuation of J has the same degree-nine and degree-three phase lengths*

$$q_9 = \alpha(\Gamma_9) = c_8(F) - r, \qquad n_3 = r + s - c_3(F).$$

They satisfy

$$q_9 \le n_3 \quad \Longleftrightarrow \quad c_8(F) + c_3(F) \le 2r + s = n.$$

This is only an inequality.

For the singleton K_v in (iii),

$$c_9(K_v) = \mathbf{1}_{\{d \ge 10\}}.$$

Restoring all $N = u - 1$ omitted original vertices gives the literal nonnegative integer debit

$$D_9 = N + c_9(K_v) = r + s - 1 + \mathbf{1}_{\{d \ge 10\}}.$$

Its difference from the actual native value is

$$D_9 - c_9(F) = s - 1 + \mathbf{1}_{\{d \ge 10\}} \ge 0.$$

Thus this particular singleton-restoration certificate gives no strict ninth-cost saving. No assertion is made that all source certificates have this debit.

This theorem supplies only the stated restrictions, realizations, masks, exchanges, matching contracts, costs, and debit. It asserts no occurrence of a surviving mask, full process-word reversal, equality $q_9 = n_3$, positive-margin degree-nine transport, exclusion of the entire strict source class, or star-forest size Ramsey formula.

Proof. The source condition is exactly that of Setup 5.5. Its first and fourth parts give the source structure, gap, low costs, and

$$\delta > 0, \qquad E \ge 10, \qquad \lambda \ge 3, \qquad p \ge 5.$$

Thus the strict-domain hypotheses used below are consequences for every source, rather than extra restrictions.

Apply Theorem 5.11. Its row definition is the present one, with the earlier q, b renamed δ, ϵ. It gives

$$0 \le \epsilon \le \delta \le \lambda + 1,$$

positive odd s, order $2r + s$, all-prefix independence, row existence, terminal row status, row edge counts and independence numbers, and the stated component counts. The sharper row theorem in Setup 5.10(i) gives well-coveredness and the seven-component list. Theorem 5.11(ii) gives the high costs and every early-phase realization, including the converse.

Since $u = n - r = r + s$, substitution yields

$$u + p = k + E - \delta, \qquad u + p + \delta - \epsilon = k + E - \epsilon.$$

These are the u-based low-cost formulas in (i). Applying Theorem 5.11(iii) to an arbitrary native A gives every assertion in (ii), with exactly the same M, J, root set, and terminal. No minimality of J or reversible process word is used.

For (iii), Setup 5.10(ii) applies to every native row and every exterior vertex. Its graph and contact number are precisely K_v, d. It gives the full singleton profile, the cubic clique, every legal one-step deletion, actual native realization, and the one-deletion length.

For (iv), twelve-regularity gives

$$d_J(v) = 12 - |N_A(v)|.$$

Hence Q_9 is the full three-contact set. We assume, established earlier in this series, the complete contact classification in Setup 5.10(iii): every $v \in Q_9$ has exactly one of the four entire masks stated in that assumption, with no independence or size restriction on Q_9. In particular, type (D) has exactly one isolate and endpoints of two distinct whole edges.

We assume, established earlier in this series, the physical exchanges in Setup 5.10(iv): the type-(A) exchange replaces $C_{2i+1} + P_{2j}$ by $P_{2i} + C_{2j+1}$, and deleting v and restoring c recovers A. The same assumption specifies the type-(B) exchange to the whole path $u_3 - u_4 - v - z$ and isolate $\{u_1\}$, and the type-(C) exchange to the whole edge u_3u_4, whole path $z - v - w - w'$, and isolate $\{u_1\}$. In the latter two types, u_2 has row degree two and is adjacent to v, so it is cubic in K_v. Part (iii) also verifies the native status of the exchanged graph. The entire masks and whole-component conditions ensure that all other components remain physically unchanged.

We assume, established earlier in this series, the triangle-contact restriction in Setup 5.10(iv): no exterior vertex has entire row neighborhood equal to a whole triangle. This assumption applies in the strict domain already established. A vertex meeting a whole triangle therefore has an additional row contact. The identity $d_J(v) = 12 - |N_A(v)|$ then gives $d_J(v) \le 8$. This argument does not exclude additional-contact triangles.

For (v), we assume, established earlier in this series, the complete-owner matching statement in Setup 5.10(iii) for the same A and its entire Q_9. The owners are pairwise distinct whole odd components, with $N_A(v) \cap V(O_v) = \{o_v\}$. For every $X \subseteq Q_9$, the assumed statement gives

$$\nu(G_X) = \nu(A) + |X|.$$

It also states that every maximum matching contains all owner edges vo_v, maximum internal matchings in every whole row component, and no other edge incident with X. Its converse reserves all owner edges simultaneously with any compatible internal matchings exposing the prescribed ports.

We also assume the edge-deletion clause of Setup 5.10(iii): deleting any actual edge set disjoint from the row edges and all prescribed owner edges preserves the entire physical maximum-matching family, and every maximum matching in the resulting graph covers all vertices of degree at least three. The permitted paid-edge set is exactly such a set. When $\Delta(K_X) \le 3$, such a matching witnesses three-goodness. Conversely, three-goodness includes that degree bound by definition.

The no-reuse clause, exact costs, independent-X specialization, and neighborhood restrictions are also explicit earlier-series assumptions in Setup 5.10(iii). In particular, the assumed cost identities are

$$c_1(G_X) = c_1(A) + |X|, \qquad c_2(G_X) = p + |X|, \qquad \mathrm{ps}(G_X) + \sum_{i \ge 3} c_i(G_X) = e(F[X]),$$

and the assumed neighborhood bound is

$$\alpha\big(\Gamma_9[N_F(w)\cap Q_9]\big)\le 3-d_A(w).$$

The independent-set specialization and the restrictions for whole isolates and odd cycles are stated in the same numbered assumption.

Finally, we assume, established earlier in this series, the phase identities in Setup 5.10(iii): every complete continuation of this J has full phase lengths $q_9=\alpha(\Gamma_9)=c_8(F)-r$ and $n_3=r+s-c_3(F)$. Rearrangement gives

$$q_9\le n_3 \quad\Longleftrightarrow\quad c_8(F)+c_3(F)\le 2r+s.$$

No equality is taken from this inequality. The singleton profile gives $c_9(K_v)=0$ for $d\le 9$ and $c_9(K_v)=1$ for $d\ge 10$. Setup 5.10(v) restores $u-1$ omitted original vertices, so its debit is

$$u-1+c_9(K_v)=r+s-1+\mathbf{1}_{\{d\ge 10\}}.$$

Subtracting $c_9(F)=r$ gives $s-1+\mathbf{1}_{\{d\ge 10\}}$, which is nonnegative because $s\ge 1$. This describes that certificate only. □

5.1.6. *The original numerical frame.* We now normalize the source costs and both coordinates of every missing budget on one fixed native row.

Theorem 5.15 (Native normalization, isolate reserve, and retained-set slack)**.** *Let $F\in\mathcal{M}_{12}$, fix any native row A, and put $J=F-V(A)$. Let z_A count the whole isolates of A, and set*

$$v_{12}=\left\lfloor\frac{2E-3}{10}\right\rfloor.$$

The parameters are nonnegative integers with

$$E\ge 10,\quad \lambda\ge 3,\quad p\ge 5,\quad \delta>0,\quad 0\le\epsilon\le\delta\le\lambda+1,$$

and s positive odd and $v_{12}\ge 1$.

(I) *The fixed source is connected and twelve-regular, with*

$$|V(F)|=2k+1=2r+s.$$

The fixed row has independence number a and components only from $K_1,P_2,P_4,C_3,C_4,C_5,C_7$. Every complete continuation on the same J is an actual source continuation whose early pool is exactly A. It has s roots, independence a at every remaining prefix, and a terminal native row.

Without changing these choices,

$$u=|V(J)|=r+s,\qquad \beta=p+\delta-\epsilon,$$
$$r=a+2p+\delta=3p+2\delta+z_A,\qquad z_A=a-p-\delta,\qquad r-\beta=a+p+\epsilon.$$

The exact original costs at indices $0,1,2,9,10,11$ are, respectively,

$$(u+p+\delta,\ u+\beta,\ u+p,\ r,\ r-\beta,\ a).$$

(II) *At least one missing ordered pair $x,y\in\mathbb{Z}_{\ge 0}$ exists at total $c_0(F)-1$. For every such pair, put*

$$j_x=u-x,\qquad j_y=u-y.$$

Then

$$s+p+\delta+v_{12}+1\le j_x,j_y\le a-\delta-v_{12},$$
$$j_x+j_y=u-p-\delta+1,$$
$$z_A\ge s+\delta+1+2v_{12},\qquad r\ge s+3p+3\delta+1+2v_{12}.$$

In particular j_x,j_y are positive integers at most u. These bounds hold for every missing pair; they do not assert that every pair in the displayed interval is missing.

(III) *Fix $h \in \mathbb{Z}_{\geq 0}$ with $h \leq u$, and any $W \subseteq V(J)$ with $|W| = h$. Put*

$$H = F[V(A) \cup W], \qquad N = |V(F) \setminus V(H)|, \qquad \rho = \beta + h.$$

Then $N = u - h$, and

$$\begin{aligned} c_0(F) - N &= p + \delta + h, & c_1(F) - N &= \rho, \\ c_2(F) - N &= p + h, & c_9(F) - N &= h - s, \\ c_{10}(F) - N &= h - s - \beta, & N - \rho + 1 &= u - \beta - 2h + 1. \end{aligned}$$

The scalar ρ is a cost difference and is not asserted to be a matching number.

If $g, e_{\mathrm{sel}} \in \mathbb{Z}_{\geq 0}$, $e_{\mathrm{sel}} \geq 1$, and

$$h = s + \beta + g + e_{\mathrm{sel}},$$

then, for every missing pair in (ii)*,*

$$j_x - h,\ j_y - h \geq \epsilon + v_{12} + 1 - g - e_{\mathrm{sel}},$$

and

$$\begin{aligned} N - (\beta + h - 1) &= z_A - s - \delta + 3\epsilon - 2g - 2e_{\mathrm{sel}} + 1 \\ &\geq 3\epsilon + 2v_{12} + 2 - 2g - 2e_{\mathrm{sel}}. \end{aligned}$$

No sign is asserted for these lower bounds without further bounds on g, e_{sel}.

The theorem uses one original source and one fixed native row throughout. It asserts no selector existence, port assignment, matching prefix, payment, or whole-degree exclusion.

Proof. **Step 1.** The native theorem fixes all source and process choices simultaneously.

Theorem 5.14(i)–(ii) gives the source structure, parameter ranges, row existence, row component list, and every continuation assertion for the same arbitrary A. Since $E \geq 10$,

$$v_{12} \geq \left\lfloor \frac{17}{10} \right\rfloor = 1.$$

The row and source orders give $u = r + s$.

Step 2. Component counts give the isolate normalization.

The parameter definitions imply

$$k - a = 2E - 3 - \lambda = 2p + \lambda + 1,$$

and hence

$$r = k - \lambda - 1 + \delta = a + 2p + \delta.$$

Subtracting $\beta = p + \delta - \epsilon$ gives

$$r - \beta = a + p + \epsilon.$$

On every permitted nonisolated component, the independence and matching numbers coincide. Their common values on $P_2, P_4, C_3, C_4, C_5, C_7$ are $1, 2, 1, 2, 2, 3$, respectively. Alternating vertices and edges attain these values. For a cycle, mapping each independent vertex to its following nonselected vertex bounds the independence number by the floor of half its order. For P_2, P_4, a perfect matching bounds an independent set by at most one endpoint per matching edge.

An isolate contributes one to independence and zero to matching number. Both parameters add over components: restriction gives the upper bound, and unions of component witnesses attain it. Therefore

$$\alpha(A) - \nu(A) = z_A,$$

so

$$z_A = a - p - \delta, \qquad r = 3p + 2\delta + z_A.$$

Theorem 5.14(i) gives the required costs. To check the zero-cost normalization directly,

$$u = r + s = k + \lambda + 2 - \delta,$$

and

$$u+p+\delta = k+\lambda+2-\delta+E-2-\lambda+\delta = k+E.$$

Substituting β and $r-\beta$ gives all six entries in (i).

Step 3. Both missing coordinates lie in the stated interval.

The cost $c_0(F) = k+E$ is positive. If every pair at total $c_0(F)-1$ admitted a two-part partition, the two zero-capacity parts could be empty and that smaller total would be universal. Thus a missing pair exists.

The connectedness, order $2k+1$, matching number k, degree bound twelve, and $E \geq 1$ verify every hypothesis of Setup 5.7(i). Its floor parameter is v_{12}. It gives

$$2E-1+v_{12} \leq x \leq k-E-v_{12}, \qquad k \geq 3E-1+2v_{12}.$$

Interchanging the two matching parts preserves feasibility, so (y,x) is missing whenever (x,y) is. Hence the same interval holds for y.

Subtracting either interval from u, the lower endpoint becomes

$$\begin{aligned} u-(k-E-v_{12}) &= r+s-k+E+v_{12} \\ &= s+E-\lambda-1+\delta+v_{12} \\ &= s+p+\delta+v_{12}+1. \end{aligned}$$

The upper endpoint is

$$u-(2E-1+v_{12}) = k+\lambda+3-\delta-2E-v_{12} = a-\delta-v_{12}.$$

The lower endpoint is positive, and $j_x, j_y \leq u$ because $x, y \geq 0$. Moreover,

$$j_x+j_y = 2u-(c_0(F)-1) = u-p-\delta+1.$$

Step 4. The rank bound gives the isolate and row-order reserves.

Rewrite the bound on k as

$$\begin{aligned} r &\geq 3E-\lambda+\delta-2+2v_{12} \\ &= 3p+2\lambda+\delta+4+2v_{12} \\ &= s+3p+3\delta+1+2v_{12}. \end{aligned}$$

Here $E = p+\lambda+2$ and $s = 2\lambda+3-2\delta$. Subtracting $3p+2\delta$ from $r = 3p+2\delta+z_A$ gives

$$z_A \geq s+\delta+1+2v_{12}.$$

Step 5. The retained-set formulas are exact source-cost differences.

Since $V(A)$ and W are disjoint,

$$N = (2r+s)-(r+h) = u-h \geq 0.$$

Subtracting N from the costs in (i) gives every displayed difference in (iii). Also

$$N-\rho+1 = (u-h)-(\beta+h)+1 = u-\beta-2h+1.$$

If $h = s+\beta+g+e_{\mathrm{sel}}$, subtracting it from the lower missing-coordinate endpoint gives

$$j_x-h,\ j_y-h \geq p+\delta+v_{12}+1-\beta-g-e_{\mathrm{sel}} = \epsilon+v_{12}+1-g-e_{\mathrm{sel}}.$$

For the omission slack,

$$\begin{aligned} N-\rho+1 &= r-s-3\beta-2g-2e_{\mathrm{sel}}+1 \\ &= z_A-s-\delta+3\epsilon-2g-2e_{\mathrm{sel}}+1. \end{aligned}$$

The isolate reserve from Step 4 gives the stated lower bound. These are integer identities and inequalities; none identifies a residual matching number or establishes a nonnegative margin without additional bounds. □

5.1.7. *Terminal odd-cycle selectors.* We retain every actual phase root and complete one odd cycle inside a maximum independent selection of terminal degree-two vertices.

Theorem 5.16 (The terminal-row selector and its exact boundary)**.** *Let $F \in \mathcal{M}_{12}$, fix any native row A, and fix any complete continuation on $F - V(A)$. Let X_{root} be its entire root set and R its actual terminal row. For $j \in \{3, \dots, 9\}$, let B_j be its full degree-j phase.*

(I) *Put*

$$D_2 = \{v \in V(R) : d_R(v) = 2\}.$$

Let $n_2, n_4, z_3, z_4, z_5, z_7$ count, respectively, the whole $P_2, P_4, C_3, C_4, C_5, C_7$ components of R. Then

$$\beta = n_4 + z_3 + 2z_4 + 2z_5 + 3z_7 = \alpha(R[D_2]).$$

The entire graph $R[D_2]$ consists of every whole cycle of R and the single middle edge of every whole P_4; there are no other degree-two vertices.

Choose any maximum independent set T_0 of $R[D_2]$. Then $|T_0| = \beta$. A whole odd cycle exists in R. Choose any such cycle C, and put

$$e_C = \frac{|V(C)| + 1}{2} \in \{2, 3, 4\}, \qquad Y = T_0 \cup V(C).$$

Then

$$|T_0 \cap V(C)| = e_C - 1, \qquad |Y| = \beta + e_C.$$

Every $y \in Y$ has $d_R(y) = 2$, and the entire induced graph $R[Y]$ is C together with isolates. Thus

$$\nu(R[Y]) = e_C - 1.$$

These conclusions hold for every indicated T_0, C in the same fixed R.

(II) *Put*

$$W = X_{\mathrm{root}} \cup Y, \qquad h = |W|, \qquad H = F[V(A) \cup W], \qquad N = |V(F) \setminus V(H)|.$$

The union defining W is disjoint, and

$$|X_{\mathrm{root}}| = s, \qquad h = s + \beta + e_C, \qquad N = r - \beta - e_C \geq 0.$$

The sets $V(A), X_{\mathrm{root}}, V(R)$ partition $V(F)$. Every phase root, including every B_3 vertex, is retained.

For every $y \in Y$, its entire original neighborhood outside R lies in the retained set $V(A) \cup X_{\mathrm{root}}$, and it has exactly ten H-neighbors there. All vertices and induced edges are the original physical ones. Moreover,

$$H[Y] = R[Y].$$

(III) *The original-source identities are*

$$r = a + 2p + \delta, \qquad r - \beta = a + p + \epsilon = c_{10}(F),$$

and

$$N + \nu(H[Y]) = r - \beta - 1 = c_{10}(F) - 1.$$

This last identity is only a matching-rank debit calculation. It asserts no matching prefix, ten sequential matchings, residual graph after such deletions, integer certificate at this debit, or source exclusion.

The selector applies to every native row and actual continuation, without a cubic catalogue, triangle reserve, positive phase-length assumption, or extra source-sector premise. If C is a triangle, then $e_C = 2$ and $|Y| = \beta + 2$.

Proof. Theorem 5.14(i)–(ii) gives twelve-regularity, source order $2r+s$, $p \geq 5$, native-row existence, and the actual realization of the fixed continuation. Its root set has size s, and its terminal R is native of order r. The sets $V(A), X_{\mathrm{root}}, V(R)$ partition $V(F)$. The row has components only from

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7,$$

with p odd cycles and ϵ positive even-order paths. Its matching number is $p+\delta$. Setup 5.5(v) identifies all full phases $B_9, \ldots, B_3$ in this same continuation, including empty phases, and their union is X_{root}.

A P_2 has matching number one and a P_4 has matching number two. A cycle of order ℓ has matching number $\lfloor \ell/2 \rfloor$: consecutive disjoint pairs attain the bound, and disjoint endpoints give the upper bound. Matching numbers add over components by restriction and union of component maxima. Hence

$$p + \delta = n_2 + 2n_4 + z_3 + 2z_4 + 2z_5 + 3z_7.$$

The component counts give

$$p = z_3 + z_5 + z_7, \qquad \epsilon = n_2 + n_4.$$

Subtracting ϵ from the matching identity yields

$$\beta = n_4 + z_3 + 2z_4 + 2z_5 + 3z_7.$$

Neither K_1 nor P_2 contributes to D_2. A P_4 contributes exactly its two internal vertices and their middle edge. Every cycle vertex has row degree two. The exhaustive component list therefore gives the stated entire graph $R[D_2]$.

An independent set in a cycle of order ℓ has size at most $\lfloor \ell/2 \rfloor$: its vertices and their cyclic successors are disjoint sets of equal size. Alternating vertices attain this bound. An edge has independence number one, and independence numbers add over components by restriction and union. Therefore

$$\alpha(R[D_2]) = n_4 + z_3 + 2z_4 + 2z_5 + 3z_7 = \beta.$$

Every maximum independent set T_0 restricts to a maximum independent set on each component. If one restriction were smaller, replacing it inside that component would increase the total without adding cross-component edges. Since $p \geq 5$, a whole odd cycle exists. It is C_3, C_5, or C_7, so its order is $2e_C - 1$ with $e_C \in \{2,3,4\}$. Consequently

$$|T_0 \cap V(C)| = e_C - 1$$

and

$$|Y| = \beta + (2e_C - 1) - (e_C - 1) = \beta + e_C.$$

Every vertex of Y lies in D_2. Outside C, the set Y is the independent set $T_0 \setminus V(C)$. No edge joins C to another row component. Thus the entire induced graph is C and isolates, with matching number $e_C - 1$.

The root set is disjoint from $V(R)$, so

$$|W| = s + \beta + e_C.$$

The omitted set is exactly $V(R) \setminus Y$, giving

$$N = r - \beta - e_C \geq 0.$$

Every B_j is contained in the retained root set. For $y \in Y$, twelve-regularity and $d_R(y) = 2$ give exactly ten original neighbors outside R. The complement of $V(R)$ is exactly $V(A) \cup X_{\mathrm{root}}$, all retained in H. Since H and R are induced in F,

$$H[Y] = F[Y] = R[Y].$$

Finally,

$$k = a + 2E - 3 - \lambda$$

implies

$$r = k - \lambda - 1 + \delta = a + 2E - 4 - 2\lambda + \delta = a + 2p + \delta.$$

Subtracting β gives $r - \beta = a + p + \epsilon$, which equals $c_{10}(F)$ by Theorem 5.14(i). Thus

$$N + \nu(H[Y]) = (r - \beta - e_C) + (e_C - 1) = r - \beta - 1 = c_{10}(F) - 1.$$

No matching edges have been deleted in this argument, so the arithmetic identity does not establish a matching prefix or certificate. □

5.1.8. *The exceptional triangle reserve.* At the specified parameter boundary, the missing allocation excludes a triangle-free native row.

Theorem 5.17 (A whole triangle at the exceptional boundary)**.** *Let $F \in \mathcal{M}_{12}$, and assume*

$$E = 11, \qquad \lambda = 4, \qquad \delta = 5, \qquad \epsilon = 0.$$

Native rows exist, and every native row contains a whole C_3 component. Equivalently, the subclass of these sources having some triangle-free native row is empty. This does not exclude the entire parameter cell or the entire degree-twelve source class.

Proof. Fix a source with these parameters. Theorem 5.14(i) gives connectedness, twelve-regularity, order $2k + 1$, $c_0(F) = k + E$, and native-row existence. Substitution gives

$$p = 11 - 2 - 4 = 5, \qquad r = k - 4 - 1 + 5 = k, \qquad s = 2 \cdot 4 + 3 - 2 \cdot 5 = 1.$$

The source-class identification in Setup 5.5(i) allows Theorem 5.13 to be applied with $E = 11 \geq 10$ and $\lambda = 4 \geq 2$. It gives

$$23k < 100 \cdot 11 - 100 - 50 \cdot 4 = 800.$$

Setup 5.7(i) applies to the same connected source of odd order and matching number k. Its floor parameter is

$$\max\left(0, \left\lfloor \frac{2E - 3}{10} \right\rfloor\right) = \left\lfloor \frac{19}{10} \right\rfloor = 1.$$

Thus $k \geq 3E - 1 + 2 = 34$. Since $23 \cdot 35 = 805 > 800$ and k is an integer, $k = 34$. The definition of λ gives

$$a = k - 2E + 3 + \lambda = 19.$$

Consequently

$$r = 34, \qquad |V(F)| = 69, \qquad c_0(F) = 45,$$

and every native row has a complement of order 35.

Let A be an arbitrary native row. Suppose it has no whole triangle. Theorem 5.14(i) gives its entire component list

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7,$$

exactly $p = 5$ odd cycles, exactly $\epsilon = 0$ positive even-order paths, and matching number $p + \delta = 10$. Hence no P_2 or P_4 occurs, and every odd cycle is C_5 or C_7.

The matching numbers of C_5, C_7, C_4 are $2, 3, 2$, respectively: consecutive disjoint pairs attain these values and order bounds them above. Matching numbers add over components by restriction and union of maxima. The five odd cycles already contribute at least ten. They must therefore all be C_5, and no C_4 occurs. The remaining components are isolates. Since the five cycles use 25 of the 34 vertices,

$$A = 5C_5 \,\dot\cup\, 9K_1.$$

Its family $\mathcal{O}$ of odd components has size $o = 14$. Put $U = V(F) \setminus V(A)$, so $|U| = 35$.

At total $44 = c_0(F) - 1$, a missing pair $x, y \in \mathbb{Z}_{\geq 0}$ exists by the least-universal-total definition. The coordinate interval in Setup 5.7(i) gives

$$22 = 2E - 1 + 1 \leq x \leq k - E - 1 = 22.$$

Thus $x = y = 22$, and

$$j = |U| - x = 13.$$

Every component satisfies the required matching-order identity: it was checked for C_5, and an isolate has matching number zero. Moreover,

$$c_0(F) = 45 = 35 + 10 = |U| + \nu(A).$$

Therefore Setup 5.7(ii) applies to this same induced row.

Choose any two distinct singleton components. Each singleton vertex is an exposed port because deleting it leaves the empty matching. Let $\mathcal{C}$ consist of these two components, so $\ell = 2$. Then

$$o - j + 1 = 14 - 13 + 1 = 2 \le \ell \le 14 = o.$$

The port expansion statement gives at least

$$|U| - o + \ell = 35 - 14 + 2 = 23$$

vertices of U adjacent to their union.

For each singleton z, all original neighbors lie in U. Twelve-regularity and simplicity give $|N_F(z)| = 12$. Thus any two distinct singleton vertices z, z' satisfy

$$|N_F(z) \cup N_F(z')| \ge 23$$

and

$$|N_F(z) \cap N_F(z')| = 24 - |N_F(z) \cup N_F(z')| \le 1.$$

Choose four of the nine singleton vertices, say z_1, z_2, z_3, z_4, and put $S_i = N_F(z_i)$ for $i \in \{1, 2, 3, 4\}$. Each $S_i \subseteq U$ has size twelve. For $i \in \{2, 3, 4\}$,

$$S_i \cap \bigcup_{\ell<i} S_\ell \subseteq \bigcup_{\ell<i} (S_i \cap S_\ell),$$

so its size is at most $i - 1$. Adding S_i therefore contributes at least $12 - (i - 1)$ new vertices. Hence

$$|S_1 \cup S_2 \cup S_3 \cup S_4| \ge 12 + 11 + 10 + 9 = 42.$$

This union lies in the 35-vertex set U, a contradiction. Since A was arbitrary, every native row contains a whole triangle. No existence assertion is made for the remaining triangle-containing class. □

5.1.9. *Shared boundary endpoints.* The next result keeps boundary collisions in the matching optimization.

Theorem 5.18 (Joint boundary rank and at most two odd-cycle roots)**.** *Let R, I be disjoint finite vertex sets, let J be a graph on R, and let B be bipartite with parts R, I. Let P have vertex set $R \cup I$ and entire edge set $E(J) \cup E(B)$, so I is independent. Put $m = \nu(J)$. For $Z \subseteq R$, define*

$$\eta_J(Z) = |Z| + \nu(J - Z) - m.$$

Let $\mathcal{Z}(B)$ consist of the sets $Z \subseteq R$ that occur as exactly the R-endpoint set of a matching in B. In particular $\varnothing \in \mathcal{Z}(B)$. Put

$$T = \{v \in R : N_B(v) \ne \varnothing\}.$$

(I) *The exact matching number is*

$$\nu(P) = \max_{Z \in \mathcal{Z}(B)} \big(|Z| + \nu(J - Z)\big) = m + \max_{Z \in \mathcal{Z}(B)} \eta_J(Z).$$

Every displayed maximum is attained. The function η_J is nondecreasing under inclusion, and

$$0 \le \eta_J(Z) \le |Z|.$$

If B has a matching saturating T, then

$$\nu(P) = |T| + \nu(J - T),$$

including $T = \varnothing$.

(II) *Suppose $T = \{u, v\}$, with $u \neq v$, and put*

$$O_u = N_B(u), \qquad O_v = N_B(v).$$

Both sets are nonempty. If $|O_u \cup O_v| \geq 2$, then

$$\nu(P) = 2 + \nu(J - \{u, v\}),$$

and $\nu(P) = m$ if and only if $\nu(J - \{u, v\}) = m - 2$.
If $|O_u \cup O_v| = 1$, then

$$\nu(P) = 1 + \max\big(\nu(J - u), \nu(J - v)\big),$$

and $\nu(P) = m$ if and only if both

$$\nu(J - u) = m - 1, \qquad \nu(J - v) = m - 1.$$

The neighborhoods may have arbitrary positive sizes; distinct outside neighbors are not assumed.

(III) *Suppose J is the vertex-disjoint union of a whole odd cycle C of order $2e - 1$, where $e \in \mathbb{Z}$ and $e \geq 2$, and an arbitrary graph J_0. Assume*

$$\varnothing \neq T \subseteq V(C), \qquad |T| \leq 2.$$

Then, regardless of shared outside endpoints or boundary multiplicities,

$$\nu(P) = \nu(J) + 1 = e + \nu(J_0).$$

If J_0 has a vertex cover of cardinality $\nu(J_0)$, then P has a vertex cover of cardinality $\nu(P)$.

(IV) *Individual root penalties and boundary degrees do not determine the joint penalty. Let J be the single edge uv. Then*

$$\nu(J - u) = \nu(J - v) = 0,$$

and attaching a nonempty boundary star at either root separately gives zero rank increase. If

$$I = \{x\}, \qquad E(B) = \{ux, vx\},$$

then P is a triangle and $\nu(P) = 1$. If

$$I = \{x, y\}, \quad x \neq y, \qquad E(B) = \{ux, vy\},$$

then P is a four-vertex path and $\nu(P) = 2$. The R-boundary degree vector is $(1, 1)$ in both cases. Thus the sum of the two individual zero penalties can underestimate the joint penalty by one.

This is a generic matching and cover interface. It asserts no occurrence as a minimal-source terminal, cost lower bound, source-degree exclusion, or Ramsey formula.

Proof. **Step 1.** Boundary endpoint sets give the exact maximum.

Let M_P be any matching in P. Its edges partition into B-edges and J-edges because those entire edge sets are disjoint. Let Z be the R-endpoint set of its B-edges. They form a boundary matching with exactly $|Z|$ edges, so $Z \in \mathcal{Z}(B)$. The internal edges avoid Z, giving

$$|M_P| \leq |Z| + \nu(J - Z).$$

Taking maxima gives the upper bound.

Conversely, for $Z \in \mathcal{Z}(B)$, choose a boundary matching with endpoint set exactly Z, and a maximum matching in $J - Z$. Their endpoints lie in $Z \cup I$ and $R \setminus Z$, respectively, so their union is a matching of size $|Z| + \nu(J - Z)$. The family $\mathcal{Z}(B)$ is finite and nonempty, and all graphs are finite. Every choice and maximum therefore exists. Subtracting m gives the second formula.

Step 2. Vertex deletion proves monotonicity and the bounds.

For $Z \subseteq Z' \subseteq R$, delete from a maximum matching of $J - Z$ the edges incident with $Z' \setminus Z$. At most $|Z' \setminus Z|$ matching edges are removed, so

$$\nu(J - Z') \geq \nu(J - Z) - |Z' \setminus Z|.$$

Thus

$$|Z'| + \nu(J - Z') \geq |Z| + \nu(J - Z),$$

which is the monotonicity of η_J. Taking $Z = \varnothing$ gives

$$\nu(J - Z') \geq m - |Z'|.$$

Every matching of $J - Z'$ is also a matching of J, so $\nu(J - Z') \leq m$. These bounds give $0 \leq \eta_J(Z') \leq |Z'|$.

Every feasible endpoint set lies in T. If T is feasible, monotonicity makes its value maximal and Step 1 gives $\nu(P) = |T| + \nu(J - T)$. If $T = \varnothing$, then B is edgeless and the same formula holds.

Step 3. Two roots have exactly the two stated feasibility cases.

Both singleton endpoint sets are feasible. The pair $\{u, v\}$ is feasible exactly when O_u, O_v have distinct representatives. For two nonempty sets, this is equivalent to $|O_u \cup O_v| \geq 2$. To see sufficiency, choose $a \in O_u$. If O_v contains $b \neq a$, use a, b. Otherwise $O_v = \{a\}$, and the union condition supplies $b \in O_u \setminus \{a\}$; use b, a. Necessity follows because a two-edge matching uses two distinct I-vertices.

When the pair is feasible, Step 2 gives

$$\nu(P) = 2 + \nu(J - \{u, v\}).$$

Otherwise the feasible sets are exactly $\varnothing, \{u\}, \{v\}$. Deleting one vertex reduces matching number by at most one, so both singleton values are at least m. The empty-set value can be omitted, yielding

$$\nu(P) = 1 + \max(\nu(J - u), \nu(J - v)).$$

The first zero-increase criterion follows by subtraction. For the second, each deletion rank is m or $m - 1$, so their maximum is $m - 1$ exactly when both are $m - 1$.

Step 4. Odd-cycle deletion ranks force one unit of increase.

A path on $h \in \mathbb{Z}_{\geq 0}$ vertices, including the empty path, has matching number $\lfloor h/2 \rfloor$: alternating consecutive edges attain that value and order gives the upper bound. The cycle of order $2e - 1$ has matching number $e - 1$ for the same reasons. Deleting one cycle vertex leaves an even path on $2e - 2$ vertices, of matching number $e - 1$.

Deleting two distinct cycle vertices leaves two possibly empty paths whose orders sum to $2e - 3$. One order is odd and the other even. Their matching numbers therefore sum to $e - 2$. Matching numbers add over vertex-disjoint components by restriction and union of maxima. Hence

$$\nu(J) = e - 1 + \nu(J_0),$$
$$\nu(J - u) = e - 1 + \nu(J_0) \quad (u \in V(C)),$$
$$\nu(J - \{u, v\}) = e - 2 + \nu(J_0) \quad (u \neq v,\ u, v \in V(C)).$$

If T is a singleton, Step 1 gives $e + \nu(J_0)$. If $|T| = 2$, either case of Step 3 gives the same value.

Step 5. An independent set avoiding the active roots gives the cover.

A path on h vertices has an independent set of size $\lceil h/2 \rceil$, obtained by choosing alternating vertices starting at an endpoint. The empty path contributes the empty set. If T is a singleton, $C - T$ is an even path and supplies an independent set U_C of size $e - 1$. If $|T| = 2$, the two remaining path orders sum to $2e - 3$, with one odd and one even. The sum of their ceiling half-orders is $e - 1$, so alternating sets again give $|U_C| = e - 1$.

The set U_C is independent in the original cycle because $C - T$ is induced. Its complement in $V(C)$ is a cover of size e containing T. It also covers every boundary edge. Adjoining the supplied cover of J_0 gives a cover of size

$$e + \nu(J_0) = \nu(P).$$

This includes empty J_0.

Step 6. The two boundary graphs verify the collision comparison.

For $J = uv$, $\nu(J) = 1$, and deleting either endpoint leaves rank zero. A boundary star attached only at u, together with uv, has all edges incident with u, so its matching number is one. The assertion at v follows by the same direct edge-incidence check.

With the shared endpoint x, the entire edge set is $\{uv, ux, vx\}$. Every pair of edges meets and at least one edge exists, so the rank is one. With distinct x, y, the entire edge set is $\{uv, ux, vy\}$. The edges ux, vy are disjoint, giving rank at least two, and order four gives rank at most two. Each root has boundary degree one in both graphs. This proves (iv). □

5.1.10. *Complete owner reservations and the loss from payment.* We reserve the entire owner family before assigning helper vertices to free odd components.

Theorem 5.19 (Source-native reservations and exact owner-edge loss)**.** *Let $F \in \mathcal{M}_{12}$, and assume*

$$\delta > 0, \qquad E \geq 10, \qquad \lambda \geq 3, \qquad p \geq 5.$$

Let A be any native row, put

$$J = F - V(A), \qquad u = |V(J)|, \qquad Q_9 = \{q \in V(J) : d_J(q) = 9\}.$$

For each $q \in Q_9$, let O_q be its unique odd whole row component containing a neighbor, and let o_q be that unique neighbor. These owners are pairwise distinct. Let

$$\mathcal{O} = \{\textit{odd whole components of } A\}, \qquad \mathcal{O}_{\text{owned}} = \{O_q : q \in Q_9\}, \qquad \mathcal{O}_{\text{free}} = \mathcal{O} \setminus \mathcal{O}_{\text{owned}}.$$

Fix any missing pair $b_r, b_s \in \mathbb{Z}_{\geq 0}$ at total $c_0(F) - 1$, and put

$$j = u - b_r, \qquad u_{10} = \left\lfloor \frac{2E - 3}{10} \right\rfloor.$$

(I) *Such missing pairs exist, and*

$$j \geq s + p + \delta + u_{10} + 1 \geq 12.$$

For every $T \subseteq V(J)$ with $|T| \leq j$, put

$$H_T = F[V(A) \cup T].$$

Then

$$\nu(H_T) = \nu(A) + |T|.$$

Every maximum matching consists of a maximum internal matching in every whole row component and one T-to-A edge for each vertex of T, ending at exposed ports in distinct odd components. No edge inside T belongs to such a maximum matching.

(II) *For any $Y \subseteq V(J) \setminus Q_9$, put $h = |Y|$ and*

$$\mathcal{O}(Y) = \{C \in \mathcal{O}_{\text{owned}} : N_F(y) \cap \text{Port}(C) \neq \varnothing \text{ for some } y \in Y\}, \qquad m(Y) = |\mathcal{O}(Y)|.$$

If $h + m(Y) \leq j$, there is one matching assigning every $y \in Y$ to an exposed port in a distinct free odd component, simultaneously containing every physical owner edge qo_q for the entire Q_9 and a maximum internal matching in every row component.

Consequently, for

$$H_Y = F[V(A) \cup Q_9 \cup Y],$$
$$\nu(H_Y) = \nu(A) + |Q_9| + h.$$

Every maximum matching has exactly this form: all owner edges, one free-component edge at each Y-vertex, distinct free destinations, and maximum internal row matchings. Conversely, every compatible such choice is maximum.

In particular every maximum matching avoids every nonowner edge incident with Q_9, every edge inside Y, and every Y-to-A edge ending in an even or owned component. Deleting any subset of these avoided edges preserves the entire physical maximum-matching family and its size. No independence of Q_9 or Y is assumed.

(III) *Every $v \in V(J) \setminus Q_9$ with $d_J(v) \geq 1$ has a free destination:*

$$N_{\text{free}}(v) = \{C \in \mathcal{O}_{\text{free}} : N_F(v) \cap \text{Port}(C) \neq \varnothing\} \neq \varnothing.$$

Now suppose x has entire row neighborhood $\{z, a_0, b_0\}$, where $\{z\}$ is a whole isolate and a_0a_1, b_0b_1 are two distinct whole edges. For every $v \neq x$ in J adjacent to all of z, a_1, b_1, we have

$$v \notin Q_9, \qquad N_{\text{free}}(v) \neq \varnothing,$$

even if $d_J(v) = 0$.

For any two distinct such vertices v, w, exactly one of the following occurs:

(a) *They have assignments to distinct free odd components. All simultaneous matching and matching-family conclusions in* (ii) *then hold for $Y = \{v, w\}$, whether or not its numerical sufficient condition holds.*

(b) *There is one $C_* \in \mathcal{O}_{\text{free}}$ with*

$$N_{\text{free}}(v) = N_{\text{free}}(w) = \{C_*\}.$$

In this case

$$m(\{v, w\}) \geq j - 1 \geq 11,$$

$$d_J(v) + d_J(w) \leq 20 - j \leq 8,$$

and

$$\nu\big(F[V(A) \cup Q_9 \cup \{v, w\}]\big) = \nu(A) + |Q_9| + 1.$$

Thus $d_J(v) + d_J(w) \geq 9$ suffices for the first alternative. The occurrence of the second alternative is neither asserted nor excluded.

(IV) *In the type-*(D) *setting of* (iii)*, the entire neighborhood $N_F(z)$ consists of twelve vertices in J. There is one maximum matching of*

$$F[V(A) \cup N_F(z)]$$

using xz and assigning its other eleven exterior vertices to exposed ports in eleven distinct odd components different from $\{z\}$, with maximum internal matchings in all row components.

More generally, for every $T \subseteq V(J)$ containing x, with $|T| \leq j$, every maximum matching of H_T contains xz. If P_z is any set of actual H_T-edges incident with z, and $xz \in P_z$, then

$$\nu(H_T - P_z) = \nu(H_T) - 1 = \nu(A) + |T| - 1.$$

This includes paying three distinct edges xz, zv, zw whenever T contains distinct $v, w \in N_F(z) \setminus \{x\}$. The set $T = \{x, v, w\}$ always lies in the allowed range. The other two paid star edges are not prescribed matching edges.

The theorem does not claim that the helpers are roots in one common process, that arbitrary neighbors v, w also meet a_1, b_1, that the displayed row-extension matching belongs to the different host consisting of J and retained row vertices, or that a positive-margin certificate or source exclusion follows.

Proof. **Step 1.** Missing budgets give the universal row-extension range.

Theorem 5.11 supplies twelve-regularity,

$$0 \le \epsilon \le \delta \le \lambda + 1,$$

positive s, order $2r+s$, and the native component list. Each listed component has matching number the floor of half its order, by alternating path or cycle edges and the disjoint-endpoint upper bound.

We assume, established earlier in this series, the complete-owner input in Setup 5.10(iii): every $q \in Q_9$ has a unique odd whole row component O_q containing a neighbor, that neighbor is o_q, and the owners O_q are pairwise distinct. Each owner is an isolate, C_3, or C_5, and q has no neighbor on any other odd whole component. An isolate exposes its vertex by the empty matching. Deleting a vertex of C_3 or C_5 leaves an even path with a perfect matching. Thus these owner neighbors are exposed ports; no exposability conclusion is taken merely from odd order for other components.

Since $c_0(F) = k + E \ge 10$, a missing pair exists at total $c_0(F) - 1$ by the least-total definition. The native parameters give

$$u = r + s, \qquad u + \nu(A) = r + s + p + \delta = k + E = c_0(F).$$

Theorem 5.12 applies to every such pair and gives

$$j \ge s + p + \delta + u_{10} + 1.$$

Using $s = 2\lambda + 3 - 2\delta$ and $\delta \le \lambda + 1$,

$$j \ge p + 2\lambda + 4 - \delta + u_{10} \ge p + \lambda + 3 + u_{10} \ge 12,$$

because $p \ge 5, \lambda \ge 3, u_{10} \ge 1$.

We assume, established earlier in this series, the exposed-port extension statement in Setup 5.7(ii). Its hypotheses hold for this actual induced row: every component has matching number the floor of half its order, $c_0(F) = u + \nu(A)$, and $j = u - b_r$ comes from the fixed missing pair. For every $T \subseteq V(J)$ with $|T| \le j$, the assumed statement assigns T to exposed ports in distinct odd components, compatibly with maximum internal matchings of all row components. The union of these port edges and internal matchings has size $\nu(A) + |T|$.

For any matching in H_T, at most $\nu(A)$ edges are internal to A. Every other edge uses at least one vertex of T, so at most $|T|$ further edges occur. This proves the upper bound. Equality forces a maximum internal matching in every row component, and exactly one crossing edge at every exterior vertex. An edge inside T would use two exterior vertices for one edge and prevent equality. Every even component is internally covered, and every odd component has one internally uncovered exposed port. Thus the crossing endpoints are ports in distinct odd components. This proves the every-maximum assertion for the whole range, including $|T| < j$.

Step 2. Reserve the owners reached by the helpers, then extend to all owners.

For Y in (ii), let

$$Q(Y) = \{q \in Q_9 : O_q \in \mathcal{O}(Y)\}.$$

Distinctness of owners gives $|Q(Y)| = m(Y)$, and $Q(Y) \cap Y = \varnothing$. Apply (i) to

$$T = Y \cup Q(Y), \qquad |T| = h + m(Y) \le j.$$

Each $q \in Q(Y)$ must use its only odd component and sole port, through qo_q. Those components are occupied.

A vertex of Y cannot be assigned to an owned component outside $\mathcal{O}(Y)$, since it has no exposed-port neighbor there. It cannot be assigned to a component inside $\mathcal{O}(Y)$, since that component is occupied by its forced owner edge. Hence all helper edges end in distinct free components.

Retain these actual helper edges and add every owner edge for the entire Q_9. Their whole owned destinations are pairwise distinct and disjoint from the free destinations. In each assigned component, choose a maximum internal matching exposing its incoming port, as permitted by the

definition of Port. In every other component, choose any maximum internal matching. The union is one matching of H_Y of size

$$\nu(A) + |Q_9| + h.$$

The same counting upper bound from Step 1 applies to the exterior set $Q_9 \cup Y$ without a premise on its size. Hence this matching is maximum.

Equality forces maximum internal matchings and one crossing edge per exterior vertex into distinct odd ports. Every $q \in Q_9$ must use qo_q, so every helper uses a free component. This proves necessity of the description. The explicit union construction proves sufficiency for every compatible choice.

Every edge declared avoided lies outside this exhaustive description. Thus all original maximum matchings survive deletion of any subset of those edges. A matching in the deleted graph is also a matching in the original graph, so the rank cannot increase. Since the original maximum size survives, any new maximum matching also has the original maximum size and belongs to the original family. The entire physical families coincide.

Step 3. Row-incidence bounds give free destinations and the two-helper obstruction.

Fix $v \notin Q_9$ with $d_J(v) \geq 1$. Twelve-regularity gives

$$|N_A(v)| = 12 - d_J(v) \leq 11.$$

Every owned component counted by $m(\{v\})$ uses at least one different row neighbor. Therefore

$$m(\{v\}) \leq 11, \qquad 1 + m(\{v\}) \leq 12 \leq j.$$

Part (ii) gives a free destination.

Now fix the type-(D) seed x. Its exact three row contacts and twelve-regularity put it in Q_9. The complete-owner assumption in Setup 5.10(iii) states that every whole isolate has at most one Q_9-neighbor. Since x is such a neighbor of z, the isolate z has no other Q_9-neighbor. Hence every other vertex adjacent to z lies outside Q_9.

A helper adjacent also to a_1, b_1 has at least two row neighbors on even components. These cannot be exposed ports of odd components. Thus

$$m(\{v\}) \leq |N_A(v)| - 2 \leq 10,$$

even when $d_J(v) = 0$. Part (ii) again gives a free destination. This applies to both v, w.

Two nonempty finite destination sets fail to have distinct representatives exactly when they are the same singleton. If one set contains at least two elements, choose an element of the other first and then a different element of that set. If both are singletons, they either differ or coincide. This proves the exhaustive alternatives.

In the first alternative, choose two actual free-port edges. Add the entire owner family and compatible maximum internal matchings as in Step 2. The construction and the counting upper bound give every conclusion of (ii), without its numerical sufficient condition.

In the second alternative, the inequality $2 + m(\{v, w\}) \leq j$ would let (ii) assign distinct free destinations, a contradiction. Since the cardinalities are integers,

$$m(\{v, w\}) \geq j - 1.$$

The total number of row incidences of v, w is

$$|N_A(v)| + |N_A(w)| = 24 - d_J(v) - d_J(w).$$

Four incidences are their contacts with a_1, b_1 on even components. After discarding these, there are at most

$$20 - d_J(v) - d_J(w)$$

incidences that could meet odd components. The owned component $\{z\}$ is counted by both helpers, so its two incidences represent only one component. Therefore

$$m(\{v,w\}) \le 19 - d_J(v) - d_J(w).$$

Additional coincidences or even contacts only reduce this upper bound. Combining it with $m(\{v,w\}) \ge j-1$ yields

$$d_J(v) + d_J(w) \le 20 - j \le 8, \qquad m(\{v,w\}) \ge 11.$$

The singleton application of (ii) gives a matching on $V(A) \cup Q_9 \cup \{v\}$ of size $\nu(A) + |Q_9| + 1$. It remains a matching after adding w. The general counting bound is $\nu(A) + |Q_9| + 2$. Attaining that value would force all owner edges and two distinct free destinations, by the equality characterization already proved. This is impossible in the singleton-collision case. Integer matching sizes therefore give exactly

$$\nu(A) + |Q_9| + 1.$$

No occurrence of this residual case has been assumed.

Step 4. The compulsory seed edge gives the exact payment loss.

Since z is a whole row isolate, all its twelve distinct original neighbors lie in J. Apply (i) to $T = N_F(z)$, using $j \ge 12$. Every maximum matching assigns these exterior vertices to distinct odd components. The seed x has only the odd contact z, so it must use xz. The other eleven exterior vertices consequently use eleven distinct odd components different from $\{z\}$.

The same argument forces xz in every maximum matching of H_T whenever $x \in T$ and $|T| \le j$. Let P_z be as stated. A matching of the old maximum size in $H_T - P_z$ would be an original maximum matching and would contain the deleted edge xz, which is impossible. Hence

$$\nu(H_T - P_z) \le \nu(H_T) - 1.$$

Choose an original maximum matching M_T. Since it contains xz, it contains no other edge incident with z. Thus

$$M_T \setminus \{xz\}$$

avoids all of P_z and survives with size $\nu(H_T) - 1$. This proves equality.

The argument permits three paid star edges, but only xz belongs to the original maximum matching. It uses neither process-root eligibility nor a change of ambient matching graph. □

Acknowledgements.

6. Degree four: saturated frames, source normalization and the exact cost interface

[addressed-partial: authorship placeholders preserved | deferred: author, affiliation, and email to the operator.] [metadata/blocker] The subject classification and keywords await operator input. [addressed-partial: metadata placeholders preserved | deferred: subject classification and keywords to the operator.] [acknowledged: these checks remain pending external verification; no compilation or pagination result is claimed.]

6.1. **Saturated quartic frames and the cost interface.**

We study saturated frames through their terminal components and the original edges joining those components to the frame. The certificate calculus and source results assumed below allow us to pass from canonical frame exchanges to exact matching families, and then to the capacity-resolved quartic cost interface.

6.1.1. *Setup.* We assume, established earlier in this series, the certificate, matching, and source statements recorded below. The uniform quartic source statements are available throughout the stated excess range by the automatic quartic minimality result of Chapter 1. No separate proper-induced or proper-subgraph minimality hypothesis is imposed here. Results from Companion I and the preceding chapters are used by their stated conclusions and are not re-proved.

[cite/blocker] [GAP: Supply the precise locations in Companion I and the preceding chapters for the prerequisites below, including the automatic quartic minimality result. The requested Companion I convention is "[I, Theorem X]"; no theorem numbers or bibliography key for that work were supplied.] [deferred: to reference_verify]

Notation 6.1. All graphs are finite and simple. For a graph G, write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for its vertex set, edge set, edge count, matching number, independence number, and maximum degree. Write $\tau(G)$ for its minimum vertex-cover cardinality. Vertex deletion always means induced vertex deletion; edge deletion retains all vertices. The symbols P_j, C_j, K_j denote the path, cycle, and complete graph on j vertices, respectively, with $j \geq 1$ for paths and complete graphs and $j \geq 3$ for cycles. A whole component means an entire connected component.

All capacities, matching budgets, least totals, cost indices, degrees, cardinalities, component counts, matching sizes, and finite choice domains take values in $\mathbb{Z}_{\geq 0}$. Differences of these quantities initially take values in $\mathbb{Z}$, until a stated bound proves nonnegativity. The closed neighborhood of a vertex set I is denoted by $N_G[I]$. A graph is well-covered when every maximal independent set is maximum. A graph is factor-critical when deletion of any one vertex leaves a perfect matching.

Definition 6.2 (Certificate costs)**.** Let $A_c, B_c \in \mathbb{Z}_{\geq 0}$. The cost $c_{(A_c,B_c)}(G)$ is the least $T \in \mathbb{Z}_{\geq 0}$ such that, for every $x_c, y_c \in \mathbb{Z}_{\geq 0}$ with $x_c + y_c = T$, the edges of G admit a partition into four parts whose corresponding spanning subgraphs have maximum degrees at most A_c, B_c and matching numbers at most x_c, y_c, respectively. For $i \in \mathbb{Z}_{\geq 0}$, put

$$c_i(G) = \max_{\substack{A_c,B_c \in \mathbb{Z}_{\geq 0} \\ A_c + B_c = i}} c_{(A_c,B_c)}(G).$$

These are the generic costs also denoted by c_i^*; in particular, $c_{00} = c_0 = c_0^*$. Put

$$\mathrm{ex}(G) = c_0(G) - \nu(G), \qquad \mathrm{ps}(G) = e(G) - \nu(G) - \sum_{i \geq 1} c_i(G), \qquad \mathrm{gap}(G) = \sum_{i \geq 0} c_i(G) - e(G).$$

The gap is a signed integer. A missing matching allocation for G is a pair $x, y \in \mathbb{Z}_{\geq 0}$ for which the edges cannot be partitioned into two parts of matching numbers at most x, y.

Setup 6.3 (Quartic source parameters)**.** When a statement concerns a source F, assume that F is connected, four-regular, and factor-critical, with

$$\mathrm{gap}(F) \geq 1, \qquad E = \mathrm{ex}(F) \in \mathbb{Z}, \qquad E \geq 5.$$

Put

$$k = \nu(F), \qquad a = \alpha(F), \qquad x = k - a, \qquad \sigma = 2E - 2 - x.$$

The uniformity of

$$b = \nu(F - S) - c_1(F - S)$$

over maximal independent sets S is proved in Theorem 6.24. Once this is established, put

$$m = E - 1 - \sigma + b, \qquad q = m - b = E - 1 - \sigma. \tag{36}$$

In particular, the complement cost interface to be proved is

$$\nu(F - S) = 3E - 2 - \sigma, \qquad c_1(F - S) = \nu(F - S) - b, \qquad c_2(F - S) = 2m - b.$$

Definition 6.4 (Saturated frames)**.** Let H be a four-regular graph, let S be a maximal independent set, and let $m \geq 1$ be an integer. A saturated frame is a pair (S, U), where U is an independent set of exactly m degree-three vertices in $H - S$. Its frame vertex set and terminal are

$$W = S \cup U, \qquad R = H - W.$$

Whenever $H[W]$ is a matching together with isolates, its matching edges are called frame edges. Their endpoints form P_{fr}, and its isolated vertices form Z.

Definition 6.5 (Canonical terminal sets)**.** Let R have components among

$$C_3, C_5, C_7, \quad P_2, P_4, P_6, \quad P_3, C_4, K_1.$$

A canonical set D is obtained independently on whole components by choosing a maximum independent set on each odd cycle, the center of each P_3, either opposite pair on each C_4, and no isolates. On a consecutively numbered even path P_{2s}, with $s \in \{1, 2, 3\}$, choose either

$$\{2, 4, \ldots, 2s-2\} \quad \text{or} \quad \{3, 5, \ldots, 2s-1\}.$$

Both choices are empty when $s = 1$.

Definition 6.6 (Exposed ports)**.** An exposed port of an odd whole component L of a terminal is a vertex $v \in V(L)$ such that

$$\nu(L - v) = \nu(L).$$

The following prerequisites state precisely the consequences of the earlier development used in the proofs.

Setup 6.7 (Certificate calculus)**.** The costs are symmetric in the degree capacities:

$$c_{(A_c,B_c)}(G) = c_{(B_c,A_c)}(G).$$

They are subadditive over vertex-disjoint components. The generic slack is nonnegative, and

$$\nu(G) \leq c_0(G) \leq \tau(G), \qquad c_1(G) \leq \nu(G).$$

Deleting one vertex decreases each of c_1, c_2 by an integer in $\{0, 1\}$. In a zero-slack subcubic graph, all costs of indices at least three vanish. A graph consisting of one edge, or of one triangle, has $c_0 = 1$. For every finite simple graph,

$$c_2(G) = c_{(1,1)}(G).$$

Setup 6.8 (Zero-slack cubic deletion)**.** A zero-slack subcubic graph has a matching covering all its degree-three vertices. Deleting an arbitrary degree-three vertex preserves zero slack and decreases its matching number by one. More generally, deleting an independent set U of degree-three vertices decreases each of ν, c_1, c_2 by $|U|$. If U is inclusion-maximal among independent sets of degree-three vertices, the remaining graph has maximum degree at most two.

Setup 6.9 (Zero-slack terminals)**.** Let $\Delta(R) \leq 2$ and $\mathrm{ps}(R) = 0$. Put

$$q_R = c_2(R), \qquad b_R = \nu(R) - c_1(R).$$

Then R has exactly q_R odd-cycle components, each in $\{C_3, C_5, C_7\}$, and exactly b_R positive even-order path components, each in $\{P_2, P_4, P_6\}$. Every other component is P_3, C_4, or K_1. For each zero-slack even-path component P,

$$c_1(P) = \nu(P) - 1.$$

Setup 6.10 (Uniform quartic source profile)**.** Under Setup 6.3, the graph F is well-covered. For every maximal independent set S, put $G = F - S$. The integer $g = c_2(G)$ is independent of S, and

$$\mathrm{ps}(G) = 0, \qquad \nu(G) = x + E, \qquad c_1(G) = 3x + 2 - E - g, \qquad c_2(G) = g,$$
$$|V(G)| = a + 2x + 1, \qquad c_0(F) = a + \nu(G),$$
$$E - 1 \leq x \leq 2E - 2, \qquad 2x + 2 - 2E \leq g \leq x, \qquad 5E - 5 \leq k \leq 4x.$$

Every independent set of degree-three vertices of G has size at most $E - x - 1 + g$. The automatic quartic minimality result makes this profile applicable at every excess $E \geq 5$ in Setup 6.3.

Setup 6.11 (Missing-allocation bound)**.** Under Setup 6.3, every missing matching allocation $(r_0, s_0) \in \mathbb{Z}^2_{\geq 0}$ with

$$r_0 + s_0 = c_0(F) - 1 = k + E - 1$$

satisfies $r_0 \leq k - 2E + 2$. This is the earlier matching-budget bound with degree parameter 4 and auxiliary integer parameter $E - 2$.

Setup 6.12 (Missing allocations and independence)**.** Let F be well-covered, let $R = F - W$ have maximum degree at most two, and let q_R count its odd-cycle components. Assume that

$$c_0(F) = |W| + \nu(R).$$

If (x_0, y_0) is a missing matching allocation of total $c_0(F) - 1$ and $j = |W| - x_0 \geq q_R + 1$, then $\alpha(R) = \alpha(F)$.

Setup 6.13 (Common exterior from a missing allocation)**.** Under the hypotheses of Prerequisite 6.12, with $j \geq q_R + 1$, let Z_{even} be the union of all even-order whole components of R. There is an independent set

$$I \subseteq V(R) \setminus Z_{\text{even}}, \qquad |I| = \alpha(R - Z_{\text{even}}),$$

such that $V(F) \setminus N_F[I] = Z_{\text{even}}$. The graph $F[Z_{\text{even}}]$ and each of its whole components are well-covered. Every maximal independent set of $F[Z_{\text{even}}]$ extends with this same I to a maximal independent set of F. A terminal P_6 component is excluded under these hypotheses.

Setup 6.14 (Saturated source allocation range)**.** For a saturated frame of a source in Setup 6.3, using the parameters of Theorem 6.24,

$$c_0(F) = |W| + \nu(R), \qquad b \geq 1, \qquad \sigma \geq 1.$$

For every missing matching allocation $x_0 + y_0 = c_0(F) - 1$,

$$|W| - x_0 \geq q + \sigma + b, \qquad q = m - b.$$

Setup 6.15 (Minimum-P_3 decomposition)**.** For a saturated source frame minimizing the number t of terminal P_3 components,

$$t \geq m - 10b.$$

For every orientation of its frame edges, with complement G', there is a union X_0 of whole G'-components such that

$$|X_0| \leq 231b.$$

The graph $G' - X_0$ has components only of types

$$D_6, \text{ diamond}, C_3, C_5, C_7, P_3, C_4, K_1,$$

including at least $m - 41b$ components of type D_6 or diamond and at least $m - 31b$ odd-cycle components. Here D_6 is the tree with two adjacent degree-three centers and two leaves at each center, and a diamond is K_4 with one edge removed. Its surviving C_5, C_7, C_4 components are unchanged whole components of the terminal; the other retained nonterminal components are the local D_6 and diamond packets.

Setup 6.16 (Local root exchange)**.** In a minimum-P_3 saturated source frame, a frame endpoint contacting a whole terminal P_3 center has two other terminal neighbors, both terminal isolates. Orient that endpoint p into the roots and let h be the contacted center. The root exchange $p \mapsto h$ deletes h from the terminal, makes its two leaves isolated, and makes p the center of a P_3 on two old isolates. Every other terminal component is unchanged.

Call a frame edge anchored if both its endpoints contact whole terminal P_3 centers. Among frames minimizing t, choose one maximizing the number u of unanchored frame edges. If pq is

anchored, p contacts h, and z_h is the second frame-set neighbor of h, the exchange replaces the frame edge pq by the anchored edge hz_h. All other frame edges retain their endpoints and anchored status, and both extrema remain attained.

Setup 6.17 (Unanchored-edge bound)**.** For the two successive frame extrema in Prerequisite 6.16,

$$u \le 2b.$$

Setup 6.18 (The excess-five profile)**.** For a source with $E = 5$ and $c_2(F-S) = 5$ for every maximal independent set S,

$$k - a = 6, \qquad \sigma = 2, \qquad \nu(F-S) = 11, \qquad c_1(F-S) = 10.$$

A saturated independent cubic triple exists, and the earlier frame conclusions apply to every such triple.

Setup 6.19 (Source parameters and a conditional cost interface)**.** For the normalized source parameters,

$$q = m - b \ge 1.$$

Writing $H = F - S$, $v = |V(H)|$, and

$$\beta_{\mathrm{prop}} = \nu(H) - b - m, \qquad \beta_{\mathrm{cost}} = \nu(H) - 2m,$$

we have

$$\beta_{\mathrm{cost}} \ge 1, \qquad \beta_{\mathrm{prop}} - \beta_{\mathrm{cost}} = q, \qquad v - \nu(H) + b = a + m.$$

The exact second-cost identity for F is the additional hypothesis

$$c_2(F) = a + 2m - b. \tag{37}$$

Under (37), we also have

$$c_2(F) = v - m - \beta_{\mathrm{cost}}.$$

Saturation alone is not used to assert (37).

[addressed-partial: the exact second-cost identity and its uses are conditional | deferred: a proof of the unrestricted source-cost lower bound, including for saturated sources, to the author.]

Setup 6.20 (Saturated source inequality)**.** Every source in Setup 6.3 with a supplied saturated frame satisfies

$$E \le 2\sigma - b + 1.$$

This is the generic saturated-source conclusion; it requires neither source minimality nor an additional special source clause.

Setup 6.21 (Core matching interface)**.** For a supplied saturated source frame, put

$$H_{\mathrm{core}} = F - Z, \qquad N = a - m, \qquad C = 3E - 2 - \sigma + m.$$

Then

$$N > 0, \qquad c_0(H_{\mathrm{core}}) = C, \qquad \mathrm{gap}(H_{\mathrm{core}}) = 0,$$

and

$$\mathrm{ex}(H_{\mathrm{core}}) = \mathrm{ps}(H_{\mathrm{core}}) = C - \nu(H_{\mathrm{core}}) \ge 0.$$

If $m \le \sigma$, then $\nu(H_{\mathrm{core}}) = C$. In this range, every maximum core matching has exactly one edge from each frame endpoint to an exposed port in a distinct odd whole terminal component, together with a maximum internal matching in every whole terminal component, and no edge within P_{fr}.

Setup 6.22 (Positive core costs)**.** For the same frame,

$$|P_{\mathrm{fr}}| = 2m, \qquad \nu(R) = 3E - 2 - \sigma - m, \qquad c_1(H_{\mathrm{core}}) = C - b, \qquad c_2(H_{\mathrm{core}}) = 3m - b.$$

The induced frame graph is mK_2 together with $a-m$ isolates, and the terminal has the path, cycle, and isolate catalogue of Prerequisite 6.9.

6.1.2. *Canonical saturated frames.* We first prove the frame theorem under a uniform well-covered profile, without imposing the source hypotheses of Setup 6.3.

Theorem 6.23 (Canonical saturated frames)**.** *Let $m, b \in \mathbb{Z}$ satisfy $m \geq 1$ and $0 \leq b \leq m$. Let $H \neq K_5$ be connected, four-regular, and well-covered, and put $a = \alpha(H)$. Assume that, for every maximal independent set S, the graph $G = H - S$ satisfies*

$$\begin{gathered} \operatorname{ps}(G) = 0, \qquad c_1(G) = \nu(G) - b, \qquad c_2(G) = 2m - b, \\ |V(G)| - \nu(G) = a + m - b, \end{gathered} \tag{38}$$

and that every independent set of degree-three vertices of G has size at most m.

For every saturated frame (S, U), put $W = S \cup U$, $R = H - W$, and $r = \nu(R)$. Then

$$H[W] = mK_2 \,\dot\cup\, (a-m)K_1, \qquad \Delta(R) \leq 2, \qquad \operatorname{ps}(R) = 0,$$
$$c_1(R) = r - b, \qquad c_2(R) = m - b, \qquad \nu(R) = \nu(G) - m, \qquad \alpha(R) = a.$$

The terminal has exactly $m - b$ odd cycles from C_3, C_5, C_7, exactly b positive even-order paths from P_2, P_4, P_6, and otherwise only P_3, C_4, and isolates.

For every canonical D from Definition 6.5, the following hold.

(I) The set D is independent, $|D| = r - b$, and

$$R_D = H[W \cup D]$$

is the terminal of another saturated frame. Every D-vertex has degree two in R_D. Each of the m designated components of R_D, namely its $m - b$ odd cycles and its b even paths, contains exactly one edge of $H[W]$. These m edges lie in distinct components.

Consequently, each frame-edge endpoint has at most one D-neighbor, and no degree-two R-vertex has neighbors at endpoints of two different frame edges. Every orientation choosing one endpoint of each frame edge as U gives another saturated frame with the same terminal R.

(II) No frame-edge endpoint has two different internal neighbors on one even-path component of R, where internal excludes the path endpoints.

(III) If a frame-edge endpoint z meets an odd-cycle component O of R, its entire R-neighborhood consists of exactly one vertex of O and both endpoints of one of the b even-path components. In particular, if $b = 0$, every frame-edge endpoint avoids every terminal odd cycle.

No existence of a saturated frame is asserted.

Proof. Fix a saturated frame. We first establish its terminal profile, then reflect it through a canonical set, and finally prove the two contact restrictions.

Step 1. Terminal identities. Since S is maximal independent, it dominates H, so $G = H - S$ is subcubic. Prerequisite 6.8 gives a matching covering all cubic vertices of G. It also says that deletion of any cubic vertex preserves zero slack and decreases the matching number by one.

By Prerequisite 6.7, costs of indices at least three vanish on these zero-slack subcubic graphs. Each of the first two costs decreases by an integer in $\{0, 1\}$ under a vertex deletion. A cubic deletion removes three edges and one matching unit. Subtracting the two zero-slack identities therefore shows that the decreases in c_1, c_2 sum to two. Hence both decreases equal one.

The vertices of U remain cubic while the other vertices of U are deleted, because U is independent. After all m deletions,

$$\begin{gathered} \operatorname{ps}(R) = 0, \qquad r = \nu(G) - m, \qquad c_1(R) = r - b, \qquad c_2(R) = m - b, \\ |V(R)| - \nu(R) = a + m - b. \end{gathered}$$

The upper bound on cubic independence makes U inclusion-maximal among independent cubic sets. Every remaining cubic vertex therefore loses a neighbor, so $\Delta(R) \leq 2$.

Prerequisite 6.9 gives the asserted component list and counts. On a path or an even cycle, independence equals order minus matching number. On an odd cycle, it is one less. Summing over the components gives

$$\alpha(R) = a + m - b - (m - b) = a.$$

Thus every maximum independent set of R is also maximum independent in H.

Each selected cubic vertex has exactly one neighbor in S. Two selected cubics cannot have the same neighbor in S: replacing that common neighbor by the two cubics would enlarge S, contrary to well-coveredness. Since S and U are independent,

$$H[W] = mK_2 \,\dot\cup\, (a - m)K_1.$$

Step 2. Canonical reflection and orientations. A canonical D is independent. It takes $\nu(C)$ vertices on each odd cycle, P_3, and C_4, and $\nu(P) - 1$ vertices on each of the b even paths. Hence $|D| = r - b$.

On each odd cycle O, the graph $O - D$ has exactly one edge. Choose one endpoint x_O of this edge and put

$$J_O = V(O) \setminus (D \cup \{x_O\}).$$

Then J_O is a maximum independent set of O, and x_O has exactly one neighbor in J_O.

On a path P_{2s}, if $D \cap V(P) = \{2, 4, \ldots, 2s - 2\}$, put

$$J_P = \{1, 3, \ldots, 2s - 1\}, \qquad x_P = 2s.$$

For the other canonical choice, put

$$J_P = \{2, 4, \ldots, 2s\}, \qquad x_P = 1.$$

Again J_P is maximum independent, and x_P has exactly one J_P-neighbor.

On each P_3, place its two leaves in J. On each C_4, use the opposite pair complementary to D, and put all terminal isolates in J. These choices together form a maximum independent set J of R, hence of H. The $m - b$ vertices x_O and the b vertices x_P form an independent set U' of size m, since they lie in different components of R. Each has exactly one neighbor in J, so four-regularity makes each cubic in $H - J$. Its terminal is

$$H - (J \cup U') = H[W \cup D] = R_D.$$

The terminal description from Step 1 therefore applies to R_D.

Every D-vertex has two neighbors in R, both outside D, and exactly two neighbors in W. Its degree in R_D is consequently two. The set D is independent, and the only edges inside W are the m frame edges.

An odd cycle of R_D uses an odd number of W-edges, because its number of W-to-D edges is even. An even-path component has no endpoint in D, since every D-vertex has degree two. If that path has $2s$ vertices and contains t W-edges, then

$$2s - 1 = 2|D \cap V(P)| + t.$$

Thus t is odd. Each of the m designated components uses at least one frame edge. Since only m frame edges exist, each designated component uses exactly one, and they exhaust all frame edges.

The degree bound two in R_D now gives at most one D-neighbor at each frame endpoint. Every degree-two vertex of R belongs to some canonical set: this follows from the choices on cycles and P_3, and each internal even-path vertex belongs to one of the two alternating choices. Such a vertex cannot meet endpoints of two different frame edges, since those edges would then belong to the same component of R_D.

For any orientation of the m frame edges, the unchosen endpoints together with the $a-m$ isolated vertices of $H[W]$ form an independent set of size a. It is maximum independent in H. Each chosen endpoint has exactly its partner on that independent side and its three other neighbors in R. The chosen endpoints therefore give another saturated frame with the same terminal.

Step 3. Internal even-path contacts. Suppose that a frame endpoint z has two internal neighbors on a terminal even path P. Orient the frame so that z is selected. Delete the other $m-1$ selected cubics from $H-S$, leaving

$$K = H[V(R) \cup \{z\}].$$

The deletion identities give

$$\nu(K) = r+1, \qquad c_1(K) = \nu(K) - b.$$

Every internal vertex of P can be selected canonically. Part (I), using independent choices on distinct components, shows that z has no degree-two neighbor on any other terminal component. It also gives at most one neighbor in each of the two alternating internal sets of P. Thus the pair is $\{2,3\}$ on P_4, and on P_6 it is one of

$$\{2,3\}, \quad \{2,5\}, \quad \{3,4\}, \quad \{4,5\}.$$

A third internal neighbor is impossible. The third R-neighbor h of z is therefore an even-path endpoint, an isolate, or a P_3-leaf.

If h is an endpoint of $P = P_{2s}$, the component containing z has $2s+1$ vertices and matching number exactly s. The original perfect matching of P attains s, and its order bounds the matching number above by s. Every other component is unchanged, contradicting $\nu(K) = r+1$.

If h is an endpoint of another even path P_{2t}, the joined component has $2s+2t+1$ vertices and matching number exactly $s+t$. The two original perfect path matchings attain this number, and the order bounds it above. This again contradicts $\nu(K) = r+1$.

It remains that h is an isolate or a P_3-leaf. Let C be the component of K containing z. If h is an isolate, take the perfect matching Q consisting of the consecutive path edges $12, 34$, and also 56 for P_6, together with zh. For P_4, the graph $C-Q$, after discarding isolates, is the triangle $z23$, whose zeroth cost is one. For P_6 with contact pair $\{2,3\}$ or $\{4,5\}$, the remainder is a triangle and a disjoint edge. For contact pair $\{2,5\}$ or $\{3,4\}$, the remainder is P_5. A triangle and an edge each have zeroth cost one, while P_5 has a two-vertex cover. Prerequisite 6.7 therefore gives, in every case,

$$c_0(C-Q) \le \nu(C) - 2.$$

If h is a P_3-leaf, write that component as $h-\ell-t$ and add ℓt to Q. This increases $\nu(C)$ by one and leaves one further disjoint edge $h\ell$ in $C-Q$. The same inequality follows. Put Q in the degree-one part of a threshold-one certificate and use a zeroth-cost certificate on $C-Q$. Capacity symmetry covers both threshold-one splits. Hence

$$c_1(C) \le \nu(C) - 2.$$

The other $b-1$ original even paths remain whole components of K. Each has first cost equal to its matching number minus one. To justify applying the terminal statement componentwise, cost subadditivity and matching additivity give

$$\mathrm{ps}(R) \ge \sum_{C'} \mathrm{ps}(C') \ge 0.$$

Since $\mathrm{ps}(R) = 0$, each component has zero slack. Every other unchanged component has first cost at most its matching number, by Prerequisite 6.7. Thus

$$c_1(K) \le \nu(K) - 2 - (b-1) = \nu(K) - b - 1,$$

contrary to the exact value above. The supposed even path requires $b \ge 1$, so this proves part (II) in every applicable case.

Step 4. Odd-cycle contacts. Suppose that a frame endpoint z meets an odd cycle O. Part (I) shows that z has no degree-two terminal neighbor outside O. Its neighbors on O must be pairwise adjacent. On C_5, a nonadjacent pair is maximum independent. On C_7, a nonadjacent pair extends to a maximum independent triple: its two closed neighborhoods cover at most six vertices, so

another vertex can be added. A canonical choice containing either such pair would contradict part (I). Consequently, the neighbors of z on O consist of one vertex, two adjacent vertices, or all three vertices when $O = C_3$. Any remaining terminal neighbors have degree at most one.

Orient z into U and choose a neighbor $t \in V(O)$. In $H - S$, the vertex t has its two cycle neighbors and z as neighbors. Since $H - S$ is subcubic, t is cubic and has no neighbor in $U \setminus \{z\}$. Replacing z by t therefore gives another saturated frame, with terminal $R - t + z$.

If z has exactly two adjacent neighbors on O and a third neighbor outside O, this exchange breaks O into a path and attaches z from one endpoint to a degree-at-most-one vertex in another component. No new cycle is created. This contradicts the unchanged number $m - b$ of odd-cycle components.

If z has exactly one neighbor t on O, the exchange breaks O, so z must create a new odd cycle with its other two terminal neighbors. These must be the endpoints of the same path component. Two P_3-leaves create an even C_4; isolates create no cycle; neighbors in different components create no cycle. Thus the two neighbors are the endpoints of an even path.

Only the possibility that z meets all three vertices of a triangle remains. Let p be its frame-edge partner. Choose a canonical set, denote its selected triangle vertex by t, and denote the other two by v, w. In R_D, the neighbors of z are p, t, and t has degree two. The edge zp lies in a designated component.

If this component is an even path, then z, t are adjacent internal vertices. In the reflected frame, v, w are the endpoints of one frame edge, and both meet z, t. This contradicts part (II) for that frame.

If zp lies on an odd cycle O', then O' contains p, z, t. Each of v, w is a new frame endpoint with two O'-neighbors, namely z, t. The preliminary contact rule just proved forces O' to be a triangle and forces both v, w to meet all its vertices. Therefore

$$H[\{p, z, t, v, w\}] = K_5.$$

Each of these vertices already has degree four in this set. Connectedness gives $H = K_5$, contrary to the hypothesis. This excludes the final alternative and proves part (III). □

6.1.3. *Source normalization.* The next theorem supplies the uniform frame parameters and separates the saturated and unsaturated possibilities.

Theorem 6.24 (Exact source normalization)**.** *Assume Setup 6.3. Then F is well-covered. For every maximal independent set S, put $G = F - S$. The integers*

$$g = c_2(G), \qquad b = \nu(G) - c_1(G), \qquad m = E - 1 - \sigma + b$$

are independent of S, and

$$\begin{aligned} 0 \le b \le \sigma &\le \left\lfloor \frac{3(E-1)}{4} \right\rfloor, \\ m &\ge \left\lceil \frac{E-1}{4} \right\rceil + b \ge 1, \\ 3E - 3 + \sigma \le a &\le 6E - 6 - 3\sigma. \end{aligned} \tag{39}$$

Moreover,

$$\begin{gathered} \mathrm{ps}(G) = 0, \qquad \nu(G) = 3E - 2 - \sigma, \qquad c_1(G) = \nu(G) - b, \qquad c_2(G) = 2m - b, \\ |V(G)| - \nu(G) = a + m - b, \end{gathered} \tag{40}$$

and every independent set of cubic vertices of G has size at most m.

Let h be the largest cubic independence number among these complements, and put $d = m - h$. An attaining pair (S, U), with U an independent cubic h-set, has terminal $R = F - (S \cup U)$ satisfying

$$\Delta(R) \le 2, \qquad \mathrm{ps}(R) = 0, \qquad \nu(R) = 3E - 2 - \sigma - h.$$

It has exactly

$$q_R = 2m - b - h = m - b + d$$

odd cycles from C_3, C_5, C_7*, exactly* b *even paths from* P_2, P_4, P_6*, and otherwise only* P_3, C_4*, and isolates. Its independence number is* $a - d$*. If* $d > 0$*, then*

$$2d \geq \sigma + b.$$

Assume now that $h = m$*, and choose a saturated frame minimizing the number* t *of terminal* P_3 *components. Let* q_5, q_7, u *count its* C_5, C_7, C_4 *components, and write* $2s_1, \ldots, 2s_b$ *for the even-path orders. Then*

$$q_5 + 2q_7 + \sum_{j=1}^{b}(s_j - 1) + t + 2u = E + \sigma - 2b, \tag{41}$$

$$t \geq m - 10b, \qquad q_5 + 2q_7 + \sum_{j=1}^{b}(s_j - 1) + 2u \leq 1 + 2\sigma + 7b. \tag{42}$$

For every orientation of this frame, with complement G'*, there is a union* X *of whole* G'*-components such that*

$$|X| \leq 276\sigma + 5$$

and $G' - X$ *has components only of types*

$$D_6,\ \text{diamond},\ C_3,\ P_3,\ K_1.$$

At least $E-1-41\sigma$ *components are* D_6 *or diamonds, and at least* $E-2-40\sigma$ *are triangles. Negative lower bounds are vacuous.*

No triangle-free, order, excess-ceiling, or additional minimality premise is imposed. The component bound is conditional on $h = m$*; it does not establish saturation or an inequality* $E \leq C_4\sigma + D_4$ *for absolute constants.*

Proof. We first normalize the source profile, then examine a largest cubic independent set, and finally apply the minimum-P_3 decomposition in the saturated case.

Step 1. Uniform parameters. Prerequisite 6.10 applies at every excess in the present range. It gives well-coveredness, uniform g, and all formulas stated there. Subtraction yields

$$b = (x + E) - (3x + 2 - E - g) = g - 2x - 2 + 2E.$$

The bounds on g give

$$0 \leq b \leq 2E - 2 - x = \sigma.$$

Since $5E - 5 \leq k \leq 4x = 8E - 8 - 4\sigma$,

$$4\sigma \leq 3E - 3.$$

Also $x \geq \lceil 5(E-1)/4 \rceil$, so

$$m = x + 1 - E + b \geq \left\lceil \frac{E-1}{4} \right\rceil + b \geq 1.$$

Substituting

$$x = 2E - 2 - \sigma, \qquad g = 2x + 2 - 2E + b$$

gives

$$m = E - 1 - \sigma + b, \qquad g = 2m - b, \qquad \nu(G) = 3E - 2 - \sigma.$$

Furthermore,

$$|V(G)| - \nu(G) = a + x + 1 - E = a + m - b.$$

The earlier cubic-independence bound $E - x - 1 + g$ is exactly m. Finally,

$$a = k - x \geq 3E - 3 + \sigma, \qquad a = k - x \leq 3x = 6E - 6 - 3\sigma.$$

This proves (39) and (40).

Step 2. The unsaturated alternative. There are finitely many maximal independent sets and at least one, so h is attained. Fix an attaining independent cubic set U of size h. It is inclusion-maximal among independent cubic sets. Prerequisites 6.8 and 6.9 give

$$\Delta(R) \le 2, \qquad \mathrm{ps}(R) = 0, \qquad \nu(R) = \nu(G) - h,$$
$$c_1(R) = c_1(G) - h, \qquad c_2(R) = c_2(G) - h.$$

Thus the number of odd cycles is $q_R = 2m-b-h$, and the number of even paths is $\nu(R)-c_1(R) = b$, with the stated component list. Summing independence numbers gives

$$\begin{aligned}\alpha(R) &= |V(R)| - \nu(R) - q_R \\ &= (a+m-b) - (2m-b-h) = a-m+h = a-d.\end{aligned}$$

Suppose $d > 0$. The definition of $c_0(F)$ supplies a missing allocation

$$r_0 + s_0 = c_0(F) - 1 = k + E - 1, \qquad r_0, s_0 \in \mathbb{Z}_{\ge 0}.$$

Prerequisite 6.11 gives $r_0 \le k - 2E + 2$. With $W = S \cup U$,

$$|W| = a + h, \qquad c_0(F) = a + \nu(G) = |W| + \nu(R).$$

Therefore, for $j = |W| - r_0$,

$$j \ge a + h - k + 2E - 2 = h + \sigma.$$

If $2d < \sigma + b$, then $h = m - d$ and $q_R = m - b + d$ give

$$j - q_R \ge \sigma + b - 2d \ge 1.$$

All hypotheses of Prerequisite 6.12 hold: F is well-covered, R is induced with maximum degree at most two, the cost equality and missing allocation hold, and $j \ge q_R + 1$. Its conclusion $\alpha(R) = a$ contradicts $\alpha(R) = a - d < a$. Hence $2d \ge \sigma + b$.

Step 3. The saturated component bound. Assume $h = m$. Factor-criticality gives $|V(F)| = 2k + 1$, and the earlier lower bound gives

$$|V(F)| \ge 10E - 9 > 5.$$

Thus $F \ne K_5$. The alternative cost check is $c_0(K_5) \le \tau(K_5) = 4$ and $\nu(K_5) = 2$, which also excludes excess at least five.

Theorem 6.23 applies with exactly the uniform data just proved. The terminal matching number is

$$r = \nu(G) - m = 2E - 1 - b.$$

It has $m - b$ odd cycles and b even paths. Summing their matching numbers and those of the remaining components gives

$$r = (m-b) + q_5 + 2q_7 + b + \sum_{j=1}^{b}(s_j - 1) + t + 2u.$$

Since $r - m = E + \sigma - 2b$, this is (41).

Prerequisite 6.15 gives $t \ge m - 10b$. Subtracting this from (41) gives

$$q_5 + 2q_7 + \sum_{j=1}^{b}(s_j - 1) + 2u \le r - 2m + 10b = 1 + 2\sigma + 7b.$$

Fix any orientation. Prerequisite 6.15 supplies X_0 with $|X_0| \le 231b$, leaving only the types stated there, at least $m - 41b$ D_6 or diamond components, and at least $m - 31b$ odd cycles. Its surviving C_5, C_7, C_4 components are unchanged whole components of R. Add all of them to X_0. Their total order is at most

$$5q_5 + 7q_7 + 4u \le 5(q_5 + 2q_7 + 2u) \le 5(1 + 2\sigma + 7b).$$

The resulting union X of whole G'-components satisfies

$$|X| \le 266b + 10\sigma + 5 \le 276\sigma + 5.$$

Only D_6, diamonds, triangles, P_3, and isolates remain.

No D_6 or diamond is removed in this last operation, so their number is at least

$$m - 41b = E - 1 - \sigma - 40b \ge E - 1 - 41\sigma.$$

At most

$$q_5 + q_7 \le 1 + 2\sigma + 7b$$

odd cycles are removed. The triangle count is therefore at least

$$m - 31b - (1 + 2\sigma + 7b) = E - 2 - 3\sigma - 37b \ge E - 2 - 40\sigma.$$

These bounds are vacuous when their right sides are negative. The argument neither bounds the number of surviving standard components nor proves saturation when $d > 0$, so it gives no further slope or degree-closure conclusion. □

6.1.4. *Terminal contacts and the common exterior.* We strengthen the contact rule of Theorem 6.23 and then use the source allocation range to remove P_6 from every saturated source terminal.

Theorem 6.25 (One degree-two contact)**.** *Let $m, b \in \mathbb{Z}$ satisfy $m \ge 1$ and $0 \le b \le m$. Let $H \ne K_5$ be connected, four-regular, and well-covered, with $a = \alpha(H)$. Assume that every maximal independent complement $G = H - S$ satisfies*

$$\mathrm{ps}(G) = 0, \qquad c_1(G) = \nu(G) - b, \qquad c_2(G) = 2m - b, \qquad |V(G)| - \nu(G) = a + m - b,$$

and has cubic independence number at most m.

For every saturated frame (S, U), each endpoint of a frame edge has at most one neighbor whose degree in $R = H - (S \cup U)$ is two. In particular, it cannot have two adjacent neighbors in a whole C_4 component of R.

No triangle-free, factor-critical, gap, excess, order, or minimality premise is imposed, and no existence of a saturated frame is asserted.

Proof. The hypotheses are those of Theorem 6.23, so its terminal, canonical, orientation, and path-contact conclusions apply. Every degree-two terminal vertex belongs to some canonical set. If a frame endpoint u had such neighbors in two distinct components, both could be selected in one canonical set, contrary to Theorem 6.23(I).

Thus all its degree-two terminal neighbors lie in one component. An odd cycle allows exactly one by Theorem 6.23(III). An even path allows at most one by Theorem 6.23(II). A P_3 has only one degree-two vertex, and an isolate has none. Only C_4 remains.

Write this cycle as $x_1x_2x_3x_4x_1$. Opposite pairs are canonical choices, so u cannot meet an opposite pair. Three cycle neighbors would contain such a pair. The only remaining possibility is exactly two adjacent neighbors, which we name x_1, x_2.

The endpoint u has exactly three terminal neighbors. Its third neighbor h lies outside the C_4, since a third cycle neighbor would create an opposite pair. It has terminal degree at most one, since otherwise it could be selected canonically together with x_1. Thus h is an isolate, a P_3-leaf, or an endpoint of P_{2s} for $s \in \{1, 2, 3\}$.

Orient the frame so that $u \in U$. In the same graph $G = H - S$, the vertex x_1 has neighbors x_2, x_4, u. Since G is subcubic, these are all its neighbors in G. Hence x_1 is cubic and has no neighbor in $U \setminus \{u\}$. Therefore

$$U' = (U \setminus \{u\}) \cup \{x_1\}$$

is another independent cubic m-set in the same G. Its terminal is

$$R' = R - x_1 + u.$$

All components except the old square and the component containing h are unchanged. The square minus x_1 is the path $x_2 - x_3 - x_4$, and u joins its endpoint x_2 to h.

If h was isolated, the resulting component is the forbidden path $x_4 - x_3 - x_2 - u - h$, of order five. If h was a P_3-leaf, the resulting component is P_7, also forbidden. If h was an endpoint of P_{2s}, the resulting component is P_{2s+4}. For $s = 2, 3$, this is P_8 or P_{10}, outside the terminal list. Thus $s = 1$. Writing the old edge as ht, the new component is

$$x_4 - x_3 - x_2 - u - h - t.$$

In the new frame, the selected root x_1 is a frame-edge endpoint. It meets both x_2 and u, which are internal vertices of this P_6. These are actual old edges x_1x_2 and x_1u. This contradicts Theorem 6.23(II) for the new frame. The last possibility is excluded. □

Theorem 6.26 (Common exterior and exclusion of the six-vertex path)**.** *Assume Setup 6.3, with the parameters of Theorem 6.24, and assume that a saturated frame exists. For any such frame (S, U), put $W = S \cup U$, $R = F - W$, and let Z_{even} be the union of all even-order whole components of R.*

There is one independent set

$$I \subseteq V(R) \setminus Z_{\text{even}}$$

such that

$$|I| = a - \frac{|Z_{\text{even}}|}{2}, \qquad V(F) \setminus N_F[I] = Z_{\text{even}}.$$

Every maximal independent set L of $F[Z_{\text{even}}]$ extends with this same I to a maximum independent set $I \cup L$ of F. Every even terminal component is well-covered, and the terminal components belong to

$$K_1,\ P_2,\ P_3,\ P_4,\ C_3,\ C_4,\ C_5,\ C_7.$$

Thus P_6 is absent.

The conclusion is uniform in excess, source order, and the saturated frame; it does not prove saturation for every source. In particular, it holds for every saturated cubic-triple frame in the profile $E = 5$, $c_2(F - S) = 5$ for all maximal independent S, and such frames exist in that profile.

Proof. Theorem 6.24 and Prerequisite 6.14 give well-coveredness,

$$c_0(F) = |W| + \nu(R),$$

and, for every missing allocation $x_0 + y_0 = c_0(F) - 1$,

$$j = |W| - x_0 \geq q + \sigma + b, \qquad q = m - b.$$

They also give $b, \sigma \geq 1$, so $j \geq q + 1$. A missing allocation exists by the definition of $c_0(F)$. The terminal has maximum degree at most two by Theorem 6.23.

Prerequisite 6.13 therefore gives an independent I outside Z_{even}, with

$$V(F) \setminus N_F[I] = Z_{\text{even}}, \qquad |I| = \alpha(R - Z_{\text{even}}).$$

It also gives well-coveredness of $F[Z_{\text{even}}]$ and of each even component, and extends every maximal independent set there using the same I.

Theorem 6.23 gives $\alpha(R) = a$ and the initial nine-component list

$$K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.$$

Each allowed even-order component has independence number half its order: alternating parity choices attain that size, and a perfect matching bounds it above. Consequently,

$$\alpha(R - Z_{\text{even}}) = \alpha(R) - \alpha(F[Z_{\text{even}}]) = a - \frac{|Z_{\text{even}}|}{2}.$$

Since F is well-covered, every maximal extension $I \cup L$ has size a. Prerequisite 6.13 excludes P_6, leaving precisely the stated eight types.

For the final profile, Prerequisite 6.18 gives

$$k-a=6,\quad \sigma=2,\quad \nu(F-S)=11,\quad c_1(F-S)=10,\quad c_2(F-S)=5.$$

Thus $b=1$ and $m=5-1-2+1=3$. That prerequisite supplies a saturated cubic triple and its universal frame properties. Every such frame therefore satisfies the result just proved. No source exclusion, general saturation assertion, or absolute parameter bound follows here. □

6.1.5. *Private odd-cycle packets.* We next use the two finite frame extrema to propagate anchored edges around fixed reflected components.

Theorem 6.27 (Private packets and independent root orbits)**.** *Assume Setup 6.3, with the normalized parameters, and assume a saturated frame exists. Choose a saturated frame (S,U) minimizing the number t of whole P_3 components of $R=F-(S\cup U)$, and among these maximizing the number u of unanchored frame edges. A frame edge is anchored when both endpoints contact whole terminal P_3 centers.*

Fix any canonical D, put $W=S\cup U$, $R_D=F[W\cup D]$, and set $A=m-u$. For every anchored frame edge e, its R_D-component O_e is an odd cycle of length $2d_e+1$, where $d_e\in\{1,2,3\}$. The set $D\cap V(O_e)$ consists of d_e whole P_3 centers of the original terminal.

The cycles O_e are pairwise disjoint and anticomplete. Actual saturated root rotations confined to each O_e preserve t,u, keep $W\cup D$ fixed with the accompanying canonical choices, and realize every maximum independent set of $F[V(O_e)]$ as the set of terminal vertices in O_e. All these terminal vertices are P_3 centers. The rotations on different cycles can be performed independently.

Every vertex of O_e has exactly two F-neighbors outside O_e. All these boundary vertices, over all anchored cycles, are distinct, lie in the original terminal outside all O_e, and are independent. Writing

$$D_\Sigma=\sum_e d_e,$$

exactly $2D_\Sigma$ boundary vertices are original P_3-leaves, and exactly $2D_\Sigma+2A$ are original terminal isolates. If z counts original terminal isolates, then

$$b\le u\le 2b,\qquad t\ge D_\Sigma,\qquad z\ge 2D_\Sigma+2A,$$

and

$$a=\nu(R)+t+z\ge \nu(R)+3D_\Sigma+2A.$$

Empty sums are allowed. No absolute bound on b,σ, or E is asserted.

Proof. Theorem 6.24 supplies the uniform profile and well-coveredness. The finite nonempty family of saturated frames has the two stated extrema. Prerequisite 6.16 supplies the anchor-preserving root exchanges.

Step 1. A fixed reflected graph. Let an anchored frame edge pq contact a terminal P_3 center h at p, and let z_h be the second W-neighbor of h. Orient p into the roots and exchange $p\mapsto h$. The new terminal is $R-h+p$. The old center h is deleted, its two leaves become isolates, and p becomes a P_3 center on two old isolates. Every other terminal component is unchanged.

Consequently,

$$D'=D-h+p$$

is canonical for the new terminal. It selects the new center p, omits the four affected leaves or isolates, and preserves every other component choice. At the same time $W'=W-p+h$, so

$$W'\cup D'=W\cup D. \tag{43}$$

The induced reflected graph stays fixed. Its only changed frame edge is pq, replaced by hz_h. The new edge is anchored, every other frame edge retains its anchored status, and both extrema remain attained.

Step 2. The orbit is an odd cycle. Fix the component C of this reflected graph containing an anchored frame edge. By Theorem 6.23, it is a designated odd cycle in $\{C_3, C_5, C_7\}$ or even path in $\{P_2, P_4, P_6\}$. It contains exactly one frame edge. All its other edges join W to D, every D-vertex has reflected degree two, and both endpoints of the anchored edge have a D-neighbor that is a terminal P_3 center.

The exchange shifts the unique frame edge two consecutive edges along C: on

$$q - p - h - z_h$$

it replaces pq by hz_h. Either endpoint can be selected, so either direction is available. The resulting edge remains anchored.

If C were a path, choose a direction toward an end. Each shift shortens the distance to that end by two. Since the path is finite and has even order, finitely many shifts make the frame edge an end edge. Its endpoint at the end has no D-neighbor, contradicting anchoredness. Hence

$$C = O_e = C_{2d_e+1}, \qquad d_e \in \{1, 2, 3\}.$$

On this fixed odd cycle, repeated shifts in one direction visit every edge. Adding two modulo an odd integer visits every residue, since an odd integer dividing $2r$ must divide r. At every state, vertices outside the frame edge alternate between D and W. The d_e terminal vertices therefore form a maximum independent set complementary to the alternating W-vertices with their single frame edge.

All original D-vertices of O_e are P_3 centers. To prove this without assuming it, move the frame edge around the fixed cycle. Before an original D-vertex is first exchanged, it has not been deleted. Every earlier exchange modifies only one P_3 and two isolates, leaving other components unchanged. An original D-vertex in another type of component would remain there until first contacted. But every visited frame edge is anchored, and the D-vertex about to be exchanged must be a P_3 center. This excludes that possibility. Every vertex inserted into the terminal is also a P_3 center by the exchange description.

For an odd cycle of order $2d_e+1$, its maximum independent sets are exactly the $2d_e+1$ alternating sets with one gap of two unselected vertices. The d_e selected vertices have d_e positive cyclic gaps of unselected vertices, totaling $d_e + 1$. Exactly one gap therefore has size two, and all others have size one. Its position specifies the independent set, and its edge is the unique frame edge. Since the orbit realizes every frame-edge position, it realizes every maximum independent set.

Every exchange is confined to O_e, so terminal membership outside O_e is unchanged.

Step 3. Independent cycle orbits. Distinct anchored edges lie in distinct designated reflected components by Theorem 6.23. Their cycles are disjoint and anticomplete in F, because the reflected graph is induced. A rotation on one cycle changes neither the other frame edges nor their anchored status, and (43) keeps the entire reflected graph fixed. The allowed rotations on another cycle therefore remain available after any sequence on the first. This proves independent realization of the orbit choices.

Step 4. Private boundary vertices. Every cycle vertex has two neighbors on its induced cycle and exactly two others by four-regularity. If it originally belongs to D, those two neighbors are its original P_3-leaves. If it originally belongs to W, its only possible W-neighbor is its frame-edge mate, already on the cycle; an originally isolated W-vertex has no W-neighbor. Thus all exterior neighbors lie in the original terminal.

No exterior neighbor lies in another O_f, since distinct reflected components are anticomplete. All boundary vertices therefore belong to the fixed set

$$R_{\text{ext}} = V(R) \setminus \bigcup_f V(O_f).$$

Their terminal membership is unchanged throughout all the orbits.

For any cycle vertex v, choose an orbit state in which v is a terminal P_3 center. Its two exterior neighbors are then leaves of that whole P_3, each of terminal degree one. Thus they have no F-edge to any vertex of R_{ext}. In particular, the union of all boundary vertices is independent.

Two distinct nonadjacent vertices v, w on one cycle cannot share an exterior neighbor. Such a pair extends to a maximum independent set: on C_5 the pair is already maximum, and on C_7 the union of its two closed neighborhoods has at most six vertices, leaving a vertex to add. There is no nonadjacent pair on C_3. In the corresponding orbit state, v, w are distinct terminal P_3 centers. A shared exterior neighbor would have terminal degree at least two, although it is a P_3-leaf.

Two adjacent cycle vertices v, w cannot share an exterior neighbor either. Choose the orbit state whose frame edge joins v to its other cycle neighbor. Then w is a terminal P_3 center adjacent to the frame endpoint v. A common exterior neighbor would be a terminal leaf at w. Prerequisite 6.16, however, says that the other two terminal neighbors of v are isolates. The common neighbor would be one of these and would not be isolated, a contradiction.

The two exterior neighbors of any one cycle vertex are distinct because F is simple. Finally, if vertices in distinct cycles shared a boundary vertex, independent orbit choices could make both cycle vertices terminal P_3 centers simultaneously. Their shared vertex stays in R_{ext}, so it would have terminal degree at least two while being a P_3-leaf. Therefore all boundary neighbors over all cycle vertices are distinct.

Step 5. Counts. There are d_e original D-vertices on O_e, all original P_3 centers. Their $2d_e$ distinct boundary neighbors are original P_3-leaves. There are $d_e + 1$ original W-vertices. Their $2d_e + 2$ boundary neighbors have no edges to R_{ext}, and boundary distinctness gives no other neighbor in the union of cycles. Their sole neighbor in that union is their original W-parent, which lies outside R. Hence these boundary vertices are isolates of the original terminal.

Summing gives exactly $2D_\Sigma$ leaves and $2D_\Sigma + 2A$ isolates, with no overlap. Thus

$$t \geq D_\Sigma, \qquad z \geq 2D_\Sigma + 2A.$$

The A anchored cycles consume A of the $m-b$ reflected odd cycles, so $A \leq m-b$, or $u = m-A \geq b$. Prerequisite 6.17 gives $u \leq 2b$.

From the terminal list, each P_3 contributes one more independence unit than matching unit, and each isolate contributes one. Each of $P_2, P_4, P_6, C_3, C_4, C_5, C_7$ has equal independence and matching numbers. Theorem 6.23 therefore gives

$$a = \alpha(R) = \nu(R) + t + z \geq \nu(R) + 3D_\Sigma + 2A.$$

The same calculation includes $A = 0$ and empty sums. □

6.1.6. *The zero-slack core matching interface.* The saturated-source inequality places every supplied frame in the exact range of the earlier core matching interface.

Theorem 6.28 (Exact matching families for retained endpoints)**.** *Assume Setup 6.3, with the normalized parameters, and fix any supplied saturated frame (S, U). Put*

$$W = S \cup U, \qquad R = F - W,$$

and let P_{fr}, Z be the endpoints and isolates of $F[W]$. Set

$$H_{\mathrm{core}} = F - Z, \qquad N = |Z| = a - m, \qquad C = 3E - 2 - \sigma + m, \qquad r = 3E - 2 - \sigma - m.$$

Then the following hold.

(I) We have

$$m \leq \sigma, \qquad N > 0, \qquad |P_{\mathrm{fr}}| = 2m, \qquad \nu(R) = r,$$

and

$$\nu(H_{\mathrm{core}}) = c_0(H_{\mathrm{core}}) = C,$$
$$\mathrm{ex}(H_{\mathrm{core}}) = \mathrm{ps}(H_{\mathrm{core}}) = \mathrm{gap}(H_{\mathrm{core}}) = 0,$$
$$c_1(H_{\mathrm{core}}) = C - b, \qquad c_2(H_{\mathrm{core}}) = 3m - b.$$

Every maximum core matching has exactly one edge from each frame endpoint to an exposed port in a distinct odd whole component of R, together with a maximum internal matching in every whole component of R, and no edge within P_{fr}.

(II) For every $T \subseteq P_{\mathrm{fr}}$, including $T = \varnothing$, put

$$G_T = F[V(R) \cup T].$$

Then

$$\nu(G_T) = r + |T|.$$

Its maximum matchings are exactly the following family. For each vertex of T, choose one actual incident source edge to an exposed port of an odd whole component of R, with all chosen components distinct. In each chosen component, choose a maximum internal matching exposing that port; in each unchosen component, choose any maximum internal matching. Take the union of these internal matchings and the chosen edges from T to R.

At least one such simultaneous choice exists for every T, and every maximum G_T-matching is obtained this way. An arbitrary partial prescription of ports is not asserted to extend.

There is no positive-core-slack alternative under these supplied-frame hypotheses. The theorem does not assert saturation for an arbitrary source, a strict source-cost saving, a quartic-source exclusion, or a Ramsey formula.

Proof. The supplied frame verifies the saturation premise of Prerequisite 6.20. Its other premises are exactly Setup 6.3. Hence

$$E \le 2\sigma - b + 1, \qquad m = E - 1 - \sigma + b \le \sigma.$$

This uses the generic conclusion, with no source minimality assumption or additional special clause.

Apply Prerequisite 6.21 to the same frame. It gives

$$N > 0, \qquad c_0(H_{\mathrm{core}}) = C, \qquad \mathrm{gap}(H_{\mathrm{core}}) = 0,$$

and

$$\mathrm{ex}(H_{\mathrm{core}}) = \mathrm{ps}(H_{\mathrm{core}}) = C - \nu(H_{\mathrm{core}}) \ge 0.$$

Since $m \le \sigma$, its exact-matching clause applies and gives $\nu(H_{\mathrm{core}}) = C$, together with the stated structure of every maximum matching. Thus excess and slack vanish. Prerequisite 6.22 gives $|P_{\mathrm{fr}}| = 2m$, $\nu(R) = r$, the induced frame matching, and the two positive core costs. This proves part (I). The definitions also give

$$C = r + 2m.$$

Fix $T \subseteq P_{\mathrm{fr}}$, and put $j = |T|$. Since the core is finite, its nonempty finite family of matchings has a maximum member M. Part (I) says that M has r internal terminal edges and $2m$ cross edges, one at each frame endpoint. Delete the cross edges at endpoints outside T. The retained $r + j$ edges lie in the actual induced graph G_T and remain a matching. Hence

$$\nu(G_T) \ge r + j.$$

This also supplies one simultaneous port choice for T, without requiring any preassigned port.

For the reverse inequality, let L be any matching of G_T. Write $e_R, e_T, e_{\mathrm{cross}}$ for its numbers of edges inside R, inside T, and between T and R. Its internal edges form a matching of R, so $e_R \le r$. Its edges meeting T use $e_{\mathrm{cross}} + 2e_T$ distinct vertices of T, so

$$e_{\mathrm{cross}} + 2e_T \le j.$$

Therefore

$$|L| = e_R + e_{\text{cross}} + e_T \le r + j - e_T \le r + j.$$

This proves the matching-number identity, including $j = 0$.

If L is maximum, equality forces

$$e_T = 0, \qquad e_{\text{cross}} = j, \qquad e_R = r.$$

Thus every vertex of T uses exactly one cross edge, and the internal matching is maximum in R. Matching number is additive over components: a matching restricts to each component, and componentwise maximum matchings can be united. Hence $e_R = r$ forces a maximum internal matching in every whole component separately.

Every even terminal path or cycle has a perfect matching and leaves no vertex for a cross edge. Every odd terminal path, cycle, or isolate has matching number one half its order minus one half, so a maximum internal matching leaves exactly one vertex exposed. Consequently, the cross edges enter distinct odd whole components. If v is such an endpoint in component K, the internal matching avoids v and has size $\nu(K)$, proving

$$\nu(K - v) = \nu(K).$$

Thus v is an exposed port, and every maximum matching has the asserted form.

Conversely, choose any simultaneous family of actual edges satisfying the stated conditions. Their endpoints in T are distinct, and their terminal endpoints lie in distinct components. For every chosen port v in a component K, the equality $\nu(K - v) = \nu(K)$ supplies a maximum internal matching avoiding v. In each unchosen component, choose a maximum matching. These choices exist because each graph is finite and its matching number is a maximum over a nonempty finite family. Componentwise matchings are vertex-disjoint, and those in chosen components avoid the cross-edge endpoints. Their union with the j cross edges is a matching of size

$$r + j = \nu(G_T).$$

This proves the exact family characterization. The inequality $m \le \sigma$ applies to the entire supplied saturated class, so the positive-slack alternative of the earlier core interface cannot occur here. □

6.1.7. *Canonical port rank.* We identify the exact rank of the canonical port graph and construct two actual matchings. Under the exact second-cost identity (37), the resulting certificate attains $c_2(F)$.

Theorem 6.29 (Port rank and a neutral two-matching certificate)**.** *Assume Setup 6.3, with the normalized parameters, and fix any saturated frame (S, U). Put*

$$H = F - S, \qquad W = S \cup U, \qquad R = F - W,$$

and let P_{fr}, Z be the endpoints and isolates of $F[W]$. Thus $F[W]$ consists of m edges and $a - m$ isolates. Set

$$q = m - b, \qquad \beta_{\text{prop}} = \nu(H) - b - m, \qquad \beta_{\text{cost}} = \nu(H) - 2m = \beta_{\text{prop}} - q.$$

For every canonical D, put

$$J_D = F[W \cup D], \qquad B_{\text{port}} = F[D \cup Z],$$

where B_{port} is regarded as bipartite with parts D, Z. Let $O_1, \dots, O_q$ be all whole odd-cycle components of J_D; these are reflected components and need not be components of R.

(I) The exact rank and left deficiency are

$$|D| = \beta_{\text{prop}}, \qquad \nu(B_{\text{port}}) = \beta_{\text{cost}}, \qquad |D| - \nu(B_{\text{port}}) = q.$$

Every maximum port matching leaves exactly one D-vertex unmatched in each O_i, and no other D-vertex unmatched. Before choosing the matching, one may independently prescribe any

$$d_i \in D \cap V(O_i), \qquad 1 \le i \le q.$$

There is a maximum port matching with unmatched D-set exactly $Q_{\mathrm{omit}} = \{d_1, \dots, d_q\}$. This is a matching of actual original edges, with no assumed Hall assignment.

(II) Fix any such prescriptions, put $D_0 = D \setminus Q_{\mathrm{omit}}$, and choose any maximum port matching L with that unmatched D-set. Let

$$J_0 = F[S \cup U \cup D_0], \qquad M_0 = E(F[W]) \cup L, \qquad M_1 = E(J_0) \setminus M_0.$$

Then M_0, M_1 are edge-disjoint matchings whose union is all of $E(J_0)$. They are sequentially maximum:

$$|M_0| = \nu(J_0) = m + \beta_{\mathrm{cost}}, \qquad |M_1| = \beta_{\mathrm{cost}}.$$

After deleting M_0, the entire remaining edge set is M_1, which is maximum there. All vertices are retained during these edge deletions, and no original edge of J_0 survives both.

(III) If $N_0 = |V(F) \setminus V(J_0)|$, then

$$N_0 = a + 2m - b.$$

Assuming the exact second-cost identity (37) for F, we have $N_0 = c_2(F)$.

For every $A_c, B_c \in \mathbb{Z}_{\ge 0}$ with $A_c + B_c = 2$, and every $x_c, y_c \in \mathbb{Z}_{\ge 0}$ with $x_c + y_c = N_0$, the entire original F has a four-part certificate whose degree-bounded parts have union exactly $M_0 \cup M_1$. Every original edge with an omitted endpoint, including every edge between two omitted vertices, lies in a matching-bounded part.

Assuming (37), the certificate attains the exact second cost and is neutral: no one-unit saving is asserted. Neither saturation existence, a quartic-source exclusion, nor an unconditional Ramsey formula is asserted.

Proof. Theorem 6.24 gives well-coveredness, $|S| = a$, uniform b, m, $m \ge 1$, $0 \le b \le m$, and the full profile of Theorem 6.23 for every maximal independent complement. It also gives

$$a \ge 3E - 3 + \sigma \ge 12,$$

so $F \ne K_5$. Therefore Theorem 6.23 applies to the supplied frame and every stated canonical choice.

Its conclusions give the frame matching, the terminal list, independence of D, and

$$|D| = \nu(R) - b, \qquad \nu(R) = \nu(H) - m.$$

Every D-vertex has degree two in J_D, which is another saturated terminal. Each of its $q = m - b$ odd cycles and b positive even paths contains exactly one frame edge, and these designated components contain all m frame edges. Every other component is P_3, C_4, or an isolate. Prerequisite 6.19 gives

$$q \ge 1, \qquad \beta_{\mathrm{cost}} \ge 1, \qquad |D| = \beta_{\mathrm{prop}}, \qquad \beta_{\mathrm{prop}} - q = \beta_{\mathrm{cost}}.$$

Step 1. Componentwise port rank. The vertices of J_D are $D \cup P_{\mathrm{fr}} \cup Z$. There are no D-to-D edges, and the only edges inside W are the frame edges, all incident with P_{fr}. Thus

$$B_{\mathrm{port}} = J_D - P_{\mathrm{fr}},$$

and each of its edges joins D to Z. Every P_{fr}-vertex lies in the designated component containing its frame edge. A component with no frame edge contains no such vertex.

Consider an odd designated cycle O_i of order $2\ell + 1$. Its unique frame edge joins two adjacent P_{fr}-vertices, while all other edges alternate between D and W. Hence O_i contains ℓ D-vertices, $\ell - 1$ Z-vertices, and its two frame endpoints. Deleting those endpoints leaves an alternating path of order $2\ell - 1$, with both endpoints in D. Its maximum matching has size $\ell - 1$: the number of Z-vertices is an upper bound, attained by alternate edges. It leaves exactly one D-vertex unmatched.

Deleting any prescribed D-vertex from this path splits it into two possibly empty even-order paths. Each has an alternating perfect matching. Their union has size $\ell - 1$ and leaves precisely the prescribed vertex unmatched. For $\ell = 1$, the port graph contribution is one isolated D-vertex, which satisfies the same assertion.

Consider a designated even path of order 2ℓ. Its endpoints lie in W, since no D-vertex has reflected degree one. Deleting the adjacent endpoints of its unique frame edge leaves two possibly empty alternating paths. Each nonempty path starts next to a removed frame endpoint with a D-vertex and ends at an original path endpoint in Z. It has equally many D- and Z-vertices and admits an alternating perfect matching. No D-vertex is lost.

A remaining whole P_3 has its center in D and its two endpoints in Z. It has no frame endpoint, and its leaves cannot lie in D because every D-vertex has degree two. It admits a matching covering its single D-vertex. Every remaining C_4 alternates between D and Z and has a perfect matching. Every isolated reflected vertex lies in Z.

These exhaust the components. Matching number adds over vertex-disjoint components by restriction and union of componentwise maximum matchings. Exactly the q odd-cycle-derived paths lose one D-vertex each, while every other D-vertex can be covered. Hence

$$\nu(B_{\mathrm{port}}) = |D| - q = \beta_{\mathrm{cost}}.$$

Every maximum matching must be maximum on every component, since otherwise its restriction on one component could be enlarged. Thus every maximum matching has exactly the unmatched-D pattern asserted. The path construction independently realizes each prescribed vertex, proving part (I), including the order of its quantifiers.

Step 2. Two sequential maximum matchings. The chosen port matching has

$$|L| = |D_0| = \beta_{\mathrm{cost}}$$

and uses only edges between D_0 and Z. All frame endpoints lie in the disjoint set P_{fr}. Therefore M_0 is a matching, with

$$|M_0| = m + \beta_{\mathrm{cost}}.$$

We check that M_0 covers every degree-two vertex of J_0. Deleting the prescribed D-vertex from a reflected odd cycle of order $2\ell + 1$ leaves an even path of order 2ℓ. On it, M_0 contains one frame edge and $\ell - 1$ port edges, so it is perfect. On each designated even path, L is perfect on the two balanced paths left after removing the frame endpoints. Adding the frame edge makes M_0 perfect on the entire path. It is perfect on every reflected C_4, and it covers the center of every remaining P_3. Isolates have no degree-two vertex.

Deleting one vertex from each separate odd cycle changes no other component, so this list includes every component of J_0. As $\Delta(J_0) \le 2$, deleting M_0 leaves maximum degree at most one. Thus M_1 is a matching.

Every D_0-vertex retains its two original reflected neighbors in W. Since D_0 is independent and $F[W]$ has exactly m edges,

$$e(J_0) = m + 2|D_0|.$$

Therefore

$$|M_1| = e(J_0) - |M_0| = |D_0| = \beta_{\mathrm{cost}}.$$

The set $U \cup D_0$ covers every edge of J_0: each frame edge meets U, and every other edge meets D_0. This cover has size $m + \beta_{\mathrm{cost}}$, so no matching is larger than M_0. After its deletion, M_1 is the entire remaining graph and is itself maximum. This proves part (II).

Step 3. The full-source certificate. Write $v = |V(H)|$. The retained sets S, U, D_0 are disjoint and have sizes $a, m, \beta_{\mathrm{cost}}$, while $|V(F)| = a + v$. Thus

$$\begin{aligned} N_0 &= v - m - \beta_{\mathrm{cost}} \\ &= v - \nu(H) + m \\ &= a + 2m - b \end{aligned}$$

by the definition of β_{cost} and the numerical identity $v - \nu(H) + b = a + m$ in Prerequisite 6.19. Under (37), this gives $N_0 = c_2(F)$.

Fix capacities $A_c, B_c \in \mathbb{Z}_{\geq 0}$ with $A_c + B_c = 2$. Put the first A_c members of the ordered pair M_0, M_1 into the first degree-bounded part, and the remaining B_c into the second. A union of A_c matchings has maximum degree at most A_c, and the other union has maximum degree at most B_c. Their union is exactly $E(J_0)$. This includes the splits with a zero capacity.

Order the N_0 omitted vertices. Assign every original edge with an omitted endpoint to its earliest omitted endpoint. The resulting N_0 labeled edge sets are stars, possibly empty. They partition every edge outside $E(J_0)$, including edges with two omitted endpoints. Each star has matching number at most one.

For any $x_c, y_c \in \mathbb{Z}_{\geq 0}$ with $x_c + y_c = N_0$, place any x_c of these stars in the first matching-bounded part and the remaining y_c in the second. A matching uses at most one edge from each assigned star, so the required bounds hold even when different stars share vertices. Every original edge is assigned exactly once. This proves the universal certificate at total $N_0 = a + 2m - b$. Under (37), this total equals $c_2(F)$, with no strict decrease. □

6.1.8. *Capacity resolution and the exact restoration debit.* The final theorem distinguishes the two degree-capacity splits and computes the exact restoration debit after arbitrary omissions and arbitrary changes of the two matchings.

Theorem 6.30 (Capacity-resolved cost and canonical restoration)**.** *Assume Setup 6.3, with the normalized parameters, and put* $q = m - b = E - 1 - \sigma$.

(I) For every maximal independent set S*, put* $H = F - S$ *and*

$$V_3 = \{v \in V(H) \mid d_H(v) = 3\}.$$

Let U_3 *be any inclusion-maximal independent set of* $H[V_3]$*, and put* $u = |U_3|$*. Such a set exists,* $u \leq m$*, and*

$$\Delta(F - (S \cup U_3)) \leq 2.$$

Moreover,

$$c_{(2,0)}(F) = c_{(0,2)}(F) \leq a + u \leq a + m.$$

The generic identity $c_{(1,1)}(F) = c_2(F)$ *holds. Assuming the exact second-cost identity* (37) *for* F*, we also have*

$$c_{(1,1)}(F) = c_2(F) = a + 2m - b = a + m + q,$$

and

$$c_{(1,1)}(F) - c_{(2,0)}(F) \geq q + (m - u) \geq q \geq 1.$$

No equality for $c_{(2,0)}(F)$*, and no saturation-existence assertion, is made.*

(II) Fix any saturated frame (S, U)*, put* $W = S \cup U$*,* $R = F - W$*, and choose any canonical* D*. Then*

$$J_D = F[W \cup D]$$

has maximum degree at most two and exactly q *whole odd-cycle components, and*

$$N_D = |V(F) \setminus V(J_D)| = a + m.$$

For any $X \subseteq V(J_D)$, put $t = |X|$, let $q_{\mathrm{hit}}(X)$ count the whole odd cycles of J_D meeting X, and set

$$q_{\mathrm{keep}} = q - q_{\mathrm{hit}}(X), \qquad K_X = J_D - X, \qquad N_X = N_D + t.$$

For any two edge-disjoint matchings M_0, M_1 in K_X, let Q_X have all vertices of K_X and edge set

$$E(Q_X) = E(K_X) \setminus (M_0 \cup M_1).$$

There is no maximum-matching or prescribed-edge assumption. Then

$$N_X + \nu(Q_X) \geq a + m + q + t - q_{\mathrm{hit}}(X) \geq a + m + q.$$

Assuming the exact second-cost identity (37) for F, this gives

$$N_X + \nu(Q_X) \geq c_2(F) + t - q_{\mathrm{hit}}(X) \geq c_2(F). \tag{44}$$

Both finite minima over all such pairs exist and satisfy

$$\begin{aligned} \min_{M_0,M_1} \big[N_X + \nu(Q_X)\big] &= \min_{M_0,M_1} c_0\big(Q_X \,\dot\cup\, N_X K_2\big) \\ &= a + m + q + t - q_{\mathrm{hit}}(X). \end{aligned} \tag{45}$$

Under (37), their common value is $c_2(F)+t-q_{\mathrm{hit}}(X)$. The added K_2 components use new vertices, mutually disjoint and disjoint from Q_X. The second minimum is the independent-unit restoration proxy, charging one separate matching-capacity unit per omitted source vertex. No equality with an optimum over all actual-source edge partitions is asserted.

(III) For every X, one attaining pair has residual graph Q_X consisting of exactly q_{keep} disjoint edges and isolates. At total

$$T_X = N_X + q_{\mathrm{keep}} = a + m + q + t - q_{\mathrm{hit}}(X),$$

this same pair supplies a four-part certificate for the entire original F, for every capacity split $A_c + B_c = 2$ and every matching-budget split $x_c + y_c = T_X$, with all four parameters in $\mathbb{Z}_{\geq 0}$. The degree-bounded parts have union exactly $M_0 \cup M_1$. Every other original edge belongs to a matching-bounded part, including every edge with two omitted endpoints.

Assuming (37), we have $T_X = c_2(F)+t-q_{\mathrm{hit}}(X)$. Under this hypothesis, the two minima in (45) equal $c_2(F)$ if and only if X contains exactly one vertex from each odd J_D-cycle it meets and no other vertex.

Under (37), neither arbitrary matching changes nor additional omissions give a strict saving within a canonical reflected terminal under this restoration debit. The statement applies separately to every supplied saturated frame and every canonical choice. It asserts neither their existence nor source exclusion or a Ramsey formula.

Proof. We first separate the capacity splits without assuming saturation. We then fix an arbitrary supplied canonical reflection and calculate both restoration minima.

Step 1. The unsaturated capacity bound. Theorem 6.24 and Prerequisite 6.19 give well-coveredness, $|S| = a$, uniform b, m, and $q \geq 1$. They also give the bound m on every independent set of degree-three vertices in $H = F - S$.

Since S is maximal independent, each vertex outside S has an S-neighbor. Four-regularity therefore gives $\Delta(H) \leq 3$. The finite graph $H[V_3]$ has an inclusion-maximal independent set, obtained by successively adjoining eligible vertices until none remains. Every such U_3 has $u \leq m$. Each vertex of $V_3 \setminus U_3$ has a neighbor in U_3, by inclusion-maximality. Thus deleting U_3 lowers the degree of every surviving cubic vertex, while originally lower-degree vertices remain of degree at most two. Hence

$$\Delta(F - (S \cup U_3)) \leq 2.$$

Order the $a+u$ vertices of $S \cup U_3$. Assign every original edge having an endpoint there to its earliest such endpoint. These labeled edge sets are stars, possibly empty, and partition all edges outside the retained induced graph, including edges with two omitted endpoints.

For any $x_c, y_c \in \mathbb{Z}_{\geq 0}$ with $x_c + y_c = a + u$, put any x_c stars in the first matching-bounded part and the remaining stars in the second. A matching in a union of j of these edge-disjoint stars has at most j edges, since its restriction to each star has at most one edge. This remains true if the stars share vertices. Put all retained edges in the degree-two part and leave the degree-zero part empty. This gives

$$c_{(2,0)}(F) \leq a + u.$$

Reversing the degree parts gives the same bound for $c_{(0,2)}$, and capacity symmetry gives equality of the two costs.

Prerequisite 6.7 gives $c_2 = c_{(1,1)}$. Assume now that F satisfies the exact second-cost identity (37). Then

$$c_{(1,1)}(F) = a + 2m - b.$$

Subtracting the preceding upper bound yields

$$c_{(1,1)}(F) - c_{(2,0)}(F) \geq 2m - b - u = q + (m - u) \geq q \geq 1.$$

This proves part (I) under its stated conditions, without saturation.

Step 2. The canonical retained graph. Now fix the stated saturated frame. We use (37) below only for comparisons with $c_2(F)$. Theorem 6.24 gives, for every maximal independent complement,

$$\mathrm{ps}(H) = 0, \quad c_1(H) = \nu(H) - b, \quad c_2(H) = 2m - b, \quad |V(H)| - \nu(H) = a + m - b,$$

and cubic independence at most m. It also gives $m \geq 1$, $0 \leq b \leq m$, and

$$a \geq 3E - 3 + \sigma \geq 12.$$

Thus $F \neq K_5$. Connectedness, four-regularity, and well-coveredness hold, so all premises of Theorem 6.23 are verified.

For every canonical D, that theorem says J_D is another saturated terminal, has maximum degree at most two, and has exactly $q = m - b$ whole odd cycles. It also gives

$$\nu(R) = \nu(H) - m, \qquad |D| = \nu(R) - b = \nu(H) - m - b.$$

Writing $v = |V(H)|$, we have $v - \nu(H) + b = a + m$. Since the retained sets S, U, D are disjoint,

$$N_D = v - m - |D| = v - \nu(H) + b = a + m.$$

Step 3. A lower bound for arbitrary matchings. Fix $X \subseteq V(J_D)$. A connected finite graph of maximum degree at most two is an isolate, a path, or a cycle. If it has a degree-one vertex, following successive unused incident edges cannot branch and ends at the other endpoint; connectedness leaves no additional vertices. If every vertex of a nontrivial component has degree two, following edges closes a cycle, and connectedness together with degree two makes it the entire component.

Deleting vertices from a path leaves paths and isolates. Deleting at least one vertex from a cycle also leaves paths and isolates, while an untouched cycle stays unchanged. Hence K_X has exactly

$$q_{\mathrm{keep}} = q - q_{\mathrm{hit}}(X)$$

whole odd cycles, namely the original odd cycles disjoint from X.

Take any edge-disjoint matchings M_0, M_1 in K_X. A surviving odd cycle of order $2\ell+1$ contributes at most ℓ edges to each matching, because those edges have disjoint endpoints. Their union therefore covers at most 2ℓ of its $2\ell + 1$ edges. At least one edge remains in Q_X.

Choose one residual edge from each surviving odd cycle. These edges belong to distinct whole components, so they form a matching of size q_{keep}. Consequently,

$$\nu(Q_X) \geq q_{\mathrm{keep}},$$

and

$$\begin{aligned} N_X + \nu(Q_X) &\geq a + m + t + q - q_{\text{hit}}(X) \\ &= a + m + q + t - q_{\text{hit}}(X). \end{aligned}$$

Every hit cycle contains a distinct vertex of X, so $q_{\text{hit}}(X) \leq t$. Hence $N_X + \nu(Q_X) \geq a + m + q$. Assuming (37), the substitution $a + m + q = c_2(F)$ proves (44) for arbitrary pairs.

Step 4. Attainment of both minima. Choose one arbitrary edge from each of the q_{keep} surviving odd cycles, and let L_{res} be the set of these edges. Temporarily delete them. The remaining graph has only paths, even cycles, and isolates.

Label successive edges of each path alternately $0, 1$. On each even cycle, apply the same labeling cyclically; its even length makes this consistent. Each label class is a matching, and unions across distinct components remain matchings. Let these two edge-disjoint matchings be M_0, M_1. They cover every edge outside L_{res}, so Q_X consists exactly of the q_{keep} residual edges and isolates. Thus the first minimum equals the stated bound. The optimization domain is nonempty, since the empty pair is allowed, and finite, since K_X is finite.

For the second minimum, Prerequisite 6.7 gives $c_0(G) \geq \nu(G)$. Matching additivity on vertex-disjoint components yields, for every pair,

$$\begin{aligned} c_0(Q_X \mathbin{\dot\cup} N_X K_2) &\geq \nu(Q_X) + N_X \\ &\geq a + m + q + t - q_{\text{hit}}(X). \end{aligned}$$

For the attaining pair, the auxiliary graph is a matching of size $N_X + q_{\text{keep}}$, together with isolates. Its zeroth cost equals that size. The matching lower bound applies, and every budget split of that total is certified by assigning the corresponding numbers of disjoint edges to the two matching-bounded parts. Therefore the second minimum attains the same bound. Its finite domain also ensures existence. This proves (45). Under (37), the common value is $c_2(F) + t - q_{\text{hit}}(X)$.

Step 5. A certificate for every split. Use the attaining pair just constructed. It does not depend on a capacity split or a matching-budget split. Fix $A_c, B_c \in \mathbb{Z}_{\geq 0}$ with $A_c + B_c = 2$. Put the first A_c members of the ordered pair M_0, M_1 in the first degree-bounded part, and the remaining B_c in the second. Their maximum degrees are at most A_c, B_c, respectively, including the cases with a zero capacity.

Order all N_X omitted source vertices. Partition the original edges with omitted endpoints into their earliest-endpoint stars, as in Step 1. These N_X labeled stars and the q_{keep} separate residual edges form

$$T_X = N_X + q_{\text{keep}}$$

edge-disjoint labeled pieces. Each has matching number at most one, and together they partition every original edge outside $M_0 \cup M_1$.

For any $x_c, y_c \in \mathbb{Z}_{\geq 0}$ with $x_c + y_c = T_X$, assign any x_c pieces to the first matching-bounded part and the rest to the second. Restricting a matching to the pieces bounds its size by the number of assigned pieces. This accounts for every original edge exactly once, including edges between two omitted vertices. It proves the full-source certificate in part (III).

Finally,

$$t - q_{\text{hit}}(X)$$

is the number of X-vertices outside the hit odd cycles, plus

$$\sum_{\substack{O \text{ an odd } J_D\text{-cycle} \\ X \cap V(O) \neq \varnothing}} \big(|X \cap V(O)| - 1\big).$$

Every term is a nonnegative integer. Their sum is zero precisely when X contains one vertex from each odd cycle it meets and no other vertex. By (45), this is precisely when the minima equal $a + m + q$. Under (37), it is precisely when they equal $c_2(F)$. This proves the conditional equality characterization.

All universal inequalities were proved for arbitrary supplied S, U, D, X and arbitrary matching pairs under the stated hypotheses. Changing the frame or the canonical choices therefore does not improve the lower bound under the stated debit. No existence assertion is made. The comparisons with $c_2(F)$ assume (37); the restoration argument does not prove that identity or an unrestricted source-cost lower bound. □

7. Degree four: parent partitions and odd-cycle contacts of saturated frames

7.1. **Parent partitions and odd-cycle contacts.** This chapter continues the analysis of saturated frames in degree four. We retain the earlier source normalization and canonical reflection results and study the parents of whole odd terminal cycles. The restrictions proved here are inputs to the subsequent exclusions of saturated pentagons and saturated heptagons.

7.1.1. *Setup.* We assume, established earlier in this series, the certificate conventions of Companion I, the universal structure of Chapter 1, and the saturated-frame, source-normalization, exact-cost-interface, and movable-selector results of Chapter 7. The statements used below are recorded explicitly as assumptions. They are not re-proved in this chapter.

[cite/blocker] [GAP: precise reference locations in Companion I and the earlier chapters have not been supplied. In particular, [I, Theorem X] remains an unresolved reference placeholder.]

[math/blocker] [GAP: the definitions of the inherited quantities $\mathrm{gap}(F)$, $\mathrm{ex}(F)$, and $c_1(H)$, and the full generic certificate definitions, have not been supplied. Their notation and uses are retained below; their definitions must be supplied from the earlier development.]

Notation 7.1 (Graph and integer conventions)**.** All graphs are finite and simple. We write $\alpha(H)$ and $\nu(H)$ for the independence number and matching number of a graph H. For a vertex set X, the graph $F - X$ is the induced graph on $V(F) \setminus X$. A whole path or cycle is a connected component of the terminal under discussion. The symbols P_k and C_k denote a path and a cycle on k vertices, respectively, with integer k in their indicated ranges.

All degree capacities, matching budgets, cost indices, least universal totals, component counts, incidence counts, and finite optimization parameters have domains in $\mathbb{Z}_{\geq 0}$. Quantities initially introduced by differences are integers; their nonnegativity is supplied by the assumed results or proved below. All probabilities are real numbers on explicitly finite probability spaces.

Setup 7.2 (Source normalization and saturated frames)**.** Let F be connected, four-regular, and factor-critical. Assume that

$$\mathrm{gap}(F) \geq 1, \qquad E = \mathrm{ex}(F) \in \mathbb{Z}_{\geq 5}.$$

The earlier source normalization gives well-coveredness. Put

$$a = \alpha(F), \qquad \sigma = 2E - 2 - \big(\nu(F) - a\big), \qquad b = \nu(F - S) - c_1(F - S), \qquad m = E - 1 - \sigma + b.$$

Here S ranges over maximal independent sets, all of which have size a, and b is independent of this choice. The quantities a, b, m are nonnegative integers, and σ is an integer.

Assume that a saturated frame exists, and fix any such frame (S, U). Thus S is maximal independent and U is an independent m-set of vertices of degree three in $F - S$. Saturation means that the global maximum size of such an independent cubic set, over the permitted maximal independent sets, equals m. In particular, every saturated frame attains this maximum.

Put

$$W = S \cup U, \qquad R = F - W.$$

The earlier saturated-frame structure gives

$$F[W] = mK_2 \,\dot\cup\, (a - m)K_1.$$

Let P be the set of endpoints of these m edges, and let Z be the set of isolates of $F[W]$. The two endpoints of a frame edge are called mates. A parent of $x \in V(R)$ is an F-neighbor of x in W. Every degree-two vertex of R has exactly two parents.

Under the same graph hypotheses, $E = 6$ guarantees the existence of a saturated frame.

Definition 7.3 (Canonical choices)**.** A canonical choice $D \subseteq V(R)$ makes the following selections, independently on the whole components of R:

(I) a maximum independent set on each odd cycle;

(II) the center of each P_3;

(III) an opposite pair on each C_4;

(IV) one internal vertex on each P_4;

(V) no vertex on a P_2 or an isolate.

The canonical state space is the finite product of these component choices. A legal single-component change replaces one permitted component choice while fixing all other choices.

Setup 7.4 (Canonical reflections and terminal components)**.** The normalization in Setup 7.2 supplies the uniform complement profile required by the earlier canonical reflection theorem. Every saturated terminal has component list

$$K_1,\ P_2,\ P_3,\ P_4,\ C_3,\ C_4,\ C_5,\ C_7. \tag{46}$$

In particular, every such terminal has maximum degree at most two, and P_6 is absent.

For every canonical choice D, the induced graph

$$R_D = F[W \cup D]$$

is the terminal of an actual saturated frame of the same graph F. Every vertex of D has degree two in R_D. Every degree-two vertex of R occurs in at least one canonical choice, and vertices of R of degree at most one occur in none.

Put $q = m - b$. There are exactly q whole odd-cycle components in R and in every R_D. Each old frame edge lies on its own designated component of R_D, which is either an odd cycle or a positive even-order path. Every designated component contains exactly one old edge of $F[W]$. For an odd cycle, the endpoints of its unique residual edge after a canonical selection form a frame edge in the reflected saturated frame.

Setup 7.5 (Earlier parent and defect restrictions)**.** The following restrictions hold in every saturated frame of Setup 7.2.

(I) A frame-edge endpoint has at most one neighbor of degree two in the terminal. When present, this neighbor is its child.

(II) There is at most one edge from P to any whole odd-cycle component of R. At such a contact, the other parent of the contacted cycle vertex belongs to Z.

(III) If two vertices are adjacent on a whole pentagon and one of them has both parents in Z, their parent sets are disjoint.

(IV) A frame-edge endpoint cannot be adjacent to both the center and a leaf of the same whole P_3.

(V) If A counts the frame edges whose two endpoints have whole P_3-center children, then

$$A - m + 2b \geq 0.$$

Setup 7.6 (Earlier root exchanges and the port rule)**.** The movable-selector result permits the relevant frame endpoint to be oriented into the cubic root set before each of the following exchanges. These statements apply to every saturated frame, including every canonical reflection.

(I) If a frame endpoint contacts a whole odd cycle, its other two terminal neighbors are the endpoints of a distinct whole positive even-order path. For a contact with a whole C_7, the permitted root exchange at the contacted cycle vertex opens that C_7 into a whole P_6 and closes the distinct even path into an odd cycle. The resulting frame is saturated.

(II) We use the following specialization of the earlier odd-component exposed-port rule. Suppose a frame endpoint has terminal neighborhood $\{z, z', t\}$, where z, z' are the leaves of one whole P_3, and t has degree two and lies on no odd-cycle component. Then t is the center of a different whole P_3. In the port terminology, the exposed ports of a P_3 are its leaves; its center is not exposed, and an even-order component contributes no odd-component exposed port.

(III) Suppose a frame endpoint v has terminal neighborhood $\{z, z', t\}$, where t is the center of a whole P_3 and z, z' are the leaves of a different whole P_3, centered at u. After orienting v into the cubic root set, replacing v by t is an actual saturated root exchange. It changes the terminal by removing t and inserting v. The former leaves of t become isolates, and the other P_3 becomes the whole square $u - z - v - z' - u$. This is the earlier exchange with the two leaves belonging to a different P_3.

7.1.2. *Odd-cycle contacts and the binary defect.* In this section, we prove Theorem 7.8 by comparing canonical reflections that differ on one old component.

Definition 7.7 (Frame-edge types)**.** A frame edge is anchored if both endpoints have whole P_3-center children. A binary component is a whole P_4 or C_4. Its two canonical classes are the two internal singletons of a P_4, or the two opposite pairs of a C_4.

An unanchored edge is aligned if both endpoints have children and all its noncenter children lie in one canonical class of one binary component. It is cross if its two children lie in distinct binary components. All other unanchored edges form the zero class.

Theorem 7.8 (Odd-cycle contacts and the exact binary defect)**.** *Use Setup 7.2, with an arbitrary saturated frame (S, U). Let A be the number of anchored edges, and put*

$$q = m - b, \qquad u = m - A, \qquad r = A - m + 2b = 2b - u.$$

Let $N_{\mathrm{half}}, N_{\mathrm{cross}}, N_{\mathrm{zero}}$ count the aligned, cross, and remaining unanchored edges, respectively. Then the following statements hold.

(I) If px is an edge from P to a whole odd-cycle component O of R, the mate p' of p has no degree-two neighbor in R. If z is the other parent of x, then, for every canonical D containing x, the component of x in R_D is the whole path

$$p' - p - x - z.$$

(II) Choose the canonical component choices independently and uniformly. The probabilities that an aligned, cross, or remaining unanchored edge lies on an odd-cycle component of R_D are, respectively,

$$\frac{1}{2}, \qquad \frac{1}{4}, \qquad 0.$$

Moreover,

$$r = N_{\mathrm{zero}} + \frac{N_{\mathrm{cross}}}{2}. \tag{47}$$

Consequently, N_{cross} is even and

$$N_{\mathrm{zero}} + N_{\mathrm{cross}} \le 2r.$$

If γ_{odd} counts the actual edges from P to old whole odd-cycle components, then

$$\gamma_{\mathrm{odd}} \le N_{\mathrm{zero}} \le r.$$

Every asserted count is a nonnegative integer.

Proof. Source normalization permits the use of Setup 7.4 for every saturated frame under consideration. Thus $F[W] = mK_2 \dot{\cup} (a-m)K_1$, and the original terminal and every canonical reflection have exactly q whole odd cycles. The earlier defect estimate in Setup 7.5 gives $r \geq 0$. The exact decomposition below also proves this nonnegativity.

Step 1. We establish an odd-count identity for every canonical choice. For an old frame edge e, put

$$I_e(D) = \begin{cases} 1, & \text{if both endpoints of } e \text{ have a neighbor in } D, \\ 0, & \text{otherwise.} \end{cases}$$

By Setup 7.4, e lies on its own designated odd-cycle or positive even-path component of R_D. On an odd cycle, both endpoints have their old mate and a D-neighbor, so $I_e(D) = 1$.

The only permitted designated even paths are P_2 and P_4, by (46). On a P_2, neither endpoint of e has a D-neighbor. On a P_4, the old edge cannot be the middle edge. If it were, both ends of that P_4 would belong to D, whereas every D-vertex has degree two in R_D. Hence the old edge is an end edge of the P_4, and one endpoint has degree one. Thus $I_e(D) = 0$ on either permitted even path.

It follows that $I_e(D) = 1$ exactly when e belongs to a reflected odd cycle. Since every reflected odd cycle contains exactly one old frame edge,

$$\sum_{e \in E(F[W])} I_e(D) = q \tag{48}$$

for every canonical D. This proves invariance under every legal single-component change, including odd-cycle changes and both choices on each binary component.

Step 2. We prove the childless-mate assertion and identify the whole reflected path. Suppose px contacts an old whole odd cycle O. By Setup 7.5, this is the unique P-incidence of O, and the other parent z of x belongs to Z. Let p' be the mate of p.

Suppose that p' has a child y. The vertex y lies outside O, since otherwise $p'y$ would be a second edge from P to O. Every degree-two terminal vertex belongs to some canonical choice. Choose the option containing y on its component, and fix arbitrary choices on all other components except O.

Write $O = C_{2s+1}$, where $s \in \{1,2,3\}$ is an integer, and label $x = 0$ cyclically. The maximum independent sets

$$\{0, 2, \ldots, 2s-2\} \quad \text{and} \quad \{1, 3, \ldots, 2s-1\}$$

contain and omit x, respectively. Complete them using the same fixed choices outside O, obtaining D_{in} and D_{out}. Since p has x as its only child,

$$I_{pp'}(D_{\text{in}}) = 1, \qquad I_{pp'}(D_{\text{out}}) = 0.$$

No other frame endpoint has a neighbor in O, by the unique-incidence restriction. Every other summand in (48) is therefore unchanged. The two sums would differ by one, contrary to their common value q. Thus p' has no child.

For every canonical D containing x, the four distinct vertices p', p, x, z form the reflected path $p' - p - x - z$. The vertex p' has reflected degree one, since it has no child. The vertices p and x have reflected degree two. The designated component of pp' is therefore a path containing these four vertices. Its permitted even types are P_2 and P_3-free P_4, namely P_2 and P_4. It must be the whole P_4 displayed in (I).

Step 3. We determine the reflection probabilities of all unanchored edges. Use the finite product distribution in which each binary choice is equiprobable and each maximum independent set of an odd cycle is equiprobable. The choices on distinct components are independent. Every whole P_3 center is always selected.

An odd-cycle child belongs to an edge whose mate is childless, by (I). Such an edge has indicator zero in every canonical state. An absent child also forces its edge indicator to vanish. Hence a nonzero indicator on an unanchored edge requires two children, neither on an odd cycle, with at least one child not a P_3 center. By (46), every such noncenter child is a binary vertex.

If one child is a center and the other is binary, simultaneous selection is the selection of that binary class. Its probability is $1/2$, and the edge is aligned. If both children lie in one binary component, they lie either in the same canonical class or in opposite classes. In the first case their simultaneous selection probability is $1/2$, including when the children coincide. In the second case the probability is zero. If the children lie in distinct binary components, their independent selection probabilities are both $1/2$. The simultaneous selection probability is $1/4$, and the edge is cross. These cases exhaust the unanchored edges and give the stated probabilities.

Step 4. We derive the exact defect identity and count actual odd-cycle contacts. Every anchored edge contributes one to (48) in every state. Taking expectations gives

$$q - A = \frac{N_{\mathrm{half}}}{2} + \frac{N_{\mathrm{cross}}}{4}.$$

The unanchored-edge partition gives

$$u = N_{\mathrm{half}} + N_{\mathrm{cross}} + N_{\mathrm{zero}}.$$

Subtracting twice the first equality from the second yields

$$u - 2(q - A) = N_{\mathrm{zero}} + \frac{N_{\mathrm{cross}}}{2}.$$

Since $u = m - A$ and $q = m - b$, the left side is $A - m + 2b = r$. This proves (47). The integrality of r and N_{zero} implies that N_{cross} is even. Also,

$$N_{\mathrm{zero}} + N_{\mathrm{cross}} = 2r - N_{\mathrm{zero}} \le 2r.$$

Each actual edge from P to an old odd cycle belongs to a frame edge of the zero class. There is at most one such incidence at each endpoint, since an endpoint has at most one child. Both endpoints of one frame edge cannot contribute, by (I). Thus sending an incidence to its frame edge is injective. It follows that

$$\gamma_{\mathrm{odd}} \le N_{\mathrm{zero}} \le r.$$

Every reflected graph used here is an actual saturated terminal of the same source. □

Remark 7.9 (Range of the contact restriction). Theorem 7.8 strengthens the earlier unique-incidence restriction and nonnegative binary-defect bound. No extremal-frame, full-center, first-deficit, or source-order hypothesis is imposed. The result is a structural restriction within a saturated frame. It neither excludes the entire quartic source class nor establishes a Ramsey formula.

7.1.3. *Pentagon parent classes and leaf families.* The goal of this section is to use Theorem 7.8 to classify pentagon parent sets and prove the two-family obstruction.

Definition 7.10 (Componentwise maximum independent choices)**.** Let H be a union of whole terminal components. A componentwise maximum independent choice on H is an independent set whose intersection with each component of H is a maximum independent set of that component. Write $\mathcal{M}(H)$ for the finite set of these choices.

Theorem 7.11 (Pentagon parents and the two-family obstruction)**.** *Use Setup 7.2, with an arbitrary saturated frame (S, U). Then the following statements hold.*

(I) For every whole pentagon O of R, the two-element parent sets

$$A_v = N_F(v) \cap W, \qquad v \in V(O),$$

are equal or disjoint. Their equality classes are singletons or nonadjacent pairs. There is at most one edge from P to O. If such an edge contacts x, then x is a singleton class, and its other parent in Z neighbors no other vertex of O.

(II) Fix a whole pentagon O and the leaf pairs X_1, X_2 of two distinct whole P_3 components of R. Let Q be the union of all whole even-order components of R, and let L be the set of endpoints of the whole P_2 and P_4 components.

There do not exist disjoint nonempty finite sets $C_1, C_2 \subseteq Z$ with all the following properties. For every integer $i \in \{1,2\}$ and every $z \in C_i$,

$$N_F(z) \subseteq X_i \cup L \cup V(O), \qquad N_F(z) \cap X_i \neq \varnothing,$$

and there is a choice $J \in \mathcal{M}(Q \cup O)$ with $N_F(z) \cap J = \varnothing$. For every $J \in \mathcal{M}(Q \cup O)$ and each $i \in \{1,2\}$,

$$\left|\{z \in C_i \mid N_F(z) \cap J = \varnothing\}\right| = 1. \tag{49}$$

All counts and finite extrema have nonnegative integer domains. No minimum-frame or avoidance assumption on edges from P to O is required.

Proof. By Setup 7.4, the frame graph is $mK_2 \dot\cup (a-m)K_1$, and every actual saturated terminal has component list (46). For a canonical choice D, every D-vertex has degree two in R_D, and each designated odd-cycle or even-path component contains exactly one old frame edge.

Step 1. We prove the parent partition in (I). By Setup 7.5, there is at most one edge from P to O. For each adjacent pair of vertices of O, at least one vertex has both parents in Z. The adjacent-parent restriction in Setup 7.5 therefore makes their parent sets disjoint.

Suppose x is contacted by an edge from P. Write its parents as $p \in P$ and $z \in Z$, and let p' be the mate of p. Theorem 7.8(I) gives the whole reflected path $p' - p - x - z$ for every canonical choice containing x. For each nonadjacent $y \in V(O)$, the set $\{x, y\}$ is a maximum independent set of O and can be selected canonically. The vertex z cannot neighbor y, since the resulting edge would extend this whole P_4. Adjacent vertices already share no parent with x, and p has no other neighbor on O. Thus the parent pair of x is disjoint from every other parent pair.

Now let v, w be distinct nonadjacent vertices of O, each with two parents in Z, and suppose they share a parent z. Select v, w on O and complete the choice canonically elsewhere. Write their other parents as z_v, z_w. If $z_v = z_w$, their parent pairs are equal.

Suppose instead that $z_v \neq z_w$. The reflected terminal contains the path

$$z_v - v - z - w - z_w$$

on five distinct vertices. The three displayed W-vertices belong to Z. No permitted path component contains five vertices. A cycle containing this path cannot be a C_3 or a C_4. If it were a whole C_5, its closing edge would be $z_w z_v$, contrary to the independence of Z. If it were a whole C_7, the remaining path from z_w to z_v would have length three. Its first interior vertex must belong to D, since z_w has no neighbor in W. Its second interior vertex must also belong to D, since z_v has no neighbor in W. These two interior vertices would be adjacent, contrary to the independence of D. This exhausts the permitted cycle lengths. Hence $z_v = z_w$.

The parent sets are therefore equal or disjoint. Adjacent vertices have disjoint parent sets, so each equality class is independent on C_5. Such a class has size at most two: three pairwise nonadjacent vertices would require three further vertices to separate them around the cycle. The contacted vertex, when present, is a singleton with the stated restriction on its other parent. This proves (I).

Step 2. We show that every candidate in either family avoids all whole P_4 components. Suppose the families in (II) exist. Their support condition already excludes internal vertices of P_4 components from every candidate neighborhood.

Fix arbitrary componentwise choices on $Q \cup O$ outside the P_4 components. On each P_4, choose both endpoints. For either fixed family, (49) gives a unique available candidate. That candidate has no neighbor among any P_4 endpoints. It has no internal P_4 neighbor either, so it remains available for every P_4 choice with the outside choices fixed.

Let z be any candidate in that family. Choose a witnessing componentwise choice avoided by z, as supplied by the liveness hypothesis. Fix its outside-P_4 choices and use the candidate obtained in the preceding paragraph. For the witnessing P_4 choices, both candidates are available. Uniqueness in (49) makes them equal. Thus z has no P_4 neighbor. This holds for both families.

Step 3. We determine the exact leaf-incidence totals in each family. Fix $i \in \{1,2\}$. For $z \in C_i$, put

$$a_z = |N_F(z) \cap X_i|, \qquad d_z = |N_F(z) \cap V(O)|.$$

Then $a_z \in \{1,2\}$. Part (I) gives $d_z \in \{0,1,2\}$, with any two O-neighbors nonadjacent. If $d_z = 2$, these two neighbors are $N_O(v)$ for one unique vertex v of the pentagon.

By Step 2 and the support condition, the remaining $4 - a_z - d_z$ neighbors of z are endpoints of whole P_2 components. They lie in distinct components. Meeting both endpoints of a P_2 would contradict the existence of a componentwise maximum independent choice avoiding z.

Choose independently and equiprobably one endpoint of each whole P_2. The probability that z avoids these choices is

$$2^{-(4-a_z-d_z)} = \frac{a_z}{2^{3-d_z}}. \tag{50}$$

The equality uses $a_z \in \{1,2\}$. The exponent $4 - a_z - d_z$ is a nonnegative integer because it counts the actual P_2-neighbors of z.

Fix a P_2 assignment and call a candidate eligible if it avoids that assignment. Choices on whole C_4 components do not affect eligibility, by the support condition. For each eligible candidate, consider the maximum independent sets of O avoiding its neighborhood. By (49), these sets partition the five maximum independent sets of O. Their cardinalities are

$$5 \quad (d_z = 0), \qquad 3 \quad (d_z = 1), \qquad 2 \quad (d_z = 2).$$

Thus the partition consists either of one set of size five, or of one set of size three and one set of size two.

In the latter case, the candidate with $d_z = 1$ has its unique O-neighbor at a vertex v, and the candidate with $d_z = 2$ has O-neighborhood $N_O(v)$ for the same v. To verify this, label O cyclically by $0,1,2,3,4$. Avoiding $N_O(0) = \{1,4\}$ permits exactly

$$\{0,2\}, \qquad \{0,3\},$$

which are the two maximum independent sets containing 0. Avoiding the vertex 0 permits the other three maximum independent sets. For a different central vertex, the two maximum independent sets containing it form a different two-element set. Hence the size-three and size-two sets are complementary only at the same central vertex. Cyclic relabeling gives the assertion for every v.

For integers $d \in \{0,1,2\}$, define

$$A_d = \sum_{\substack{z \in C_i \\ d_z = d}} a_z.$$

In every P_2 assignment, the number of eligible candidates with $d_z = 1$ equals the number with $d_z = 2$. Also, the number with $d_z = 0$ plus the number with $d_z = 1$ equals one. Taking expectations and using (50) gives

$$\frac{A_1}{4} = \frac{A_2}{2}, \qquad \frac{A_0}{8} + \frac{A_1}{4} = 1.$$

Therefore

$$A_1 = 2A_2, \qquad A_0 + 4A_2 = 8. \tag{51}$$

Each vertex of X_i is a terminal leaf. It has one neighbor in R and exactly three neighbors in W. Since every candidate belongs to $Z \subseteq W$,

$$A_0 + A_1 + A_2 \leq 6.$$

Substitution from (51) gives

$$8 - A_2 \leq 6,$$

so $A_2 \geq 2$. The nonnegativity of A_0 gives $A_2 \leq 2$. Consequently,

$$(A_0, A_1, A_2) = (0, 4, 2). \tag{52}$$

Each family therefore contains at least two distinct candidates with $d_z = 1$, since each contributes at most two to A_1. Every such candidate is eligible at some P_2 assignment, because its P_2 contacts lie in distinct components. At that assignment, the partition argument pairs it with a $d_z = 2$ candidate whose O-neighborhood is the neighbor pair of its unique O-neighbor.

Step 4. We show that the two families would require four parents at one pentagon vertex. Call a vertex $v \in V(O)$ admissible if, among the two families, there is a $d_z = 1$ candidate with O-neighborhood $\{v\}$ and a $d_z = 2$ candidate with O-neighborhood $N_O(v)$.

Suppose distinct vertices u, v are both admissible. By (I), the two vertices in $N_O(u)$ have equal parent pairs. The two vertices in $N_O(v)$ also have equal parent pairs. If these neighbor pairs intersected, the equal-or-disjoint rule would give a parent class of at least three vertices, contrary to (I). Thus the two neighbor pairs are disjoint.

On a pentagon, the neighbor pairs of distinct central vertices are disjoint only when the central vertices are adjacent. Nonadjacent central vertices have their intermediate vertex in both neighbor pairs. Hence u and v are adjacent. Then $u \in N_O(v)$, whose two vertices have the same full parent pair. Every parent of u therefore neighbors another vertex of O. This contradicts the $d_z = 1$ candidate with sole O-neighbor u. There is at most one admissible vertex.

By Step 3, each family's at least two distinct $d_z = 1$ candidates has its O-neighbor at an admissible vertex. All these candidates therefore neighbor the same unique vertex of O. Since C_1 and C_2 are disjoint, that vertex has at least four distinct parents in W. A vertex of a whole terminal pentagon already has two neighbors in R and degree four in F, so it has exactly two parents. This contradiction proves (II).

Part (I) permits a singleton contacted from P. The parent-class argument and the final degree count include that case. No source-minimum hypothesis was used, and every candidate was an actual vertex of the given graph. □

Remark 7.12 (Comparison and scope). Theorem 7.11(I) removes the avoidance premise from the earlier pentagon parent-partition statement. Part (II) gives the corresponding uniform obstruction without the fixed-cell premise of the earlier leaf-family statement. Those earlier statements are comparison results; their additional hypotheses are not assumed here. The conclusion excludes the specified conditional family configuration. It does not exclude an entire source cell or the quartic source class, and it does not establish a Ramsey formula.

7.1.4. *Heptagon parent classes and a square exchange.* In this section, we prove Theorem 7.13 from the absence of P_6 and the root exchanges in Setup 7.6.

Theorem 7.13 (Heptagon parents and a boundary-preserving square exchange)**.** *Use Setup 7.2, with an arbitrary saturated frame (S, U). Then the following statements hold.*

- *(I) Every frame-edge endpoint avoids every whole C_7 component of R.*
- *(II) Let O be a whole heptagon of R. Each vertex of O has exactly two parents, both in Z. Adjacent vertices of O share no parent. Each parent has at most two neighbors on O. Whenever two distinct vertices of O share a parent, they are nonadjacent and share both parents.*

Consequently, O is partitioned into singleton classes and disjoint nonadjacent pairs with equal parent pairs. Parent pairs belonging to different classes are disjoint. For a two-vertex class $\{u, v\}$ with parents $\{z, z'\}$, the other two neighbors of each of z, z' have degree at most one in R.

(III) *Every such two-vertex class admits the following actual saturated exchange. There is a canonical choice on O, specifiable independently of the choices on all other components, whose every canonical completion D has these properties:*

$$R_D \text{ contains the whole } P_3 \; z - u - z'$$

and a distinct whole P_3 centered at a vertex t, while v is a reflected frame-edge endpoint with

$$N_{R_D}(v) = \{z, z', t\}.$$

Orient v into the reflected cubic root set and replace v by t. The resulting frame is saturated and has terminal

$$R_D - t + v = F\big[((W \cup D) \setminus \{t\}) \cup \{v\}\big].$$

Its only component changes are the removal of those two P_3 components, the addition of the whole square

$$u - z - v - z' - u,$$

and two new isolates at the former leaves of t. In particular, it has two fewer P_3 components than R_D. The ambient graph and all its exterior contacts remain fixed.

When $E = 6$, the initial saturation hypothesis is automatic.

Proof. The normalization in Setup 7.2 supplies well-coveredness and the uniform complement profile needed for canonical reflection and the child bound. Since the global cubic maximum equals m, every saturated frame attains it. Setup 7.4 applies to each such frame. It gives actual saturated reflections, degree two at every selected vertex, and one old frame edge in every designated odd-cycle or positive even-path component. All these terminals have component list (46).

Step 1. We exclude frame-endpoint contacts with a whole heptagon. Suppose a frame-edge endpoint p touches a whole C_7. By Setup 7.6, orienting p into the cubic root set permits an actual root exchange that opens the C_7 into a whole P_6 and closes a distinct even path into an odd cycle. The new frame is saturated. This contradicts (46). Thus (I) holds in every saturated frame of this source.

Each vertex of a whole heptagon has two cycle neighbors and degree four in F. It therefore has exactly two parents. Part (I) places both parents in Z.

Step 2. We identify the reflected component of a selected heptagon vertex next to a residual frame edge. Choose a canonical D containing a vertex u of O such that u is adjacent on O to an endpoint w of the unique residual edge of $O - D$. The endpoints of this residual edge form a frame edge in the reflected saturated frame, by Setup 7.4.

The vertex u has degree two in R_D, and its two neighbors are its old parents, both isolates of $F[W]$. Its component is neither K_1 nor P_2. It cannot be P_4. Such a designated component would contain an old frame edge, whose two endpoints are distinct from the two isolate parents of u, requiring at least five vertices including u.

The component cannot be C_3 or C_5 either. An odd cycle containing exactly one old frame edge alternates between W and D on all other edges. Thus a C_3 contains two W-vertices and a C_3-free C_5, namely a C_5, contains three. The two isolate parents of u and the two endpoints of the required old frame edge would give at least four distinct W-vertices.

Finally, the component cannot be C_7. The reflected frame endpoint w is adjacent to u, which would contradict (I) in the reflected saturated frame. The list (46) now implies that u is the center

of a whole P_3 or a vertex of a whole C_4. This observation holds for every canonical choice outside O.

Step 3. We prove the parent classes in (II). Suppose adjacent vertices u, w of O share a parent z. Label O cyclically by $0, \dots, 6$, with $u = 0$ and $w = 1$. Choose

$$D \cap V(O) = \{0, 3, 5\}.$$

The residual edge is 1–2, so Step 2 applies to u and w.

If the component of u in R_D is a P_3, then z is one of its leaves. The reflected frame endpoint w is adjacent to both the center u and the leaf z. This contradicts Setup 7.5. If the component is a C_4, then both u and z have degree two in R_D, and w is adjacent to both. This contradicts the bound of one child at a frame endpoint. Hence adjacent heptagon vertices share no parent.

For $z \in Z$, its neighborhood on O is now independent. An independent set on C_7 has size at most three, since four selected vertices would require four additional vertices separating consecutive selections around the cycle. If z had three neighbors on O, they would form a maximum independent set of O. Selecting them canonically would give z degree at least three in R_D, contrary to the terminal degree bound. Thus every parent meets at most two vertices of O.

Suppose distinct nonadjacent vertices u, v share a parent z. By a cyclic labeling and reversal, take $u = 0$ and $v \in \{2, 3\}$. If $v = 2$, choose

$$D \cap V(O) = \{0, 2, 4\}.$$

Its residual edge is 5–6, and u is adjacent to 6. If $v = 3$, choose

$$D \cap V(O) = \{0, 3, 5\}.$$

Its residual edge is 1–2, and u is adjacent to 1. In either case, $u, v \in D$, and Step 2 applies to u.

The common parent z has neighbors u, v in R_D, so it has degree two. The component of u cannot be a P_3, whose center has leaf neighbors. It is therefore a whole C_4. Since D is independent and z has neighbors u, v, this square has the form

$$u - z - v - z' - u.$$

Thus z' is the second parent of both u and v.

Every heptagon vertex has exactly two parents, and every parent meets at most two heptagon vertices. The shared pairs therefore cannot overlap. Singletons and these nonadjacent pairs form the required classes, and parent pairs from distinct classes are disjoint.

Step 4. We bound the exterior terminal degrees of the parents of a two-vertex class. Let $\{u, v\}$ have parents $\{z, z'\}$, and let y be a neighbor of z other than u, v. Since z is isolated in $F[W]$, all its neighbors belong to R. The vertex y lies outside O, by the bound of two O-neighbors per parent.

Suppose y has degree two in R. By Setup 7.4, a canonical choice on its component can include y. This component differs from O. Combine that choice with a maximum independent set of O containing u, v, and choose canonically elsewhere. Then z has the three neighbors u, v, y in D. This contradicts the degree bound in R_D. Hence both remaining neighbors of z have degree at most one in R. Applying this argument to each remaining neighbor of z' gives the same conclusion for z'. This completes (II).

Step 5. We construct the square exchange while retaining all exterior contacts. Again label $u = 0$ and take $v \in \{2, 3\}$. If $v = 2$, choose

$$D \cap V(O) = \{0, 3, 5\}.$$

The residual frame edge is 1–2, and the selected cycle neighbor of v is $t = 3$. If $v = 3$, choose

$$D \cap V(O) = \{0, 2, 5\}.$$

The residual frame edge is 3–4, and the selected cycle neighbor of v is $t = 2$. Complete D on every other component in any canonical way.

Both u and t are selected, and v is a reflected frame endpoint. By (II), the other neighbors of z, z' outside $\{u, v\}$ have degree at most one in R. No such vertex belongs to D. Also, $v \notin D$, and z, z' were isolates of $F[W]$. Thus each of z, z' has exactly one neighbor in R_D, namely u. The path

$$z - u - z'$$

is therefore a whole P_3.

The four F-neighbors of v are its two original cycle neighbors and z, z'. One cycle neighbor is its reflected frame mate, and the other is t. Consequently,

$$N_{R_D}(v) = \{z, z', t\}.$$

Since $t \in D$, it has degree two in R_D.

Suppose t lies on an odd-cycle component of R_D. The odd-cycle contact rule in Setup 7.6, applied to this reflected frame and the endpoint v, would make z, z' the endpoints of a whole positive even-order path. They are instead the leaves of the whole P_3 centered at u. This is a contradiction. Thus t lies on no odd-cycle component.

By (46), a degree-two vertex outside the odd cycles is a P_3 center, a C_4 vertex, or an internal P_4 vertex. None is an exposed port of an odd terminal component. For a P_3, only its leaves are exposed ports. Hence the odd-component exposed ports adjacent to v are precisely z, z', both in the P_3 centered at u. The port rule in Setup 7.6 forces t to be the center of a different whole P_3.

The root exchange in Setup 7.6 now applies. Orient v into the reflected cubic root set and replace it by t. The resulting frame is saturated. Removing t leaves its former two leaves as isolates. Inserting v next to z, z' closes $z - u - z'$ into the whole square

$$u - z - v - z' - u.$$

The vertex v has no further terminal neighbors, and the old component of t was whole. These are therefore all component changes. Exactly two P_3 components disappear. The graph F and all its exterior contacts remain fixed throughout.

Finally, the excess-six assertion follows from the last statement of Setup 7.2. □

Remark 7.14 (Range of the heptagon exchange). Theorem 7.13 strengthens the boundary classification and gives an exchange in an actual saturated frame. It does not exclude all heptagons, all excess-six sources, or the whole quartic source class. Literature priority for this strengthening, for Theorem 7.8, and for Theorem 7.11 has not been determined. No external theorem or computation is used in the three proofs.

an automated proof-search and verification system.

Remark 7.15 (Automated assistance). The main results of this chapter were obtained with the assistance of the Ansatz system, an automated proof-search and verification system. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts.

[metadata/blocker] Mathematics Subject Classification codes remain to be supplied.

8. Assembly of the structural propositions

We assemble the fifteen supplied statements in dependency order, with each absent technical premise located in Section 3.

8.1. **Universal source restrictions.**

Proposition 8.1 (Source structure, numerical domains, and sufficient reduction)**.** *Use Definitions 2.2–2.5 and Setup 2.4.*

(I) *Every $F \in \mathcal{M}_d$, $d \in \{4,6,8,10,12\}$, is connected, d-regular, factor-critical, and well-covered. Moreover,*

$$n = 2k+1, \qquad \operatorname{gap}(F) = 1,$$

and

$$\begin{aligned} c_0(F) &= k + E, \\ c_1(F) &= k + E - \epsilon = n - r + p + q_{\mathrm{def}}, \\ c_2(F) &= k + E - \delta = n - r + p. \end{aligned}$$

For $d = 8$, $c_4(F) = r + q_8$. The necessary lower bounds on E, in degree order $4,6,8,10,12$, are $14,6,10,12,10$. All following restrictions are simultaneous necessary conditions; no realization is asserted.

(II) *If $d = 4$, the source is strict. Put $\sigma = \lambda + 1$. For every maximal independent set S_0,*

$$b_4 = \nu(F - S_0) - c_1(F - S_0)$$

has the same value. Put $m_4 = E - 1 - \sigma + b_4$. Then

$$E \geq 14, \qquad 9 \leq \sigma \leq \left\lfloor \frac{3E-5}{4} \right\rfloor, \qquad 3 \leq b_4 \leq \left\lfloor \frac{2\sigma}{3} \right\rfloor,$$
$$E \leq 2\sigma - b_4 + 1, \qquad 20E \leq 43\sigma + 11, \qquad 24E \leq 51\sigma + 28,$$
$$E \leq 3\sigma - 2b_4 + 2, \qquad 6b_4 \leq 5\sigma + 3.$$

Saturated frames exist: S is maximal independent and U is an independent set of exactly m_4 vertices of degree three in $F - S$. The entire classes with $b_4 = 2$, with $k = 8$, and with $E = 13$ are empty. No additional frame optimization or exchange assertion is included.

(III) *If $d = 6$, the source is strict and*

$$E \geq 6, \quad 2 \leq \lambda \leq E - 4, \quad p \geq 2, \quad 2 \leq \epsilon < \delta \leq \lambda + 1,$$
$$\delta - 1 \leq \lambda \leq \min\left\{ \delta + \epsilon - 2, \ \left\lfloor \frac{3\delta + 1369 q_{\mathrm{def}} - 3}{2} \right\rfloor, \ \left\lfloor \frac{2\delta + \epsilon - 3}{2} \right\rfloor \right\}.$$

The integer s is positive and odd, with $1 \leq s \leq \epsilon$. Exactly one of the following disjoint necessary cells contains F, with all general restrictions still imposed:

U2 :	$q_{\mathrm{def}} = 1,\ \epsilon = 2,\ \lambda = 2,\ s = 1;$
WINDOW :	$q_{\mathrm{def}} = 1,\ \epsilon \geq 3,\ \epsilon \leq \lambda \leq \min\{E-4, \lfloor (3\epsilon - 2)/2 \rfloor\},\ 1 \leq s \leq \epsilon - 1;$
Q2 :	$q_{\mathrm{def}} = 2,\ \epsilon \geq 2;$
HIGHDEF :	$q_{\mathrm{def}} \geq 3,\ \epsilon \geq 2.$

For every $q_{\mathrm{def}} = 1$ source,

$$1 \leq s \leq \epsilon - 1, \qquad 2\lambda \leq 3\epsilon - 2.$$

Thus the entire $q_{\mathrm{def}} = 1, \epsilon = s$ class is empty. If $\epsilon = s$ at any deficit, then

$$q_{\mathrm{def}} \geq 2, \qquad 3 \leq s \leq 2q_{\mathrm{def}} - 1.$$

The entire $\epsilon = 1$ class is empty at every admitted order, rank, excess, deficit, and process. Universally,

$$\lambda \geq q_{\mathrm{def}} + 1, \qquad E \geq q_{\mathrm{def}} + 5,$$

and every actual prefix has independence number a.

Throughout the entire $\epsilon = 2$ sector,

$$s = 1, \qquad \lambda = \delta - 1 = q_{\mathrm{def}} + 1, \qquad r = k.$$

If $E = 6$, then necessarily

$$(\delta, \epsilon, q_{\mathrm{def}}) = (3,2,1), \quad \lambda = 2, \quad p = 2, \quad s = 1, \quad r = k, \quad k - a = 7.$$

For every source with $\epsilon = 2, \delta = 3, \lambda = 2$, some actual native row has entire form

$$pC_3 + P_2 + P_4 + zK_1, \qquad z = a - p - 3 \geq 0.$$

This existence is not coupled to an independently minimizing prefix. No entire surviving cell or degree is excluded.

(IV) *If $d = 8$, the source is strict and*

$$E \geq 10, \quad 2 \leq \epsilon \leq \delta \leq \lambda \leq \left\lfloor \frac{E-4}{2} \right\rfloor, \quad \lambda \geq 3, \quad p \geq \lambda + 2 \geq 5,$$
$$q_8 \geq 1, \qquad \eta = \lambda - \delta \geq 0, \qquad p \geq q_8 + \delta + 1.$$

The entire $\epsilon = 0$ and $\epsilon = 1$ classes are empty at every q_8, order, rank, excess, deficit, component inventory, and actual process. Exactly one of the following disjoint necessary families contains F:

$$\begin{aligned}
&\mathrm{B}^+ : && \delta = \lambda \geq 3, \quad q_8 = 1, \quad 2 \leq \epsilon \leq \delta; \\
&\mathrm{D}_2 = \mathrm{I}_2^+ : && \delta = \epsilon = 2 < \lambda, \quad q_8 = 1, \quad 3 \leq \lambda \leq 9, \\
& && E \geq 2\lambda + 5, \quad p \geq \lambda + 3, \quad p \geq q_8 + 4; \\
&\mathrm{I}_3^+ : && 3 \leq \delta < \lambda, \quad 1 \leq q_8 \leq \eta + 1, \quad 2 \leq \epsilon \leq \delta, \quad E \geq 12.
\end{aligned}$$

The labels D_2 and I_2^+ denote the same family. In I_3^+, the signatures $q_8 = 1$ and $q_8 \geq 2$ remain distinct necessary alternatives. Every interior source $\delta < \lambda$ has $E \geq 11$, with $E \geq 12$ in I_3^+.

The entire boundary $\lambda = \delta = 2$, the entire class $\delta = \epsilon = 2, q_8 \geq 2$, and the entire $E = 8, 9$ classes are empty. At $E = 10$, only the necessary alternative

$$\mathrm{B}^+, \quad \delta = \lambda = 3, \quad q_8 = 1, \quad \epsilon \in \{2, 3\}, \quad p = 5, \quad r = k - 1, \quad k - a = 14$$

remains.

For every native row A, every complete continuation on $F - V(A)$ has word

$$5^{q_8} 4^{s-2q_8} 3^{q_8}, \qquad s = 2\eta + 3,$$

where $s - 2q_8$ is positive and odd. In B^+, the word is 543. Every actual source prefix has independence number a. No surviving family, signature, or entire degree is excluded.

(V) *Every degree-ten source satisfies*

$$E \geq 12, \qquad \lambda \geq 3, \qquad p \geq 5$$

in both sectors $\delta = 0$ and $\delta > 0$. Every degree-twelve source is strict and satisfies

$$E \geq 10, \qquad \lambda \geq 3, \qquad p \geq 5, \qquad \delta > 0.$$

The entire degree-twelve equality sector is empty. These numerical restrictions give no whole-degree exclusion.

(VI) *The conclusions are exactly the numerical domains, source structure, costs, saturation existence, row existence, and process assertions stated above. They do not import omitted structural conclusions, prove reachability for an additional configuration, or give a strict source-cost certificate. In particular, no degree-ten supported-triangle transfer theorem is asserted.*

(VII) *Under exactly $N_4, N_6, N_8, N_{10}, N_{12}$, the equality in Theorem A holds for all positive integer lengths and nonincreasing positive integer component-size lists specified there. No converse is asserted, and none of these five hypotheses is proved here.*

Proof. The missing universal input is identified in Section 3.1. It has the same four-part certificate definitions, integer domains, trace convention, property T_-, and entire proper-subgraph-minimal source classes. No class is replaced by an induced-minimal class, a chosen process, or an additional configuration.

For (I), that input supplies connectedness, regularity, factor-criticality, well-coveredness, odd order, gap one, and the five excess floors. The definition of excess gives $c_0(F) = k + E$. The definitions of the two deficits give the first expressions for $c_1(F)$ and $c_2(F)$. Moreover,

$$\begin{aligned} n - r + p &= (2k + 1) - (k - \lambda - 1 + \delta) + (E - 2 - \lambda) \\ &= k + E - \delta. \end{aligned}$$

Adding $q_{\mathrm{def}} = \delta - \epsilon$ gives $n - r + p + q_{\mathrm{def}} = k + E - \epsilon$. The fourth octic cost is the rearranged definition of q_8.

For (II), the required quartic part supplies strictness, uniformity of b_4, the definition of m_4, all eight inequalities, saturated-frame existence, and the three complete numerical class exclusions. The definition of saturation here asks for exactly the independent cubic m_4-set furnished by that input. No frame extrema or additional labels enter the argument.

For (III), the required sextic part supplies every general inequality and all four alternatives. They are disjoint because the first two have $q_{\mathrm{def}} = 1$ with different ϵ-ranges, while the last two have $q_{\mathrm{def}} = 2$ and $q_{\mathrm{def}} \geq 3$. Their exhaustiveness is a full-source assertion of the missing input. The universal boundary restrictions, the $\epsilon = 1$ exclusion, the bounds on λ, E, and every-prefix independence have the same quantifiers. The $\epsilon = 2$ specialization and the $E = 6$ list are retained without additional restrictions. In the asserted P_4-row, $q_{\mathrm{def}} = 1$; its matching number and first two costs are $p + 3, p + 1, p$, respectively. These agree with the native-row definition. No surviving cell has thereby been excluded.

For (IV), the required octic part supplies the simultaneous inequalities and the exhaustive families. The identification $\mathrm{D}_2 = \mathrm{I}_2^+$ creates no extra family. The stronger bounds in D_2, the η, q_8, p restrictions in I_3^+, the interior floor, the boundary exclusions, and the $E = 10$ alternatives are all retained. Its process theorem uses the original-source value q_8 and gives the positive odd quartic phase length. Its prefix independence assertion is unchanged. No statement conditioned on the excluded $\epsilon = 1$ class is transferred to a surviving class.

For (V), the required input gives the displayed degree-ten restrictions in both sectors and the degree-twelve restrictions together with the equality-sector exclusion. There is no transfer from degree-twelve equality to strictness or to the unresolved degree-ten supported-triangle problem. Part (VI) records the limits of these conclusions.

For (VII), the entire classes in Definition 2.3 are exactly the five hypotheses of the required sufficient reduction. That reduction gives the antidiagonal maximum formula of Theorem A for all stated integer lists. It gives a sufficient implication, not a converse. Parts (I)–(V) have not established any N_d. □

Proposition 8.2 (Induced-minimal failures in degrees six and eight)**.** *Use the generic certificate quantities* $c_{00}, c_\theta^*, \nu, \mathrm{ex}, \mathrm{ps}, \mathrm{gap}, \mathrm{tr}$ *whose missing definitions are identified in Section 3.1; put* $c_0^* = c_{00}$*. Let* H *have maximum degree* $d \in \{6, 8\}$*, violate the corresponding property* T_-*, and have every proper induced subgraph satisfy* T_-*.*

Then H *is a totally critical closed* d*-lump: it is connected,* d*-regular, of odd order, and factor-critical. It has* $\mathrm{gap}(H) = 1$*. Put* $|V(H)| = 2k + 1$ *and* $E = \mathrm{ex}(H)$*. Then*

$$E \geq 4, \qquad \alpha(H) \geq \left\lceil \frac{3E + 5}{4} \right\rceil \geq 5.$$

For every vertex v *and every integer* $0 \leq \theta \leq d$*,*

$$c_\theta^*(H - v) = c_\theta^*(H) - 1, \qquad \mathrm{ex}(H - v) = E - 1, \qquad \mathrm{gap}(H - v) = 0.$$

Put

$$u = \max\left\{0, \left\lfloor \frac{2E - 3}{d - 2} \right\rfloor\right\}.$$

Then

$$k \geq h_-(d) + 2E - 1 + 2u.$$

For every maximal independent set S, write

$$a = |S|, \qquad G = H - S, \qquad \eta = \nu(G) - (k + 1 - a).$$

Then

$$a \geq 2, \qquad \Delta(G) \leq d - 1, \qquad 0 \leq \eta \leq \min\left\{E - 1, \left\lfloor \frac{a-1}{2} \right\rfloor\right\},$$

and

$$\mathrm{ps}(G) = E - 1 - \eta, \qquad c^*_\theta(H) = a + c^*_\theta(G) \quad (1 \leq \theta \leq d - 1),$$
$$c^*_d(H) = 1, \qquad c_{00}(H) = a + \nu(G) + \mathrm{ps}(G), \qquad \mathrm{gap}(G) \in \{0, 1\}.$$

These are necessary conditions only. No full degree-six or degree-eight closure is asserted.

Proof. The generic induced closed-lump reduction from Section 3.1 gives an induced totally critical closed even-degree lump F that violates T_- and has gap one. Induced minimality forces $F = H$. Hence

$$\nu(H) = k, \qquad \mathrm{tr}(H) + E \geq h_-(d) + 1.$$

We first exclude $E \leq 3$. In degree six, the required low-excess theorem says that every closed six-lump with excess at most two satisfies the stronger bound $\mathrm{tr} + \mathrm{ex} \leq 4 = h_-(6)$ whenever its gap is positive. Thus $E \leq 2$ is impossible. If $E = 3$, failure requires $\mathrm{tr}(H) \geq 2$; the required excess-three exclusion applies because every proper induced subgraph satisfies T_-.

In degree eight, every proper induced subgraph has generic gap at most one. The required low-excess result makes a closed eight-lump of excess at most two LP-good, excluding $E \leq 2$. At $E = 3$, the required excess-three result gives $\mathrm{tr}(H) = 1$. Thus

$$\mathrm{tr}(H) + E = 4 \leq h_-(8) = 5,$$

again a contradiction. Consequently $E \geq 4$.

Positive gap makes H not LP-good. The contrapositive of the required independence criterion, valid for $E \geq 4$, gives

$$4\alpha(H) > 3E + 4.$$

Integrality gives the stated lower bound.

Put $B = h_-(d) \in \{4, 5\}$. Every proper induced subgraph J satisfies

$$\mathrm{gap}(J) > 0 \Longrightarrow \mathrm{gap}(J) = 1, \quad \mathrm{tr}(J) + \mathrm{ex}(J) \leq h_-(\Delta(J)) \leq B.$$

Thus it satisfies the constant-bound property required by the criticality theorem. The graph H is totally critical, has gap one, and has $E \geq 4 \geq B - 1$. That theorem gives all vertex-deletion identities.

The required rank bound applies because H is connected, has order $2k + 1$, matching number k, maximum degree d, positive excess, and $\mathrm{tr}(H) + E \geq h_-(d) + 1$. Its parameter is exactly

$$u = \max\{0, \lfloor (2E - 3)/(d - 2) \rfloor\},$$

so it yields the displayed inequality for k.

Finally, the required maximal-independent-cut theorem gives

$$a \geq 2, \quad \Delta(G) \leq d - 1, \quad 0 \leq \eta \leq \lfloor (a - 1)/2 \rfloor.$$

Since h_- is nondecreasing and

$$\mathrm{tr}(H) + E \geq h_-(d) + 1 > h_-(d - 1),$$

its exact-profile clause applies. It gives

$$\mathrm{ps}(H) = \eta + \mathrm{ps}(G),$$

all the cost identities, and $\mathrm{gap}(G) \in \{0,1\}$. The generic identity $\mathrm{gap} = \mathrm{ex} - \mathrm{ps}$ gives $\mathrm{ps}(H) = E-1$. Therefore $\mathrm{ps}(G) = E - 1 - \eta$, whose nonnegativity also gives $\eta \le E - 1$. □

8.2. **Sextic rows, selectors, and four compatible matchings.**

Proposition 8.3 (Universal sextic native rows and complete processes)**.** *Let $F \in \mathcal{M}_6$, with the notation of Setup 2.4, and put*

$$M = k + E - r, \qquad u_6 = \left\lfloor \frac{2E-3}{4} \right\rfloor.$$

All assertions below hold throughout the entire source class.

(I) *The source is connected, six-regular, factor-critical, and well-covered, with*

$$|V(F)| = 2k+1 = 2r+s, \qquad \mathrm{gap}(F) = 1,$$

and

$$E \ge 6, \quad 2 \le \lambda \le E-4, \quad p \ge 2, \quad 2 \le \epsilon < \delta \le \lambda + 1.$$

The integer s is positive and odd. Native rows exist. Every native row is well-covered, has independence number a, and has entire form

$$A = zK_1 + bP_2 + tP_4 + n_3C_3 + n_4C_4 + n_5C_5 + n_7C_7,$$

with nonnegative integer counts satisfying

$$\begin{gathered} n_3 + n_5 + n_7 = p, \qquad b + t = \epsilon, \\ b + 2t + 2n_4 + n_5 + 2n_7 = \delta, \qquad z = a - p - \delta, \\ q_{\mathrm{def}} = t + 2n_4 + n_5 + 2n_7. \end{gathered}$$

In particular,

$$e(A) = 3p + 2\delta - \epsilon.$$

Every maximal independent set of A is a maximum independent set of F. No whole P_3 or P_6 occurs.

The exact costs are

$$(c_0, c_1, c_2, c_3, c_4, c_5)(F) = (k+E, k+E-\epsilon, k+E-\delta, r, a+p+\epsilon, a).$$

For any native A, put $J = F - V(A)$ and $u = |V(J)| = r+s$. Then

$$\begin{gathered} \Delta(J) \le 3, \quad \mathrm{ps}(J) = 0, \quad \alpha(J) = a, \quad \nu(J) = M = p + s + \delta, \\ c_1(J) = M - \epsilon, \quad c_2(J) = M - \delta, \quad c_i(J) = 0 \quad (i \ge 3), \\ c_0(F) = u + \nu(A), \quad c_1(F) = u + p + q_{\mathrm{def}}, \quad c_2(F) = u + p. \end{gathered}$$

(II) *Every actual prefix and terminal of every complete source process has independence number a. If S, T_5, T_4 are its initial maximal independent set and entire degree-five and degree-four phases, then*

$$|S| = a, \qquad |T_5| = p + \epsilon, \qquad |T_4| = p + \delta - \epsilon,$$

and $F[S \cup T_5 \cup T_4]$ is native. Every complete process has word

$$5^{p+\epsilon}4^{p+\delta-\epsilon}3^{s}.$$

Conversely, fix any native A. Label each whole path consecutively and each whole cycle cyclically, independently. Choose componentwise

component	S_A	T_5
K_1	$\{1\}$	$\varnothing$
P_2	$\{1\}$	$\{2\}$
P_4	$\{1,3\}$	$\{4\}$
C_3	$\{1\}$	$\{2\}$
C_4	$\{1,3\}$	$\varnothing$
C_5	$\{1,3\}$	$\{4\}$
C_7	$\{1,3,5\}$	$\{6\}$

and put $T_4 = V(A) \setminus (S_A \cup T_5)$. Then S_A is maximum independent in F, and T_5, T_4 are independent. Deleting S_A, then T_5 in any order, then T_4 in any order gives the complete degree-five and degree-four phases and leaves the same J. Any prescribed complete continuation on J can be appended. Every later physical prefix, root, and terminal remains unchanged. No preservation of unrelated earlier owner labels or certificate colors is asserted.

(III) *Put*

$$C_A = \{v \in V(J) : d_J(v) = 3\}, \qquad \Gamma = J[C_A].$$

The vertex sets of complete cubic phases are exactly the maximal independent sets of Γ. Every ordering of each such set is legal. Every such set has size s, so Γ is well-covered with $\alpha(\Gamma) = s$. Every prescribed independent subset extends to a complete cubic phase.

For each such set X, put $R = J - X$. Then R is native, and $V(A), V(R), X$ partition $V(F)$. The set X is independent, and every $x \in X$ has exactly three source neighbors in A and three in R. The graph $F - V(R)$ is subcubic with zero slack. Deleting X in any order from that same graph is a complete cubic continuation ending in the original A. Canonical replay on R realizes it in a complete process of the same source.

(IV) *An odd whole A-component is owned if it has an original edge to C_A, and free otherwise. Each cubic vertex has exactly one odd whole component neighbor, and every owned component has exactly one cubic owner. Put $L_A = V(J) \setminus C_A$.*

Missing pairs exist. For every missing pair $b_r + b_s = c_0(F) - 1$, put $j = u - b_r$. Then

$$j \geq p + s + \delta + 1 + u_6 = E + \lambda + 2 - \delta + u_6 \geq E + 1 + u_6 \geq 7.$$

For $Y \subseteq L_A$, let $\mathcal{O}(Y)$ be the owned odd components touched by Y, and put $m(Y) = |\mathcal{O}(Y)|$. If

$$|Y| + m(Y) \leq j,$$

one actual matching assigns every vertex of Y to an exposed port in a distinct free odd component, assigns the unique owner of every member of $\mathcal{O}(Y)$ to that component, and contains a maximum internal matching in every whole A-component. No independence of Y is required. Every noncubic vertex of J therefore has a neighbor in a free odd component. This is simultaneous existence, not preservation of arbitrary earlier assignments.

These assertions impose no deficit, rank, order, or independently minimizing-process restriction. They assert no source realization, strict cost saving, whole-degree exclusion, or unconditional Ramsey formula.

Proof. The additional sextic inputs are identified in Section 3.2.

Step 1. Source and row identities. Proposition 8.1(I),(III) gives the source structure, numerical restrictions, and costs through index two. The definitions give $2r + s = 2k + 1$. The required native-row theorem gives existence, the remaining three costs, replay, and the complement identities. The required row classification gives the seven whole types, their relations, well-coveredness, independence number a, and the maximum-independent-set conclusion.

Counting edges gives

$$\begin{aligned} e(A) &= b + 3t + 3n_3 + 4n_4 + 5n_5 + 7n_7 \\ &= 3(n_3 + n_5 + n_7) + 2(b + 2t + 2n_4 + n_5 + 2n_7) - (b + t) \\ &= 3p + 2\delta - \epsilon. \end{aligned}$$

The absence of P_3, P_6 is part of that classification. Also

$$u = 2k + 1 - r = r + s, \qquad M = k + E - r = p + s + \delta.$$

Substituting $u = k + \lambda + 2 - \delta$, $p = E - 2 - \lambda$, and $q_{\mathrm{def}} = \delta - \epsilon$ proves the three source/complement identities. Replaying A makes J an actual prefix, so Proposition 8.1(III) gives $\alpha(J) = a$.

Step 2. Complete processes and replay. The universal prefix assertion gives independence number a at every prefix and terminal. The required early-phase theorem gives the row and the two phase lengths. Before imposing prefix independence, its terminal formulas have the form

$$h = a - \alpha(R), \quad |R| = r + h, \quad \nu(R) = p + \delta + h, \quad c_1(R) = p + \delta + h - \epsilon, \quad c_2(R) = p + h,$$

with zero terminal slack and cubic length $s - h$. Here $h = 0$, so the terminal is native and the word is the displayed one.

The canonical construction in the required replay theorem includes precisely the seven table rows after P_3, P_6 have been excluded. The choices are independent across disjoint whole components. Their union S_A has size a and is independent, hence maximum in F. That construction verifies independence of T_5, T_4, their sizes, and completeness of the two phases in every ordering. The remaining induced graph is exactly J. Appending the prescribed deletion sequence on this identical graph preserves each later graph and vertex.

Step 3. Cubic phases and reversal. In the fixed subcubic graph J, every vertex deleted at current degree three was originally cubic. Two such roots cannot be adjacent: deletion of the earlier one would lower the later one's degree below three. Thus the root set is independent in Γ. Completeness makes it maximal, since an original cubic vertex with no deleted neighbor would retain degree three.

Conversely, let X be maximal independent in Γ. Every member retains degree three while other members are deleted, in any order. After all are deleted, every unselected original cubic has lost a neighbor, and every original noncubic began with degree at most two. The continuation is therefore complete. Every independent subset extends by repeatedly adding an available vertex until none remains; the graph is finite.

Append this continuation to canonical replay. Step 2 gives $|X| = s$ and a native terminal R. Thus every maximal independent set of Γ has size s, including its maximum independent sets. The required terminal/root theorem gives the partition, the three neighbors on each side, and the zero-slack subcubic opposite complement. In $F - V(R)$, each $x \in X$ has its three A-neighbors and no X-neighbor. Every ordering is therefore a legal cubic phase ending at A. Canonical replay of R realizes this reversal in the same source.

Step 4. Owners and free assignments. The required owner theorem applies to the same arbitrary row, with complementary graph J and cubic/noncubic sets C_A, L_A. It supplies uniqueness, the bound for every ordered missing pair, and the simultaneous matching for exactly $|Y| + m(Y) \le j$. Its statement includes exposed ports, the whole touched owned family, and maximum internal matchings, without independence of Y. Its one-vertex consequence gives the last assertion.

Finally, $c_0(F) = k + E \ge 6$. By minimality of this universal total, not every allocation at $c_0(F) - 1$ succeeds. Hence a missing pair exists. No ordering of its coordinates is imposed. □

Proposition 8.4 (Physical sextic selector and completed odd-cycle residual)**.** *Let $F \in \mathcal{M}_6$, fix any native A, and put*

$$J = F - V(A), \quad U = \{v : d_J(v) = 3\}, \quad \Gamma = J[U], \quad U_0 = \{v \in U : d_\Gamma(v) = 0\},$$

$$b = |U_0|, \quad \mu = \nu(\Gamma), \quad u = |V(J)|, \quad t = (s-1)/2, \quad u_6 = \lfloor (2E-3)/4 \rfloor.$$

Let $n(T)$ count whole J-components of type T, and define

$$D = \{v \in V(J) : d_J(v) = 2,\ N_J(v) \cap U = \varnothing\},$$
$$g = n(\mathrm{cor}(P_6)) + 2n(\mathrm{cor}(C_4)),$$
$$L = n(B_{33}) + n(B_{35}) + n(B_{55}) + n(\mathrm{cor}(P_4)) + n(\mathrm{cor}(P_6)) + n(S_6).$$

(I) *The entire $J[D]$ consists of every whole ordinary C_3, C_4, C_5, C_7, one internal P_2 from every whole ordinary P_4, and one P_2 from each pentagon shore of B_{35}, B_{55}. On a shore $v_0v_1v_2v_3v_4v_0$ with bridge endpoint v_0, this edge is v_2v_3. There are no other D-vertices.*

Choose any maximum independent set T_0 of $J[D]$, and put $\kappa = |T_0|$. Then T_0 is independent in J, has no neighbor in U, and

$$\kappa = n(P_4) + n(C_3) + 2n(C_4) + 2n(C_5) + 3n(C_7) + n(B_{35}) + 2n(B_{55}),$$
$$\mu - g = L, \qquad \kappa + L = p + q_{\mathrm{def}}, \qquad 0 \le \kappa \le p + q_{\mathrm{def}},$$
$$p = n(C_3) + n(C_5) + n(C_7) + n(B_{33}) + n(B_{35}) + n(B_{55}) \ge 2.$$

(II) *Choose any original cycle C that is a whole ordinary odd cycle or one shore of B_{33}, B_{35}, B_{55}. Such a cycle exists. Put*

$$\ell = |V(C)|, \quad Z = V(C) \setminus (U \cup T_0), \quad e_C = |Z|, \quad Y = T_0 \cup Z, \quad d = |Y|.$$

Then $\ell \in \{3, 5, 7\}$, with barbell shores only of order three or five, and

$$e_C = \lfloor \ell/2 \rfloor + 1 \le 4, \qquad d = \kappa + e_C \le p + q_{\mathrm{def}} + 4.$$

Every $y \in Y$ has J-degree two and no neighbor in U_0.

For every missing pair $b_r + b_s = c_0(F) - 1$, put $j = u - b_r$, and let $m(Y)$ count the owned odd A-components touched by Y, with ownership defined using U. Then

$$d + m(Y) \le d + |U| \le j.$$

Thus the actual set Y satisfies the full assignment and degree hypotheses for simultaneous retention of all U and Y.

(III) *For any maximum matching M_Γ of Γ, let Q be the spanning graph on $V(A) \cup U \cup Y$ with edge set*

$$E(J[U \cup Y]) \setminus M_\Gamma.$$

The entire Q is the disjoint union of the chosen physical cycle C, g copies of P_2, and isolates, including all vertices of A. Hence

$$\nu(Q) = g + \lfloor \ell/2 \rfloor = g + e_C - 1.$$

Every maximum Γ-matching leaves exactly g pairwise disjoint Γ-edges. If C is a barbell shore, its bridge is a whole K_2 of Γ, belongs to every maximum matching, and leaves no surviving attachment to the other shore.

All choices occur in the same source, uniformly over component multiplicities. No whole-source exclusion or Ramsey formula is asserted.

Proof. The catalogue and odd-corona inputs required here are identified in Section 3.2.

Step 1. Components and cubic pieces. Proposition 8.3 gives native existence, six-regularity, $p \ge 2, \epsilon \ge 2, q_{\mathrm{def}} \ge 1, E \ge 6$, and $u = r + s$. The required catalogue has eighteen initial types and exact local parameter columns. The odd-corona exclusions remove $\mathrm{cor}(C_3), \mathrm{cor}(C_5), \mathrm{cor}(C_7)$, leaving

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7, B_{33}, B_{35}, B_{55}, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(P_6), \mathrm{cor}(C_4), S_6.$$

The same input gives

$$\Gamma \text{ a union of } K_1, K_2, P_4, C_4, \qquad |U| = 2s - b, \qquad \mu = s - b,$$

with U_0 exactly the cubic middles of whole $\mathrm{cor}(P_3)$'s.

Each barbell, $\mathrm{cor}(P_4)$, and S_6 contributes K_2 to Γ; $\mathrm{cor}(P_3)$ contributes K_1; $\mathrm{cor}(P_6)$ contributes P_4; and $\mathrm{cor}(C_4)$ contributes C_4. Ordinary degree-at-most-two types contribute none. A maximum matching is perfect on every nontrivial piece. Removing it leaves zero edges from K_2, the middle edge from P_4, two disjoint edges from C_4, and none from K_1. Thus every choice leaves g disjoint edges. The matching rank minus the residual-edge count is one on each barbell, $\mathrm{cor}(P_4), \mathrm{cor}(P_6), S_6$, and zero otherwise. Hence $\mu - g = L \geq 0$.

Step 2. The entire degree-two selector. A whole K_1 or P_2 has no degree-two vertex. The two internal vertices of an ordinary P_4 form its contribution. All vertices of an ordinary cycle qualify. In a barbell, the bridge endpoints are cubic. Every nonbridge triangle vertex touches its bridge endpoint. On a pentagon shore, v_1, v_4 touch U, while precisely v_2, v_3 qualify. A path corona has degree-one leaves, degree-two endpoint bases adjacent to cubic internal bases, and no further degree-two vertices. A cycle corona has only cubic bases and leaves. In S_6, each unleafed square vertex touches a cubic square vertex. These checks exhaust the fifteen types. Distinct whole components have no joining edges; the two pentagon interiors in B_{55} also have no joining edge.

A maximum independent set of finite $J[D]$ exists. Its restriction to each component is maximum, since a deficient restriction can be replaced independently. An edge contributes one, and C_3, C_4, C_5, C_7 contribute $1, 2, 2, 3$. For a cycle of order ℓ, pairing each independent vertex with its next cyclic neighbor gives the upper bound $\lfloor \ell/2 \rfloor$; alternating vertices attain it. This proves the formula for κ and its independence and anticompleteness assertions.

In the displayed fifteen-type order, the pairs (κ_T, L_T) are

$$(0,0),(0,0),(1,0),(1,0),(2,0),(2,0),(3,0),$$
$$(0,1),(1,1),(2,1),(0,0),(0,1),(0,1),(0,0),(0,1).$$

Their sums are

$$0,0,1,1,2,2,3,1,2,3,0,1,1,0,1,$$

the corresponding $p_T + q_{\mathrm{def}\,T}$ columns of the required catalogue. Summing gives $\kappa + L = p + q_{\mathrm{def}}$. The p-column is one precisely on the three ordinary odd cycles and the three barbells. Its sum gives the stated identity for p, so an eligible C exists.

Step 3. Completing an arbitrary eligible cycle. For an ordinary C_{2v+1}, the set T_0 uses exactly v vertices and U uses none, giving $e_C = v + 1$. A triangle shore has one cubic vertex and no T_0-vertex, giving $e_C = 2$. A pentagon shore has one cubic vertex and one selected interior vertex, giving $e_C = 3$. Thus the asserted counts hold for every eligible choice.

Every selected or added vertex has J-degree two. Vertices of T_0 have no U-neighbor. Added ordinary-cycle vertices have none because their cycle is whole. An added barbell-shore vertex can touch only its shore's bridge endpoint in U; that endpoint is nonisolated in Γ. Hence no member of Y touches U_0.

Step 4. Uniform assignment range. Proposition 8.3(IV) gives

$$j \geq p + s + \delta + 1 + u_6.$$

Distinct cubics own distinct odd components, so $m(Y) \leq |U|$. Using $|U| = 2s - b$, $\kappa = p + q_{\mathrm{def}} - L$, and $\delta = \epsilon + q_{\mathrm{def}}$, we obtain

$$j - |U| - \kappa \geq \epsilon - s + 1 + u_6 + b + L.$$

Here $u_6 \geq 2$, $\epsilon \geq s$, and $b, L \geq 0$, so the right side is at least three. This suffices when $e_C \leq 3$.

If $e_C = 4$, the cycle is an ordinary C_7. When $\epsilon > s$, the lower bound is at least $1 + 1 + 2 = 4$. When $\epsilon = s$, oddness and $\epsilon \geq 2$ give $s \geq 3$ and $t \geq 1$. The parameter identities give

$$\lambda = q_{\mathrm{def}} + \epsilon + t - 1, \qquad E = p + q_{\mathrm{def}} + \epsilon + t + 1 \geq 8.$$

Thus $u_6 \geq 3$, and the same bound is again at least four. Therefore

$$d + m(Y) \leq d + |U| \leq j$$

for every missing pair.

Step 5. The whole residual. All vertices of C are retained. No cycle edge belongs to Γ. An ordinary cycle has no exterior attachment. A barbell shore has just its bridge attachment; its whole K_2 cubic piece is removed by every maximum matching.

Every retained vertex outside $C \cup U$ lies in $T_0 \setminus V(C)$. It has no neighbor in T_0 or U. It has no neighbor in Z: different whole components are disjoint, and between shores the only edge is the bridge, whose endpoints are cubic. These selected vertices are therefore isolated. The remaining edges outside C are exactly the g residual cubic edges. They are disjoint from C, since a barbell shore's only cubic vertex loses its bridge. The A-vertices are isolated by definition of Q.

A cycle of order $2v + 1$ has matching number v, by vertex counting and alternating edges. Matching numbers add over disjoint components, by restriction for the upper bound and union of maximum matchings for attainment. Hence $\nu(Q) = g + e_C - 1$. □

Proposition 8.5 (Four compatible sextic matchings and exact fourth remainder)**.** *Let $F, A, J, U, \Gamma, U_0, b, \mu, u$ be as in Proposition 8.4. Put*

$$z = a - p - \delta, \qquad t = (s-1)/2, \qquad u_6 = \lfloor (2E-3)/4 \rfloor.$$

Choose any missing pair $b_r + b_s = c_0(F) - 1$, and put $j = u - b_r$. For $Y \subseteq V(J) \setminus U$, let $m(Y)$ count the owned odd A-components touched by Y. Assume

$$d = |Y| \leq p + q_{\mathrm{def}} + 4, \qquad d + m(Y) \leq j.$$

Retain

$$H = F[V(A) \cup U \cup Y], \qquad h = |U| + d, \qquad N = u - h.$$

(I) *For every prescription of one original edge from each cubic vertex to its unique owned odd component, there exists a matching M_0 consisting of those owner edges, one edge from each Y-vertex to an exposed port of a distinct free odd component, and a maximum internal matching in every whole A-component. All exterior destinations are distinct odd components.*

Fix any such M_0. Define

$$P_i = \left(V(H), E(H) \setminus \bigcup_{\ell=0}^{i-1} M_\ell \right), \qquad i = 1, 2, 3, 4,$$

choosing successively any maximum M_1 in P_1, any maximum M_2 in P_2, and any maximum M_3 in P_3. Then

$$\Gamma \text{ is a union of } K_1, K_2, P_4, C_4, \quad |U| = 2s - b, \quad \mu = s - b, \quad u = r + s,$$

and its isolates are precisely the middles of whole $\mathrm{cor}(P_3)$'s. The omission margin is

$$N \geq p + q_{\mathrm{def}} + h - 1.$$

The exact ranks and restored totals are

	$\nu(P_i)$	$N + \nu(P_i)$
$i = 1$	$p + q_{\mathrm{def}} + h$	$u + p + q_{\mathrm{def}} = c_1(F)$
$i = 2$	$p + h$	$u + p = c_2(F)$
$i = 3$	$\mu + d = h - s$	$r = c_3(F)$.

The entire first row residual is $pP_3 + q_{\mathrm{def}}P_2$ plus isolates. The matching M_1 is maximum on that row and has exactly h crossing edges saturating the exterior. The entire second row residual is pP_2 plus isolates, all of whose edges belong to M_2; M_2 has exactly h further crossing edges. Neither M_1 nor M_2 uses an exterior-internal edge. The first three matchings contain every original A-edge and every original U-to-A edge. The entire Γ survives in P_3.

(II) *Every maximum M_3 of this same P_3 consists of a maximum Γ-matching M_Γ of size μ and exactly d edges saturating Y into $V(A) \cup U_0$. It has no edge within Y and no edge from Y to $U \setminus U_0$. Its Γ-part can be replaced by any maximum Γ-matching while retaining all other edges.*

If every $y \in Y$ has J-degree two and no J-neighbor in U_0, then each of M_0, M_1, M_2, M_3 matches y into A, using four distinct original edges. Their union contains every Y-to-A edge. The entire P_4 then has edge set

$$E(J[U \cup Y]) \setminus M_\Gamma,$$

with all A-vertices isolated.

(III) *For $i = 1, 2, 3, 4$, put*

$$K_i = \left(V(H), \bigcup_{\ell=0}^{i-1} M_\ell \right), \qquad \rho_4 = \nu(P_4).$$

The graph K_i is a union of i edge-disjoint matchings, has maximum degree at most i, and M_0 covers every degree-i vertex. For every nonnegative integer capacity split totaling i and budget split totaling $N + \nu(P_i)$, there is a full F-certificate leaving exactly K_i in the degree-bounded parts inside H. Every source edge, including edges between omitted vertices, is included. All chosen matching and owner edges remain in those degree-bounded parts. In particular,

$$\rho_4 \le \mu + d, \qquad N \ge \max\{0, \rho_4 - 1\}, \qquad c_4(F) \le N + \rho_4.$$

These are conditional matching and payment interfaces at every sextic deficit. They give no surviving-cell or whole-degree exclusion.

Proof. The catalogue order bound and the two restoration inputs are missing as recorded in Section 3.2.

Step 1. Numerical margin. Proposition 8.3 supplies the source and row data, ownership, and

$$c_1(F) = u + p + q_{\mathrm{def}}, \quad c_2(F) = u + p, \quad c_3(F) = r.$$

The required catalogue bound gives positive odd $s \le \epsilon$ and

$$z \ge \epsilon + q_{\mathrm{def}} + 2 + 2(\lambda - \delta + 1) + 2u_6 = \epsilon + q_{\mathrm{def}} + s + 1 + 2u_6.$$

The odd-corona input gives the asserted $\Gamma, U_0, |U|, \mu$.

Since $s = 2t + 1$ and $\delta = \epsilon + q_{\mathrm{def}}$,

$$\lambda = q_{\mathrm{def}} + \epsilon + t - 1, \qquad E = p + q_{\mathrm{def}} + \epsilon + t + 1.$$

The row inventory gives

$$r = 3p + 2\delta + z = 3p + 2\epsilon + 2q_{\mathrm{def}} + z.$$

Indeed, the component vertex count

$$z + 2b_A + 4t_A + 3n_3 + 4n_4 + 5n_5 + 7n_7$$

equals $3p + 2\delta + z$ by the row identities.

For integer E,

$$2\left\lfloor \frac{2E - 3}{4} \right\rfloor \ge E - 3 :$$

the left side is $E-2$ for even E and $E-3$ for odd E. Hence

$$z \geq p + 2\epsilon + 2q_{\mathrm{def}} + s + t - 1.$$

Using $h = 2s - b + d$, $N = r + s - h$, and $d \leq p + q_{\mathrm{def}} + 4$, we obtain

$$\begin{aligned} N - (p + q_{\mathrm{def}} + h - 1) &= 2p + 2\epsilon + q_{\mathrm{def}} + z - 3s + 1 + 2b - 2d \\ &\geq 3p + 4\epsilon + 3q_{\mathrm{def}} - 2s + t + 2b - 2d \\ &\geq p + 4\epsilon + q_{\mathrm{def}} - 2s + t - 8 + 2b. \end{aligned}$$

If $\epsilon \geq 3$, this is at least

$$p + 2\epsilon + q_{\mathrm{def}} + t - 8 + 2b \geq 2 + 6 + 1 - 8 = 1.$$

If $\epsilon = 2$, positive odd $s \leq 2$ gives $s = 1, t = 0$, and the lower bound is $p + q_{\mathrm{def}} - 2 + 2b \geq 1$. This proves the margin. Independently, $N \geq 0$ because it counts actual omitted vertices.

Step 2. All prescribed owners in one matching. Proposition 8.3(IV) assigns Y to distinct free odd components under $d + m(Y) \leq j$. Different cubics have different owned components. Every vertex of an odd native component is exposable: an isolate remains unmatched, and deleting a vertex of an odd cycle leaves an even path with a perfect matching. Choose the prescribed cubic edges and maximum internal matchings avoiding all chosen ports. Free and owned destinations are disjoint, and distinct components separate all crossing edges. Choose arbitrary maximum internal matchings in unassigned components. Their union is the required M_0.

Step 3. The first two tight costs. Removing a maximum matching from P_2 leaves isolates; from P_4, its middle edge; and from C_4, two disjoint edges. On C_{2v+1}, $v \in \{1, 2, 3\}$, it leaves one P_3 at the exposed vertex and $v - 1$ disjoint edges. To check this, remove the exposed vertex first; the remaining even path has its unique alternating perfect matching. The row relations therefore give

$$P_1[V(A)] = pP_3 + q_{\mathrm{def}}P_2 + \text{isolates},$$

with matching number $p + q_{\mathrm{def}}$.

Apply the required fixed-first-matching rank theorem with both equal-order rows chosen to be A, fixed matching M_0, exterior size h, and row rank $p + q_{\mathrm{def}}$. Distinct rows are not required. Step 1 verifies its margin and exact cost hypothesis. It gives

$$\nu(P_1) = p + q_{\mathrm{def}} + h$$

and the asserted form of every maximum M_1. Removing its internal maximum matching leaves exactly p row edges.

Every P_2-matching has at most p internal row edges and at most h other edges. Thus $\nu(P_2) \leq p + h$. The union $K_2 = M_0 \cup M_1$ has degree at most two, and M_0 covers every degree-two vertex. Step 1 gives $N \geq \nu(P_2) - 1$. The required full restoration theorem yields

$$c_2(F) \leq N + \nu(P_2) \leq u - h + p + h = u + p = c_2(F).$$

Equality forces $\nu(P_2) = p + h$. Every maximum M_2 attains both component bounds. Its p internal edges are all row edges. The other h edges use the h exterior vertices exactly once; an exterior-to-exterior edge would consume two and prevent equality. Hence all are crossing edges.

Step 4. The third cost and the fourth matching. All row edges have been removed. Each cubic has exactly three original A-neighbors, since F is six-regular. The first three matchings use three distinct such edges at every cubic, exhausting them. No exterior-to-exterior edge has been removed, so Γ survives. Every other P_3-edge meets Y. Consequently

$$\nu(P_3) \leq \mu + d = |U| - s + d = h - s.$$

The union K_3 has degree at most three, with M_0 covering every degree-three vertex. The margin gives $N \geq \mu + d - 1$. Full restoration gives

$$r = c_3(F) \leq N + \nu(P_3) \leq u - h + h - s = u - s = r.$$

Thus $\nu(P_3) = \mu + d$.

Every maximum M_3 attains the μ cubic-edge bound and the d remaining-edge bound. Its cubic part is perfect on each nontrivial K_2, P_4, C_4, covering exactly $U \setminus U_0$. Its other d edges saturate Y, with one Y-endpoint each and their other endpoints in $A \cup U_0$. This excludes edges inside Y and into already covered nonisolated cubics. Replacing the cubic matching preserves exactly its covered set, so the other edges remain disjoint.

Under the additional degree hypothesis, no Y-vertex can use U_0, so its fourth edge also goes into A. Six-regularity and $d_J(y) = 2$ give exactly four original A-neighbors. The four edge-disjoint matchings exhaust these edges; simplicity gives distinct endpoints. Only M_Γ is removed inside the exterior. This proves the exact P_4 formula.

Step 5. Full restoration. A degree-i vertex of K_i is incident with an edge of each of its i matchings, including M_0. The paid and degree-bounded parts partition $E(H)$. Also

$$\rho_4 \le \nu(P_3) = \mu + d.$$

The margin from Step 1 gives $N \ge \max\{0, \nu(P_i) - 1\}$ at all four indices. Apply the required restoration theorem to each fixed partition.

For the omitted-edge accounting, order the N omitted vertices and assign each edge meeting that set to its first omitted endpoint. This gives N possibly empty star pieces, each edge appearing once, including edges with two omitted endpoints. All other edges lie in the induced H and in exactly one of K_i, P_i. The restoration construction distributes the star pieces and the paid graph within every budget split of total $N + \nu(P_i)$, retaining K_i in the degree-bounded parts. No selected edge is charged again. □

8.3. **Octic selectors and matching prefixes.**

Proposition 8.6 (Actual octic triangle selector)**.** *Let $F \in \mathcal{M}_8$, put $q = q_8$, $\zeta = \delta - \epsilon$, $\beta = p + \zeta$, and $u = r + s$. Fix any native row A and any complete continuation on $J_5 = F - V(A)$. This continuation is part of an actual complete source process obtained by deleting a maximum independent set and the full degree-seven and degree-six phases inside A, with the prescribed later continuation unchanged.*

Let B_5, B_4 be the complete degree-five and degree-four root sets, and put

$$K = J_5 - (B_5 \cup B_4).$$

Then

$$|B_5| = q, \quad |B_4| = s - 2q, \quad |V(K)| = r + q, \quad \alpha(K) = a.$$

Put

$$U = \{v \in V(K) : d_K(v) = 3\}, \quad \Gamma = K[U], \quad U_0 = \{v \in U : d_\Gamma(v) = 0\}, \quad \mu = \nu(\Gamma).$$

Let $n(T)$ count whole K-components, and define

$$\begin{aligned} g &= n(\mathrm{cor}(P_6)) + 2n(\mathrm{cor}(C_4)), \\ L &= n(B_{33}) + n(B_{35}) + n(B_{55}) + n(L_\square) + n(\mathrm{cor}(P_4)) + n(\mathrm{cor}(P_6)), \\ D &= \{v : d_K(v) = 2,\ N_K(v) \cap U = \varnothing\}. \end{aligned}$$

(I) *The complete necessary whole-component list is*

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7, B_{33}, B_{35}, B_{55}, L_\square, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(P_6), \mathrm{cor}(C_4).$$

The graph Γ is a union of K_1, K_2, P_4, C_4, and every maximum matching covers every nonisolated vertex. Moreover,

$$|U| = q + \mu, \qquad \mu - g = L, \qquad 0 \le g \le q \le \epsilon.$$

The entire $K[D]$ consists of every whole ordinary C_3, C_4, C_5, C_7, the middle edge of every whole ordinary P_4, and one edge from each pentagon shore of a barbell. With bridge endpoint v_0 on $v_0v_1v_2v_3v_4v_0$, that edge is v_2v_3. There are no other D-vertices.

(II) *Choose any maximum independent set T_0 of $K[D]$, and put $\kappa = |T_0|$. It is independent in K and has no neighbor in U. Moreover,*

$$\kappa = n(P_4) + n(C_3) + 2n(C_4) + 2n(C_5) + 3n(C_7) + n(B_{35}) + 2n(B_{55}), \qquad \kappa + L = \beta.$$

There are at least $p - \delta \geq 2$ whole ordinary triangles. Choose any one, C, and put

$$Z = V(C) \setminus (U \cup T_0), \qquad e_C = |Z|, \qquad Y = T_0 \cup Z.$$

Then

$$e_C = 2 = (|V(C)| + 1)/2, \qquad |Y| = \kappa + 2.$$

Every $y \in Y$ has K-degree two and no neighbor in U, including U_0. Every U-vertex has exactly five original neighbors outside K; every Y-vertex has exactly six. The graph $K[Y]$ is the whole triangle C and isolates.

(III) *For any maximum M_Γ, let Q be the spanning graph on*

$$V(A) \cup B_5 \cup B_4 \cup U \cup Y$$

with edge set $E(K[U \cup Y]) \setminus M_\Gamma$. Its entire graph is C, exactly g disjoint edges, and isolates. All vertices of A, B_5, B_4 are isolated. Thus

$$\nu(Q) = g + 1 = g + e_C - 1.$$

Put

$$W = B_5 \cup B_4 \cup U \cup Y, \quad H = F[V(A) \cup W], \quad h = |W|, \quad N = |V(F) \setminus V(H)|.$$

Then

$$h = s + \mu + \kappa + 2 = s + \beta + g + 2, \qquad N = u - h,$$

and

$$N + \nu(Q) = r - \beta - 1 = c_6(F) - 1.$$

Every choice concerns the same arbitrary original source. This statement does not construct six sequential matchings, identify their residual with Q, assign compatible original-row owner ports, supply a certificate for Q and the omitted edges, or exclude any source class.

Proof. The octic row and catalogue inputs are identified in Section 3.3.

Step 1. Source and actual cubic entry. Proposition 8.1(IV) gives eight-regularity and

$$E \geq 10, \quad 2 \leq \epsilon \leq \delta \leq \lambda \leq \lfloor (E-4)/2 \rfloor, \quad \lambda \geq 3, \quad p \geq \lambda + 2, \quad q \geq 1.$$

The additional required row input gives $\delta \geq 3$. Thus

$$E \geq 8, \qquad 1 \leq \delta \leq \lambda \leq E - 3, \qquad p - \delta \geq 2.$$

The inequality $(E-4)/2 \leq E - 3$ holds for $E \geq 2$, checking the stated range.

The required replay theorem realizes any native A through the initial independent set and the complete degree-seven and degree-six phases, leaving every prescribed later graph unchanged. The continuation word is

$$5^q 4^{s-2q} 3^q.$$

Therefore B_5, B_4, K are actual, and

$$|K| = (r + s) - q - (s - 2q) = r + q.$$

Prefix independence gives $\alpha(K) = a$. Every complete cubic phase has a native terminal with the original p, δ, ϵ. The required local-phase theorem permits simultaneous maximal independent selections in separate cubic components.

Step 2. Complete catalogue and cubic residual. The required catalogue excludes whole $\mathrm{cor}(C_5), \mathrm{cor}(C_7)$ and initially permits K_4. For $q \geq 2$, its exclusion clause applies because $\epsilon \geq 2$.

For $q = 1$, the separate required exclusion applies with the numerical hypotheses checked in Step 1. These alternatives exhaust integer $q \geq 1$. Removing K_4 gives exactly the fifteen stated types, as a necessary list only.

The cubic graph is then a union of K_1, K_2, P_4, C_4. Its isolates are the middles of whole $\mathrm{cor}(P_3)$'s. If $b = n(\mathrm{cor}(P_3))$, the required count gives

$$|U| = 2q - b, \qquad \mu = q - b,$$

hence $|U| = q + \mu$. Each barbell, $L_\square$, and $\mathrm{cor}(P_4)$ contributes K_2; $\mathrm{cor}(P_6)$ contributes P_4; and $\mathrm{cor}(C_4)$ contributes C_4. Every maximum matching is perfect on these nontrivial pieces. Removing it leaves one edge from P_4, two disjoint edges from C_4, and none from K_2. Thus it leaves g disjoint edges, and subtracting this count from the local matching ranks gives $\mu - g = L$.

A $\mathrm{cor}(P_6)$ contributes two roots to a full cubic phase and one to g; a $\mathrm{cor}(C_4)$ contributes two to each. Other types contribute zero to g and nonnegative phase counts. Hence $g \leq q$. The required local identity is

$$\epsilon - q = n(P_2) + n(P_4) + n(\mathrm{cor}(P_3)) + n(\mathrm{cor}(P_4)) + n(\mathrm{cor}(P_6)),$$

so $q \leq \epsilon$.

Step 3. The degree-two selector and its original parameters. Inspecting all fifteen types gives the stated D. The types K_1, P_2 contribute none. An ordinary P_4 contributes its internal edge; ordinary cycles contribute all vertices. In a barbell, the bridge endpoints are cubic; triangle-shore vertices touch them, while precisely the two inner pentagon vertices avoid them. In $L_\square$, each remaining degree-two square vertex touches a cubic vertex. In path coronas, each degree-two endpoint base touches an internal cubic base; cycle-corona bases are cubic and leaves have degree one. This exhausts the list.

The maximum set T_0 exists by finiteness and is maximum on every component of $K[D]$. An edge contributes one; a cycle contributes $\lfloor j/2 \rfloor$, attained by alternating vertices and bounded by pairing selected vertices with their clockwise successors. This gives the formula for κ.

To establish $\kappa + L = \beta$ with the original parameters, choose an actual complete cubic phase X, and put $R = K - X$. For each whole K-component T, let o_T, m_T, f_T be its terminal odd-cycle count, terminal matching number, and terminal positive-even-path count. Then

$$\sum_T o_T = p, \qquad \sum_T m_T = p + \delta, \qquad \sum_T f_T = \epsilon.$$

The complete local table is

T	o_T	m_T	f_T	κ_T	L_T
K_1	0	0	0	0	0
P_2	0	1	1	0	0
P_4	0	2	1	1	0
C_3	1	1	0	1	0
C_4	0	2	0	2	0
C_5	1	2	0	2	0
C_7	1	3	0	3	0
B_{33}	1	2	1	0	1
B_{35}	1	3	1	1	1
B_{55}	1	4	1	2	1
$L_\square$	0	2	1	0	1
$\mathrm{cor}(P_3)$	0	2	2	0	0
$\mathrm{cor}(P_4)$	0	3	2	0	1
$\mathrm{cor}(P_6)$	0	4	3	0	1
$\mathrm{cor}(C_4)$	0	2	2	0	0

Here are the terminal checks. The first seven types have no cubic vertices and remain unchanged; alternating edges and vertex counting give their matching numbers. A barbell phase deletes one bridge endpoint, leaving the opposite odd cycle and an even path of order one less than the deleted endpoint's shore. Its matching number is $2, 3, 4$, respectively, independently of the endpoint. Deleting either cubic vertex of $L_{\square}$ leaves P_4 and an isolate.

For $\mathrm{cor}(P_t)$, $t = 3, 4, 6$, a local cubic phase selects a maximal independent set of internal bases of sizes $b_T = 1, 1, 2$. The selected bases' leaves become isolates. Each of the $t - b_T$ unselected bases keeps its leaf, giving a matching of that size. Every remaining edge meets an unselected base, proving maximal rank. The remaining base graph has $t-1-2b_T$ edges: each selected base has base degree two, and selected bases are independent. Its maximum degree is at most one, because every unselected internal base has a selected neighbor. Its number of components is

$$(t - b_T) - (t - 1 - 2b_T) = b_T + 1.$$

Their coronas are P_2's or P_4's. This gives the listed even-path counts and no odd cycles. In $\mathrm{cor}(C_4)$, opposite selected bases leave two P_2's and two isolates. These checks cover every local phase choice.

Each row of the table satisfies $m_T - f_T = \kappa_T + L_T$. Additivity of matching number follows by restricting any matching to components and taking the union of component maxima. Summing therefore gives

$$\kappa + L = p + \delta - \epsilon = \beta.$$

Step 4. A whole ordinary triangle. Put $\delta_T = m_T - o_T$. All δ_T are nonnegative. Except for ordinary C_3, every type satisfies $\delta_T \geq o_T$. Among types with nonzero o_T, the values for $C_5, C_7, B_{33}, B_{35}, B_{55}$ are $1, 2, 1, 2, 3$. For ordinary C_3, $o_T = 1, \delta_T = 0$. Hence

$$\delta \geq p - n(C_3), \qquad n(C_3) \geq p - \delta \geq 2.$$

This concerns the same actual K, with no phase normalization.

Every maximum T_0 meets a whole ordinary triangle once. Completing any chosen triangle adds exactly two vertices, all in D. Thus all Y-vertices have degree two and no cubic neighbor. The whole-triangle condition and independence of T_0 give $K[Y] = C$ plus isolates. Eight-regularity gives the outside-neighbor counts five and six.

Step 5. Exact residual count. The sets U, Y are disjoint and anticomplete. Removing M_Γ leaves g disjoint edges on U, while $K[Y]$ is unchanged. Adjoining A, B_5, B_4 as isolates gives exactly Q. Its matching number is $g + 1$, attained by all g edges and one triangle edge, and bounded componentwise.

The phase and selector sets are disjoint, so

$$h = q + (s - 2q) + (q + \mu) + (\kappa + 2) = s + \beta + g + 2.$$

Since W is actual, $N = u - h \geq 0$. Therefore

$$N + \nu(Q) = r + s - (s + \beta + g + 2) + (g + 1) = r - \beta - 1.$$

Finally,

$$r - a = (k - a) - \lambda - 1 + \delta = 2p + \delta,$$

so $r - \beta = a + p + \epsilon$. The required exact cost identity is $c_6(F) = a + p + \epsilon$. This proves the numerical debit identity without a sequential-matching or certificate conclusion. □

Proposition 8.7 (Octic three-matching prefix with prescribed retained owners)**.** *Let F be an eight-regular member of $\mathcal{M}_8$, and put*

$$q = q_8, \qquad \zeta = \delta - \epsilon, \qquad \beta = p + \zeta.$$

Native rows exist. Fix any native A, any integers

$$0 \leq g \leq \min\{q, \epsilon\}, \qquad 1 \leq e_{\mathrm{sel}} \leq 4,$$

and put

$$J = F - V(A), \quad u = |V(J)|, \quad h = s + \beta + g + e_{\mathrm{sel}}.$$

Every h-set $W \subseteq V(J)$ is permitted. Such sets exist. For

$$H = F[V(A) \cup W], \qquad N = u - h,$$

one has $N \geq \max\{0, \beta + h - 1\}$.

Put $Q_5 = \{v \in V(J) : d_J(v) = 5\}$ and $Q_W = Q_5 \cap W$. Every $c \in Q_W$ has a unique odd whole A-component $O_A(c)$ with a source edge to c, and these components are distinct. Before choosing other destinations, prescribe any actual edge ct_c into each retained owner, any maximum internal matching of that owner exposing t_c, any perfect matching in every even whole component, and any maximum internal matching in each odd whole component with no W-neighbor. One maximum matching M_0 of the full H contains all prescriptions, is internally maximum in every whole A-component, and matches all W into exposed ports of distinct odd whole components. Nonquintics use no retained quintic owner. Owners of omitted quintics are not reserved.

Fix any such M_0. Put $P_0 = H$, and choose any successive maximum matchings M_1 in $P_1 = P_0 - M_0$ and M_2 in $P_2 = P_1 - M_1$; put $P_3 = P_2 - M_2$.

(I) *The exact ranks are*

$$\nu(P_0) = p + \delta + h, \qquad \nu(P_1) = \beta + h, \qquad \nu(P_2) = p + h.$$

For every native row R contained in H, the same three matchings each saturate $W_R = V(H) \setminus V(R)$ into R, with no edge inside W_R. The restriction of M_0 is maximum on every whole R-component. The entire first row residual is

$$pP_3 + \zeta P_2 + \mathit{isolates},$$

and M_1 is maximum there. The entire second row residual consists of p edges and isolates, and M_2 contains all those edges. The matching union contains every edge of every such R. Every original $H[W_R]$-edge survives in P_3.

(II) *For every such R and every retained $v \in W_R$ of degree five in $F - V(R)$, all three original v-to-R edges belong to the same matching union. In particular, all retained original quintic-to-A edges are cleared.*

(III) *For $i = 1, 2$, put*

$$K_i = \left(V(H), \bigcup_{j=0}^{i-1} M_j \right), \qquad \widehat{D}_i = N + \nu(P_i).$$

Every capacity split totaling i and every budget split totaling $\widehat{D}_i$ admits a certificate on the entire F, with the degree-bounded union inside H exactly $E(K_i)$. All omitted-incident edges, including those with two omitted endpoints, are included. Every fixed owner or internal M_0-edge remains in the degree-bounded union. The totals are neutral:

$$\widehat{D}_1 = u + \beta = c_1(F), \qquad \widehat{D}_2 = u + p = c_2(F).$$

No independent matching, Hall inequality, omission premise, independence of W, or retention of all Q_5 is assumed. The only condition on W is its stated size. No later matching, strict payment, whole-source exclusion, or Ramsey formula is asserted.

Proof. The three required inputs are identified in Section 3.4. They use the same source class and certificate definitions; their first-deficit notation is $\zeta = q_{\mathrm{def}}$.

Step 1. Numerical hypotheses on the fixed graph. The required retained-size theorem supplies native rows and a missing pair $x + y = c_0(F) - 1$. Fix one such pair. For the stated

g, e_{sel}, A, it gives $h \le u$. Thus h-subsets exist. For every such W, it gives the actual omission $N = u - h \ge 0$, the assignment bound

$$h \le j_x = u - x,$$

and $N \ge \beta + h - 1$. These bounds precede any matching choice. Fix the arbitrary W; it will not change.

Step 2. A prescribed first matching. Apply the required original-row matching theorem to the same A, W and missing pair. Its threshold is the same j_x, so its size hypothesis holds. It gives unique distinct retained owners and permits exactly the prescribed edges and internal matchings before any remaining destination is chosen. It supplies one maximum M_0 of size

$$\nu(A) + h = p + \delta + h,$$

with the asserted internal and crossing structure. Retained nonquintics avoid all retained quintic owners. The stronger version requiring all quintics is not used, and no omitted owner is reserved.

Step 3. Sequential ranks on this graph. Fix any M_0 with those properties. The required rank theorem assumes its size and the omission margin, both already proved on this H. It permits arbitrary successive maxima on the actual edge residuals. It gives the three ranks, simultaneous saturation relative to every contained native row, and the entire two residual rows. It also gives clearance of every retained degree-five boundary and preservation of all exterior-internal edges. Thus (I) and (II) concern one sequence, not separately chosen witnesses.

Step 4. Original-source certificates. The same rank theorem's restoration clause gives every integer capacity and budget split in (III), with the exact first-i matching union. It includes all omitted-incident edges and keeps every M_0-edge in the degree-bounded parts. The exact source identities give

$$N + \beta + h = u + \beta = c_1(F), \qquad N + p + h = u + p = c_2(F).$$

The order of choices is source, row, permitted parameters, retained set, prescriptions, one M_0, then arbitrary successive maxima. No additional restriction on the quintic, quartic, or other parts of W is introduced. □

8.4. **Degree-ten strict and equality prefixes.**

Proposition 8.8 (Strict degree-ten all-root terminal-cycle prefix)**.** *Let $F \in \mathcal{M}_{10}$ with $\delta > 0$, and put $u = r + s$, $m = p + \delta$, $\beta = m - \epsilon$. Fix any native row A and any complete continuation on $F - V(A)$. Let B_j, $3 \le j \le 7$, be its entire degree-j phases, allowing empty phases. Put*

$$X_{\mathrm{root}} = \bigcup_{j=3}^{7} B_j,$$

and let R be its terminal. Let $D_2 = \{v \in V(R) : d_R(v) = 2\}$. Choose any maximum independent set T_0 of $R[D_2]$ and any shortest whole odd-cycle component C of R. These choices and native rows exist. Define

$$e_C = (|V(C)| + 1)/2, \quad Y = T_0 \cup V(C), \quad W = X_{\mathrm{root}} \cup Y,$$
$$h = |W|, \quad H = F[V(A) \cup W], \quad S = V(F) \setminus V(H), \quad N = |S|.$$

(I) *One has*

$$p \ge 5, \quad \beta \ge p, \quad |X_{\mathrm{root}}| = s, \quad |T_0| = \beta, \quad 2 \le e_C \le 4,$$

and

$$|Y| = \beta + e_C, \quad h = s + \beta + e_C, \quad N = u - h = r - \beta - e_C.$$

Moreover,

$$N \ge \max\{0, \beta + h - 1\}, \qquad N \ge e_C - 2, \qquad N \ge \max\{0, h - |B_7| - 2\}.$$

Missing pairs exist, and every missing pair $b_R + b_B = c_0(F) - 1$ satisfies

$$h \le u - \max\{b_R, b_B\}.$$

(II) *Starting from $P_0 = H$, choose any successive maximum matching M_i of P_i, and put $P_{i+1} = P_i - M_i$, $i = 0, 1, 2$. Then*

$$\nu(P_0) = |M_0| = m + h, \quad \nu(P_1) = |M_1| = \beta + h, \quad \nu(P_2) = |M_2| = p + h.$$

Each matches every W-vertex into A and uses no edge within W. The first is internally maximum on every whole component, with crossing endpoints exposed in distinct odd whole components. The entire first row residual is

$$pP_3 + (\beta - p)P_2 + \textit{isolates},$$

and M_1 is internally maximum there. The second is pP_2 plus isolates, all those edges belong to M_2, and $P_3[V(A)]$ is edgeless. Every original $H[W]$-edge survives.

(III) *Every B_j is independent in F. Every $v \in B_7$ has exactly three original A-neighbors, and their physical edges occur one in each of M_0, M_1, M_2. Hence $V(A) \cup B_7$ is independent in P_3, and $W \setminus B_7$ is a vertex cover of size $h - |B_7|$.*

For

$$A_j = V(A) \cup \bigcup_{j<\ell\le 7} B_\ell,$$

every $v \in B_j$ has exactly $10 - j$ original neighbors in A_j and exactly $7 - j$ neighbors there in the same P_3. Every $y \in Y$ has exactly eight original neighbors in $V(A) \cup X_{\mathrm{root}}$, and exactly five there in P_3. If a retained $w \in W$ has exactly one A-neighbor exposed in an odd whole component, its physical edge belongs to every permitted M_0.

(IV) *For $i = 1, 2$, put $L_i = \bigcup_{j=0}^{i-1} M_j$. Every capacity split totaling i and budget split totaling $N + \nu(P_i)$ admits a certificate on the entire F, with $Q_R \cup Q_B = L_i$. Every edge meeting S, including one with two endpoints in S, is paid. The restored totals are*

$$N + \nu(P_1) = u + \beta = c_1(F), \qquad N + \nu(P_2) = u + p = c_2(F).$$

The statement adds no initial-matching, margin, positive-phase, or owner-selection premise. It asserts no strict payment or degree-ten exclusion.

Proof. The three technical inputs are identified in Section 3.5.

Step 1. All original choices and margins. The required strict-source theorem gives ten-regularity, $p \ge 5$, $0 \le \epsilon \le \delta$, native existence, and native components

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

There are exactly p odd cycles and ϵ positive even-order paths, with matching number $m = p + \delta$. Thus $\beta = m - \epsilon \ge p$. The same theorem gives maximum degree at most seven in $F - V(A)$, actuality of every prescribed continuation, $|X_{\mathrm{root}}| = s$, and the partition into A, X_{root}, R, with R native. Nonincreasing maximum degree ensures the stated phases exhaust all roots.

The required terminal-selector theorem applies to these same arbitrary objects and supplies the selector, all cardinalities, the cycle bound, the first two omission bounds, and the bound for every missing pair. It also says that every $y \in Y$ has R-degree two and eight original neighbors outside R, all in retained $A \cup X_{\mathrm{root}}$. Since $\beta \ge 5$,

$$\beta + h - 1 \ge h - |B_7| - 2.$$

Together with $N \ge 0$, this gives the remaining omission bound.

Step 2. One generic prefix application. The source identities are

$$c_0(F) = u + m, \qquad c_1(F) = u + \beta, \qquad c_2(F) = u + p.$$

For the first,

$$u + m = (k - \lambda - 1 + \delta) + (2\lambda + 3 - 2\delta) + (E - 2 - \lambda + \delta) = k + E.$$

The generic theorem permits all listed native component types, including original whole P_3's. Its row order, matching number, positive-even-path count, odd-cycle count, and residual rank are r, m, ϵ, p, β.

Fix one original missing pair. The theorem's threshold $j_{\mathrm{hi}} = u - \min\{b_R, b_B\}$ is at least $u - \max\{b_R, b_B\}$, so the retained-size hypothesis holds. Also

$$N = u - h \geq \beta + h - 1 \quad \Longrightarrow \quad 2h \leq u - \beta + 1 \leq u - \beta + 2.$$

Thus both generic retained-range hypotheses hold on this H, with omitted set exactly S. Its conclusion gives the ranks for every successive maximum choice and the row structure with compatible row A. Finite graphs have maximum matchings, and the residual definitions make the three chosen edge sets disjoint. The unique exposed-neighbor clause gives the last assertion of (III). Its restoration clause gives (IV), and $N + h = u$ yields the two exact totals.

Step 3. Physical phase counts. At the entry to a nonempty degree-j phase, every root later deleted in that phase has degree j: its degree cannot increase, is at most j at entry, and equals j at deletion. If two roots were adjacent, deleting the earlier one would lower the later one below j. Hence B_j is independent. Empty phases satisfy this too.

The vertices deleted before that phase are exactly A_j. Ten-regularity gives $10 - j$ original neighbors there for every $v \in B_j$. For $j = 7$, this is three neighbors in A. Each of the first three matchings uses one distinct v-to-A edge, exhausting those three. Since the row is edgeless afterward and B_7 was independent, $A \cup B_7$ is independent in P_3; its complement is the asserted cover.

For arbitrary j, all earlier-side edges are retained in H, and precisely three, all into A, are removed at v. No edge within W is removed. The remaining count is $10 - j - 3 = 7 - j$. Each $y \in Y$ likewise begins with eight retained earlier-side edges and loses three into A, leaving five. These counts allow zero-length phases and every maximum-matching choice. □

Proposition 8.9 (Degree-ten equality prefix)**.** *Let $F \in \mathcal{M}_{10}$ satisfy $c_2(F) = c_0(F)$. Put*

$$p = E - 2 - \lambda, \quad z = a - p, \quad r = k - \lambda - 1, \quad s = 2\lambda + 3, \quad u = r + s, \quad u_{10} = \left\lfloor \frac{2E - 3}{8} \right\rfloor.$$

Here a native row is an induced copy of p whole triangles and z isolates.

(I) *Native rows exist. The source is ten-regular, connected, factor-critical, and well-covered, with*

$$E \geq 12, \quad p \geq 5, \quad u_{10} \geq 2, \quad r = 3p + z = a + 2p, \quad |V(F)| = 2r + s.$$

Fix any native A and any complete continuation on its complement. Its terminal R is native. Write B_6, B_5, B_4 for the complete root sets and $X = B_6 \cup B_5 \cup B_4$. For some nonnegative integer t,

$$|B_6| = |B_4| = t, \qquad |B_5| = s - 2t > 0$$

with $s - 2t$ odd. Each block is independent, and $B_7 = B_3 = \varnothing$. The sets $V(A), X, V(R)$ partition $V(F)$. For $j \in \{6, 5, 4\}$, each $v \in B_j$ has exactly $10 - j$ original neighbors in $V(A) \cup \bigcup_{\ell > j} B_\ell$.

The exact costs are

$$c_0 = c_1 = c_2 = u + p, \quad c_3 = u, \quad c_4 = u - t, \quad c_5 = r + t,$$
$$c_6 = c_7 = r, \quad c_8 = a + p = r - p, \quad c_9 = a.$$

(II) *Choose any T_0 containing one vertex of each whole R-triangle and no isolate, and any whole triangle C. Put*

$$Y = T_0 \cup V(C), \quad W = X \cup Y, \quad h = s + p + 2, \quad H = F[V(A) \cup W], \quad N = r - p - 2.$$

These are actual choices, $|Y| = p + 2$, every Y-vertex has R-degree two, and $R[Y]$ is C plus isolates.

Missing pairs exist. For every missing pair $b_{\mathrm{red}} + b_{\mathrm{blue}} = c_0(F) - 1$, put $j_{\mathrm{req}} = u - b_{\mathrm{red}}$. Then

$$j_{\mathrm{req}} \geq s + p + u_{10} + 1 \geq h + 1, \qquad N - (p + h - 1) \geq 2u_{10} - 2 \geq 2.$$

One matching of H is maximum internally in every whole A-component and assigns all W-vertices to exposed ports in distinct odd whole components.

(III) *From $P_0 = H$, choose any successive maxima M_0, M_1, M_2, with $P_{i+1} = P_i - M_i$. Then*

$$\nu(P_0) = \nu(P_1) = \nu(P_2) = p + h.$$

Each M_i saturates W into A, uses no edge inside W, and is internally maximum on $P_i[V(A)]$. The entire rows are

$$P_1[V(A)] = pP_3 + zK_1, \qquad P_2[V(A)] = pP_2 + (p + z)K_1,$$

and $P_3[V(A)]$ is edgeless. Every original row edge is removed, and every original $H[W]$-edge survives. The row is independent in P_3, but is not asserted isolated.

The remaining earlier-side counts at B_6, B_5, B_4 are $1, 2, 3$, respectively. Every $y \in Y$ has exactly five remaining edges into $V(A) \cup X$.

(IV) *For $i = 1, 2$, put $L_i = \bigcup_{j=0}^{i-1} M_j$ and $\widehat{D}_i = N + \nu(P_i)$. Every capacity split totaling i and budget split totaling $\widehat{D}_i$ admits a full F-certificate with $Q_R \cup Q_B = L_i$. Every omitted-incident edge, including one with two omitted endpoints, is paid exactly once. The totals are neutral:*

$$\widehat{D}_1 = c_1(F) = u + p, \qquad \widehat{D}_2 = c_2(F) = u + p.$$

No further port assignment, positive-phase, strict-sector, or omission hypothesis is imposed. No later matching, strict saving, sector exclusion, or whole-degree exclusion is asserted.

Proof. The equality-frame, missing-budget, and exposed-port inputs are identified in Section 3.5.
Step 1. Physical equality frame. The required equality theorem applies to precisely $c_2(F) = c_0(F)$ and proper-subgraph minimality. It gives all structure, native rows, parameter identities, the word

$$6^t 5^{s-2t} 4^t,$$

the native terminal, block independence, and exact costs. Thus B_7, B_3 are physically empty, rather than positive phases on which an argument divides. Substitution gives $c_0 = k + E = u + p$ and $c_8 = a + p = r - p$. Since $E \geq 12$,

$$u_{10} \geq \lfloor 21/8 \rfloor = 2.$$

At deletion of $v \in B_j$, earlier roots in the same block are nonneighbors by independence. Ten-regularity therefore gives exactly $10 - j$ original neighbors in A and the higher blocks, including when other blocks are empty.
Step 2. Selector and original assignment range. The set T_0 is independent. Since $p \geq 5$, a triangle can be chosen. Completing its selected vertex adds two, so $|Y| = p + 2$. Whole components have no joining edges, giving $R[Y] = C$ plus isolates. Every chosen vertex has its two triangle neighbors in R. The set X has s vertices and is disjoint from R, so

$$h = s + p + 2, \qquad N = 2r + s - (r + h) = r - p - 2.$$

Since $c_0(F) > 0$, its least-total definition gives a missing pair. For any ordered missing pair, the required generic bound applies with degree bound ten, connected odd order $2k + 1$, and matching number k. It gives

$$b_{\mathrm{red}} \leq k - E - u_{10}, \qquad k \geq 3E - 1 + 2u_{10}.$$

Therefore

$$j_{\mathrm{req}} \geq u - k + E + u_{10} = s + p + u_{10} + 1 = h + u_{10} - 1 \geq h + 1.$$

Also

$$r \geq 3E - \lambda - 2 + 2u_{10} = s + 3p + 1 + 2u_{10},$$

so

$$N - (p + h - 1) = r - s - 3p - 3 \geq 2u_{10} - 2.$$

No coordinate ordering was used.

Apply the required exposed-port theorem with row A, exterior $V(F) \setminus V(A)$, and this same missing pair. Every component is a triangle or isolate, its matching number is the floor of half its order, and

$$\nu(A) = p, \qquad c_0(F) = u + p.$$

The bound $h \leq j_{\mathrm{req}}$ gives one matching with p internal edges and h exposed-port crossing edges into distinct odd components.

Step 3. Direct full-source restoration. Suppose $i \in \{1, 2\}$, and i edge-disjoint matchings have already been chosen in H. Let P_i be the residual and $m' = \nu(P_i) \leq p + h$. Step 2 gives

$$N \geq \max\{0, m' - 1\}.$$

For any capacities $A_c + B_c = i$, put the first A_c matchings into Q_R and the remaining B_c into Q_B. Their union is exactly L_i, and their degrees satisfy the capacities.

Order the omitted vertices and assign each edge meeting them to its earliest omitted endpoint. The resulting N possibly empty stars partition every edge outside the induced H, including edges with two omitted ends. Each star has matching number at most one.

Fix budgets $x + y = N + m'$. If $m' = 0$, the residual is edgeless, and x stars may be assigned to the first paid part and the rest to the second. If $m' \geq 1$, then $N + m' \geq 2m' - 1$, so at least one budget is at least m'. If $x \geq m'$, place all residual edges and $x - m'$ stars into P_R, and the remaining y stars into P_B. The chosen number lies between zero and N. Restricting a matching to each edge piece bounds its size by $m' + (x - m') = x$. The other part has bound y. If $x < m'$, use P_B for the residual instead. This includes zero budgets and capacities. Shared vertices do not affect the upper bounds, since the pieces partition edges. Consequently

$$c_i(F) \leq N + \nu(P_i).$$

Step 4. Every sequential maximum. A matching in H has at most p internal row edges and at most h other edges. The witness in Step 2 attains $p + h$. Every maximum M_0 must attain both bounds. Its h exterior-meeting edges consume exactly all h exterior vertices, so each has one exterior endpoint and none lies within W. The internal matching is maximum on every triangle. Each triangle exposes one vertex and each isolate exposes itself, so crossing destinations are exposed ports in distinct components.

Deleting M_0 leaves $pP_3 + zK_1$, with row rank p. Thus $\nu(P_1) \leq p + h$. Step 3 and $c_1(F) = N + p + h$ give

$$N + p + h \leq N + \nu(P_1) \leq N + p + h.$$

Hence equality holds. Every maximum M_1 again attains both bounds and saturates W into the row. Removing one internal edge per P_3 leaves $pP_2 + (p + z)K_1$.

The same row bound gives $\nu(P_2) \leq p + h$. Step 3 at index two and $c_2(F) = N + p + h$ force equality. Every maximum M_2 includes all p row edges and saturates W into A. The row is now edgeless. No original $H[W]$-edge was removed. Each exterior vertex has lost exactly three distinct A-edges.

Step 5. Boundary counts and neutral totals. Every earlier side is fully retained. The original count $10 - j$ loses precisely three edges into A, leaving $1, 2, 3$ for $j = 6, 5, 4$. Each Y-vertex has two

neighbors in R, hence eight outside R, all in $A \cup X$. It loses three and retains five. Step 3 gives the certificates for the actual prefixes, whose ranks yield

$$N + \nu(P_i) = N + p + h = u + p, \qquad i = 1, 2.$$

All constructions allow B_6, B_4 to be empty and remain in the one source and one retained graph. □

8.5. **Degree-twelve reserve and matching prefix.**

Proposition 8.10 (Shortest-terminal-cycle reserve in degree twelve)**.** *Let $F \in \mathcal{M}_{12}$, put $u = r + s$, $\beta = p + \delta - \epsilon$, and*

$$v_{12} = \left\lfloor \frac{2E - 3}{10} \right\rfloor.$$

Native rows exist. Fix any native A and any complete continuation on $F - V(A)$, with entire root set X_{root} and actual terminal R. Put $D_2 = \{v \in V(R) : d_R(v) = 2\}$, and choose any maximum independent set T_0 of $R[D_2]$. Choose any shortest whole odd-cycle component C of this same R; such a component exists. Define

$$e_C = (|V(C)| + 1)/2, \quad Y = T_0 \cup V(C), \quad W = X_{\mathrm{root}} \cup Y,$$
$$h = |W|, \quad H = F[V(A) \cup W], \quad N = |V(F) \setminus V(H)|.$$

Then

$$2 \le e_C \le 4, \qquad e_C \le \epsilon + v_{12} + 1,$$

and

$$h = s + \beta + e_C, \qquad N = u - h = r - \beta - e_C.$$

Moreover,

$$\begin{aligned} N - (\beta + h - 1) &\ge 3\epsilon + 2v_{12} + 2 - 2e_C \\ &= \epsilon + 2(\epsilon + v_{12} + 1 - e_C) \ge 0. \end{aligned}$$

In particular,

$$N \ge \max\{0, \beta + h - 1\}, \qquad N \ge e_C - 2.$$

Missing pairs exist. For every ordered missing pair $x + y = c_0(F) - 1$, put $j_x = u - x, j_y = u - y$. Without exchanging the coordinates,

$$j_x - h \ge \epsilon + v_{12} + 1 - e_C \ge 0, \qquad j_y - h \ge \epsilon + v_{12} + 1 - e_C \ge 0.$$

Thus the same W fits both actual missing-allocation ranges. No triangle-only, positive-phase, extra order, or sampled-parameter hypothesis is imposed. No matching prefix, strict payment, or degree exclusion is asserted.

Proof. The required degree-twelve inputs are identified in Section 3.6.

Step 1. Actual objects and the remaining scalar inequality. The native theorem gives

$$E \ge 10, \quad \lambda \ge 3, \quad p \ge 5, \quad 0 < \delta \le \lambda + 1, \quad 0 \le \epsilon \le \delta,$$

positive odd s, twelve-regularity, $|V(F)| = 2k + 1 = 2r + s$, and native rows. Every prescribed continuation has exactly s roots and native terminal R. Its components are

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7,$$

with exactly p odd cycles and ϵ positive even-order paths. Since $p \ge 5$, shortest odd cycles exist, and $e_C \in \{2, 3, 4\}$. Also $v_{12} \ge 1$.

The required selector theorem applies to every T_0, C, even without shortestness. It gives the actual disjoint sets and

$$h = s + \beta + e_C, \qquad N = r - \beta - e_C \ge 0.$$

The required scalar theorem, with its additional selector count set to zero and its completion size set to e_C, gives missing pairs and

$$j_x - h,\ j_y - h \geq \epsilon + v_{12} + 1 - e_C,$$

together with the stated omission lower bound. Its hypotheses $W \subseteq V(F) \setminus V(A)$ and $h \leq u$ follow from the actual construction. It remains to prove the reserve.

Step 2. Exclude a reserve failure. Suppose

$$e_C > \epsilon + v_{12} + 1.$$

Integer domains, $\epsilon \geq 0$, and $v_{12} \geq 1$ exclude $e_C = 2$.

If $e_C = 3$, then $\epsilon = 0, v_{12} = 1$, so $E \in \{10, 11\}$. Shortestness excludes whole triangles in R. Every odd cycle is C_5 or C_7, with matching number at least two. Each positive even path contributes at least one. Therefore

$$p + \delta = \nu(R) \geq 2p + \epsilon, \qquad \delta \geq p + \epsilon.$$

Since $\delta \leq \lambda + 1 = E - 1 - p$,

$$E \geq 2p + \epsilon + 1 \geq 11.$$

Thus

$$E = 11, \quad p = 5, \quad \epsilon = 0, \quad \delta = 5, \quad \lambda = 4.$$

The required boundary theorem forces a whole triangle in every native row at these exact parameters, contradicting the choice of C.

If $e_C = 4$, then $\epsilon + v_{12} \leq 2$. The possibilities are $v_{12} = 1, \epsilon \in \{0, 1\}$, or $v_{12} = 2, \epsilon = 0$. Shortestness makes all p odd cycles heptagons. Hence

$$p + \delta = \nu(R) \geq 3p + \epsilon, \qquad \delta \geq 2p + \epsilon,$$

and

$$E \geq 3p + \epsilon + 1 \geq 16.$$

The case $v_{12} = 1$, which gives $E \leq 11$, is impossible. The case $v_{12} = 2$ gives $12 \leq E \leq 16$, so

$$E = 16, \quad p = 5, \quad \delta = 10, \quad \lambda = 9.$$

The required source-class identification permits the required bound

$$23k < 100E - 100 - 50\lambda = 1050.$$

But now $r = k, s = 1, v_{12} = 2$, and the required row-order inequality gives

$$k = r \geq s + 3p + 3\delta + 1 + 2v_{12} = 1 + 15 + 30 + 1 + 4 = 51.$$

Thus $23k \geq 1173 > 1050$, a contradiction.

These matching estimates use only whole-component additivity. Alternating edges attain ranks two and three on C_5, C_7, and vertex counting bounds them. Each P_2, P_4 has an edge. The case split uses integer inequalities, not sampled parameters. The reserve follows.

Step 3. Finish all margins. Insert the reserve into the inequalities from Step 1. Both missing-coordinate margins are nonnegative, and the omission lower bound becomes

$$\epsilon + 2(\epsilon + v_{12} + 1 - e_C) \geq 0.$$

The actual cardinality also gives $N \geq 0$. Finally, $\beta \geq p \geq 5$ and $s \geq 1$, so

$$\beta + h - 1 = s + 2\beta + e_C - 1 \geq e_C.$$

Hence $N \geq e_C - 2$. Every original choice was arbitrary within the stated range. □

Proposition 8.11 (Degree-twelve three-matching prefix and ninth-owner consumption). *Let $F \in \mathcal{M}_{12}$, put*

$$m = p + \delta, \quad \beta = m - \epsilon, \quad v_{12} = \lfloor (2E - 3)/10 \rfloor,$$

and fix any native row A. Put $U = V(F) \setminus V(A)$, $u = |U| = r + s$. For any missing pair $x + y = c_0(F) - 1$, define

$$j_* = u - \max\{x, y\}, \qquad L_{12} = s + p + \delta + v_{12} + 1.$$

Native rows and missing pairs exist, and

$$1 \le L_{12} \le j_* \le u.$$

Let $W \subseteq U$ be any set of size $h \le j_*$, and put

$$H = F[V(A) \cup W], \qquad N = u - h.$$

In particular every W of size at most L_{12} is permitted.

Choose any maximum M_0 of $P_0 = H$, any maximum M_1 of $P_1 = P_0 - M_0$, and any maximum M_2 of $P_2 = P_1 - M_1$; put $P_3 = P_2 - M_2$.

(I) *All choices exist, and*

$$\nu(P_0) = m + h, \qquad \nu(P_1) = \beta + h, \qquad \nu(P_2) = p + h.$$

The exact omission identity is

$$N - (\beta + h - 1) = |x - y| + 2(j_* - h) + \epsilon \ge 0.$$

Thus $N \ge \max\{0, \beta + h - 1\}$.

(II) *Each of the same three matchings saturates W into A and uses no W-internal edge. The matching M_0 is maximum in every whole component, and its crossing endpoints are exposed ports in distinct odd whole components. The entire first row residual is*

$$pP_3 + (\delta - \epsilon)P_2 + \textit{isolates},$$

with M_1 internally maximum. The entire second row residual is pP_2 plus isolates, all its edges belong to M_2, and every original row edge is removed. Moreover,

$$P_3[W] = H[W],$$

and each W-vertex uses three distinct original A-neighbors.

(III) *Put $Q_9 = \{v \in U : d_{F - V(A)}(v) = 9\}$. Each $v \in W \cap Q_9$ has a unique odd whole component O_v with an A-neighbor, and exactly one neighbor o_v in it. The physical edge vo_v belongs to every M_0. The two other original v-to-A edges occur one in M_1, one in M_2, in some order. None survives in P_3.*

Every prescription in the fixed M_0, including these owner edges, remains literally in that matching and in the fixed degree-bounded prefix. Later matchings use its actual edge residuals. For every independent $X \subseteq W \cap Q_9$, the set $V(A) \cup X$ is independent in P_3, and $W \setminus X$ is a vertex cover.

(IV) *For $i = 1, 2$, put*

$$L_i = \left(V(H), \bigcup_{0 \le t < i} M_t \right), \qquad \rho_i = \nu(P_i).$$

Every capacity split totaling i and budget split totaling $N + \rho_i$ admits a certificate on the entire original F, with $Q_R \cup Q_B = E(L_i)$. Every edge with an omitted endpoint, including two omitted endpoints, is paid. The exact totals are neutral:

$$N + \rho_1 = c_1(F), \qquad N + \rho_2 = c_2(F).$$

The only retained-set condition is the displayed range. This proposition does not assert that a specified selector fits it, construct later matchings, give a strict saving, exclude a degree, or prove the Ramsey formula.

Proof. The native, missing-budget, and generic-prefix inputs are identified in Section 3.6.

Step 1. Source data and both missing coordinates. The native theorem gives twelve-regularity, order $2r+s$, $E \geq 10, \lambda \geq 3, p \geq 5, \delta > 0$, and $0 \leq \epsilon \leq \delta$. Every row has only

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7,$$

with exactly ϵ positive even paths, p odd cycles, and matching number m. Its source costs are

$$c_1(F) = u + \beta, \qquad c_2(F) = u + p, \qquad c_0(F) = u + m.$$

The last follows by substituting the parameters.

The required missing-budget theorem gives $u - x \geq L_{12}$ for every ordered missing pair. A swapped pair is also missing, since exchanging the two edge parts would otherwise solve the original pair. Applying the same bound to it gives $u - y \geq L_{12}$. Hence $j_* \geq L_{12}$. Since $v_{12} \geq 1$, this lower bound is positive; nonnegative budgets give $j_* \leq u$.

Put $x_+ = \max\{x, y\}$, $x_- = \min\{x, y\}$. Then

$$x_+ + x_- = u + m - 1, \qquad j_* = u - x_+, \qquad \beta = m - \epsilon.$$

Consequently

$$\begin{aligned} N - (\beta + h - 1) &= u - \beta - 2h + 1 \\ &= u - m - 2h + 1 + \epsilon \\ &= (x_+ - x_-) + 2(u - x_+ - h) + \epsilon. \end{aligned}$$

This proves the exact identity, including equal budgets and $h = j_*$. Since N is an actual cardinality, it proves the full omission bound.

Step 2. Sequential ranks on the retained graph. The ordered missing pair (x_+, x_-) and $h \leq u - x_+$ give

$$\nu(H) = \nu(A) + h = m + h$$

by the required original matching theorem. Thus no initial matching premise is added.

Apply the required generic three-stage theorem to this F, H, A. Its row order and rank are r, m, its positive-even-path count is ϵ, its odd-cycle count is p, and its first residual rank is $m - \epsilon = \beta$. All row types are permitted. The matching and omission hypotheses have just been proved, and

$$c_1(F) = N + h + \beta, \qquad c_2(F) = N + h + p$$

verify its source-cost bounds with equality. It gives the ranks and entire row restrictions for every successive maximum choice. No graph or matching is replaced.

Step 3. Literal ninth-owner edges. A retained complement-degree-nine vertex has exactly three original A-neighbors by twelve-regularity. The required owner theorem gives its unique odd component and sole port there. Part (II) forces M_0 to send it to an exposed port in an odd component, so its only possible edge is vo_v. The next two matchings use one further A-edge each. Their residual definitions make all three edges distinct; they therefore exhaust the original three. The first matching remains fixed throughout.

For independent $X \subseteq W \cap Q_9$, all A-internal and X-to-A edges have been removed. No X-internal edge existed. Thus $A \cup X$ is independent in P_3, and its complement covers every edge.

Step 4. Neutral restoration. The generic theorem supplies full certificates on the same source and prefix. Its totals are

$$N + \beta + h = u + \beta = c_1(F), \qquad N + p + h = u + p = c_2(F).$$

It covers every nonnegative integer split, including zero budgets, and assigns each omitted-incident edge once. No owner or other fixed prefix edge is charged again. □

8.6. **The quartic cost interface and movable selectors.**

Proposition 8.12 (Exact quartic costs and the positive correction). *Assume Setup 2.7, without assuming a saturated frame for (I) or (III). For any maximal independent S, define*

$$q_4 = m - b, \quad \rho_{\mathrm{cost},4} = v - m, \quad \beta_{\mathrm{proposed},4} = \nu(H) - b - m, \quad \beta_{\mathrm{cost},4} = \nu(H) - 2m.$$

(I) *The graph F is well-covered, $|S| = a$, and all displayed quantities $b, m, q_4, \rho_{\mathrm{cost},4}, \beta_{\mathrm{proposed},4}, \beta_{\mathrm{cost},4}$ are independent of S. Moreover,*

$$q_4 = E - 1 - \sigma \geq \left\lceil \frac{E-1}{4} \right\rceil \geq 1,$$

$$\beta_{\mathrm{cost},4} = E + \sigma - 2b \geq E - \sigma \geq \left\lceil \frac{E+3}{4} \right\rceil \geq 1,$$

and

$$\beta_{\mathrm{proposed},4} = \beta_{\mathrm{cost},4} + q_4, \qquad v = a + \nu(H) + q_4.$$

The exact costs are

$$\begin{aligned} c_0(F) &= a + \nu(H) = v - q_4, \\ c_1(F) &= a + \nu(H) - b = \rho_{\mathrm{cost},4}, \\ c_2(F) &= a + 2m - b = \rho_{\mathrm{cost},4} - \beta_{\mathrm{cost},4}. \end{aligned}$$

In particular,

$$c_2(F) = [v - \nu(H) + b] + q_4 = \rho_{\mathrm{cost},4} - \beta_{\mathrm{proposed},4} + q_4, \qquad v - \nu(H) + b = a + m.$$

Thus $c_2(F) = v - \nu(H) + b$ is incompatible with these hypotheses by the positive integer q_4. No existence of a graph satisfying the hypotheses is asserted.

(II) *If (S, U) is any supplied saturated frame and $R = F - (S \cup U)$, then*

$$c_1(R) = \beta_{\mathrm{proposed},4}, \qquad c_2(R) = q_4, \qquad \beta_{\mathrm{cost},4} = c_1(R) - c_2(R).$$

The integer q_4 counts exactly the whole odd-cycle components of R, each of order three, five, or seven.

(III) *For any $W \subseteq V(H)$, put $h = |W|$, $N = v - h$. The retained graph $F[S \cup W]$ has exactly N omitted source vertices, and*

$$c_0(F) - N = h - q_4, \qquad c_1(F) - N = h - m, \qquad c_2(F) - N = h - m - \beta_{\mathrm{cost},4}.$$

For any integer $g \geq 0$ and integer $e \geq 2$, if

$$h = m + \beta_{\mathrm{cost},4} + g + e,$$

then

$$N + g + e - 1 = c_2(F) - 1.$$

This is arithmetic only; it asserts no matching rank, compatible sequence, residual odd cycle, or certificate.

(IV) *Parts (I) and (III) apply to every member of the entire class $\mathcal{M}_4$. In that class $E \geq 14$, and consequently $q_4 \geq 4$. No quartic exclusion, degree-twelve statement, or unconditional Ramsey formula is asserted.*

Proof. The normalization, independent-cut, canonical-frame, and source-floor inputs are identified in Section 3.7.

The required normalization gives well-coveredness, uniform b, m, and

$$\nu(H) = 3E - 2 - \sigma, \quad c_1(H) = \nu(H) - b, \quad c_2(H) = 2m - b,$$
$$v - \nu(H) = a + m - b, \qquad 0 \le b \le \sigma \le \lfloor 3(E-1)/4 \rfloor,$$
$$m \ge \lceil (E-1)/4 \rceil + b.$$

The set S is nonempty because F is nonempty and four-regular. The required independent-cut theorem applies for $E \ge 5 \ge 4$, giving

$$c_i(F) = a + c_i(H) \quad (0 \le i \le 3), \qquad \mathrm{ex}(H) = 0.$$

Thus $c_0(H) = \nu(H)$, proving the first forms of the three costs.

The definition gives $q_4 = m - b = E - 1 - \sigma$, and the lower bound on m proves its positivity. The order identity becomes $v = a + \nu(H) + q_4$. Since $m = q_4 + b$,

$$v - m = a + \nu(H) - b = c_1(F).$$

Also

$$\beta_{\mathrm{cost},4} = (3E - 2 - \sigma) - 2(E - 1 - \sigma + b) = E + \sigma - 2b.$$

The bounds $b \le \sigma \le \lfloor 3(E-1)/4 \rfloor$ give the stated positive lower bound. Direct subtraction gives $\beta_{\mathrm{proposed},4} - \beta_{\mathrm{cost},4} = m - b = q_4$. The cost identity makes $\rho_{\mathrm{cost},4}$ nonnegative. Uniformity follows from that of $b, m, \nu(H), a$, and $v = |V(F)| - a$.

Next,

$$\rho_{\mathrm{cost},4} - \beta_{\mathrm{cost},4} = v - \nu(H) + m = a + 2m - b = c_2(F),$$

whereas

$$\rho_{\mathrm{cost},4} - \beta_{\mathrm{proposed},4} = v - \nu(H) + b = a + m.$$

Their difference is $q_4 > 0$, proving (I).

For (II), the normalization also gives

$$a \ge 3E - 3 + \sigma \ge 12,$$

so $F \ne K_5$. It gives $m \ge 1$, $0 \le b \le m$, $\mathrm{ps}(H) = 0$, the displayed cost/order identities, and independence number at most m in the graph induced by the cubic vertices of H. These verify the required canonical-frame theorem for the supplied frame. It gives

$$\nu(R) = \nu(H) - m, \quad c_1(R) = \nu(R) - b, \quad c_2(R) = m - b,$$

and exactly $m - b$ whole odd cycles, each C_3, C_5, C_7. Substitution proves (II), without asserting saturation for every graph under the broader hypothesis.

For (III), disjoint S, W retain $a + h$ of the $a + v$ vertices, so $N = v - h$. Subtracting this from the three costs gives the comparison values. Under the additional size identity,

$$N + g + e - 1 = v - h + g + e - 1 = v - m - \beta_{\mathrm{cost},4} - 1 = c_2(F) - 1.$$

No edge partition is involved.

Finally, the required quartic-source theorem gives connectedness, four-regularity, factor-criticality, gap one, and $E \ge 14$ for every $F \in \mathcal{M}_4$. Thus (I),(III) apply, and

$$q_4 \ge \lceil (E-1)/4 \rceil \ge \lceil 13/4 \rceil = 4.$$

□

Proposition 8.13 (Movable quartic selectors and deficient port paths)**.** *Assume Setup 2.7 with any supplied saturated frame. Put*

$$q_4 = m - b, \qquad \beta = \nu(H) - 2m.$$

Then

$$F[W_{\mathrm{fr}}] = mK_2 + (a - m)K_1, \quad |S| = a, \quad |P_{\mathrm{fr}}| = 2m, \quad |Z| = a - m,$$
$$\Delta(R) \le 2, \qquad q_4 \ge 1, \qquad \beta \ge 1, \qquad m \le \sigma.$$

Thus the proposed boundary $m = \sigma + 1$ does not occur for supplied frames.

For any $X \subseteq D_2(R) = \{x : d_R(x) = 2\}$, including the empty set, put

$$J_X = R[X], \quad K_X = F[W_{\mathrm{fr}} \cup X], \quad N_X = v - m - |X|, \quad \mathrm{gain}_X = |X| - \nu(J_X) - \beta.$$

Let G_X be the spanning graph on $V(K_X)$ with exactly the frame edges L_{fr} and all original X-to-W_{fr} edges, excluding all J_X-edges. Let B_X be the bipartite graph with parts X, Z and exactly the original X-to-Z edges; it need not be induced in F.

(I) *The following are equivalent:*

(a) *There are edge-disjoint matchings M_0, M_1 of K_X with*

$$L_{\mathrm{fr}} \subseteq M_0, \qquad M_0 \cup M_1 = E(G_X).$$

(b) *Every $p \in P_{\mathrm{fr}}$ has at most one X-neighbor, every $z \in Z$ has at most two, and B_X has a matching saturating X.*

A pair is constructed by choosing an X-saturating B_X-matching that also covers every degree-two Z-vertex, adjoining L_{fr} to form M_0, and taking $M_1 = E(G_X) \setminus M_0$. Every pair in (a) is sequentially maximum in the actual $K_X, K_X - M_0$, with

$$\nu(K_X) = |M_0| = m + |X|, \quad \nu(K_X - M_0) = |M_1| = |X|,$$

and

$$E(K_X - M_0 - M_1) = E(J_X).$$

(II) *Under (I),*

$$N_X \geq \nu(H) + q_4 \geq \nu(J_X) + 1,$$

$$N_X + \nu(K_X - M_0) = c_1(F), \qquad N_X + \nu(J_X) = c_2(F) - \mathrm{gain}_X\,.$$

For the same pair and every capacity split totaling two and budget split totaling $N_X + \nu(J_X)$, there is a certificate on the entire F, whose degree-bounded union is exactly $M_0 \cup M_1$. All omitted-incident edges, including those with two omitted endpoints, are included. Consequently $\mathrm{gain}_X \leq 0$.

(III) *Assume only the two load bounds in (I). Then $\Delta(B_X) \leq 2$. Let $\mathcal{D}_X$ be its whole path components with both endpoints in X, including an isolated X-vertex as a path of order one. Put $d_X = |\mathcal{D}_X|$. For $T \in \mathcal{D}_X$, put $X_T = X \cap V(T)$, $Z_T = Z \cap V(T)$. Then*

$$|X_T| = |Z_T| + 1, \qquad N_{B_X}(X_T) = Z_T.$$

These X_T are exactly the inclusion-minimal nonempty X-subsets with smaller neighborhoods. The graph B_X saturates X if and only if $d_X = 0$.

Choose any one $x_T \in X_T$ for each T, and put $A = \{x_T : T \in \mathcal{D}_X\}$, empty when $d_X = 0$. Then $X \setminus A$ satisfies (I), and its full restored total is

$$N_X + d_X + \nu(J_X - A) = c_2(F) - \mathrm{gain}_X + \eta_{J_X}(A),$$

where

$$\eta_{J_X}(A) = |A| + \nu(J_X - A) - \nu(J_X).$$

For every simultaneous choice,

$$\mathrm{gain}_X \leq \eta_{J_X}(A) \leq d_X.$$

If $d_X = 1$ and $\mathrm{gain}_X \geq 1$, then $\mathrm{gain}_X = 1$, and

$$\nu(J_X - x) = \nu(J_X)$$

for every x in the unique deficient side. Equivalently, each can be exposed by a maximum matching. Each lies on a whole odd J_X-cycle or in an odd-numbered position of a whole odd-order J_X-path, numbered from either endpoint. Isolates count as paths of order one. None lies on a whole even path or even cycle.

If

$$T = x_0 - z_1 - x_1 - \cdots - z_\ell - x_\ell, \qquad \ell \geq 0,$$

is deficient, exactly two original edges join T to P_{fr}, one at each X-endpoint, both at x_0 if $\ell = 0$. Their endpoints p_T, q_T are distinct. They are the only G_X-edges leaving $V(T)$. If $p_T q_T$ is a frame edge, these edges, T, and that frame edge form a whole odd G_X-cycle. Otherwise the boundary path joins endpoints of two distinct frame edges, with no whole-component assertion.

(IV) *Choose a canonical independent set D in R by selecting a maximum independent set on each whole odd cycle, the center of each whole P_3, either opposite pair on each whole C_4, no isolates, and on each consecutively numbered whole P_{2s}, $s \geq 1$, either*

$$\{2, 4, \ldots, 2s-2\} \quad \textit{or} \quad \{3, 5, \ldots, 2s-1\}.$$

Both are empty for $s = 1$. In the reflected terminal $F[W_{\mathrm{fr}} \cup D]$, choose one D-vertex from each whole odd cycle and delete precisely these choices from D, obtaining D_0. Then

$$D_0 \subseteq D_2(R), \quad |D_0| = \beta, \quad R[D_0] \textit{ is edgeless}, \quad N_{D_0} = c_2(F).$$

For any $Z_- \subseteq D_0$ and $Z_+ \subseteq D_2(R) \setminus D_0$, put

$$X = (D_0 \setminus Z_-) \cup Z_+, \qquad d_- = |Z_-|, \quad d_+ = |Z_+|.$$

Then

$$N_X = c_2(F) + d_- - d_+, \qquad \mathrm{gain}_X = d_+ - d_- - \nu(J_X).$$

No retention of all old D_0-vertices is required. If J_X is a whole odd cycle of order $2e-1$, $e \geq 2$, disjoint from an arbitrary graph B, and $g = \nu(B)$, then

$$\mathrm{gain}_X = d_+ - d_- - g - e + 1.$$

The proposed unit-payment condition $d_+ - d_- = g + e$ forces a load violation or at least one actual deficient alternating port path. A unique such path obeys (III)'s exposure restriction.

This concerns the explicit template retaining every frame vertex, putting every frame edge in M_0, and clearing every selected boundary edge with two matchings. It asserts no positive-gain feasible selector, obstruction for every other layout, saturation in an unsupplied generic source, whole quartic exclusion, or Ramsey formula.

Proof. The frame, cubic-independent-set, and neutral-selector inputs are identified in Section 3.7.
Step 1. Source data. Proposition 8.12 gives

$$|S| = a, \quad q_4 \geq 1, \quad \beta \geq 1, \quad v = a + \nu(H) + q_4, \quad c_1(F) = v - m, \quad c_2(F) = v - m - \beta.$$

The required frame theorem gives the induced matching structure and $m \leq \sigma$. Extend U to a maximal independent set U_3 in the graph induced by the cubic vertices of H. The required cubic bound gives $|U_3| \leq m$. Since $|U| = m$, $U_3 = U$, and the same input gives $\Delta(R) \leq 2$. Disjointness gives $N_X = v - m - |X|$. Every $x \in X$ has degree two in R and degree four in F, hence exactly two distinct frame neighbors, even when X is not independent.
Step 2. Necessary loads and ports. If two matchings partition $E(G_X)$, all G_X-degrees are at most two. At a frame endpoint this degree is one plus its X-load; at an isolate it is its X-load. Thus both load bounds hold. Each x has degree two and uses one edge from each matching. Since M_0 already covers all P_{fr} through L_{fr}, its edge at x goes to Z. These edges saturate X in B_X.
Step 3. Components and minimal deficient sets. Under the loads, both bipartite sides have degree at most two. A finite connected graph of maximum degree two is an isolate, path, or cycle: following edges from a degree-one vertex gives an unbranched whole path; if every degree is two, following edges closes a whole cycle. Bipartite cycles are even.

An X-ended path has one more X-vertex than Z-vertex and the stated whole neighborhood. Removing any one X-vertex splits it into two possibly empty balanced even paths. Alternating

edges perfectly match those paths. For any proper subset of its X-side, choose a deleted vertex outside that subset and restrict this matching, proving no proper subset is deficient.

Every other component saturates its X-side: use alternating perfect matchings on balanced paths and even cycles; on a Z-ended path leave one endpoint unmatched and match the remaining even path; an isolated Z-vertex needs no edge. These choices cover every degree-two Z-vertex. Component neighborhoods are disjoint, so any deficient subset is deficient in at least one component. The preceding matching constructions show that the inclusion-minimal deficient subsets are exactly the whole sides X_T. This proves the criterion without importing Hall's theorem.

Step 4. Construction and every ordered maximum. If B_X saturates X, Step 3 supplies a matching L that also covers every degree-two Z-vertex. The union

$$M_0 = L_{\mathrm{fr}} \cup L$$

is a matching because its two parts use disjoint vertex sets. It has size $m + |X|$ and covers every degree-two vertex of G_X. Hence $M_1 = E(G_X) \setminus M_0$ is a matching. Since $e(G_X) = m + 2|X|$, its size is $|X|$.

The set $U \cup X$ is a vertex cover of K_X: it meets every frame edge, every boundary edge, and every internal J_X-edge. A matching uses distinct cover vertices, so its size is at most $m+|X|$. Thus M_0 is maximum. After it is deleted, all remaining edges meet X, a cover of size $|X|$; M_1 attains that bound. The final residual is exactly J_X, with the frame vertices isolated.

For any pair in (I)(a), Step 2 forces exactly one M_0-boundary edge at each X-vertex and none at P_{fr}. Its two sizes are therefore the same. The two covers prove sequential maximum size for every such pair.

Step 5. Omission margin and certificate. Port injectivity gives $|X| \le |Z| = a - m$. Hence

$$N_X = v - m - |X| \ge v - a = \nu(H) + q_4 \ge \nu(J_X) + 1.$$

The last inequality uses that J_X is a subgraph of H. Exact costs and matching ranks give

$$N_X + |X| = c_1(F), \qquad N_X + \nu(J_X) = c_2(F) - \mathrm{gain}_X\,.$$

Put $t = \nu(J_X)$, $N = N_X$. For any capacities totaling two, put exactly A_c of the two matchings into the first degree part and the rest into the second. Order the omitted vertices and assign every edge outside induced K_X to its earliest omitted endpoint. This partitions those edges into N labeled stars, possibly empty, including all edges between omitted vertices.

For budgets $x_c + y_c = N + t$, if $t = 0$, distribute x_c stars to the first paid part and the others to the second. If $t \ge 1$, $N \ge t-1$ gives total at least $2t-1$, so one budget is at least t. If $x_c \ge t$, place every residual edge and $x_c - t$ stars into the first paid part, and the other y_c stars into the second. The number chosen lies in $[0, N]$. Matching restriction to pieces gives bounds x_c, y_c. If the second budget is used instead, exchange the part names. Shared vertices do not affect these bounds. Every edge is assigned once and every zero capacity or budget is covered. Thus

$$c_2(F) \le N_X + \nu(J_X) = c_2(F) - \mathrm{gain}_X,$$

so $\mathrm{gain}_X \le 0$.

Step 6. Simultaneous repair and exposure. Select one vertex from each deficient component. These components are disjoint, so $|A| = d_X$. Each becomes two balanced paths; all other components remain unchanged. Loads cannot increase. Thus $X \setminus A$ satisfies (I), its residual is $J_X - A$, and its omitted count is $N_X + d_X$.

Deleting $|A|$ vertices can remove at most $|A|$ edges from a matching. Also every matching in $J_X - A$ is a matching in J_X. Therefore

$$0 \le |A| + \nu(J_X - A) - \nu(J_X) \le |A| = d_X.$$

The exact debit identity follows by substitution. Its certificate total is at least $c_2(F)$, giving $\mathrm{gain}_X \le \eta_{J_X}(A) \le d_X$ for every choice.

If $d_X = 1$ and $\mathrm{gain}_X \geq 1$, integrality forces $\mathrm{gain}_X = 1$. Every vertex of the unique deficient side is an allowed repair, so its deletion preserves matching number. This is equivalent to exposure by a maximum matching.

For the position description, paths have rank $\lfloor n/2 \rfloor$, by alternating edges and vertex counting. Deleting a vertex of an even path leaves two paths of total odd order and decreases rank by one. In P_{2k+1}, deleting position i leaves lengths $i-1$ and $2k+1-i$. If i is odd, both are even and their ranks sum to k; if i is even, both are odd and the sum is $k-1$. Deleting a vertex of an even cycle lowers rank by one, whereas deleting one from an odd cycle preserves it. Additivity over whole components proves the asserted classification, including isolates.

Step 7. Literal frame boundary. In a deficient path, each internal X-vertex has two Z-neighbors and no frame-endpoint neighbor. When $\ell \geq 1$, each X-endpoint has one Z-neighbor and one P_{fr}-neighbor. For $\ell = 0$, the isolated X-vertex has two distinct P_{fr}-neighbors. The two endpoint destinations are distinct: simplicity proves this for $\ell = 0$, and otherwise a common destination violates its load bound. All Z-to-X edges stay in the whole bipartite component, so these are its only leaving G_X-edges. If their endpoints are frame partners, the resulting cycle is odd and every vertex has its full degree two inside it, making it whole. Otherwise the path joins distinct frame edges, with no further conclusion.

Step 8. Movable neutral sets. The required canonical neutral-selector theorem applies to the same supplied frame and $\beta = \beta_{\mathrm{cost},4}$. It gives $|D_0| = \beta$, $N_{D_0} = c_2(F)$, and independence for every stated choice. The explicit canonical choices use only degree-two vertices: cycle vertices, P_3-centers, and internal even-path vertices. Hence $D_0 \subseteq D_2(R)$.

The exchange domains are disjoint, so $|X| = \beta - d_- + d_+$. Substitution proves both identities. If $J_X = C_{2e-1} \dot\cup B$, then $\nu(J_X) = e - 1 + g$, proving the last formula. The unit-payment condition makes $\mathrm{gain}_X = 1$, contradicting (II) when all port conditions hold. If the loads hold, (III) therefore forces a deficient path, with the stated additional restriction when unique. □

8.7. **Exclusion of saturated pentagons and heptagons.**

Proposition 8.14 (Exclusion of whole saturated pentagons)**.** *Assume Setup 2.7 with any supplied saturated frame. Put*

$$q_4 = m - b, \qquad \beta = \nu(H) - 2m.$$

Then

$$F[W_{\mathrm{fr}}] = mK_2 + (a-m)K_1, \qquad q_4, \beta \geq 1, \qquad c_1(F) = v - m, \quad c_2(F) = v - m - \beta.$$

The allowed whole terminal components are

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

A canonical D selects a maximum independent set on every whole odd cycle, the center of each P_3, either opposite pair on each C_4, either internal vertex on each P_4, and nothing on P_2's or isolates.

Put $R_D = F[W_{\mathrm{fr}} \cup D]$, and let B_D have parts D, Z and exactly the original edges between them. Let $T_1, \ldots, T_{q_4}$ be its whole D-ended path components, including isolated D-vertices.

If C were a whole pentagon of R, the following would hold for every canonical D:

(I) *No T_i meets C. Choose any $d_i \in D \cap V(T_i)$, independently, and put*

$$A = \{d_1, \ldots, d_{q_4}\}, \quad D_0 = D \setminus A, \quad Y = D_0 \cup V(C),$$

$$K = F[W_{\mathrm{fr}} \cup Y], \qquad N = |V(F) \setminus V(K)|.$$

Then $|D_0| = \beta$, $|Y| = \beta + 3$, every P_{fr}-vertex has at most one Y-neighbor, every Z-vertex at most two, and the original Y-to-Z graph has a matching saturating Y.

(II) *There are two edge-disjoint sequential maximum matchings M_0, M_1 in $K, K - M_0$, with $L_{\mathrm{fr}} \subseteq M_0$,*

$$\nu(K) = |M_0| = m + \beta + 3, \qquad \nu(K - M_0) = |M_1| = \beta + 3,$$

and

$$E(K - M_0 - M_1) = E(C).$$

All other retained vertices are isolated. Moreover,

$$N = c_2(F) - 3, \quad N \geq 3, \quad N + \nu(K - M_0) = c_1(F),$$

$$N + \nu(K - M_0 - M_1) = N + 2 = c_2(F) - 1.$$

For the same pair and every nonnegative integer split

$$A_c + B_c = 2, \qquad x_c + y_c = N + 2,$$

the entire original F has the defining four-part certificate. Every omitted-incident edge, including two omitted endpoints, is assigned exactly once.

Consequently no supplied saturated frame has a whole pentagon terminal component. This excludes the entire saturated-pentagon subclass without frame extremality, an excess ceiling, or a condition on its possible single frame contact. It asserts neither saturation in an unsupplied generic source nor exclusion of every quartic source.

Proof. The reflection and terminal inputs are identified in Section 3.7, and the parent/contact inputs in Section 3.8.

Step 1. Source and every reflection. The required normalization gives well-coveredness, common b, m, $0 \leq b \leq \sigma$, $m \geq 1$, and the complement identities required by the canonical-frame theorem. Its bound

$$a \geq 3E - 3 + \sigma \geq 12$$

excludes K_5. Thus that theorem applies to every supplied frame and canonical choice. The required P_6 exclusion leaves exactly the listed terminal types in the original and every reflected saturated frame.

The canonical theorem says that D is independent, R_D is an actual saturated terminal, every D-vertex has degree two there, and each designated odd cycle or positive even path contains exactly one original frame edge. These designated components exhaust all original frame edges. The neutral-selector theorem gives

$$|D| = \beta + q_4$$

and exactly q_4 deficient paths in B_D, one from each reflected odd cycle after its two frame endpoints are removed. Any one vertex per path can be deleted to leave size β. Proposition 8.13 gives the exact costs and ordered matching interface.

Assume a whole pentagon C exists. For $x \in V(C)$, put $A_x = N_F(x) \cap W_{\mathrm{fr}}$. The required pentagon parent theorem gives two parents per vertex, with pairs equal or disjoint and each equality class a singleton or two nonadjacent vertices. At most one edge joins P_{fr} to C. If present, its cycle vertex is a singleton class whose Z-parent meets no other cycle vertex.

Step 2. Selected pentagon vertices avoid reflected odd cycles. Fix canonical D and $x \in D \cap V(C)$. After cyclic relabeling, its selected pair is $\{0, 2\}$ on $0, 1, 2, 3, 4$. The unique remaining edge is $3\,4$, and the selected vertices touch its endpoints $4, 3$, respectively. In the actual reflected frame, the complementary frame vertex set is $V(R) \setminus D$. Thus $3\,4$ is a frame edge there, and x has an actual reflected frame-endpoint neighbor w.

First assume $A_x \subseteq Z$, and let Q be the whole component of x in R_D. Degree two excludes K_1, P_2. A whole P_4 must contain an original frame edge, whose two endpoints, together with x and its two distinct Z-parents, would require five vertices. Thus $Q \neq P_4$. A reflected C_{2j+1}, $j = 1, 2$, contains its unique frame edge and otherwise alternating D-frame edges, hence $j + 1 \leq 3$ frame vertices. The

two Z-parents and two frame-edge endpoints require four distinct frame vertices, excluding C_3, C_5. A whole C_7 is excluded by the required prohibition on a heptagon's contact with a frame endpoint in every actual reflected frame, using w. Only a whole P_3 centered at x or a whole C_4 remains.

If $A_x = \{p, z\}$, $p \in P_{\mathrm{fr}}, z \in Z$, the required contacted-vertex theorem gives the whole reflected path $p' - p - x - z$, where p' is the frame mate of p. Thus no selected pentagon vertex lies on a reflected odd cycle. No deficient T_i meets C.

Step 3. Parents shared with arbitrary outside choices. Hold $D \setminus V(C)$ fixed. Let y belong to this outside set and share a parent with $x \in C$. If $A_x \subseteq Z$, choose a maximum independent pair on C containing x, with all outside choices unchanged, giving canonical D'. Step 2 applies to x. The shared parent touches both x, y, so it cannot be a leaf of a whole P_3 centered at x. Therefore the whole component is

$$x - z - y - z' - x,$$

and $A_y = A_x = \{z, z'\} \subseteq Z$.

If $A_x = \{p, z\}$, the whole reflected P_4 excludes any outside selected neighbor at p or z. Thus the contacted pair meets no outside selected vertex.

For each two-Z pair Q on C, let $k_Q \in \{1, 2\}$ count its cycle occurrences and l_Q its occurrences among fixed outside choices. A maximum independent pair on C can contain the entire class: extend a singleton to a nonadjacent pair, or take the nonadjacent two-vertex class. In that actual reflected terminal each parent has at least $k_Q + l_Q$ neighbors, so

$$k_Q + l_Q \leq 2.$$

Every outside selected vertex touching either parent has both parents equal to Q, by the preceding argument.

Step 4. Whole-cycle retention preserves all deficient paths. Put $X = D \cup V(C)$. Each two-Z pair Q on C, its $k_Q + l_Q$ selected vertices, and its two parents form a whole component

$$K_{k_Q + l_Q, 2}$$

of the original X-to-Z graph. Its left size is one or two, so it is a right-ended P_3 or a balanced C_4, and saturates its left side. If a contacted vertex exists, its X-to-Z component is the single edge to its private Z-parent. The parent partition excludes other cycle neighbors there, and Step 3 excludes outside selected neighbors.

These components exhaust the parents of added cycle vertices. Their restrictions to B_D have left sizes zero, one, or two against the corresponding pair, or are restrictions of the contacted edge; all saturate their D-sides. None contains a deficient path. Outside them, the graphs before and after adding C are identical. Hence the entire deficient family is still the same physical $T_1, \ldots, T_{q_4}$.

The loads also hold. Parents on the cycle pairs have load at most two; the contacted Z-parent has load one. Other isolates retain their canonical load at most two. A frame endpoint without a cycle contact retains load at most one. The possible contacted endpoint has its one cycle neighbor and no outside selected neighbor, so also has load one.

Step 5. Arbitrary repair and exact sequential maxima. Choose one d_i from every unchanged deficient path. The paths are disjoint and avoid C, so $|A| = q_4$, $|D_0| = \beta$, and $|Y| = \beta + 3$. Proposition 8.13(III) repairs precisely these paths for every choice. Loads persist, and the Y-to-Z graph saturates Y.

Since C is whole and D is independent, $R[Y]$ is C plus isolates. Its matching number is two, attained by two disjoint cycle edges and bounded by its five vertices. Proposition 8.13(I),(II) constructs the same-graph sequential maxima with the stated ranks and residual. It gives

$$N \geq \nu(R[Y]) + 1 = 3.$$

The exact size gives

$$N = v - m - \beta - 3 = c_2(F) - 3.$$

The first restored total is $v - m = c_1(F)$, and the second is $N + 2 = c_2(F) - 1$.

Step 6. Full certificate and contradiction. For any capacity split totaling two, put the first A_c matchings into one degree part and the rest into the other. Order the omitted vertices and assign every edge outside K to its earliest omitted endpoint. This gives N labeled stars, including every edge with two omitted endpoints exactly once.

For budgets totaling $N + 2$, at least one is at least two: otherwise the total is at most two, contrary to $N \geq 3$. If $x_c \geq 2$, put all five cycle edges and $x_c - 2$ stars into the first paid part, and the remaining y_c stars into the second. The number selected lies between zero and N. Matching restriction gives bounds $2 + (x_c - 2) = x_c$ and y_c. If the second budget is used, exchange colors. Shared vertices cannot increase these bounds. Every original edge is assigned once.

Thus every required split succeeds at the nonnegative total

$$N + 2 = c_2(F) - 1.$$

The definition of $c_2(F)$ gives $c_2(F) \leq c_2(F) - 1$, a contradiction. All reflections, repairs, retained edges, and certificates lie in the same arbitrary source. □

Proposition 8.15 (Exclusion of whole saturated heptagons)**.** *Assume the same hypotheses, frame notation, canonical choices, and terminal types as in Proposition 8.14. Let $T_1, \dots, T_{q_4}$ be the whole D-ended path components of the original D-to-Z graph, including isolates.*

If C_{old} were a whole heptagon of R, then for every canonical D:

(I) *No T_i meets C_{old}. For $X = D \cup V(C_{\mathrm{old}})$, the entire deficient X-ended path family is exactly the same physical $T_1, \dots, T_{q_4}$. Every P_{fr}-vertex has at most one X-neighbor, and every Z-vertex at most two.*

(II) *Choose any $d_i \in D \cap V(T_i)$, independently, and put*

$$A = \{d_i : 1 \leq i \leq q_4\}, \quad D_0 = D \setminus A, \quad Y = D_0 \cup V(C_{\mathrm{old}}),$$
$$K = F[W_{\mathrm{fr}} \cup Y], \qquad N = |V(F) \setminus V(K)|.$$

Then $|D_0| = \beta$, $|Y| = \beta + 4$, and the original Y-to-Z graph saturates Y. There are two edge-disjoint sequential maximum matchings M_0, M_1, with $L_{\mathrm{fr}} \subseteq M_0$,

$$|M_0| = m + \beta + 4, \qquad |M_1| = \beta + 4,$$

and

$$E(K - M_0 - M_1) = E(C_{\mathrm{old}}).$$

All other retained vertices are isolated. Moreover,

$$N = c_2(F) - 4, \qquad N \geq 4, \qquad N + \nu(K - M_0) = c_1(F),$$
$$N + \nu(K - M_0 - M_1) = N + 3 = c_2(F) - 1.$$

For the same pair and every capacity split totaling two and budget split totaling $N + 3$, the entire original F has the defining four-part certificate. Every omitted-incident edge, including two omitted endpoints, is assigned exactly once.

Consequently no supplied saturated frame has a whole heptagon terminal component. This excludes the entire saturated-heptagon subclass without an extremal-frame, order, excess-ceiling, or parent-type assumption. It asserts no saturation in an unsupplied generic source, whole quartic exclusion, or Ramsey formula.

Proof. The frame inputs are identified in Section 3.7, and the heptagon contact inputs in Section 3.8.

Step 1. Every auxiliary reflection is covered. The normalization gives well-coveredness, uniform b, m, $m \geq 1$, $0 \leq b \leq m$, and all complement identities for the canonical theorem. Its lower bound $a \geq 3E - 3 + \sigma \geq 12$ excludes K_5. The canonical theorem therefore gives the frame matching, actual saturated reflected terminals

$$R_D = F[W_{\mathrm{fr}} \cup D],$$

independence of D, and degree two at each D-vertex. Every designated odd cycle or positive even path contains exactly one original frame edge, and these components exhaust all m frame edges.

The required terminal restriction applies to every saturated frame, original and reflected, giving exactly

$$K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7.$$

The neutral-selector and movable-selector results give the exact costs, $q_4, \beta \geq 1$, $|D| = \beta + q_4$, and exactly q_4 deficient paths in B_D, obtained by removing frame endpoints from reflected odd cycles.

Assume a whole heptagon C_{old} exists.

Step 2. Test each cycle vertex by an actual reflection. For $x \in V(C_{\text{old}})$, put $J_x = N_F(x) \cap W_{\text{fr}}$. The required heptagon parent theorem gives two parents in Z, equal-or-disjoint pairs, and equality classes of size one or two nonadjacent vertices. In particular, no P_{fr}-vertex touches the heptagon.

Fix arbitrary canonical D, and hold $D_{\text{out}} = D \setminus V(C_{\text{old}})$ fixed. For a prescribed x, label the cycle cyclically $0, \ldots, 6$ with $x = 0$, and choose the independent triple $\{0, 2, 4\}$, together with D_{out}, to obtain canonical D'. The remaining vertices $1, 3, 5, 6$ induce exactly the edge $5\,6$. In the actual reflected frame, this is a frame edge, and its endpoint 6 touches x.

Let Q_x be the whole component containing x in $R_{D'}$. Degree two excludes K_1, P_2. A designated P_4 would require the two endpoints of its original frame edge, the two distinct Z-parents of x, and x, five vertices. Thus it is not P_4. A reflected C_3 or C_5 has respectively two or three frame vertices, from its unique frame edge and otherwise alternating edges. The two Z-parents and the two frame endpoints require four distinct frame vertices, excluding both cycles. A whole C_7 is impossible because its vertex x touches the reflected frame endpoint 6, contrary to the required contact prohibition in that reflected frame. The only remaining possibilities are a whole P_3 centered at x or a whole C_4.

This test is available for every prescribed x while all outside choices stay fixed. It does not assert that every selected heptagon vertex touches the residual frame edge under one common triple.

Step 3. Shared outside parents and all loads. Suppose $y \in D_{\text{out}}$ shares a parent $z \in J_x$. Use the actual D' from Step 2 for this x, with the same outside choices. The parent touches both selected vertices, so it is not a leaf of a whole P_3 centered at x. Thus Q_x is a whole C_4,

$$x - z - y - z' - x.$$

It contains no old frame edge, and consequently

$$J_y = J_x = \{z, z'\} \subseteq Z.$$

Here y has exactly two frame neighbors because it has original terminal degree two.

For a distinct parent pair Q on the heptagon, let $k_Q \in \{1, 2\}$ count its heptagon occurrences and l_Q its occurrences among the fixed outside choices. The entire cycle class fits in a maximum independent triple. For a singleton, take an alternating triple through it. For two nonadjacent vertices, their closed neighborhoods cover at most six of the seven vertices; add a vertex outside that union. With outside choices fixed, the resulting actual reflected terminal has maximum degree two, so

$$k_Q + l_Q \leq 2.$$

Every outside selected vertex touching either parent has that entire pair, by the preceding argument.

Step 4. Whole-cycle retention preserves precisely the old deficient paths. Put $X = D \cup V(C_{\text{old}})$, and let Z_C be the union of all cycle parent pairs. Distinct pairs are disjoint. Each pair Q, its $k_Q + l_Q$ left vertices, and its two parents form a whole bipartite component

$$K_{k_Q + l_Q, 2}.$$

There are no further edges in it: all selected neighbors of the parents have been counted, and each selected vertex has both frame neighbors in that pair. Its left size is one or two, so it is a right-ended P_3 or balanced C_4, saturating its left side.

Restricting to B_D gives left size at most two against two parents, with isolated Z-vertices if the left side is empty. These restrictions also saturate their selected sides. None is deficient. Outside these components, B_X and B_D are identical, since all added vertices lie on the heptagon and use parents in Z_C. Therefore all old deficient paths remain whole and unchanged, no new one appears, and none meets the heptagon.

Frame endpoints have no cycle neighbor, so retain their canonical load at most one. Each parent in Z_C has load $k_Q + l_Q \le 2$; every other isolate has unchanged load at most two. This proves (I).

Step 5. Every repair and the actual ordered maxima. Choose one d_i from each unchanged deficient path. Their union avoids the heptagon and has size q_4. Thus $|D_0| = \beta$. The original D used three heptagon vertices, none removed, so $|Y| = \beta + 4$.

Proposition 8.13(III) repairs every such simultaneous choice. The loads persist and the actual Y-to-Z graph saturates Y. Parts (I),(II) of that proposition construct two sequential maxima in this K, with $L_{\mathrm{fr}} \subseteq M_0$, ranks $m + |Y|, |Y|$, and residual exactly $R[Y]$.

The heptagon is whole and D is independent, so $R[Y]$ is that heptagon plus isolates. Its rank is three, attained by alternating edges and bounded by its seven vertices. Hence

$$N = v - m - |Y| = c_2(F) - 4, \qquad N \ge \nu(R[Y]) + 1 = 4.$$

The same proposition gives the first restored total $c_1(F)$, while the final total is $N + 3 = c_2(F) - 1$.

Step 6. Full restoration and contradiction. For capacities totaling two, assign the first A_c matchings to the first degree part and the remaining B_c to the second. Their degrees are bounded by the numbers of matchings. Order the omitted vertices and assign each edge outside K to its earliest omitted endpoint. This partitions all such edges into N labeled stars, including every omitted-to-omitted edge exactly once.

Fix budgets $x_c + y_c = N + 3$. Since $N \ge 4$, the total is at least seven, so one budget is at least three. If $x_c \ge 3$, put all heptagon edges and $x_c - 3$ stars into the first paid part, and the remaining y_c stars into the second. The number chosen is an integer between zero and N. Restricting matchings to the heptagon and individual stars gives bounds $3 + (x_c - 3) = x_c$ and y_c. If $x_c < 3$, use the second part for the cycle. Shared vertices do not invalidate these bounds.

Every original edge is assigned once. Thus every capacity and budget split succeeds at the nonnegative total

$$N + 3 = c_2(F) - 1.$$

The definition of $c_2(F)$ gives $c_2(F) \le c_2(F) - 1$, a contradiction. Therefore no whole heptagon occurs in any supplied saturated frame. □

9. Use of the structural results

The assembly separates the inputs needed for a degree exclusion from the exclusion itself. Proposition 8.1 describes entire source classes and retains the sufficient five-degree reduction. Its necessary domains do not prove that any surviving alternative is realized, and do not prove that the corresponding class is empty.

The matching-prefix propositions concern one original source and one retained induced graph at a time. Exact equalities at the initial costs force successive matching ranks, preserve fixed owner edges, and account for every omitted-incident edge. A residual matching-number identity alone does not certify a source cost: the required compatible matchings and the restoration step must also be present.

The quartic comparison contains the positive term q_4. Retaining it is necessary for the exact second-cost accounting. The movable-selector criterion then identifies the actual deficient paths for

the stated two-matching template. The pentagon and heptagon arguments preserve those paths during whole-cycle retention and repair them independently, giving the strict index-two contradiction under their supplied-frame hypotheses.

The intended use in the main paper is to combine these structural statements with five uniform source exclusions and then apply Theorem A. The missing technical developments and the series references must be supplied before this draft is a complete verified proof of that structural input. No claim of literature priority is made for the assembled statements.

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